

Textbook of Computable General Equilibrium Modelling

Programming and Simulations

Nobuhiro Hosoe,
Kenji Gasawa and
Hideo Hashimoto



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Abbreviations

Abbreviations of general terms

AGE	applied general equilibrium
ASEAN	Association of Southeast Asian Nations
CES	constant elasticity of substitution
CET	constant elasticity of transformation
CGE	computable general equilibrium
CSV	comma-separated value
EV	equivalent variation
GAMS	General Algebraic Modeling System
GAMS IDE	GAMS Integrated Development Environment
GATT	General Agreement on Tariffs and Trade
GDP	gross domestic product
GDx	GAMS Data Exchange
GTAP	Global Trade Analysis Project
IFPRI	International Food Policy Research Institute
IO	input-output
IRTS	increasing returns to scale
MDGs	millennium development goals
MFA	Multi-Fibre Arrangement
ROW	rest of the world
SAM	social accounting matrix
TOT	terms of trade
WTO	World Trade Organization

Abbreviations in social accounting matrices and GAMS input and output files

AGR	agriculture
BRD	bread
CAP	capital
EXT	external
EPS	epsilon, i.e., a small value close to zero
EQU	equation
GOV	government
HOH	household

IDT	indirect tax
INF	infinity
INV	investment
JPN	Japan
LAB	labour
MAN	manufacturing
MLK	milk
SRV	service
TRF	tariff
USA	the United States
VAR	variable

Symbols in CGE Models

Symbols indicating indices

i, j :	goods and firms (BRD, MLK),
h, k :	factors (CAP, LAB),
l :	households (HOH1, HOH2),
r, rr :	regions (JPN, USA).

Symbols in Chapter 2

[Endogenous variables]

UU :	utility,
X_i :	consumption of the i -th good,
Z_j :	output of the j -th firm,
$F_{h,j}$:	the h -th factor used by the j -th firm,
p_i^x :	demand price of the i -th good,
p_j^z :	supply price of the j -th good,
p_h^f :	price of the h -th factor.

[Exogenous variables and constants]

FF_h :	endowments of the h -th factor for the household,
α_i :	share coefficient for the i -th good consumption in the utility function,
$\beta_{h,j}$:	share coefficient for the h -th factor used by the j -th firm in the production function,
b_j :	scaling coefficient in the production function.

Symbols in Chapter 6

[Endogenous variables]

UU :	utility,
Y_j :	composite factor, produced in the first stage and used in the second stage by the j -th firm,
$F_{h,j}$:	the h -th factor used by the j -th firm in the first stage,
$X_{i,j}$:	intermediate input of the i -th good used by the j -th firm,
Z_j :	gross domestic output of the j -th firm,
X_i^p :	household consumption of the i -th good,

X_i^g :	government consumption of the i-th good,
X_i^v :	demand for the i-th investment good,
E_i :	exports of the i-th good,
M_i :	imports of the i-th good,
Q_i :	the i-th Armington composite good,
D_i :	the i-th domestic good,
p_j^y :	price of the j-th composite factor,
p_h^f :	price of the h-th factor,
p_j^z :	price of the j-th gross domestic output,
p_i^e :	price of the i-th exported good,
p_i^m :	price of the i-th imported good,
p_i^q :	price of the i-th composite good,
p_i^d :	price of the i-th domestic good,
ε :	foreign exchange rate (domestic currency/foreign currency),
S^p :	household savings,
S^g :	government savings,
T^d :	direct tax,
T_j^z :	production tax on the j-th good,
T_i^m :	import tariff on the i-th good.

[Exogenous variables and constants]

FF_h :	endowments of the h-th factor for the household,
p_i^{We} :	price of the i-th exported good in terms of foreign currency,
p_i^{Wm} :	price of the i-th imported good in terms of foreign currency,
S^f :	current account deficits in foreign currency terms (or equivalently foreign savings),
τ^d :	direct tax rate,
τ_j^z :	production tax rate on the j-th good,
τ_i^m :	import tariff rate on the i-th good,
$ax_{i,j}$:	input requirement coefficient of the i-th intermediate input for a unit output of the j-th good,
ay_j :	input requirement coefficient of the j-th composite good for a unit output of the j-th good,
α_i :	share coefficient for the i-th good consumption in the utility function,
$\beta_{h,j}$:	share coefficient for the h-th factor used by the j-th firm in the composite factor production function,
b_j :	scaling coefficient in the j-th composite factor production function,
μ_i :	share of the i-th good in government expenditure,

λ_i :	expenditure share of the i-th good in total investment,
ss^p :	average propensity for savings by the household,
ss^g :	average propensity for savings by the government,
γ_i :	scaling coefficient in the i-th Armington composite good production function,
$\delta m_i, \delta d_i$:	input share coefficients in the i-th Armington composite good production function,
η_i :	parameter defined by the elasticity of substitution,
σ_i :	elasticity of substitution in the i-th Armington composite good production function,
θ_i :	scaling coefficient in the i-th good transformation function,
$\xi e_i, \xi d_i$:	share coefficients in the i-th good transformation function,
ϕ_i :	parameter defined by the elasticity of transformation,
ψ_i :	elasticity of transformation in the i-th good transformation function.

Symbols in Chapter 10 (selective)

SW :	social welfare,
RT_i :	monopoly or quota rents in the i-th sector,
χ_i :	quota rent rate in the i-th sector,
M_i^{quota} :	quota ceiling on the i-th good imports,
FC_j :	fixed production costs of the j-th firm,
ν_j :	the share of the fixed cost payment to the household in total capital service payment by the j-th firm.

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Preface

Computable general equilibrium (CGE) models are well grounded in standard microeconomic theory, where price is an important signal that drives agents in an economy. The modern economy cannot be examined without considering the role of the price mechanism. CGE models featuring the price mechanism are suitable for analysing contemporary policy issues in market economies and thus are used widely in various policy analyses, such as economic integration, global warming problems, tax reform, assistance for developing countries and so on. The potential of CGE models has encouraged many researchers and practitioners to cooperate in the development and use of these models; however, many of them have had difficulty in fully understanding CGE models, because of the complexity of the underlying economic theory, estimation techniques and computational requirements – this is why CGE models are sometimes called a ‘black box’.

Despite the usefulness of CGE models, they have been used only infrequently in Japan. This is partly because, although many useful books and articles on CGE modelling are available in English, there were no comprehensive textbooks in Japanese, particularly for newcomers to CGE modelling. Therefore, we decided to write this textbook on CGE modelling in Japanese several years ago. Our endeavour resulted in the publication of *Textbook of Computable General Equilibrium Modeling – Programming and Simulations* by the University of Tokyo Press in 2004.

In the Japanese version of the textbook, we made considerable effort to help beginners to understand and build CGE models by themselves. The textbook was designed to be self-inclusive, proceeding in a step-by-step manner. We covered the A-to-Z of CGE modelling, starting from a basic setup of economic agents’ behaviour, construction of databases, estimation of model parameters, computer programs and interpretation of simulation results. In each step, sample models with computer programs were presented.

The computer programs were written using the GAMS (General Algebraic Modeling System) software, whose trial version is publicly accessible on the GAMS web site free of charge. Assuming that most readers of this book were completely new to GAMS, we presented a detailed explanation of its use in the book. It was intended that, as readers become familiar with GAMS, they would be able to use the sample models on

the GAMS web site and to become acquainted with CGE models by experimenting with these sample models.

After the publication of our textbook in Japanese, we received requests from both international students studying in Japan and researchers in other countries to publish an English version. These requests motivated us to publish this book. In this English version, we retain the features of the Japanese version outlined above. At the same time, the English version is not a mere English translation. On the one hand, considerable effort has been made to improve the clarity of the explanation and to include recent changes to GAMS and data for CGE modelling. On the other hand, we omitted several sections that were specifically written for Japanese readers; in particular, actual examples of CGE analyses.

Partly because this textbook is written for beginners, and partly because of space limitations, this textbook covers only the basic features of CGE modelling. For example, we have not discussed dynamics or the monetary economy, which are discussed in other advanced books and journal articles about CGE models. After reading through this textbook, readers can start to explore such frontiers of CGE modelling. While we are confident that this book provides readers with all the essential knowledge and techniques of CGE modelling, we would appreciate any comments and suggestions that may improve this book.

We gratefully acknowledge useful comments and suggestions by Professors Kanemi Ban and Tatsuo Hatta, and by many friends, as well as readers of the Japanese version. In the publication of the Japanese version, we acknowledged Messrs Takuya Kuroda and Toshihiro Ikeda at the University of Tokyo Press for their assistance. We thank the University of Tokyo Press for providing permission to publish this English version. Regarding the software used for the modelling discussed in this book, we are indebted to Mr Alex Meeraus and his associates in GAMS Development Corporation for their generous support of our use of GAMS in both the Japanese and English versions of our book. The use of GAMS throughout this book allows us to provide a practical introduction to CGE modelling. We gratefully acknowledge the financial support of a Grant-in-Aid for Scientific Research (No. 20330053, 21730222) by the Ministry of Education, Culture, Sports, Science and Technology of Japan and the Japan Society for the Promotion of Science. We also acknowledge MIT Press and Elsevier for granting us permission to reprint certain figures originally contained in their publications, and Microsoft Corporation and GAMS Development Corporation for granting us permission to use screen shots of their products.

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Overview

1.1 Economic analysis with computable general equilibrium models

Efficiency of resource allocation has become increasingly important in recent years. In the 1960s and 1970s, the Japanese economy grew so rapidly that efficiency of resource allocation was a secondary issue; however, this has changed. Growth in Japan slowed from the 1980s onward to the more moderate speeds experienced in other developed economies. Most developing economies cannot expect large aid inflows and commercial financing from developed economies any more. The focus of economists has returned to an important issue in economics; namely, efficiency of scarce resource allocation.

Resource allocation can be viewed as a trade-off issue. Output can be either exported or set aside for domestic supply. Exports earn hard currency to support imports. Imports and domestic supply can be either consumed by households or used as intermediate inputs. Household consumption determines society's economic welfare, while intermediate inputs contribute to a further increase of output. As an economy has a number of agents, such as households, firms and a government, as well as various industrial sectors, it is difficult to solve these trade-off problems or to find an efficient allocation of resources among those agents and sectors under given resource and technology constraints.

The price mechanism is a powerful device capable of solving the complicated trade-off problems stated above. Economic agents, such as households and firms, make their decisions about their economic activities using price information prevailing in the markets. Many economic theories involve optimization behaviour of economic agents under given resource and technology constraints under signalling from market prices. Households maximize their utility subject to their budget constraints,

and firms maximize their profits subject to their production technology constraints. Solutions of these optimization problems yield the demand and supply schedules respectively. Markets equilibrate demand and supply by adjusting prices. The prices of goods with excess supply fall, and those of goods with excess demand rise, as occurs in markets such as antique auctions and daily fish markets. This is the price mechanism. Computable general equilibrium (CGE) models can depict such market economies in a quantitative manner. Incidentally, CGE models are also sometimes called applied general equilibrium (AGE) models.

The efficiency of resource allocation is not a purely theoretical issue. Hardly a day goes by when we do not hear about 'structural reforms' in Japan. Examples of public utility reforms are deregulation of telecommunications, electric power, city gas and water supply. While the regulatory reforms are designed to improve the efficiency of resource allocation, support for the reforms among stakeholders is required. This can be done only by showing the effects of these reforms not only qualitatively but also quantitatively; that is, the economy-wide impact of the reforms and their effect on individual industrial sectors, income group, and so on. Similar arguments can be applied to so-called structural adjustment policies led by the World Bank and others in developing countries. Suppose you are an economic adviser for a developing country. Even though it is trivial to show qualitatively that import tariff reduction leads to better resource allocation, it cannot simply be recommended to the organisations responsible for the country's macroeconomic management. You have to provide concrete advice about tariff reduction in percentage terms to achieve certain policy goals such as economic welfare and GDP. At the same time, you are required to indicate its impact on various aspects of the economy numerically. The CGE models are useful for quantitative analysis and evaluation of these economic policies.

The general equilibrium theory of the competitive market economy was originated by Léon Walras. His theory was further extended to proofs of existence and stability of the equilibrium by Kenneth Arrow and Gérard Debreu.¹ These studies are of a general, abstract and rigorous nature and do not include numerical analysis. In contrast, CGE models are designed to establish a numerical framework for empirical analysis and evaluation of economic policies. This is why they are called *computable* general equilibrium models.

The pioneering CGE model was the Norwegian multisectoral growth model developed by Johansen (1960). Since then, many CGE models have been developed to analyse, for example, development issues by Dervis et al. (1982) and taxation and international trade issues by Shoven

and Whalley (1992). A model of the Australian economy, known as ORANI with its variants, was built by Dixon et al. (1992). CGE modelling has become popular because of increasing needs for analysis of policies related to resource-allocation issues. At the same time, it must be noted that CGE modelling has been made possible by the algorithm introduced by Scarf (1973), in combination with rapid improvements in computer technology. Modern computers have enabled the development and solution of large-scale world trade CGE models that can analyse the impact of a series of General Agreement on Tariffs and Trade (GATT) and World Trade Organization (WTO) negotiations. The worldwide input-output (IO) tables and trade database prepared by the Global Trade Analysis Project (GTAP) at Purdue University provide the data for these analyses.²

1.2 Framework of CGE models

CGE models can depict numerically a ‘world’, where a general equilibrium is attained by the price mechanism, as stated above. A simple economy with one household and one firm is represented in Figure 1.1. In this economy, the representative household determines its consumption bundle to maximize its utility subject to its budget constraint. The firm is supposed to maximize its profits by managing its inputs and outputs subject to its production technology. This optimization behaviour yields the demand and supply curves of goods and factors of production, which are equilibrated in the markets by flexible price adjustments.

In this book, we first develop a ‘simple CGE model’ with two goods, two factors, one household and two firms. As it is necessary to include more detail in the model to analyse the real-world economic issues empirically, we gradually develop a more realistic model; that is, a ‘standard CGE model’, with a government, an external sector, investment and savings, and intermediate inputs. Once an operational knowledge of the standard CGE model is obtained, most standard economic problems can be analysed.

Both the simple CGE model and the standard CGE model primarily rely on the basic assumptions of standard microeconomics as their foundations. These models assume one representative household, which consumes goods, and two representative firms, each of which produces a certain good. The household is supposed to maximize its utility subject to its budget constraint, while the firms maximize their profits subject to given constraints on production technology. The household and all the firms are assumed to be price takers. In other words, the markets are perfectly competitive. However, in contrast to household consumption and

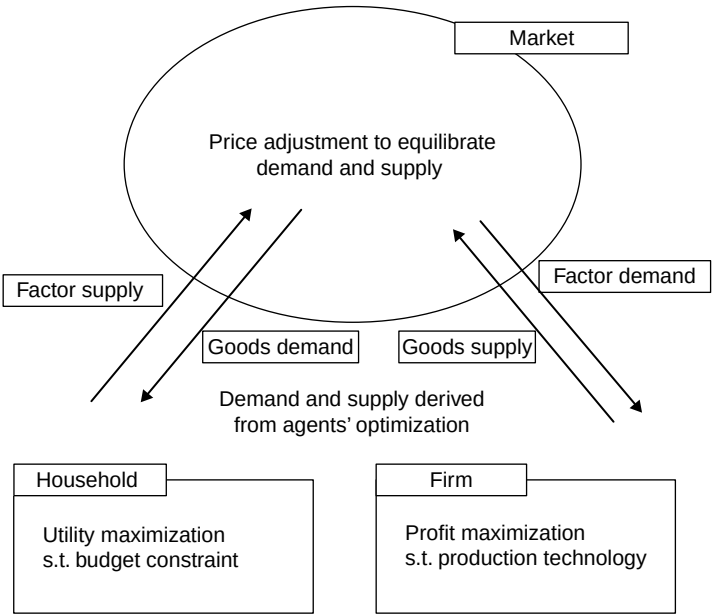


Figure 1.1 Structure of an economy

firm production, there are elements that lack microfoundations, including the behaviour of investment, savings and a government added in the standard CGE model. The problems associated with introducing investment, savings and a government into CGE models are discussed in detail, as are possible solutions.

To incorporate more complicated economic phenomena in CGE models, some of the basic assumptions stated above are replaced in the latter part of this book. First, heterogeneous households are introduced to investigate the changes in income distribution caused by certain policies or extraneous shocks. Modelling multihouseholds provides a basis for a model extension toward multiple-country models. Second, the basic assumptions regarding market structure and production technology are replaced by imperfect competition and by increasing-returns-to-scale technology. These extended CGE models are explained there.

All the CGE models dealt with in this book – simple, standard or extended – follow the Walrasian tradition. In other words, we present each of our CGE models as a system of simultaneous equations, which are derived from the agents' optimization behaviour. By Walras's law, one of the equations, against the same number of endogenous variables,

is redundant; thus, (absolute) prices cannot be determined. All prices are expressed relative to a chosen numeraire.

1.3 Advantages and shortcomings of CGE models

A major advantage of CGE models is the relatively small data requirements considering the model size. This has enabled wider use of CGE analysis. As explained below in detail, most CGE models are developed with macroeconomic data, such as IO tables, national accounts tables and trade statistics, for *one year*. This advantage makes CGE models preferable to standard econometric models, which require observations for several years to estimate parameters with sufficient degrees of freedom. This is particularly advantageous for developing economies, for which sufficient statistical data are not available or where the social and economic systems are susceptible to drastic changes such as a coup d'état. In addition, CGE models can easily incorporate dozens of industrial sectors, which would require a large dataset in econometric models.

The CGE models presented in this book have some shortcomings. First, as stated above, the parameters of CGE models are estimated on the basis of data from a single reference year. This means that the parameter estimates may be extremely sensitive to the choice of the reference year. In the case of an economy that suffers significant fluctuations, the reference-year data may not provide a good reference point for our empirical analysis.

Second, estimating the CGE models with a one-year dataset can be compared to taking a still picture of a dynamically evolving reality. That is, inclusion of dynamic components of an economy, such as investment and savings, in a static model based on a static estimation procedure is theoretically inconsistent. In consideration of this shortcoming, dynamic CGE models have been developed where intertemporal resource allocation, such as investment and savings, are established fully on the basis of microfoundations. These dynamic models are discussed in other articles.³ Our remedy for this shortcoming is discussed in terms of macro closure.

Third, financial/monetary aspects are rarely incorporated into CGE models. As stated above, following the original Walrasian general equilibrium model, most CGE models focus on the real side of the economy; thus, they can deal with economies in terms of only relative prices, not absolute prices.⁴ As a result, CGE models cannot deal with monetary phenomena such as inflation or (nominal) foreign exchange rate policy. To overcome this difficulty, a few so-called financial CGE models have been

developed; however, these models tend to be so large that they cannot be solved easily, and their simulation results are difficult to interpret. Such financial/monetary CGE models are also referred to in other articles and books.⁵

1.4 Applications of CGE models

Here, we present a list of typical research topics using CGE models, which will help readers to make their own research plans using CGE models.

- General macroeconomic issues: public expenditure cuts, impact of trade and tax reforms on income distribution
- Fiscal policy issues: introduction of value-added taxes, commodity tax reforms
- International trade policy: impact of WTO negotiations, Association of Southeast Asian Nations (ASEAN) free trade area formation, tariffication of rice import barriers
- Regional and transport policy: impact of high-speed rail and highway investment
- Environmental policy: introduction of eco-taxes, implementing a tradable CO₂ emissions permit system
- Industrial and labour policies: deregulation of electric power industry, impact of foreign workforce inflows

Some of these topics may be of interest not only to economists but also to social scientists and even engineers.

1.5 Aims of this book

In this book, we aim to develop the CGE modelling skills of readers so that they can develop and operate CGE models by themselves. In this sense, the book is intended to be a 'how-to'-type textbook. By following the book, readers will learn in a step-by-step manner how to develop CGE models, to compute these models with a PC and to interpret simulation results. The focus on the practical skills and techniques of CGE modelling is a unique feature of this book compared with others.

To accomplish our aim, we develop CGE models by explaining:

- (1) How to collect and compile data for construction of CGE models
- (2) How to estimate the parameters and coefficients of the models
- (3) How to prepare computer programs for numerical simulations

- (4) How to conduct policy simulations with the models
- (5) How to interpret the simulation results

In particular, for (2), the estimation methods of the model parameters and coefficients will be explained in great detail, because we employ a unique technique called 'calibration'. As for (3), readers are shown how to use the GAMS (General Algebraic Modeling System) software. The reason that we use GAMS will be explained in detail in the next section. As for (5), to interpret the simulation results, it is important not only to rely on numerical results but also to use intuitive economic interpretations. The simulation results of large-scale CGE models with many agents and sectors are sometimes difficult to interpret. Unless we have a good understanding of the structure, the data, the simulation scenarios and the behaviour of CGE models, there is often a tendency to be bogged down in a 'black box'. To avoid such a case, interpretation of the simulation results will be discussed in detail.

Our potential readers are presumed to be senior undergraduates (in economics) and/or junior graduates as well as experts who are actually engaged in economic analysis. Furthermore, CGE models have been increasingly applied to fields beyond pure economics, such as urban and transportation planning, and environmental analysis, as discussed in the previous section. Considering that all potential readers have not necessarily had training in economics, we will move slowly in this book – starting from the simple CGE model and moving to the standard CGE model and then to some extended models. Detailed computer input and output of many sample problems will be shown.

Knowledge in the following three areas will help readers to understand this book fully. The first is basic economics: micro- and macroeconomic theory at the undergraduate level.⁶ Some understanding of general equilibrium theory, national accounts statistics and IO tables is of great help.⁷ The second is mathematical skills, in particular partial derivatives of simple functions. Knowledge of the Lagrange multiplier method is recommended.⁸ The third is general PC skills, particularly the handling of files and the use of spreadsheet software.

We expect two types of readers for this book. The first type, beginners in CGE modelling, should read this book in order from the beginning to the end. The second type, those who have some knowledge and experience with CGE models, can use this book as a kind of dictionary by selecting the sections where they do not feel confident and studying these intensively. For the convenience of the latter type of readers, certain topics and suggestions are presented repeatedly.

1.6 Software and simulations on the PC

This section discusses the GAMS software and its graphical user interface GAMS IDE (Integrated Development Environment). To solve CGE models, which are formulated as a system of nonlinear simultaneous equations, software for numerical computations is needed. For such computations, we use GAMS throughout this book. GAMS was originally developed for analyses of developing economies by the World Bank, and now it is updated from time to time and distributed commercially by GAMS Development Corporation.⁹

The major merits of this software are: 1) its powerful algorithm of numerical calculation; 2) its algebraically expressed programming syntax; and 3) good program transparency with use of comments and notes. Thanks to the latter two merits, ordinary economists who are not experts in computer programming languages can easily write and read programs, and share them with their colleagues. Even though we use GAMS in the book exclusively, CGE modelling can be done with other software. At the same time, GAMS can be used not only for CGE models but also for other numerical computations.¹⁰

In developing and operating programs for GAMS on MS Windows, we use the GAMS IDE software, equipped with a programming editor.¹¹ GAMS IDE allows us to prepare input files, to run programs, and to show the results. That is, GAMS IDE supports all the processes for numerical computation.

While GAMS is commercially distributed by GAMS Development Corporation, its trial version is available on the GAMS web site free of charge, although there are some limitations in terms of the model size that the trial version can solve. All the sample models in this book are compiled in the GAMS Model Library, which can also be accessed on the web site free of charge.¹²

Assuming that most readers of this book are completely new to GAMS, we provide examples to illustrate the use of GAMS throughout the book, and in a particularly comprehensive manner in Chapter 3. After readers acquire an understanding of GAMS, they can download the GAMS system and the sample models, and can simulate various economic situations by changing coefficients and exogenous variables. In this manner, readers will quickly become not only accustomed to the use of GAMS but also acquainted with CGE models. Table 1.1 summarizes where the sample models appear in this book.

Table 1.1 List of CGE models developed in this book

Household utility maximization model		CGE model		Multihousehold Large-country model		Two-country model		Monopoly model		Import quota model		Increasing-returns-to-scale model	
Description of 2.2, 3.1		2.1–2.4		6.1–6.6		10.1		10.2		10.3.1		10.4.1	
the model												10.6	
Equation		3.1.1		2.5		6.7		10.1		–		–	
listing								–					
Social		–		4.1.1, 5.3.1, 4.1.2, 6.8		10.1		4.1.2, 6.8		10.3.1		4.1.2, 6.8	
accounting				(Tables 4.1 and 6.1)		(Table 10.1)		(Tables 4.2 and 6.1)		(Table 10.2)		(Tables 4.2 and 6.1)	
matrix													
Explanation of calibration		–		5.4.5		6.8.2		–		10.4.1		–	
Computer		3.2		5.4.1		6.8		10.2		10.3.1		10.6	
Input program		(List 3.1)		(List 5.1)		(List 6.1)		(List 10.1)		(List 10.2)		(List 10.4)	
filename		hmax.gms		splcge.gms		stdcge.gms		lrgcge.gms		twocge.gms		monocge.gms	
												quocge.gms	
												lrsge.gms	

1.7 Structure of this textbook

The subsequent chapters are structured as follows. Chapter 2 discusses a simple CGE model with two goods, two factors, one household and two firms. This CGE model consists of demand and supply functions, which are derived as the solution of the optimization behaviour of the household and the firms, and the market-clearing conditions. The model is formulated as a system of nonlinear simultaneous equations. Although this simple CGE model is small, it contains the essential components of fully fledged CGE models. Chapter 3 presents a very detailed explanation of the basic uses of GAMS for the CGE modelling, with an example of the household utility maximization problem, which is part of the simple CGE model. Its input and output files are shown to demonstrate the GAMS programming.

Chapter 4 covers the structure of a typical social accounting matrix (SAM) and its construction. A SAM records all the economic activities and flows of funds among agents in a base year, and it is used as a database for estimation of coefficients and exogenous variables of CGE models. Chapter 5 outlines techniques for estimating the coefficients and exogenous variables of the model, on the basis of a SAM. This technique is called 'calibration'. The input and output files of the simple CGE model, including the SAM construction and calibration processes, are presented. Chapter 6 extends the simple CGE model to the standard CGE model by incorporating intermediate inputs, a government (covering its expenditure and revenues from direct and indirect taxes), investment and savings, and international trade. A computer program for this standard CGE model is also presented. After finishing this chapter, readers are expected to be well capable of conducting most empirical CGE analysis by themselves.

Chapter 7 is devoted to the discussion of the issue of macro closure of CGE models. Although the CGE models presented in this book are all static models, often it is necessary to include dynamic aspects such as investment and savings for practical purposes. This causes potential conflicts between our model framework and its theoretical foundations. The macro closure rules provide some methods for coping with this problem. Chapter 8 provides techniques for efficient simulations to conduct comparative statics for policy analysis. CGE models are primarily used to compare and evaluate alternative policies. For this purpose, after developing and solving a CGE model to depict one equilibrium situation, we simulate equilibria under various alternative scenarios. Sensitivity analysis for CGE models will also be discussed. Chapter 9 is devoted to

demonstrating the interpretation of simulation results. As the model size increases, the model is likely to become a ‘black box’ even for the original author. We provide some hints for maintaining the transparency of CGE models.

In Chapter 10 we present, mainly for advanced CGE modellers, various extended CGE models. The extensions are of two types. The first is to introduce heterogeneous households into the CGE models. This allows more elaborate models of international trade analyses. The other is to relax our original assumptions of perfect competition and constant-returns-to-scale in production technology, which leads to monopoly/oligopoly and increasing-returns-to-scale technology models. In the concluding Chapter 11, we discuss some frontiers of CGE analyses and state our philosophy on CGE modelling that underlies this book.

Five appendices attached to the main text present technical discussions about the CGE models. The subsequent three annexes discuss how to use GAMS. More precisely, the annexes cover: (A) high-level uses of GAMS; (B) hints for debugging computer programs using the error messages in output files; and (C) information sources on the web (including the way to download GAMS and the sample models shown in this book).

As the book was initially published in Japanese for a Japanese audience, monetary figures in this book are still expressed in Japan’s currency unit, the yen. Needless to say, it can be replaced by any currency unit, say, the US dollar, the pound sterling, the euro or others, without any loss of logic. Furthermore, we use Japan’s IO tables to explain how to construct a SAM from IO tables. However, the same technique can apply to any country’s IO tables with minor adjustments.

Notes

1. Refer to Debreu (1959) for a discussion of general equilibrium theory.
2. For information about the GTAP database, see www.gtap.org.
3. For example, Ginsburgh and Keyzer (1997, Chs 6–7) and Harrison et al. (2000).
4. This is because of the homogeneity of degree zero in prices. Homogeneity is explained in detail in a footnote to Section 6.2.
5. See, for example, de Janvry et al. (1991) for a discussion of financial CGE models. Real business cycle models or dynamic stochastic general equilibrium models, where sectoral details are omitted by aggregation, may be more suitable for analysis encompassing asset markets under uncertainty. See McCandless (2008).
6. See, for example, Varian (2006) for microeconomics and Mankiw (2006) for macroeconomics.

7. National accounts summarize the activities – production, income distribution and expenditure – of agents such as households, firms and the central and local governments in an economy, in the form of flow of funds among them. IO tables within the national accounts are discussed in Chapter 4.
8. See, for example, Chiang and Wainwright (2005).
9. See www.gams.com. GAMS consists of the base module and solvers as discussed in Chapter 3. We employ the CONOPT solver, which can solve nonlinear programming problems, for our CGE models.
10. There is other software available for CGE modelling. For example, GEMPACK is the most popular software. Refer to www.monash.edu.au/policy/gempack.htm. MPSGE, another module of the GAMS system equipped with CGE modelling-focused syntax, is also widely employed. Even without such software specially prepared for CGE modelling, we can employ general computer languages, such as FORTRAN, for numerical computations.
11. GAMS IDE is included in the base module of GAMS. The use of GAMS IDE is explained in Section 3.2.
12. The GAMS Model Library provides many sample models, not only CGE models. Indices by model name, author, date and subject allow users to locate models of interest to them.

2

The Simple CGE Model

In this chapter, we develop the simplest possible CGE model, hereafter called the ‘simple CGE model’, with one household, two firms, two goods (bread and milk) and two factors (capital and labour).^{1,2} As all the CGE models used for empirical policy analysis are extensions of this simple CGE model, it is vitally important to understand how the model is built and what implications can be derived from the equilibrium of the model.

We outline the setup of the economy in Section 2.1 and explain the household utility maximization problem in Section 2.2, firms’ profit-maximization problems in Section 2.3 and the market-clearing conditions in Section 2.4. In Section 2.5, the whole model system is formulated as a system of simultaneous equations with 14 equations and the same number of endogenous variables.

Following this chapter, Chapter 3 shows how to write and read the GAMS computer program, by using an example of the household utility maximization, which is part of the simple CGE model. Chapter 4 describes the social accounting matrix (SAM) database used to construct the simple CGE model. Chapter 5 shows how to estimate the coefficients and constants of the model by means of the so-called ‘calibration’ method. Having read through Chapters 1 to 5, readers can develop a full understanding of the simple CGE model. Then, as an extension of the simple CGE model, Chapter 6 presents a ‘standard CGE model’, into which intermediate inputs, a government (including its expenditure, income tax, production tax and import tariffs), investment and savings, and international trade are incorporated. The computer input and output files of the standard CGE model are also shown.

2.1 Setup of the economy

The basic assumption of the simple CGE model is twofold. The first is a static economy in the sense that no time-related elements such as investment and savings are included. The second is a closed economy; that is, no international trade is included. Then, it is assumed that two goods are produced, bread and milk (denoted with indices i or j interchangeably) and that two factors exist, capital and labour (denoted with indices h or k interchangeably), in this economy. One representative household exists and consumes two kinds of goods to maximize its utility. There are two representative firms, each of which produces one commodity, either bread or milk. The household, endowed with two factors, provides them to the firms in return for income payments. The firms employ these factors in their production. Household and firm demand and supply of these goods and factors are equilibrated in the markets with flexible price adjustments. We assume that the markets are perfectly competitive. That is, no agent has any market power to control prices; in other words, all the agents are ‘price takers’.

Figure 2.1 summarizes the commodity flows in the model economy. (1) The goods Z_j are produced by the firms with factor inputs $F_{h,j}$. Then, (2) they are shipped to the goods markets, where they are sold to the household for their consumption X_j . (3) At the household, the goods are consumed and generate utility UU . The payments occur in the opposite direction. The factor income is generated by the firms and paid back to

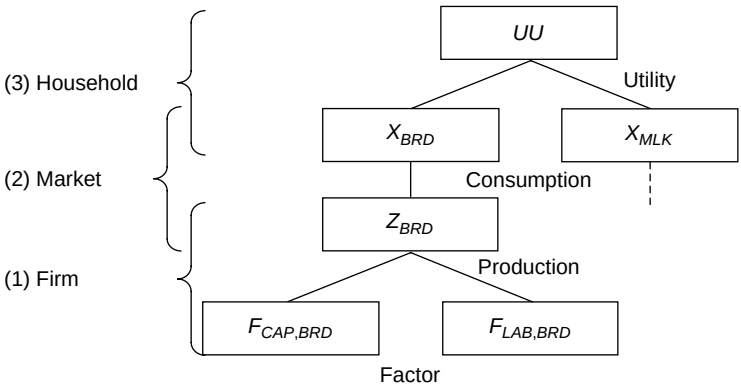


Figure 2.1 Model structure

the provider of the factors; i.e., the household. The household uses the received factor income to purchase the goods. In the following sections, household optimization behaviour, firm optimization behaviour and the market-clearing conditions are presented.

2.2 Household behaviour

The household aims to sell all of its endowed factors to the firms to earn income. (We do not assume self-consumption of factors such as consumption of leisure or voluntary unemployment.) To simplify the notation, capital and labour are abbreviated as CAP and LAB. The household expends its factor income on the consumption of bread and milk, which are also abbreviated as BRD and MLK. The household is assumed to choose the consumption of bread and milk to maximize its utility. It is assumed that the utility function is the Cobb–Douglas type. At this stage, prices of goods and factors are assumed to be given in the household utility maximization problem.

The household maximizes its utility subject to its budget constraint in the following manner:³

$$\underset{X_i}{\text{maximize}} \quad UU = \prod_i X_i^{\alpha_i} \quad (2.a)$$

subject to

$$\sum_i p_i^x X_i = \sum_h p_h^f FF_h \quad (2.b)$$

where:

i, j : goods (BRD, MLK),

h, k : factors (CAP, LAB),

UU : utility,

X_i : consumption of the i -th good ($X_i \geq 0$),

FF_h : endowments of the h -th factor for the household,

p_i^x : demand price of the i -th good ($p_i^x \geq 0$),

p_h^f : price of the h -th factor ($p_h^f \geq 0$),

α_i : share parameter in the utility function ($0 \leq \alpha_i \leq 1$, $\sum_i \alpha_i = 1$).

The first equation (2.a) is the utility function to be maximized. The second one (2.b) is the budget constraint to equilibrate total expenditure and total income.

To solve this problem analytically, let us employ the Lagrange multiplier method. With a Lagrange multiplier φ , the Lagrangian is defined as follows:

$$L(X_i; \varphi) \equiv \prod_i X_i^{\alpha_i} + \varphi \left(\sum_h p_h^f FF_h - \sum_i p_i^x X_i \right)$$

As this problem with a Cobb–Douglas specification usually has an interior solution, the first-order optimality conditions are:

$$\frac{\partial L}{\partial X_i} = \alpha_i \frac{\prod_j X_j^{\alpha_j}}{X_i} - \varphi p_i^x = 0 \quad \forall i \quad (2.c)$$

$$\frac{\partial L}{\partial \varphi} = \sum_h p_h^f FF_h - \sum_i p_i^x X_i = 0 \quad (2.d)$$

Eliminating the Lagrange multiplier φ by solving the system (2.c)–(2.d) for the demand X_i , we obtain the demand function for the i -th good:⁴

$$X_i = \frac{\alpha_i}{p_i^x} \sum_h p_h^f FF_h \quad \forall i \quad (2.1)$$

The derived demand function (2.1) implies that the demand for the i -th good X_i increases with a decline of its price p_i^x (i.e., a downward-sloping demand function) or with an increase of income $\sum_h p_h^f FF_h$.

2.3 Firm behaviour

This economy has two representative firms; one produces bread and the other milk. Each firm uses only capital and labour to produce bread or milk, and is supposed to maximize its profits subject to the given production technology.

The above-mentioned setup for firm behaviour may appear too simple to capture reality in the following two ways. First, in addition to capital and labour, the bread-producing firm might need to use intermediate inputs such as butter, salt and yeast. The same is also the case for the milk-producing firm. Such intermediate inputs in the production process are only used in Chapter 6 onward. Second, each of these firms is assumed to produce only one product with no by-products. That is, the bread firm produces only bread, and not milk as a by-product. Although by-products are important in the real world, they are not dealt with in this book.

Let us formulate the j -th firm's behaviour. It is assumed to maximize its profits subject to its production technology constraint under given output and input prices as follows:

$$\text{maximize } \pi_j = p_j^z Z_j - \sum_h p_h^f F_{h,j}$$

subject to

$$Z_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad (2.2)$$

where:

i, j : firm (BRD, MLK),

h, k : factor (CAP, LAB),

π_j : profit of the j -th firm,

Z_j : output of the j -th firm,

$F_{h,j}$: the h -th factor used by the j -th firm,

p_j^z : supply price of the j -th good,

p_h^f : price of the h -th factor,

$\beta_{h,j}$: share coefficient in the production function ($0 \leq \beta_{h,j} \leq 1$, $\sum_h \beta_{h,j} = 1$),

b_j : scaling coefficient in the production function.

The j -th firm determines the amounts of factor input $F_{h,j}$ and its output Z_j that maximize its profits. The first term on the right-hand side of the profit function represents its revenues accruing from sales of the j -th good. The second term corresponds to its payments for employed factors. The constraint (2.2) is the production function, representing its production technology, which links factor $F_{h,j}$ with output Z_j . We employ a Cobb–Douglas-type function. It should be noted that, at this stage, the quantity of factors used by the j -th firm is not constrained by their availability but solved endogenously.

To solve this optimization problem for the j -th firm, we again employ the Lagrange multiplier method with the Lagrange multiplier ω_j as follows:

$$L_j(Z_j, F_{h,j}; \omega_j) = \left(p_j^z Z_j - \sum_h p_h^f F_{h,j} \right) + \omega_j \left(b_j \prod_h F_{h,j}^{\beta_{h,j}} - Z_j \right)$$

Usually, this problem with a Cobb–Douglas specification has an interior solution. The first-order conditions for the j -th firm optimization

problem are:

$$\begin{aligned}\frac{\partial L_j}{\partial Z_j} &= p_j^z - \omega_j = 0 \quad \forall j \\ \frac{\partial L_j}{\partial F_{h,j}} &= -p_h^f + \omega_j \beta_{h,j} \frac{b_j \prod_k F_{k,j}^{\beta_{k,j}}}{F_{h,j}} = 0 \quad \forall h, j \\ \frac{\partial L_j}{\partial \omega_j} &= b_j \prod_h F_{h,j}^{\beta_{h,j}} - Z_j = 0 \quad \forall j\end{aligned}$$

Eliminating the Lagrange multiplier ω_j by solving this system for the demand for the h -th factor by the j -th firm $F_{h,j}$, we obtain the following demand function:

$$F_{h,j} = \frac{\beta_{h,j}}{p_h^f} p_j^z Z_j \quad \forall h, j \quad (2.3)$$

as well as the original production function (2.2).

The interpretation of this factor demand function is similar to that of the household demand function for goods in (2.3). Demand for the h -th factor input by the j -th firm increases when either the h -th factor price p_h^f falls, the supply price of the j -th good rises, or the production of the j -th good Z_j increases. The share parameter $\beta_{h,j}$ plays an important role in determining demand for the h -th factor input for production of the j -th good; in other words, the larger the parameter $\beta_{h,j}$ is, the more sensitive demand is to changes in the other variables.

2.4 Market-clearing conditions

Sections 2.2 and 2.3 explained how the household and firms determine their demand and supply of goods and factors as a result of their optimization behaviour. Those optimization problems are not dependent on other agents' decisions but only on the given good and factor prices. In other words, the optimization problems of the three agents (one household and two firms) have so far been solved separately. Therefore, there is no guarantee that the prices assumed by the household are the same as those assumed by the firms. More precisely, for the i -th good, the household assumes the demand price p_i^x , while the firms assume the supply price p_i^z , and these two prices are not necessarily the same in general. Furthermore, even if those prices are identical, supply is not necessarily equal to demand for each good and for each factor. In addition, the total demand for each factor by the two firms does not necessarily match its

endowments. In sum, to ensure the market equilibrium of each good and factor in terms of quantity and price, we need to impose the following market-clearing conditions:

$$X_i = Z_i \quad \forall i \quad (2.4)$$

$$\sum_j F_{h,j} = FF_h \quad \forall h \quad (2.5)$$

$$p_i^z = p_i^x \quad \forall i \quad (2.6)$$

Equation (2.4) is the market-clearing condition for the i -th good, which ensures equality of its demand and supply quantities. As discussed in Section 2.1, one representative firm produces only one good, without any by-product. Therefore, the suffix j in Z_j , which stands for the j -th firm as in Section 2.3, also represents its product; namely, the i -th good without any confusion. Consequently, the suffix i for Z_i , instead of Z_j , is used in Equation (2.4).

Equation (2.5) is the market-clearing condition for factors indicating that the total demand for each factor must be equal to its given endowments. The left-hand side of (2.5) represents the sum of demand quantities for the h -th factor by both firms. The right-hand side stands for the total endowments of each factor, assumed to be given in the economy.

The last equation (2.6) is the market-clearing condition that equates the firm's supply price of the i -th good p_i^z to the corresponding demand price for the household p_i^x . In this chapter, no indirect tax is assumed; thus, there should be no gap between the supply and demand prices. In contrast to these goods' prices, the same variable p_h^f is used for both the supply and demand prices of the factors; thus, the equality condition (2.6) is not needed for factor prices.

2.5 Model system

The demand and supply equations of goods and factors, and the market-clearing conditions discussed above, yield the following system of simultaneous equations (2.1)–(2.6):

$$X_i = \frac{\alpha_i}{p_i^x} \sum_h p_h^f FF_h \quad \forall i \quad (2.1)$$

$$Z_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j \quad (2.2)$$

$$F_{h,j} = \frac{\beta_{h,j}}{p_h^f} p_j^z Z_j \quad \forall h, j \quad (2.3)$$

$$X_i = Z_i \quad \forall i \quad (2.4)$$

$$\sum_j F_{h,j} = F F_h \quad \forall h \quad (2.5)$$

$$p_i^z = p_i^x \quad \forall i \quad (2.6)$$

In this system, (2.1) is the demand function for the i -th good derived from the household utility maximization problem; (2.2) is the production function of the j -th good, set in the j -th firm profit-maximization problem; (2.3) is the demand function for the h -th factor by the j -th firm, derived from the j -th firm profit-maximization problem; (2.4)–(2.6) are the market-clearing conditions.

Solving the above system of simultaneous equations (2.1)–(2.6), we obtain a general equilibrium of this economy.⁵ This system consists of 6 sets/14 equations (generally speaking $4i + h \cdot j + h$ equations) and the same number of endogenous variables.⁶ It is notable that this system is homogeneous of degree zero in prices. As Walras's law always holds (even when some of the market-clearing conditions are not satisfied), one of the equations in the system is redundant.⁷ Thus, we must choose one good or one factor as a numeraire and fix its price. Then, all other prices are expressed as relative prices in terms of the numeraire. Therefore, it should be noted that we cannot solve absolute prices but only relative prices as in other CGE and general equilibrium models with zero homogeneity in prices.

In concluding our discussion of the simple CGE model, we confirm the zero-profit condition for competitive firms in the following manner. Multiplying both sides of the factor demand function (2.3) by p_h^f and summing them with respect to the suffix h , we can derive the equation shown in (2.7). (In this mathematical manipulation, note that a constant-returns-to-scale production function (2.2) implies that the sum of the share coefficient $\beta_{h,j}$ with respect to the suffix h is equal to unity.)

$$\sum_h p_h^f F_{h,j} = p_j^z Z_j \quad \forall j \quad (2.7)$$

The left-hand side of (2.7) shows the cost incurred to the j -th firm, while the right-hand side represents its sales of the good. Equality of the total cost and sales in this simple CGE model proves that a firm cannot earn either excess profits nor suffer from excess losses in the equilibrium.⁸ That is, the zero-profit condition for any firm is assured.

Notes

1. Although the real economy has multiple households, this fictitious one representative household is adequate as long as all households have homogeneous preferences. Section 10.1 introduces heterogeneous multiple households into the CGE model.
2. In this example, the names of the goods are set as bread and milk. However, these goods do not imply they are complements or substitutes, or essentials or luxuries.
3. We can rewrite the utility function without the product symbol Π as follows:

$$UU = \prod_i X_i^{\alpha_i} = X_{BRD}^{\alpha_{BRD}} \cdot X_{MLK}^{\alpha_{MLK}}$$

Similarly, the budget constraint can be rewritten without the summation symbol Σ as follows:

$$p_{BRD}^x X_{BRD} + p_{MLK}^x X_{MLK} = p_{CAP}^f FF_{CAP} + p_{LAB}^f FF_{LAB}$$

The symbol $\max_{X_i}$ means to maximize the following objective function with respect to the variable X_i .

4. Detailed derivation of this household demand function (2.1) is shown in Appendix I.
5. Instead of directly solving this system of simultaneous equations, we can reformulate it into a social welfare maximization problem and solve it to obtain the same general equilibrium solution. Details are discussed in Appendix II.
6. We often verify whether a model has a solution by examining whether the number of equations is exactly equal to the number of endogenous variables. This rule is often used as a quick method of verifying the solvability of a model. However, it is not a necessary condition or a sufficient condition for the solvability of nonlinear systems either.
7. Walras's law implies that the sum of excess demand in all the markets is always equal to zero; that is:

$$p_{BRD}^x (X_{BRD} - Z_{BRD}) + p_{MLK}^x (X_{MLK} - Z_{MLK}) + p_{CAP}^f \left(\sum_j F_{CAP,j} - FF_{CAP} \right) + p_{LAB}^f \left(\sum_j F_{LAB,j} - FF_{LAB} \right) = 0.$$

When the bread, milk and capital markets are in equilibrium (i.e., the first three terms on the left-hand side are zero), Walras's law implies that the labour market is also in equilibrium (i.e., the fourth term is zero). For details of Walras's law, see Varian (2006).

8. If the right-hand side of (2.7) exceeds its left-hand side (i.e., excess profits exist), an entry into the market takes place to reduce profits per firm. In contrast, excess losses lead to the exit of firms to increase profits per firm. Finally, entries and exits make profits per firm converge to zero as (2.7) indicates. The excess profit is different from normal profit (i.e., payments for capital $p_{CAP}^f F_{CAP,j}$ in our model). Positive normal profits accrue if the marginal productivity of capital is greater than zero.

3

Computation

Chapter 2 provided a mathematical presentation of the ‘simple CGE model’ with its theoretical underpinnings based in standard micro-economics. However, the general equilibrium of an economy cannot be ‘computed’ only with this mathematical model. The CGE modelling needs two more steps. The first step is to prepare a computer program of the CGE model, using GAMS syntax. This step is covered in this chapter. The second step is to construct a social accounting matrix (SAM) on the basis of empirical data and then to estimate coefficients and exogenous variables such as the share coefficient of the utility function α_i and the factor endowments FF_h . This second step is discussed in Chapters 4 and 5.

In this chapter, the household utility maximization problem, which is part of the simple CGE model explained in Chapter 2, is used to demonstrate GAMS programming. By reading this chapter, readers will become familiar with the use of GAMS for CGE modelling.¹ The entire computer program for the simple CGE model will be explained in Section 5.4. After that, Section 6.8 presents the computer program for the ‘standard CGE model’, which includes a government, investment and savings, intermediate inputs and outputs, indirect taxes and the external sector.

3.1 Example: the ‘household utility maximization model’

To explain GAMS programming, we consider the household utility maximization problem (in Section 2.2), which is part of the simple CGE model. (Hereinafter, we call it the ‘household utility maximization model’.) In this model, two goods (bread and milk) as well as two factors (capital and labour) are considered. As assumed in the previous chapter, this household does not have any power to affect market prices. That

is, markets are perfectly competitive, and thus all the prices are given for this household. This household is endowed with a certain amount of factors, which are provided to firms to earn income. The household determines demand for goods so as to maximize its utility defined by the Cobb–Douglas-type function (2.a), subject to the budget constraint (2.b). The budget constraint equates household total expenses with household income.

3.1.1 Formulation of the model

The household utility maximization model is formulated as follows:

$$\underset{X_i}{\text{maximize}} \quad UU = \prod_i X_i^{\alpha_i} \quad (2.a)$$

subject to

$$\sum_i p_i^x X_i = \sum_h p_h^f FF_h \quad (2.b)$$

In this maximization problem, factor endowments FF_h , good prices p_i^x , factor prices p_h^f and the share coefficients of the utility function α_i are assumed to be given.

Analytically solving this maximization problem, we obtain the following demand function for each good, as explained in Section 2.2:

$$X_i = \frac{\alpha_i}{p_i^x} \sum_h p_h^f FF_h \quad \forall i \quad (2.1)$$

In other words, solving the household utility maximization model consisting of (2.a) and (2.b) is equivalent to solving the system of simultaneous equations (2.1).

Here, we are going to show how to solve numerically this household utility maximization problem expressed as a system of simultaneous equations (2.1). For this purpose, we use GAMS with its solver CONOPT. This solver, however, was developed for solving nonlinear programming problems; it is not capable of solving directly a system of simultaneous equations like that consisting of (2.1). Therefore, we employ a mathematical expedient. More precisely, adding a fictitious objective function to the original problem while using all the equations of the original model as constraints, we reformulate the system of simultaneous equations into a nonlinear programming problem. Consistency between the original and reformulated models is discussed in Appendix IV.²

In this reformulation, any fictitious objective function – except for obviously problematic ones like those that end up diverging because of inadequate constraints imposed by other equations – can be employed. Here, we employ the household utility function (2.a) presented in Section 2.2. With this fictitious objective function, our problem to be solved numerically with GAMS becomes:

$$\underset{X_i}{\text{maximize}} \quad UU = \prod_i X_i^{\alpha_i} \quad (2.a)$$

subject to

$$X_i = \frac{\alpha_i}{p_i^x} \sum_h p_h^f FF_h \quad \forall i \quad (2.1)$$

3.1.2 Specifying coefficients and exogenous variables

In programming the household utility maximization model, we need to specify values of the coefficients and exogenous variables in the model. Although these values must be estimated empirically in the actual model, we simply assume the following values for the time being:

$$\text{share coefficients in the utility function: } \begin{pmatrix} \alpha_{BRD} \\ \alpha_{MLK} \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix}$$

$$\text{good prices: } \begin{pmatrix} p_{BRD}^x \\ p_{MLK}^x \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\text{factor prices: } \begin{pmatrix} p_{CAP}^f \\ p_{LAB}^f \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\text{factor endowments: } \begin{pmatrix} FF_{CAP} \\ FF_{LAB} \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

3.2 Computational procedures

In this section, we explain the use of GAMS using the section related to CGE modelling from *GAMS – A User's Guide* by Brooke et al. (2008). Any questions can be answered by referring to *GAMS – A User's Guide*.

The standard GAMS computation procedures are as follows:

- (1) Prepare an input file of the model using the editor software.
- (2) Solve the model with GAMS.
- (3) Interpret the result shown in the output file generated by GAMS.

More precisely, the first task is to prepare an input file using an editor. The input file of the present model, named ‘hhmax.gms’, is shown in List 3.1.³ The line numbers in the first column were inserted by the authors to make the file easy to read; these line numbers should not be included in the actual file, otherwise an error message will appear. Detailed explanations about the contents of List 3.1 are given in the next section. To prepare the input file, either the GAMS IDE editor or the modeller’s favourite editor can be used.⁴

The second task is to solve this model. If the input file has been written using GAMS IDE, then click on the ‘Run GAMS’ icon . The model is solved by GAMS, and its output file automatically appears on the screen.⁵ The output file is shown in List 3.2 in Section 3.4. The third task, interpretation of the output file, is also explained there.

List 3.1 Input file for the household utility maximization model (hhmax.gms)

```
1 $ Title A Household's Utility Max. Model in Ch. 3 (HHMAX,SEQ=274)
2
3 * Definition of the Index Sets -----
4 Set      i      goods                /BRD   bread,
5                                     MLK   milk/
6          h      factors              /CAP   capital,
7                                     LAB   labor/;
8
9 * Definition of Parameters -----
10 Parameter      alpha(i)              share parameter in utility function
11                /BRD   0.2
12                MLK   0.8/;
13
14 Parameter      px(i)                  price of the i-th good
15                /BRD   1
16                MLK   2/;
17
18 Parameter      pf(h)                  price of the h-th factor
19                /CAP   2
20                LAB   1/;
21
22 Parameter      FF(h)                  factor endowment
23                /CAP   10
24                LAB   20/;
25
26 * Definition of Variables -----
```

List 3.1 (Continued)

```

27 Positive Variable X(i)          consumption of the i-th good
28 ;
29 Variable          UU            utility
30 ;
31 Equation          eqX(i)        household demand function
32                  obj            utility function
33 ;
34 * Specification of Equations -----
35
36 eqX(i)..          X(i)          =e= alpha(i)*sum(h, pf(h)*FF(h))/px(i);
37 obj..            UU            =e= prod(i, X(i)**alpha(i));
38
39 * Setting Lower Bounds on Variables to Avoid Division by Zero -----
40 X.lo(i)=0.001;
41
42 * Defining the Model -----
43 Model HHmax /all/;
44
45 * Solving the Model -----
46 Solve HHmax maximizing UU using NLP;
47 * -----
48 * end of model -----
49 * -----

```

3.3 Preparing input files

3.3.1 Structure of input files and general syntax of GAMS

To solve the household utility maximization model using GAMS, we have to include everything shown in the original model in the input file. That is, we have to declare constants and variables, assign values to constants, define the model equations (i.e., constraints and the objective function), and choose whether to maximize or minimize the objective function.⁶

Input files are usually prepared in the following manner. Typical directives used in the GAMS syntax are shown in parentheses (). Text in italics can be assigned freely by the modeller. Directives and other symbols are explained one by one in Subsection 3.3.2.

- (1) Declaration and definition of symbols
 - sets (Set)
 - constants (Scalar, Parameter, Table)

- endogenous variables (Variable, Positive Variable, Negative Variable)
constraints/objective function (Equation)
- (2) Definition of equations (*equation-name..*)
 - (3) Declaration of model name and definition of equations of the model (Model *model-name* /all/;)
 - (4) Statement to solve the model (Solve *model-name* maximizing *objective-variable-name* using NLP;)⁷

In Chapter 1, it was suggested that GAMS is convenient software because equations can be written in a common mathematical form exactly as they look. However, we need to consider some unique rules in the GAMS syntax.

- (1) Symbols can only be used if they have been declared *previously* in the input file. This is in contrast with common mathematical expressions where a symbol can be used before it is declared. For example, we can use X_i before declaring the suffix i to indicate a certain item such as a good. However, in GAMS, the symbol i must be declared first, then it can be used in defining X_i . This rule makes the order of declarations important.
- (2) Any alphabetical letters, numbers and `_` (underscore) can be used for symbols (including indices, constants, equation names, model names, etc.) with some exceptions. The exceptions are as follows. First, symbols can start only with alphabetical letters. Second, symbols must be no longer than 63 characters.⁸ Third, a symbol can only have one definition. (This point is explained in detail below.) Fourth, there are some reserved keywords in GAMS.⁹ Incidentally, although Greek letters such as α and β are often used as coefficients in mathematical expressions, they are not accepted by GAMS and thus must be spelled out in English.
- (3) The same symbol cannot be distinguished by GAMS even if it has different indices. For example, while we can distinguish x_i from $x_{i,j}$ in mathematical expressions, GAMS does not allow the use of such symbols. We have to declare a different symbol such as $y_{i,j}$, instead of $x_{i,j}$, so that GAMS distinguishes it from x_i . In addition, superscripts and subscripts cannot be used in GAMS; thus, indices must be expressed in parentheses; e.g., $x(i)$.
- (4) Symbols are case insensitive. While X is distinguished from x in mathematical expressions, X and x can be used interchangeably in GAMS. GAMS does not distinguish between the singular and

plural forms of directives. For example, both 'Parameter' and 'Parameters' are recognized by GAMS as the directive to declare constants.

- (5) Equation names must be declared, and one such name must be assigned to each equation (i.e., constraints and the objective function).
- (6) Spacing of elements in a line or splitting of one line into multiple lines is flexible, as long as there is no possible confusion, and this relies on the commonsense of the modeller. In the case of matrices, however, column and row labels and values must be placed properly to avoid confusion.¹⁰
- (7) All statements must end with a semicolon ';'.
- (8) Using one directive, we can declare several symbols of the same category (indices, constants, equations and models). In this case, we have to use a comma or carriage return as a delimiter between them. An example is lines 4–7 of List 3.1.
- (9) Lines starting with an asterisk '*' are recognized as memos by GAMS; thus, they are not included in any computation. The use of memos is highly recommended so that the meanings of symbols, equations, etc., can be recalled later or shared with other people easily.

3.3.2 Directives

We next explain the input file of the household utility maximization model in List 3.1. Before proceeding to the input file, we briefly discuss the suffixes and indices used in this book. In mathematical models, X_i , $Y_{i,j}$, etc. are used frequently, and i , j , etc. are commonly called suffixes. In GAMS, they must be expressed as $X(i)$ and $Y(i, j)$; thus, it is appropriate to call (i) , (i, j) , etc. indices.

Declaration and definition of sets

The first task is to declare sets. In the household utility maximization model, the optimal consumption bundle of goods (bread and milk) is to be determined, given the endowments of factors (capital and labour). Therefore, we must declare two sets to denote goods and factors. The directive of 'Set' is used to declare sets. Texts in brackets [] shown in the following examples are optional and thus can be omitted in terms of the GAMS syntax.

Syntax of the Set directive

Set *name-of-set-1* [*memo*] /*element-a* [*memo*],

```

                                element-b [memo],
                                ...         /
name-of-set-2 [memo] /element-g [memo],
                                element-h [memo],
                                ...         /;

```

Lines 4–5 of List 3.1 declare a set of goods using the symbol ‘i’. (This part corresponds to ‘name-of-set-1’ in the above syntax table.) The symbol ‘i’ is the name of this set, and it also functions as an index (equivalent to a suffix in the original model). The symbol ‘i’ is followed by a memo of ‘goods’, meaning that the set named ‘i’ represents goods. (This memo is not a mandatory entry but is highly recommended for better readability of input and output files.)

Then, the memo is followed by the elements of the set ‘i’: bread (BRD) and milk (MLK). The elements are enclosed with slashes ‘/’. (This part corresponds to ‘element-a’ and ‘element-b’ in the above syntax table.) The GAMS syntax:

```

Set      i      goods          /BRD bread,
                                MLK  milk/

```

means:

$$i \in \{BRD, MLK\}$$

in common mathematical notation. Among the entities enclosed by slashes ‘/’, only ‘BRD’ and ‘MLK’ are used in GAMS computations, while the following texts ‘bread’ and ‘milk’ are memos, which do not affect the computation results. Memos can be omitted or written in any manner. The GAMS compiler does not care whether the symbol ‘BRD’ implies ‘bread’ or ‘cake’ in reality. However, these memos remind the modeller of the meaning of ‘BRD’ and ‘MLK’. Lines 6–7 similarly declare a set of factors with an index of h , which means:

$$h \in \{CAP, LAB\}$$

Declaration and definition of constants

The next step is the declaration and definition of constants (i.e., exogenous variables and coefficients). Lines 10–24 are devoted to this procedure. There are three directives of `Scalar`, `Parameter` and `Table` to

Table 3.1 Mathematical expressions and GAMS syntax (1): constants

Mathematical expression	GAMS syntax	
x	Scalar	<code>x</code>
x_i	Parameter	<code>x(i)</code>
$x_{i,j}$	Table	<code>x(i,j)</code>

declare and define constants, depending upon their dimensions (i.e., the number of indices) and the ways to set their value(s) (Table 3.1).

Syntax of the Scalar directive

`Scalar name-of-constant-1 [memo] /value-of-constant-1/;`

Syntax of the Parameter directive

`Parameter name-of-constant-2(index-1) [memo]
 /element-a value-a
 element-b value-b
 ... /;`

Syntax of the Table directive

`Table name-of-constant-3(index-1, index-2) [memo]
 element-g element-h ...
 element-a value-ag value-ah ...
 element-b value-bg value-bh ...
 ...
 ;`

Note: $index-1 \in \{element-a, element-b\}$, $index-2 \in \{element-g, element-h\}$.

Consider the declaration and definition of the symbol for the coefficient α_i in the utility function (2.a) as an example. Because this coefficient has one suffix in the original model, the Parameter directive is chosen and placed in line 10. This directive is followed by the name of coefficient ‘alpha(i)’, which corresponds to ‘name-of-constant-2 (index-1)’ as shown in the syntax of the Parameter directive. As the values assigned to these coefficients are:

$$\begin{pmatrix} \alpha_{BRD} \\ \alpha_{MLK} \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix}$$

we write these values in the following GAMS format:

```
Parameter    alpha(i)   share parameter in utility function
             /BRD   0.2
             MLK    0.8/;
```

The text ‘share parameter in utility function’ shown after ‘alpha(i)’ is a memo for this coefficient. The other constants p_i^x , p_h^f and FF_h are declared and defined in the same manner.

As the `Scalar` and `Table` directives do not appear in List 3.1, we provide examples here. The `Scalar` directive is used for the declaration and definition of constants without a suffix. For example, we can define a constant ‘a’ with a value of ‘2’ as follows:

```
Scalar       a           the slope coefficient       /2/;
```

(The text ‘the slope coefficient’ is a memo.) When we use the `Parameter` and `Scalar` directives, we can adopt an alternative format, which is explained in Section A.2 in Annex A.

The `Table` directive is generally used to declare and define constants with multiple suffixes, whose values can be expressed conveniently in matrix form. When we declare and define the share coefficient $\beta_{h,j}$ in the production function (2.2) in Section 2.3, we use codes in the following manner:

```
Table    beta(h,j)   share parameter in production function
         BRD       MLK
CAP      0.3       0.6
LAB      0.7       0.4
;
```

(Note that the values appearing above are hypothetical ones.)

In using the `Table` directive, we have to note the following two points. First, the order of indices of the set written after the `Table` directive dictates the placement of the row and column labels. In the above example, as index ‘h’ precedes ‘j’ in $\beta(h,j)$, the elements in the set ‘h’, that is, factors ‘CAP’ and ‘LAB’, are placed as the row labels. The elements in the set ‘j’, namely goods ‘BRD’ and ‘MLK’, are placed as the column labels. Second, we have to place values properly under the respective column labels; that is, they must be under one column label, not spread under two or more column labels. Figure 3.1 provides some examples of errors. In the first column, the values of 0.3 and 0.7 are placed under

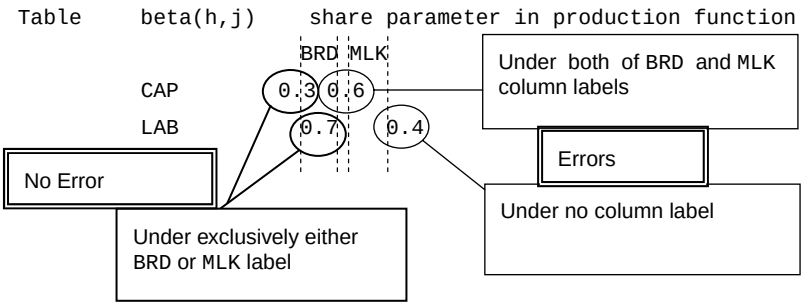


Figure 3.1 Placement of values in the Table directives

Table 3.2 Mathematical expressions and GAMS syntax (2): endogenous variables

Mathematical expression		GAMS syntax	
$x \geq 0$	nonnegative	Positive Variable	x
$x \leq 0$	nonpositive	Negative Variable	x
x	free	Variable	x

the column label BRD, and no error is caused. In contrast, the value of 0.6 is spread under the two column labels, and the value of 0.4 does not have any column label above it. This latter example causes errors.

Declaration of endogenous variables

Endogenous variables are declared in lines 27–30. There are three types of endogenous variables commonly used in GAMS programming (Table 3.2).¹¹ For example, the GAMS syntax to declare a nonnegative variable is as follows:

Syntax of the Positive Variable directive

```
Positive Variable  name-of-endogenous-variable-1 [memo]
                  name-of-endogenous-variable-2 [memo]
                  ... ;
```

Note that the Positive Variable directive is used to declare a non-negative variable, not a strictly positive variable.

The endogenous variable used in the example problem is the amount of consumption X_i . This variable must be nonnegative; thus, it is declared with the Positive Variable directive. In lines 27–28, the Positive Variable directive is placed first, followed by the symbol

'X(i)', a memo of 'consumption of the i-th good' and a semi-colon ';'. The next endogenous variable, which is declared in lines 29–30, is the objective variable of *UU*. GAMS requires this objective variable to be declared as a free (i.e., neither positive nor negative) variable by the *Variable* directive because the range of the objective variable cannot be determined before solving the problem.

Declaration of equation names

Equation names are declared in lines 31–33. In Chapter 2, we used mechanical equation numbers such as (2.1) to indicate the household demand function; however, in our GAMS programs, we usually give each equation a friendlier name. (The form of the equation is defined later, in line 36 in this case.) The *Equation* directive is for equation names. In line 31, the equation name 'eqX(i)' is declared to indicate the household demand function. Of course, other names can be chosen, if the modeller prefers. Similarly, line 32 declares 'obj' as the equation name for the objective function.

Definition of equations

Line 36 defines the functional form of the demand function of the i-th good. In this line, the equation name 'eqX(i)' is placed first; then, it is followed by the double dot ' . .' and the equation. Only a double dot can be used to indicate the beginning of each equation. Line 37 defines the formula of the objective function. The strict equality symbol '=e=' and the inequality symbols '=g=' and '=l=' are used only for equations that function as constraints or an objective function. Otherwise, the usual equality symbol '=' is used to calculate values assigned to constants.¹²

As stated above, the GAMS syntax allows us to write equations in the input files in a format similar to that used in common mathematical expressions. Using Table 3.3, we can compare the program in lines 36–37 with the original equations (2.1) and (2.a).

Lower bounds on endogenous variables

Line 40 illustrates the way to set the lower bounds of the endogenous variable X_i . Lower bounds are used to avoid problems such as undefined mathematical operations (e.g., division by zero) or problems caused by other computational difficulties.¹³ We have to pay attention to the order of ' . lo' and '(i)'. In the same manner, we can also impose upper bounds on endogenous variables. Similarly, we can fix the values of endogenous variables with ' . fx' (Table 3.4). Furthermore, to assign certain values to

Table 3.3 Mathematical expressions and GAMS syntax (3): operators and miscellaneous symbols

Mathematical expressions	GAMS syntax
$\geq, \leq, =, \neq$	inequalities and equality
$\sum_i x_i, \prod_i x_i$	summation and product
x^n	power
$+, -, *, /$	arithmetic operators
	$=g=, =l=, =e=, =n=$
	$\text{sum}(i, x(i)), \text{prod}(i, x(i))$
	$x^{**}n$
	$+, -, *, /$

Note: See Chapter 6 of Brooke et al., *GAMS – A User's Guide* (2008) for other symbols.

Table 3.4 Mathematical expressions and GAMS syntax (4): endogenous variables

Mathematical expression	GAMS syntax
$x \geq n, x \leq n$	$x.lo=n, x.up=n$
To fix an endogenous variable x at n	$x.fx=n$
To set the LEVEL value of an endogenous variable x at n	$x.l=n$

endogenous variables to initialize computation, we can use `'l'`.¹⁴ (Note that the letter 'l' follows the period, not the numeral '1'.)

Declaration of model name and definition of model

Line 43 declares the model name 'HHmax' with the `Model` directive.

Syntax of the Model directive

<To include all the equations defined before this statement >

```
Model model-name /all/;
```

<To include selected equations declared before this statement>

```
Model model-name /equation-1, equation-2, .../;
```

In the present example, we have arbitrarily chosen the model name 'HHmax', which corresponds to '*model-name*' above. Furthermore, we define the model to include all the equations declared above (i.e., `eqX` and `obj`); thus, we write the following:

```
Model HHmax /all/;
```

This can be written equivalently as follows:¹⁵

```
Model HHmax /eqX, obj/;
```


If we want to include only one (or some) equations, say, ‘obj’ in the model, we write the following:¹⁶

```
Model HHmax /obj/;
```

Solving the model

Line 46 instructs GAMS to solve the ‘HHmax’ model by maximizing ‘UU’. To solve the model, we use the `Solve` directive.

Syntax of the Solve directive

<maximization problems>

```
Solve model-name maximizing objective-variable using NLP;
```

<minimization problems>

```
Solve model-name minimizing objective-variable using NLP;
```

In the present example, we write the solving directive as follows:

```
Solve HHmax maximizing UU using NLP;
```

(In the case of minimization problems, ‘maximizing’ must be replaced by ‘minimizing’.)

The last part of this statement ‘using NLP’ commands that this problem be solved using a solver for nonlinear programming problems. As we develop and solve the type of models presented in this book, we always use the statement ‘using NLP’.

3.4 Results of computation in the output file

Solving the program in List 3.1, we obtain an output file, an excerpt of which is shown in List 3.2. As the input filename is ‘hhmax.gms’, the corresponding output filename is ‘hhmax.lst’. In general, the extension for input files is ‘gms’ and that of their output files is ‘lst’.¹⁷

List 3.2 Output file of the household utility maximization model (hhmax.lst)

GAMS Rev 116 Windows NT/95/98				11/11/03 11:52:50		PAGE	1
A Household's Utility Max. Model in Ch. 3 (HHMAX,SEQ=274)							
2							
3	* Definition of the Index Sets -----						
4	Set	i	goods	/BRD	bread,		
5				MLK	milk/		
6		h	factors	/CAP	capital,		
7				LAB	labor/;		

List 3.2 (Continued)

```

8
9 * Definition of Parameters -----
10 Parameter      alpha(i)          share parameter in utility function
11                /BRD    0.2
12                MLK    0.8/;

...

          S O L V E      S U M M A R Y

MODEL  HHmax              OBJECTIVE  UU
TYPE   NLP                DIRECTION MAXIMIZE
SOLVER CONOPT             FROM LINE 46

**** SOLVER STATUS      1 NORMAL COMPLETION
**** MODEL STATUS      1 OPTIMAL
**** OBJECTIVE VALUE                13.9288

RESOURCE USAGE, LIMIT      0.031      1000.000
ITERATION COUNT, LIMIT    4          10000
EVALUATION ERRORS         0          0

C O N O P T 3  MS Windows version 3.12A-015-049
Copyright (C)  ARKI Consulting and Development A/S
                Bagsvaerdvej 246 A
                DK-2880 Bagsvaerd, Denmark

Using default options.

** Optimal solution. There are no superbasic variables.

CONOPT time Total          0.016 seconds
  of which: Function evaluations      0.000 = 0.0%
           1st Derivative evaluations 0.000 = 0.0%

Work length =    0.02 Mbytes
Estimate =    0.02 Mbytes
Max used =    0.01 Mbytes

---- EQU eqX  household demand function

      LOWER    LEVEL    UPPER    MARGINAL
BRD    8.000    8.000    8.000    0.348
MLK   16.000   16.000   16.000    0.696

      LOWER    LEVEL    UPPER    MARGINAL
---- EQU obj      .      .      .      1.000

obj  utility function

---- VAR X  consumption of the i-th good

      LOWER    LEVEL    UPPER    MARGINAL
BRD    0.001    8.000    +INF     .
MLK    0.001   16.000    +INF     .

      LOWER    LEVEL    UPPER    MARGINAL
---- VAR UU      -INF    13.929    +INF     .
UU  utility

```

The output file shows the original input file (List 3.1) headed by line numbers in the initial section, which is called an 'echo print' in GAMS. To save space, only the first 12 lines of the output file are shown in List 3.2.¹⁸ This part is useful for checking whether the input file is the right one and for debugging syntax errors, if any.¹⁹

If there are no syntax errors, move to the solution status block shown under the heading of 'SOLVE SUMMARY'. First, we should check whether the message of '**Optimal solution' appears in the SOLVE SUMMARY. If GAMS fails to find an optimal solution, a message such as '**Infeasible solution' or '**Unbounded solution' appears there instead.

After checking the solution status block, move to the solution block under the heading 'EQU' and 'VAR'. There are two types of solutions: equations (EQU) and variables (VAR). Each has four subfields: LOWER, LEVEL, UPPER and MARGINAL. Among these, the important indicators are the MARGINAL values printed in the EQU block and the LEVEL values in the VAR block. The former are the solved Lagrange multipliers of the equations (constraints), and the latter are the solved values of the endogenous variables.

The 'household utility maximization model' is developed to determine the amounts of bread and milk consumption. They are shown in the LEVEL column of the VAR X block. The output file shows that eight units of bread (BRD) and 16 units of milk (MLK) are consumed. This consumption bundle results in 13.929 units of utility shown in the LEVEL column of the VAR UU block.

There are symbols other than numbers: '+INF', '-INF', '.' (period character) and 'EPS'. They mean 'positive infinity', 'negative infinity', 'zero' and 'small value close to zero' respectively. The LOWER and UPPER values in the VAR block are the lower and upper bounds of endogenous variables, imposed by '.lo' and '.up' suffixes.²⁰

The MARGINAL values in the EQU block indicate the solved Lagrange multipliers of the equations. They are significant only when we solve (real) constrained optimization problems. As the household utility maximization model is essentially a system of simultaneous equations, the MARGINAL values in the EQU block in List 3.2 have no significance. In contrast, the same household utility maximization model is genuinely formulated as an optimization problem in Appendix III. In that output file, the MARGINAL value in the EQU block for the budget constraint can be interpreted as the 'marginal utility of income'.

Further details of GAMS's uses are shown in Annex A of this book as well as in *GAMS – A User's Guide*. Besides general uses of GAMS, a

particularly important issue is how to deal with errors. During programming, errors are often encountered. When a model fails to solve, we have to locate and determine the reasons for the errors by making use of the error messages in the output file. Knowledge about the meaning of these messages is important for debugging and greatly helps the modelling process. Some common errors with hints about how to fix them are discussed in Annex B.

Notes

1. GAMS includes several solvers for nonlinear programming problems, nonlinear complementarity problems and so on. In this textbook, we use a solver called CONOPT.
2. Instead of maximizing, we can minimize the objective function to get the same solution when the original model has a unique equilibrium. Standard CGE models like ours rarely have multiple equilibria. Readers who are interested in the issue of multiple equilibria can refer to Mercenier (1995).
3. Filenames are divided into two parts using the period symbol. The former part is called the base name, and the latter part is called the extension. Generally, any extension can be used in filenames; however, we have to use “gms” for GAMS input file extensions to solve them using GAMS IDE.
4. Information about the use of GAMS IDE can be obtained from www.gams.com/dd/docs/tools/gamside.pdf.
5. When GAMS IDE is not used, it is necessary to type:
`gams hhmax.gms`
 at the command prompt. (For those who are not accustomed to such an operation, we recommend the use of GAMDI IDE.)
6. In GAMS, ‘declaration’ means declaring the existence of something and giving it a name, while ‘assignment’ or ‘definition’ means giving something a specific value or form.
7. In the case of the minimization problem, we replace ‘maximizing’ with ‘minimizing’ in this statement.
8. This length depends on the version of the GAMS system. We presume Release 22.7 in this book.
9. A trivial example will clarify this point. If we declare the symbol ‘Set’, it causes a conflict with the ‘Set’ directive. Another example is the error that occurs by using the symbol ‘EPS’, because GAMS reserves this symbol to indicate a positive number close to but not equal to zero. Details are explained in the section ‘Reserved words’ in Chapter 3 of *GAMS – A User’s Guide*.
10. When we insert a new line (line feeding), we use the return key, not the space key. If we use the space key, errors may occur.
11. Strictly speaking, even though variables are declared either with the ‘Positive Variable’ or the ‘Negative Variable’ directive, numerical software such as GAMS cannot handle values infinitesimally close but not equal to zero, and instead approximate these values with a finite number of decimal places. However, an approximation works adequately in most cases.

12. Line 28 in List 5.1 provides this example.
13. When we assume the Cobb–Douglas-type utility function as in this example, we know that the solutions of all the endogenous variables are strictly positive as long as the share parameters are strictly positive. Thus, we can impose such a priori lower bounds. Without these lower bounds, we might not be able to obtain solutions of the problem. Detailed discussions on this issue are provided in Annex B.
14. While the values of endogenous variables pinned down using ‘.fx’ do not change during the GAMS computation, ones whose LEVEL values are set using ‘.l’ change to satisfy constraints and/or to optimize an objective function during computation.
15. When we use the Model directive, we do not add any indices to equation names, such as ‘eqX(i)’, so it is written simply as ‘eqX’.
16. This technique may be useful when we develop several different models by making appropriate combinations of already-defined equations and solve those models in one run.
17. The extension is a part of the filename after the period. The operating system (for example, MS Windows) identifies file types by the extension.
18. Although the first line in the input file includes optional directives starting with the ‘\$’ symbol (called dollar control options), GAMS does not show them in the output file.
19. Syntax errors are indicated with the ‘*****’ symbol followed by error codes in the output file. Error codes appear below the (seemingly) erroneous parts. The meaning of each error code is shown after the echo print. The error messages and techniques for debugging are discussed in Annex B.
20. Using the Positive Variable directive or the Negative Variable directive is tantamount to the imposition of lower or upper bounds of zero on a variable, respectively.

4

The Social Accounting Matrix

In the ‘household utility maximization model’ presented in the last chapter, we used values of coefficients for the utility function and for the factor endowments, assumed on an a priori basis. However, to develop an empirical CGE model, various coefficients and exogenous variables of the model based on real data must be estimated. The estimation process consists of two steps. First, we collect data of the base year and construct a comprehensive and consistent macroeconomic database, called a social accounting matrix (SAM). Next, by using the SAM, we estimate the coefficients and exogenous variables of the model. The estimation method of the latter step is called ‘calibration’. This chapter explains first the structure of a SAM, then how to construct a SAM. The calibration procedure is explained in Chapter 5.

Section 4.1 is devoted to an explanation of the structure of a SAM. For this purpose, two SAMs are used as examples. First, we explain the structure of the SAM for the ‘simple CGE model’ shown in Chapter 2, where the household is the only user of goods in a closed economy. This SAM will be used to calibrate the coefficients and exogenous variables of the simple CGE model in Chapter 5. Second, we explain the structure of the SAM used for the ‘standard CGE model’ in advance. The standard CGE model includes an intermediate input, a government (including government consumption, direct taxes, production taxes and import tariffs) and investment and savings, as well as exports and imports. This standard CGE model is fully explained in Chapter 6, where the SAM constructed in this chapter is used for calibration of the coefficients and exogenous variables of that model.

In this chapter, an explanation of the structure of the SAMs in Section 4.1 is followed by Section 4.2, which explains how to construct the SAM based on input–output (IO) tables and other macroeconomic data. The

SAM for the standard CGE model is used as an example. Section 4.3 demonstrates the actual construction of a SAM for Japan based on the IO tables for 2000.

Section 4.4 deals with the problems often encountered in the construction of a SAM. There are two problems. The first problem is caused by missing data because of statistical imperfection or statistical discrepancy. The second problem is related to the timing of construction of the IO tables, which are the main components of a SAM. As the IO tables are constructed, at most, once every several years, we often have to use IO tables that are several years old. To fix statistical discrepancies in old SAMs or to update them, it is necessary sometimes to conduct matrix adjustments for SAMs, whose methods are explained in Section 4.4.

4.1 Structure of social accounting matrix

As discussed above, CGE models are used to analyse economic activities, which are transactions involving goods and factors, and the concurrent flows of funds between agents in an economy. The latter can be considered as a mirror image of the former (or, vice versa).

The knots of the transactions, or their starting points and ending points, can be categorized as 'Activity', 'Factor', 'Indirect Tax', 'Final Demand' and 'External'. These knots are termed agents in the following explanation.¹ A SAM describes transactions by these agents. In this categorization, 'External' (i.e., exports and imports) is conveniently treated as separated from the domestic final users in 'Final Demand', although the former is part of the latter. Depending on the needs of the analysis, these agents can be merged into broader ones or divided into narrower ones.

The SAM is written in a matrix-form table. The agents specified above are used as both row labels and column labels. (Thus, the matrix is square.) The entries in a SAM indicate flows of goods and services from the agents listed in the rows to the counterpart agents listed in the columns. The corresponding payments (or the flows of funds) are made in the opposite direction. Hereinafter, the names of each entry in the table are referred to as 'row label'–'column label' in this order.

4.1.1 Social accounting matrix for the simple CGE model

As the simple CGE model shown in Chapter 2 consists of one household and two firms with two goods and two factors, its SAM has only 'Activity', 'Factor' and 'Final Demand'.² The SAM of the simple CGE model is shown in Table 4.1.³ In this SAM, 'Activity' is subdivided into the bread sector

Table 4.1 SAM for the simple CGE model

		Activity		Factor		Final Demand	Total
		BRD	MLK	CAP	LAB	HOH	
Activity	BRD					15	15
	MLK					35	35
Factor	CAP	5	20				25
	LAB	10	15				25
Final Demand	HOH			25	25		50
Total		15	35	25	25	50	

Unit: yen

and the milk sector; 'Factor' is subdivided into capital and labour. In this simple model, we do not include a government, investment, indirect taxes and foreign trade; thus, only household consumption constitutes 'Final Demand'.

Activity and final demand (household)

The cell labelled 'BRD'-'HOH' in Table 4.1 shows that the bread producer receives 15 yen from the household. In return, the bread producer supplies bread worth 15 yen to the household. The other side of the coin is that the household spends 15 yen to purchase bread worth 15 yen. Similarly, the cell 'MLK'-'HOH' implies that there is a transaction involving the exchange of 35 yen and milk worth 35 yen between the milk producer and the household.

Factors and activity

'Factor' consists of capital and labour, which are used for 'Activity'. These transactions are shown in the cells in the 'Factor' row and the 'Activity' column. For example, the cell 'CAP'-'BRD' shows that the capital holder earns five yen in exchange for capital services provided to the bread sector. The 'LAB'-'BRD' cell shows a similar exchange of labour services for 10 yen between the labour holder and the bread sector.

Final demand (by the household) and factors

It will be easy to understand if we interpret the 'Factor' rows/columns as a virtual management agent of factors. This factor management agent is commissioned by other agents (in this case, the household) to provide their factors to 'Activity' (in this case, bread and milk) and to receive the factor income on behalf of the household (in this case). The former part

of the factor payment process is shown, for example, in the 'CAP'–'BRD' cell, which has already been explained. The latter part of the commission process is represented by the 'HOH'–'CAP' and 'HOH'–'LAB' cells. These cells mean that the household receives 25 yen for capital and 25 yen for labour from the virtual factor management agent (i.e., 'Factor'). In return, the 'Final Demand' sector provides those factors to the management agent. (This example shows that the capital payments are equal to the labour payments; however, this is coincidental.)

Row sums and column sums

The row sums and the column sums in the SAM show the total receipts and the total payments of funds respectively, and each row sum must match its corresponding column sum. For example, the row sum of 'HOH' is the total amount of funds received from the factor management agent by the household; the column sum of 'HOH' is the total amount of funds paid by the household for its purchase of goods. Both totals sum to 50 yen.

Returning to the 'Activity' block, the row sum of 'BRD', i.e., 15 yen, is the money the bread producer receives from the household to the bread producer. Conversely, the column sum of 'BRD', i.e., 15 yen, is the total production cost of the bread producer. This is also the case for the milk producer. The equality between each row sum and its corresponding column sum suggests that all the goods produced and all the factors endowed in this economy are used up.

As stated previously, the IO tables are the main components of a SAM. The shadowed cells of the SAM in Table 4.1 can be found in the corresponding IO tables. How to derive data for the cells other than the shadowed ones is explained in detail in Section 4.2.

Concerning the composition of a SAM, the order of row and column entries can be freely arranged, and row/column entries can be added depending on the purpose of analysis and data availability. We often develop a SAM and consequently a CGE model with 10 or more sectors in 'Activity' and several household types in 'Final Demand'. We can also incorporate a government and taxes, and international trade with import tariffs and export subsidies. In the next subsection, the SAM for the standard CGE model is explained.

4.1.2 SAM for the standard CGE model

Here we present the SAM for the standard CGE model, which is an extension of the simple CGE model. The standard CGE model, into which a government, investment and savings, international trade and

Table 4.2 SAM for the standard CGE model

		Activity		Factor		Indirect Tax		Final Demand			External	Total
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3	23				35
	INV							17	2		12	31
External	EXT	13	11									24
Total		92	89	50	40	9	3	90	35	31	24	

Unit: yen

intermediate inputs are incorporated, will be developed fully in Chapter 6; however, its SAM is discussed here in advance for convenience of explanation. This SAM is shown in Table 4.2.

To accommodate a government, and investment and savings, we create two more virtual agents, similar to the virtual factor management agent, explained in Subsection 4.1.1. The first virtual agent, similar to tax and customs offices, is termed a virtual tax and customs agent, which is separate from the government itself. This virtual agent collects indirect taxes (production taxes in this section) and import tariffs, and transfers the tax revenue to the government. The second virtual agent is similar to an investment agent, which accepts savings from the household, the government and the external sector to purchase goods for domestic investment. In the following part, we explain Table 4.2, focusing on these new entries in the SAM of the standard CGE model.

Activity

‘Activity’ includes intermediate inputs in addition to the capital and labour input originally considered in Table 4.1. The intermediate input of bread and milk production appears in the ‘Activity’–‘Activity’ cells. This example shows that bread worth 21 yen and milk worth 17 yen are used as intermediate inputs for the bread production. The intermediate inputs for the milk production can be interpreted in the same manner.

We omit an explanation of the capital and labour inputs here, as they are identical with those in Table 4.1. Imports from, and exports to, the external sector will be discussed later.

Indirect tax

Production taxes and import tariffs are new features in this framework. 'Indirect Tax' can be regarded as the virtual tax and customs agent that collects indirect taxes and tariffs from other agents. The amounts of indirect taxes and tariffs and their sources are usually reported in the IO tables. The (domestic) indirect taxes such as production taxes collected by this virtual agent are put in the 'IDT'-'Activity' cells (for example, five yen from the bread sector), and the import tariffs are in the 'TRF'-'Activity' cells (for example, one yen from bread imports). The indirect taxes and tariffs are then transferred to the government, shown in the 'GOV'-'IDT' cell (nine yen), and in the 'GOV'-'TRF' cell (three yen).

Final demand

In addition to the household, the government and the virtual investment agent are now included as final users. Their revenues are shown in the 'Final Demand' rows in Table 4.2. The household earns income from the virtual factor management agent by providing capital and labour as explained above. The government revenue consists of production taxes, import tariffs and direct taxes imposed on the household. The amounts of these revenues are shown in the 'GOV'-'IDT' (nine yen), 'GOV'-'TRF' (three yen) and 'GOV'-'HOH' (23 yen) cells respectively.

The revenues of the virtual investment agent, shown in the 'INV' row, consist of savings by the household (17 yen), the government (two yen) and the external sector (12 yen). In Table 4.2, we assume that investment is made only by the virtual investment agent, but we can alternatively assume that the household and the government are engaged in investment activities.

These revenues stated above are used to support their final demand. The 'HOH' column in the 'Final Demand' block indicates household expenses for purchases of goods (20 yen for bread and 30 yen for milk), the payments of direct taxes to the government (23 yen) and savings (17 yen). The 'GOV' column shows government purchases of goods (19 yen for bread and 14 yen for milk) as well as its savings (two yen). The 'INV' column indicates the virtual investment agent's expenses for the purchases of goods (16 yen for bread and 15 yen for milk) for its investment.

The external sector

The external sector can be included in 'Final Demand' like other domestic agents; however, we deal with it separately for the convenience of our analysis. The 'EXT'-'Activity' cells indicate imports (13 yen for bread and 11 yen for milk), while the 'Activity'-'EXT' cells show exports (eight yen for bread and four yen for milk). These imported goods, combined with domestic supply, are used for intermediate input, final use and exports.⁴ Total imports, indicated by the row sum of 'EXT', are 24 yen, while total exports are 12 yen; there is a gap of 12 yen, which is the trade deficit. The sum of total exports and the trade deficit is shown in the column sum of 'EXT'.

In addition to the transactions stated above, we can take account of various non-goods trade, if necessary. Workers' remittances to foreign countries appear in the 'EXT'-'LAB' cell. Payments for foreign capital services appear in the 'EXT'-'CAP' cell. Summing these with the trade deficit, we obtain the current account deficit. This deficit equals foreign savings, appearing in the 'INV'-'EXT' cell (12 yen).⁵ In this manner, we can incorporate not only transactions of goods but also factor services and transfers of capital to/from foreign countries in the SAM. Because the 'External' block summarizes such transactions between the domestic and the foreign agents, its row and column correspond to the balance of payments tables.

4.2 Construction of social accounting matrix

In the previous section, we discussed how to interpret the SAMs shown in Tables 4.1 and 4.2 as examples. These example SAMs are those that have already been constructed. However, for empirical CGE analysis, we have to construct our own SAM. This section explains how to construct the SAM for the standard CGE model – in other words, how to construct Table 4.2.⁶

Almost all the data included in the SAM are provided in the IO tables. In Table 4.2, the shadowed cells can be obtained from the IO tables. Transferring data in the IO tables into the correct cells of the SAM is rather straightforward. Therefore, the core issue of SAM construction is how to fill the cells that cannot be derived from the IO tables. To deal with this issue, we rely heavily on the property of the SAM that column sums are equal to the corresponding row sums. Let us call this property the row-sum and column-sum equality rule in the following.

The shadowed cells in Table 4.2, which are available from the IO tables, are copied into Table 4.3.A. Because some cells necessary for a complete

Table 4.3.A SAM for the standard CGE model – using data from IO tables

		Activity		Factor		Indirect Tax		Final Demand			External	Total
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect	IDT	5	4									9
Tax	TRF	1	2									3
Final	HOH											0
Demand	GOV											0
	INV											0
External	EXT	13	11									24
Total		92	89	0	0	0	0	50	33	31	12	

Unit: yen

SAM are empty, the row-sum and column-sum equality rule does not hold for the rows and the columns other than those located in the ‘Activity’ block.

Let us consider the cells that can be filled immediately by applying the row-sum and column-sum equality rule. First, as the row sums of capital and labour are 50 yen and 40 yen, the corresponding blank cells of ‘HOH’–‘CAP’ and ‘HOH’–‘LAB’ indicating factor income must be 50 yen and 40 yen (Table 4.3.B).⁷ In the same way, we can determine the total production tax and import tariff revenues transferred from the tax and customs agent to the government. Total production tax revenues appear in the row sum of ‘IDT’ (nine yen), which is transferred only to the government; thus, this receipt of production tax revenues of nine yen should be put in the ‘GOV’–‘IDT’ cell. Similarly, total import tariff revenues appear in the row sum of ‘TRF’ (three yen), which is transferred only to the government; thus, three yen should be put in the ‘GOV’–‘TRF’ cell. By following this procedure we obtain Table 4.3.B.

In the same manner, we can compute the current account deficit by subtracting total exports shown in Table 4.3.A (12 yen, shown by the sum of the shadowed cells in the ‘EXT’ column) from total imports (24 yen, shown by the sum of the shadowed cells in the ‘EXT’ rows). The current account deficit equals foreign savings, which are put in the ‘INV’–‘EXT’ cell. Then, Table 4.3.C is attained.

Table 4.3.B SAM for the standard CGE model – filling the cells for household factor income and indirect tax revenues

		Activity		Factor		Indirect Tax		Final Demand			External	Total
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3					12
	INV											0
External	EXT	13	11									24
Total		92	89	50	40	9	3	50	33	31	12	

Unit: yen

Table 4.3.C SAM for the standard CGE model – filling the cells for the current account deficits

		Activity		Factor		Indirect Tax		Final Demand			External	Total
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3					12
	INV										12	12
External	EXT	13	11									24
Total		92	89	50	40	9	3	50	33	31	24	

Unit: yen

There are three accounts that show imbalances between their receipts and payments of funds in Table 4.3.C: the household, the government and the investment. The flows of funds involving these agents are typically supposed to appear in the following three cells: direct tax payments by the household to the government ('GOV'–'HOH') and

Table 4.3.D SAM for the standard CGE model – three cells left unfilled

		Activity		Factor		Indirect Tax		Final Demand			External	Total
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3					12
	INV										12	12
External	EXT	13	11									24
Total		92	89	50	40	9	3	50	33	31	24	

Unit: yen

transfers of funds by the household and the government to the investment (‘INV’–‘HOH’ and ‘INV’–‘GOV’). (These cells are shaded in Table 4.3.D). However, these cells cannot be filled using only the row-sum and column-sum equality rule. We need an additional technique.

If one of these three unknown cells can be filled by some means, the other two cells are filled by applying the same technique employed so far. Here, by seeking data sources other than the IO tables (for example, the national account tables), we may be able to determine direct tax revenues. If that value can be determined, it is entered into the ‘GOV’–‘HOH’ cell. (In addition to the value of direct taxes, values for one of the other two cells may be available. In such a case, the selection of the first datum to be filled should be made in consideration of the reliability of the data sources.) Assuming that direct tax revenue is 23 yen, the SAM is shown in Table 4.3.E. The other two cells are mechanically filled as previously demonstrated. Finally, we complete the compilation of our SAM as shown in Table 4.2.

4.3 Example: Social accounting matrix for Japan

To demonstrate how the above-mentioned process can be applied to construction of a SAM with actual data, we construct a SAM for Japan.⁸ In empirical studies, the objective of the study is usually decided first and then the specification of the most suitable model is determined,

Table 4.3.E SAM for the standard CGE model – filling a cell using external data sources

		Activity		Factor		Indirect Tax		Final Demand			External Total	
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3	23				35
	INV										12	12
External	EXT	13	11									24
Total		92	89	50	40	9	3	73	33	31	24	

Unit: yen

encompassing sectoral aggregation, agents, policy devices and so on. However, just for demonstration purposes, we skip these processes and simply concentrate upon the construction of a SAM.

To simplify the model, we distinguish between three production activities: agriculture, manufacturing and services (hereinafter, they are abbreviated as AGR, MAN and SRV in the SAM); and two production factors: capital and labour. There is only one representative household as well as the government, the investment and the external sector. The base year is set at 2000.⁹ In addition to the IO tables, the national account tables for 2000 are also available.¹⁰ Among the IO tables with various sectoral aggregation levels – 517 × 405-sector tables and consolidated square tables with 13, 32, 104 and 188 sectors – we use the 13-sector table in this example.

The first task is to aggregate the 13-sector table into a 3-sector table for simplicity (Table 4.4). The IO table consists of the square part indicating intermediate inputs and outputs by agriculture (AGR), manufacturing (MAN) and service (SRV) sectors and the so-called overhangs; i.e., the entries outside of the square part. The lower overhang shows payments for factor uses (i.e., generation of value added), while the right-side overhang indicates final demand. While the aggregation of the square part is straightforward, special care is needed for the overhang parts. The

Table 4.4 Japan's three-sector IO table for 2000

	Agriculture	Manufacturing	Service	Household	Government
Agriculture	1,558,469	8,427,693	1,496,991	3,965,927	0
Manufacturing	2,462,949	130,321,011	68,646,492	64,927,161	459,179
Service	2,273,437	63,503,715	160,713,811	231,268,309	85,247,038
Capital	6,167,952	33,816,669	149,889,160		
Labour	1,372,650	59,034,654	234,353,029		
Indirect Tax	534,232	14,436,136	19,877,410		
Total	14,369,689	309,539,878	634,976,893		
	Investment	Export	Import	Import Tariff	Total
Agriculture	967,198	72,018	-1,974,129	-144,478	14,369,689
Manufacturing	39,070,695	46,597,315	-39,254,377	-3,690,547	309,539,878
Service	90,250,845	10,817,384	-9,096,735	-911	634,976,893
Capital					
Labour					
Indirect Tax					
Total					

Unit: million yen

Source: Compiled by the authors using the input-output tables by the Statistics Bureau, the Ministry of Internal Affairs and Communications (www.stat.go.jp/english/data/io/io00.htm)

following items (1) to (3) relate to the lower overhang, and items (4) to (9) relate to the right-side overhang.

- (1) The 'operating surplus' and the 'depreciation of fixed capital' are aggregated into one account as the payments for capital use.
- (2) The 'compensation of employees' and the 'consumption expenditure outside households (row)' are aggregated into one account as payment for labour use.¹¹
- (3) The 'indirect taxes (excluding custom duties and commodity taxes on imported goods)' and the '(less) current subsidies' are aggregated into indirect taxes.
- (4) The 'consumption expenditure outside households (column)' and the 'consumption expenditure (private)' are aggregated into the household account.
- (5) The 'consumption expenditure of general government' is used as the government account.
- (6) The 'gross domestic fixed capital formation (public)', the 'gross domestic fixed capital formation (private)' and the 'increase in stocks' are aggregated into the investment account.

Table 4.5.A Japan's three-sector SAM for 2000 – filling cells using the IO tables

	AGR	MAN	SRV	CAP	LAB	IDT
AGR	1,558,469	8,427,693	1,496,991			
MAN	2,462,949	130,321,011	68,646,492			
SRV	2,273,437	63,503,715	160,713,811			
CAP	6,167,952	33,816,669	149,889,160			
LAB	1,372,650	59,034,654	234,353,029			
IDT	534,232	14,436,136	19,877,410			
TRF	144,478	3,690,547	911			
HOH						
GOV						
INV						
EXT	1,974,129	39,254,377	9,096,735			
Total	16,488,296	352,484,802	644,074,539	0	0	0
	TRF	HOH	GOV	INV	EXT	Total
AGR		3,965,927	0	967,198	72,018	16,488,296
MAN		64,927,161	459,179	39,070,695	46,597,315	352,484,802
SRV		231,268,309	85,247,038	90,250,845	10,817,384	644,074,539
CAP						189,873,781
LAB						294,760,333
IDT						34,847,778
TRF						3,835,936
HOH						0
GOV						0
INV						0
EXT						50,325,241
Total	0	300,161,397	85,706,217	130,288,738	57,486,717	

Unit: million yen

- (7) The 'exports' and the 'balancing sector' accounts are aggregated into exports.
- (8) The '(less) imports' account is used as imports.
- (9) The '(less) custom duties' and the '(less) commodity taxes on imported goods' accounts are aggregated into import tariffs.

Further care is needed for imports and import tariffs, which are shown as negative values in the right-side overhang of the IO table. Their signs have to be changed and they have to be transposed when we put them in the SAM.

After these operations, Table 4.4 is converted into Table 4.5.A. The following should be noted. As the above operations are simply merging

rows/columns in the 13-sector table into a 3-sector table without any deletion or additions of entries, the row sums must be equal to their corresponding column sums of AGR, MAN and SRV in the aggregated 3-sector table shown in Table 4.5.A. A gap between the row sum and the corresponding column sum of these production sectors' accounts, if any, indicates that an erroneous operation was made.

The second task is to fill in the empty cells in Table 4.5.A by exploiting the row-sum and column-sum equality rule. We use the following methods in the construction of the SAM:

- (1) If we can assume that the household is the only earner of income from labour and capital, their flow of funds to be placed in the 'HOH'-'CAP' and 'HOH'-'LAB' cells can be identified.
- (2) If we need to consider the receipt of transfers from abroad (such as remittances) in addition to trade of goods, we can alternatively compute the current account deficit by subtracting the sum of incoming transfers and total exports from total imports. In this extension, we have to identify which agent receives (or pays) these transfers. The SAM differs depending on whether the recipient is the household or the government.
 - (2-a) If the household is the recipient of the transfers, the value of transfers appears in the 'HOH'-'EXT' cell.
 - (2-b) If the government is the recipient, the value of transfers appears in the 'GOV'-'EXT' cell.
- (3) If we assume that this country exports and imports only goods (including nonfactor services), its current account balance can be computed as the difference between total exports and total imports as well as miscellaneous transfers mentioned in (2), if any. This value should be entered in the 'INV'-'EXT' cell. In the case of Japan, the current account balance is positive, indicating negative foreign savings (i.e., capital outflow). This leads to a negative number in the 'INV'-'EXT' cell.¹²
- (4) The row-sum and column-sum equality rule also helps us to estimate the cells related to indirect taxes such as production taxes and import tariffs. Their total revenues are in the 'GOV'-'IDT' and 'GOV'-'TRF' cells, which indicate transfers from these indirect tax agents to the government.

Using these methods, we obtain Table 4.5.B. The last task is to fill in the remaining blank cells, indicating the flow of funds from the household to the government and to the investment agent, and that from the

Table 4.5.B Japan's three-sector SAM for 2000 – filling cells using the row-sum and column-sum equality rule

	AGR	MAN	SRV	CAP	LAB	IDT
AGR	1,558,469	8,427,693	1,496,991			
MAN	2,462,949	130,321,011	68,646,492			
SRV	2,273,437	63,503,715	160,713,811			
CAP	6,167,952	33,816,669	149,889,160			
LAB	1,372,650	59,034,654	234,353,029			
IDT	534,232	14,436,136	19,877,410			
TRF	144,478	3,690,547	911			
HOH				189,873,781	294,760,333	
GOV						34,847,778
INV						
EXT	1,974,129	39,254,377	9,096,735			
Total	16,488,296	352,484,802	644,074,539	189,873,781	294,760,333	34,847,778
	TRF	HOH	GOV	INV	EXT	Total
AGR		3,965,927	0	967,198	72,018	16,488,296
MAN		64,927,161	459,179	39,070,695	46,597,315	352,484,802
SRV		231,268,309	85,247,038	90,250,845	10,817,384	644,074,539
CAP						189,873,781
LAB						294,760,333
IDT						34,847,778
TRF						3,835,936
HOH						484,634,114
GOV	3,835,936					38,683,714
INV					-7,161,476	-7,161,476
EXT						50,325,241
Total	3,835,936	300,161,397	85,706,217	130,288,738	50,325,241	

Unit: million yen

government to the investment agent. As these cells cannot be filled in only by exploiting the row-sum and column-sum equality rule, we have to rely on other data. For example, based on prior information about the direct tax revenue (44,207,200 million yen) reported by 'the general government in the income and outlay accounts classified by institutional sectors' in the national account tables, we can fill in the 'GOV'-'HOH' cell. Once the 'GOV'-'HOH' cell is determined, the other two cells can be filled in by using the row-sum and column-sum equality rule. Finally, we obtain the SAM as shown in Table 4.5.C, where all the accounts are balanced.¹³

Table 4.5.C Japan's three-sector SAM for 2000 – completed

	AGR	MAN	SRV	CAP	LAB	IDT
AGR	1,558,469	8,427,693	1,496,991			
MAN	2,462,949	130,321,011	68,646,492			
SRV	2,273,437	63,503,715	160,713,811			
CAP	6,167,952	33,816,669	149,889,160			
LAB	1,372,650	59,034,654	234,353,029			
IDT	534,232	14,436,136	19,877,410			
TRF	144,478	3,690,547	911			
HOH				189,873,781	294,760,333	
GOV						34,847,778
INV						
EXT	1,974,129	39,254,377	9,096,735			
Total	16,488,296	352,484,802	644,074,539	189,873,781	294,760,333	34,847,778
	TRF	HOH	GOV	INV	EXT	Total
AGR		3,965,927	0	967,198	72,018	16,488,296
MAN		64,927,161	459,179	39,070,695	46,597,315	352,484,802
SRV		231,268,309	85,247,038	90,250,845	10,817,384	644,074,539
CAP						189,873,781
LAB						294,760,333
IDT						34,847,778
TRF						3,835,936
HOH						484,634,114
GOV	3,835,936	44,207,200				82,890,914
INV		140,265,517	-2,815,303		-7,161,476	130,288,738
EXT						50,325,241
Total	3,835,936	484,634,114	82,890,914	130,288,738	50,325,241	

Unit: million yen

4.4 Consistency among various databases and matrix adjustment for a SAM

The row-sum and column-sum equality is required in the SAM because the SAM (or the flow-of-funds table) is supposed to describe an ex post equilibrium status of an economy, where all the markets are cleared, and receipts and payments of funds are balanced for every agent. However, in dealing with actual data to construct a SAM, we encounter two kinds of statistical problems. The first problem is discrepancies among data derived from different sources. For example, the amount of transactions with foreign agents reported in the IO tables could be different from that reported in the balance of payments tables.

The second problem is related to the timing of the construction of the IO tables. As it takes considerable time and cost, the IO tables cannot be prepared frequently. For example, the Japanese IO tables are constructed only once every five years.¹⁴ Therefore, if we want to develop a model for a more recent year for which no IO tables are available, we must update the SAM constructed using the old IO tables by replacing the old values in the original SAM with more recent data. Replacements of the original data, however, would make the SAM unbalanced.

An unbalanced SAM, whatever the cause, cannot be used for CGE modelling. We have to adjust the data cells in the unbalanced SAM to achieve equality of the row sums with their corresponding column sums. While there are various options for this data adjustment process, the following constrained matrix problem is commonly suggested:¹⁵

$$\underset{x_{i,j}}{\text{minimize}} \quad \varpi = \sum_i \sum_j \left(\frac{x_{i,j} - x_{i,j}^0}{x_{i,j}^0} \right)^2$$

subject to

$$\sum_j x_{i,j} = \sum_j x_{j,i} \quad \forall i$$

where:

i, j : row and column labels in the SAM,

ϖ : objective value,

$x_{i,j}$: value in i -th row and the j -th column in the adjusted SAM,

$x_{i,j}^0$: value in i -th row and the j -th column in the original SAM (given).

The objective function in this minimization problem is the sum of the square of 'distance' ϖ , defined by the sum of the deviation rate of an adjusted cell value from the original one, which is analogous to the one used in econometric estimation with the (weighted) least squares method. In the constraint of the minimization problem above, the left-hand side represents the sum of the i -th row; the right-hand side represents the sum of the i -th column. That is, this constraint requires that the row sum in the adjusted SAM be equal to its column sum, to ensure the above-mentioned basic property of the SAM. Subject to this row-sum and column-sum equality, we can construct a new matrix that deviates the least from the original one.

Incidentally, in the above objective function, the distance between the original cell value and the adjusted one is normalized by the original cell value $x_{i,j}^0$. It is highly probable that some cells are zero in the SAM.

In such a case, a term such as $(x_{i,j} - x_{i,j}^0) / x_{i,j}^0$ has to be ignored for our computation; the estimated $x_{i,j}$ should be set at zero.

Depending on the availability of empirical data, we can add other constraints to this constrained matrix problem to increase the accuracy of our matrix estimates. If the total value of exports is available – even though the value of exports for each sector is not available – we can add a new constraint to bind the total value of export cells. We can also constrain the signs of certain cell entries if they are supposed to be positive or negative. For example, it is reasonable to expect that the cells in the adjusted and old SAMs should have the same signs; thus, such constraints can be imposed. However, particular care is needed for investment. The investment consists of fixed capital formation and changes in stocks. Although the former is always positive, the latter can be either positive or negative. As a result, when the negative impact of the changes in stocks dominates, the cell indicating investment is negative. The sign of investment may change over time.

These matrix adjustment problems generally have two arguable issues. First, the solution of this optimization problem satisfies only the accounting principle that the total receipts of funds must be equal to the total payments of funds for each agent (represented as the row-sum and column-sum equality). In essence, the adjustment process involves simply dividing statistical discrepancies proportionately among the cells in the SAM. This adjustment is a mere mathematical adjustment, which is not well grounded in any economic theory.¹⁶

Second, the significance of matrix adjustments is sometimes dubious. The original IO tables must have been constructed with the help of matrix adjustment methods. The matrix adjustment stated above involves simply adding another round of matrix adjustment to the already adjusted IO tables for the sake of row-sum and column-sum equality. Such costly reconstruction of the SAM may not seem beneficial. Instead, it might be productive for us to spend more time conducting sensitivity analysis with respect to the critical parameters assumed in the model to examine the robustness of simulation results.¹⁷

Considering these two problems, we can recommend a simple alternative method for updating a SAM. This method is applicable, and does not require complicated matrix adjustments, if an old SAM is available and balanced. That is, it entails the scaling up (or down) of all the cells in the original SAM in proportion to the changes in some representative macroeconomic indicator, such as GDP, between the base year of the original SAM and the target year of the new SAM. As this scaling up

(or down) method preserves the basic structure of the original SAM, we can easily compute the new matrix and do not have to take care to trace changes made in this database reconstruction process.

Notes

1. Note that the terms of agents here may sound different from those in Chapters 1 and 2, but both are closely related.
2. The agents discussed here – for example, HOH in Final Demand, and BRD and MLK in Activity – correspond to a household and firms in Chapter 2. When we follow the setup of an economy shown in Chapter 6, the government, the investment agent and the external sector are added.
3. Needless to say, although the unit of measurement is Japanese yen in this example, we may construct SAMs in any currency units.
4. Table 4.2 indicates that the same good is exported and imported simultaneously. This seems a curious situation. We will discuss how to deal with this issue using the Armington assumption in Section 6.5.
5. Strictly speaking, as the model developed in Chapter 6 is a static model, we cannot interpret these deficits as savings. We discuss this matter in connection with the macro closure issues in Chapter 7.
6. The following process can be completed using spreadsheet software, such as Excel. After completing the construction of a SAM, we put it into a GAMS program as shown in List 6.1 in Section 6.8.
7. If there are other agents eligible to earn income from their endowed factors, the above calculation requires modification. For example, if there is a state-owned enterprise, its profits (i.e., capital income) are captured by the government. In this case, we have to find data of such profits in the government budget statement, place them in the 'GOV'–'CAP' cell first, and then allocate the residual capital income to the household.
8. Although Japan's IO tables are used as an example to explain how to construct a SAM from IO tables, the same technique explained here applies to any other countries' IO tables with minimal adjustment.
9. In 2009, the IO tables for 2005 were published by the Ministry of Internal Affairs and Communications.
10. They are provided online by the Statistics Bureau of the Ministry of Internal Affairs and Communications (www.stat.go.jp) and the Economic and Social Research Institute, Cabinet Office (www.esri.go.jp).
11. The 'consumption expenditure outside households (row)' consists of entertainment and travel expenses (accommodation and subsistence) for business purposes.
12. Instead of entering a negative number in the 'INV'–'EXT' cell, we can enter a positive number in the 'EXT'–'INV' cell. There is no significant difference between these two options.
13. By using the SAM constructed here and installing it into the input file shown in List 6.1 with slight modifications, we can develop a CGE model for Japan.
14. The Statistics Bureau of the Ministry of Internal Affairs and Communications constructs Japan's IO tables once every five years jointly with other ministries.

The Ministry of Economy, Trade and Industry constructs IO tables every year, which are known as the 'extended input-output tables', by reconstructing the above-mentioned original IO tables in combination with more recently available data.

15. For further discussions about constrained matrix problems, see Nagurney and Eydeland (1992).
16. In addition to the constrained matrix problem, we can use iterative adjustment methods such as the RAS methods. However, the RAS methods also have the same problem as stated above. We cannot determine what method yields the best estimate of a SAM.
17. Sensitivity analysis is discussed in Section 8.3.

5

Calibration and Computational Strategy for General Equilibrium

Chapter 4 detailed the construction of a social accounting matrix (SAM). This chapter is devoted to explaining the method of estimation of coefficients and exogenous variables in the CGE model, based on the SAM. In estimating these, we cannot apply standard econometric methods. This is because we usually have a limited number of observations but a large number of coefficients and exogenous variables to estimate in the CGE model. This leads to insufficient degrees of freedom if econometric methods are applied.¹ To overcome this problem, we employ a unique estimation method called ‘calibration’.

Sections 5.1 and 5.2 discuss the concept and data requirements of calibration. Section 5.3 presents mathematical formulae for calibration, taking the ‘simple CGE model’ developed in Chapter 2 as an example. In Section 5.4, also using the simple CGE model as an example, we develop a GAMS program, which covers the whole process from installing the SAM, through automatically calibrating coefficients and exogenous variables from the installed SAM, to solving the CGE model with the calibrated coefficients and exogenous variables. The input file of this model written in GAMS is shown. In Section 5.5, the output file of the example CGE model stated above is presented. In sum, reading through Section 5.5, readers will understand the programming required for the simple CGE model shown in Chapter 2.

5.1 The basic concept of calibration

Calibration involves ‘solving’ unknown parameters in the model system by pinning down the endogenous variables at the values in some known equilibrium observed in the SAM. (Let us call the equilibrium depicted by

the SAM the initial equilibrium, as opposed to the base run equilibrium, explained later.)

Suppose that a CGE model is a system of simultaneous equations (expressed in vector form):²

$$\text{CGE}(\mathbf{x}, \mathbf{y}, \mathbf{a}) = \mathbf{0}$$

where $\mathbf{x}$ denotes the endogenous variable vector, $\mathbf{y}$ the exogenous variable vector, and $\mathbf{a}$ the coefficient vector. A common practice is to solve the model system $\text{CGE}(\cdot)$ for the (unknown) endogenous variable vector $\mathbf{x}$, given $\mathbf{y}$ and $\mathbf{a}$.³ In calibration, however, given the exogenous variable vector $\mathbf{y}$ and the model system $\text{CGE}(\cdot)$, we solve the coefficient vector $\mathbf{a}$ instead of $\mathbf{x}$. We denote the initial equilibrium value of $\mathbf{x}$ as $\mathbf{x}^0$, which must be a solution of the equation above by definition (if there is no shock given to $\mathbf{y}$ and $\mathbf{a}$). The following equation holds:

$$\text{CGE}(\mathbf{x}^0, \mathbf{y}, \mathbf{a}) = \mathbf{0}$$

As the initial equilibrium value vector $\mathbf{x}^0$ and the exogenous variable vector $\mathbf{y}$ can be derived from the SAM, we can solve the unknown coefficient vector $\mathbf{a}$, given the exogenous variable vector $\mathbf{y}$ and the model system $\text{CGE}(\cdot)$.

This estimation method is known as calibration.⁴ Using calibration, we estimate values of the coefficients and exogenous variables based on one observation of an economy in equilibrium; thus, it does not allow us to test the calibrated values statistically. There is another possible problem that this technique cannot be adopted if the number of constants in $\mathbf{a}$ exceeds the number of equations in the model system. In this case, we have to reduce the number of unknowns by assuming values for some of them with a priori information. This issue is discussed in Subsection 6.8.2.

A simple illustration may help readers to understand the essence of the calibration method. Assume a supply function (although such a functional form is not used for supply functions in the CGE models) as follows:

$$S = aP$$

where S denotes the supply, P denotes its price, and a denotes the coefficient to be estimated. Following these notations, let the variables with the superscript 0 denote their initial equilibrium values. Suppose that the initial equilibrium values of these quantity and price variables are

known; then the coefficient can be calibrated as follows:

$$a = \frac{S^0}{P^0}$$

The next point regarding this process is whether we can obtain information about quantity and price separately, while the SAM reports only value figures. We need a method for separating them, which is discussed in the next section.

5.2 Value, price and quantity

Almost all data needed in calibration are retrieved from the SAM. As data in the SAM are expressed only in terms of values, we have to separate the value data of the SAM into quantity data and price data for the purpose of calibration. Consider the following self-evident equation:

$$\text{Value} = \text{Price} \times \text{Quantity}$$

If price data are available, quantity data can be extracted immediately. However, we cannot collect price data for all the sectors that the SAM covers.

Consider this issue from another viewpoint. Price is the unit value of goods, while conventional units are defined in various ways, such as yen per 100g, US dollars per car, and so on. In place of these conventional units of measurement, we define a new physical unit that is worth one yen or dollar. The physical unit worth one yen differs by good, such as 20ml of orange juice, a one two-thousandth piece of pizza, or 1.5g of rice. Redefining the physical unit of goods in this manner, all the prices become unity. By setting all the prices at unity, the following holds:

$$\text{Value} = 1 \times \text{Quantity}$$

Given the value data of the SAM, we can immediately obtain quantity data.⁵ In other words, values in the SAM can be now considered as quantity figures.

To understand better the redefinition of the physical units, consider the following example. Suppose that a packet containing 200g of rice is priced at 100 yen. The price of a packet of rice is, of course, 100 yen. Because rice can be traded in packets of different size, we can use a packet size of 2g of rice. The price of this new packet is one yen (ignoring the

cost of the packet itself and miscellaneous costs for repacking, etc.). It is obvious that whatever packet size is used in trade, actual economic activity is not affected at all. Adjusting the packet size of rice, we can set the price of a packet of rice at unity. While the price does not have to be unity, we choose this for computational simplicity.

A little attention is needed when indirect taxes are incorporated in the model. We have to modify the above-mentioned process slightly. Having computed the indirect tax rates beforehand, the equation becomes:

$$\text{Tax-inclusive Value} = (1 + \text{Indirect Tax Rate}) \times \text{Tax-exclusive Price} \\ \times \text{Quantity}$$

When the tax-exclusive price is set at unity using the same logic, we can compute the quantity.⁶

5.3 Calibration procedure – mathematical manipulation

5.3.1 The case without indirect taxes

To demonstrate the calibration process, we use the simple CGE model consisting of (2.1)–(2.6) shown in Chapter 2. There are three coefficients and an exogenous variable to be estimated: the expenditure share coefficient in the utility function α_i , the input share coefficient in the production function $\beta_{h,j}$, the scaling coefficient in the production function b_j , and factor endowments FF_h .

As discussed in Chapter 4, these coefficients and exogenous variables must be estimated by exploiting data in the SAM in Table 4.1, which is presented again in Table 5.1. Symbols with superscripts 0, for example X_i^0 , denote the initial equilibrium values of the corresponding endogenous variables such as X_i . These initial equilibrium values are used for calibration purposes and also for computation of equilibria, as we explain later.

Estimating expenditure share coefficients in the utility function

Let us first estimate the expenditure share coefficient in the utility function α_i . Recall the utility maximization problem consisting of (2.a) and (2.b) shown in Section 2.2. Its first-order conditions lead to the demand function:

$$X_i = \frac{\alpha_i}{p_i^x} \sum_h p_h^f FF_h \quad \forall i \quad (2.1)$$

Table 5.1 SAM for the simple CGE model in Chapter 2

	Activity		Factor		Final Demand	Total
	BRD	MLK	CAP	LAB	HOH	
Activity	BRD	MLK			15	15
Factor	CAP	5	20		35	35
	LAB	10	15			25
Final Demand	HOH		25	25		50
Total	15	35	25	25	50	

i, j h, k $p_h^{f0} F_{h,j}^0$ $p_h^{f0} FF_h$ $p_i^{x0} X_i^0$

Unit: yen

As this equation is also supposed to hold in the initial equilibrium, we also have:

$$X_i^0 = \frac{\alpha_i}{p_i^{x0}} \sum_h p_h^{f0} FF_h \quad \forall i$$

Rearranging it for α_i , we get:

$$\alpha_i = \frac{p_i^{x0} X_i^0}{\sum_h p_h^{f0} FF_h} \quad \forall i$$

Replacing the denominator on the right-hand side with total expenditure using the household budget constraint (2.b) evaluated at the initial equilibrium as follows:

$$\sum_i p_i^{x0} X_i^0 = \sum_h p_h^{f0} FF_h$$

we get:⁷

$$\alpha_i = \frac{p_i^{x0} X_i^0}{\sum_j p_j^{x0} X_j^0} \quad (5.1)$$

As the initial equilibrium values of all the endogenous variables on the right-hand side in (5.1) can be retrieved from the 'Activity'-'HOH' cells in the SAM (Table 5.1), we can calibrate the expenditure share coefficient α_i .

We illustrate how to calibrate numerically the expenditure share coefficients for bread α_{BRD} and milk α_{MLK} using Table 5.1. Because all the

prices are set at unity as discussed in Section 5.2, the initial equilibrium prices for household consumption p_{BRD}^{x0} and p_{MLK}^{x0} can be set at unity. With these prices, we can immediately obtain the quantities of bread and milk consumed; i.e., X_{BRD}^0 and X_{MLK}^0 from the values in 'BRD'-'HOH' and 'MLK'-'HOH'. This process is written as follows:

$$\alpha_{BRD} = \frac{1 \cdot 15}{1 \cdot 15 + 1 \cdot 35} = \frac{15}{50} = 0.300$$

$$\alpha_{MLK} = \frac{1 \cdot 35}{1 \cdot 15 + 1 \cdot 35} = \frac{35}{50} = 0.700$$

Estimating input share and scaling coefficients in the production function

Next, let us estimate the unknown coefficients in the Cobb–Douglas-type production function $\beta_{h,j}$ and b_j . Recall the production function (2.2) and the factor demand function (2.3):

$$Z_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j \quad (2.2)$$

$$F_{h,j} = \frac{\beta_{h,j}}{p_h^f} p_j^z Z_j \quad \forall h, j \quad (2.3)$$

These equations contain endogenous variables in the model (Z_j , $F_{h,j}$, p_j^z , and p_h^f) and coefficients ($\beta_{h,j}$ and b_j). As the initial equilibrium values for these endogenous variables can be retrieved from the SAM, this system:

$$Z_j^0 = b_j \prod_h F_{h,j}^{0\beta_{h,j}} \quad \forall j$$

$$F_{h,j}^0 = \frac{\beta_{h,j}}{p_h^{f0}} p_j^{z0} Z_j^0 \quad \forall h, j$$

can be regarded as a system of simultaneous equations with unknown coefficients $\beta_{h,j}$ and b_j . Calibrating these coefficients means solving this system with respect to these unknown coefficients.

The calibration process will proceed as follows. Rearranging the factor demand function (2.3) with respect to $\beta_{h,j}$ and evaluating the function at the initial equilibrium, we get:

$$\beta_{h,j} = \frac{p_h^{f0} F_{h,j}^0}{p_j^{z0} Z_j^0} \quad \forall h, j$$

The value of $p_j^{z0} Z_j^0$ is not explicitly shown in the SAM. However, considering the zero-profit condition (2.7) also evaluated at the initial equilibrium:

$$p_j^{z0} Z_j^0 = \sum_h p_h^{f0} F_{h,j}^0 \quad \forall j \quad (5.2)$$

we can replace the denominator of that equation with $\sum_h p_h^{f0} F_{h,j}^0$ to obtain:

$$\beta_{h,j} = \frac{p_h^{f0} F_{h,j}^0}{\sum_k p_k^{f0} F_{k,j}^0} \quad \forall h, j \quad (5.3)$$

As p_h^{f0} can be set at unity, and $F_{h,j}^0$ can be retrieved from the 'Factor'-'Activity' cells in Table 5.1, we can calibrate $\beta_{h,j}$ using (5.3).

After estimating the input share coefficient $\beta_{h,j}$ by (5.3), we can estimate the scaling coefficient b_j . From the production function (2.2), we immediately obtain:

$$b_j = \frac{Z_j^0}{\prod_h F_{h,j}^{\beta_{h,j}}} \quad \forall j \quad (5.4)$$

Using the initial equilibrium values $F_{h,j}^0$ in the 'Factor'-'Activity' cells of the SAM, Z_j^0 estimated by (5.2) and $\beta_{h,j}$ calibrated by (5.3), we can now calibrate b_j from (5.4). A little attention to the order of the calibration process of $\beta_{h,j}$ and b_j is needed. As the input share coefficient $\beta_{h,j}$ is required in (5.4), $\beta_{h,j}$ must be estimated prior to the estimation of b_j .

The following is a numerical calibration exercise for the input share coefficient of capital for production in the bread sector $\beta_{CAP,BRD}$. Starting from the initial equilibrium values on the right-hand side of (5.3) shown in Table 5.1, we set all the prices p_{CAP}^{f0} and p_{LAB}^{f0} at unity and replace $F_{CAP,BRD}^0$ and $F_{LAB,BRD}^0$ with the values in the 'CAP'-'BRD' and 'LAB'-'BRD' cells. The input share coefficient for capital $\beta_{CAP,BRD}$ is computed as follows:

$$\beta_{CAP,BRD} = \frac{1 \cdot 5}{1 \cdot 5 + 1 \cdot 10} = \frac{1}{3} = 0.333$$

Similarly, we can compute the input share coefficient for labour for this sector $\beta_{LAB,BRD}$ as follows:

$$\beta_{LAB,BRD} = \frac{1 \cdot 10}{1 \cdot 5 + 1 \cdot 10} = \frac{2}{3} = 0.667$$

In the same way, we can compute the input share coefficients for the milk sector as follows:

$$\beta_{CAP,MLK} = \frac{1 \cdot 20}{1 \cdot 20 + 1 \cdot 15} = \frac{4}{7} = 0.571$$

$$\beta_{LAB,MLK} = \frac{1 \cdot 15}{1 \cdot 20 + 1 \cdot 15} = \frac{3}{7} = 0.429$$

The next step is estimation of the scaling coefficient b_j . Z_{BRD}^0 is needed in the calibration process (5.4) for b_j but is not explicitly shown in the SAM as discussed above. Using the zero-profit condition (5.2), we get:

$$Z_{BRD}^0 = \frac{\sum_h p_h^{f0} F_{h,BRD}^0}{p_{BRD}^{z0}} = \frac{1 \cdot 5 + 1 \cdot 10}{1} = 15$$

Inserting the figures of $F_{CAP,BRD}^0$ and $F_{LAB,BRD}^0$ as well as the estimated coefficients $\beta_{CAP,BRD}$ and $\beta_{LAB,BRD}$ into (5.4), we can estimate the scaling coefficient for the bread sector as follows:

$$b_{BRD} = \frac{15}{5^{1/3} \cdot 10^{2/3}} = 1.890$$

In the same manner, we can compute the initial equilibrium value of milk production Z_{MLK}^0 using the zero-profit condition (5.2) as follows:

$$Z_{MLK}^0 = \frac{\sum_h p_h^{f0} F_{h,MLK}^0}{p_{MLK}^{z0}} = \frac{1 \cdot 20 + 1 \cdot 15}{1} = 35$$

With this estimated Z_{MLK}^0 in combination with the calibrated share coefficients $\beta_{CAP,MLK}$ and $\beta_{LAB,MLK}$, we can finally calibrate the scaling coefficient for the milk sector b_{MLK} as follows:

$$b_{MLK} = \frac{35}{20^{4/7} \cdot 15^{3/7}} = 1.980$$

Estimating factor endowments

The factor endowments are exogenous variables, which can be retrieved directly from the SAM in Table 5.1. The endowed capital FF_{CAP} is in the 'HOH'-'CAP' cell, 25 units. In the same manner, the endowed labour FF_{LAB} is in the 'HOH'-'LAB' cell, 25 units.

Instead of following the above-stated procedures, we can alternatively compute them using the market-clearing condition for the factors (2.5) evaluated at the initial equilibrium as follows:

$$FF_h = \sum_j F_{h,j}^0 \quad \forall h$$

As the uses of factors by sectors $F_{h,j}^0$ are in the 'Factor'-'Activity' cells of the SAM, the endowments of capital and labour are:

$$FF_{CAP} = 5 + 20 = 25$$

$$FF_{LAB} = 10 + 15 = 25$$

The values of the factor endowments computed by the latter method are consistent with those computed by the former method.

5.3.2 The case with indirect taxes

When we incorporate indirect taxes in a CGE model, how do we have to modify the model and its calibration procedures? Temporarily deviating from our present objective of developing the simple CGE model, we extend the model by incorporating an ad valorem indirect tax on the production of the i -th good with a tax rate τ_i^z . It is also assumed that all its tax revenue is refunded to the household in a lump-sum fashion; no government consumption is assumed.

To incorporate an indirect tax in the above-mentioned manner, we have to make two modifications to the original model. The first modification is to modify the household budget constraint by adding the second term on the right-hand side, such that the lump-sum tax is written as follows:

$$\sum_i p_i^x X_i = \sum_h p_h^f FF_h + \sum_j \tau_j^z p_j^z Z_j \quad (2.b')$$

The resulting demand function should be (2.1'), shown below. The second modification is to modify the price equalization condition of the market-clearing conditions (2.6) in such a way that there is a gap between the supply price p_i^z and the demand price p_i^x , caused by the indirect tax rate τ_i^z . The new price equalization condition is (2.6').

The new model system becomes:

$$X_i = \frac{\alpha_i}{p_i^x} \left(\sum_h p_h^f FF_h + \sum_j \tau_j^z p_j^z Z_j \right) \quad \forall i \quad (2.1')$$

$$Z_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j \quad (2.2)$$

$$F_{h,j} = \frac{\beta_{h,j}}{p_h^z} p_j^z Z_j \quad \forall h, j \quad (2.3)$$

$$X_i = Z_i \quad \forall i \quad (2.4)$$

$$\sum_j F_{h,j} = F F_h \quad \forall h \quad (2.5)$$

$$(1 + \tau_i^z) p_i^z = p_i^x \quad \forall i \quad (2.6')$$

The calibration process in the case of an indirect tax is identical to that in the case without it. However, care must be taken in dealing with the calibration formula that includes goods prices. Because the indirect tax creates a gap between the demand and the supply prices, we cannot set both prices at unity. Recall the formula decomposing values into prices and quantities presented in Section 5.2:

$$\begin{aligned} \text{Tax-inclusive Value} &= (1 + \text{Indirect Tax Rate}) \times \text{Tax-exclusive Price} \\ &\times \text{Quantity} \end{aligned}$$

In this indirect-tax model, we set the supply price of the i -th goods for the household p_i^{z0} at unity; thus, the corresponding demand price p_i^{x0} must be $1 + \tau_i^z$, rather than unity, as (2.6') suggests. The demand price p_i^{x0} appears only in (5.1) of the calibration formulae presented above; thus, (5.1) has to be modified as follows:⁸

$$\alpha_i = \frac{p_i^{x0} X_i^0}{\sum_j p_j^{x0} X_j^0} = \frac{(1 + \tau_i^z) p_i^{z0} X_i^0}{\sum_j (1 + \tau_j^z) p_j^{z0} X_j^0} = \frac{(1 + \tau_i^z) \cdot 1 \cdot X_i^0}{\sum_j (1 + \tau_j^z) \cdot 1 \cdot X_j^0} \quad \forall i$$

5.4 GAMS programming

5.4.1 Programming procedures and an input file

When the number of coefficients and exogenous variables to be calibrated is limited, we can easily compute them manually with a calculator and put their estimates in the input file. However, as the model size increases, the calibration procedures become too complicated to do manually. We necessarily rely on a PC for calibration. In this section, we develop a program that covers the entire modelling process from installing the SAM in an input file, through automatically calibrating coefficients and exogenous variables from the installed SAM, to solving

the CGE model with the calibrated coefficients and exogenous variables. The input file of this program is explained in this section, and the output file in the next section.

Now that we understand that the model with indirect taxes can be constructed by making slight modifications to the model without indirect taxes, we deal only with the latter model in this section, by taking the simple CGE model as an example. Readers are expected to build the model with indirect taxes, following the explanation given in this section.

Programming usually occurs in the following manner:

- (1) Declaration and definition of sets
- (2) Installation of the SAM, together with assumed key elasticity parameters⁹
- (3) Retrieval of the data from the SAM
- (4) Calibration
- (5) Specifying and solving the CGE model
- (6) Aftermath computation for various useful indicators.

Among the above six steps, the matters related to the SAM and calibration such as (2), (3) and (4) are new, compared with the 'household utility maximization model' in Chapter 3. There are also some new matters in the specifying and solving step (5). The following explanation focuses on these new matters. Aftermath computation is discussed in Section 8.2 in detail.

In the present programming exercise, we use the SAM originally shown in Table 4.1 in Chapter 4 and also in Table 5.1 in Section 5.3. That SAM is displayed again as Table 5.1 in this section for readers' convenience.

Table 5.1 SAM for the simple CGE model in Chapter 2 (*redisplayed*)

	Activity		Factor		Final Demand	Total
	BRD	MLK	CAP	LAB	HOH	
Activity	BRD				15	15
	MLK				35	35
Factor	CAP	5	20			25
	LAB	10	15			25
Final Demand	HOH		25	25		50
Total	15	35	25	25	50	

i, j

h, k

$p_h^{f0} F_{h,j}^0$

$p_h^{f0} F F_h$

$p_i^{x0} X_i^0$

Unit: yen

The GAMS input file for the simple CGE model is shown in List 5.1 (splcge.gms). The line numbers, which appear in the first column of List 5.1, are included only for convenience in our explanation of the program and must not be typed in the actual input files.¹⁰

List 5.1 Input file for the simple CGE model in Chapter 2 (splcge.gms)

```

1 $ Title A Simple CGE Model in Ch. 5 (SPLCGE,SEQ=275)
2
3 * Definition of sets for suffix -----
4 Set      u          SAM entry      /BRD, MLK, CAP, LAB, HOH/
5         i(u)        goods          /BRD, MLK/
6         h(u)        factor         /CAP, LAB/;
7 Alias (u,v), (i,j), (h,k);
8 * -----
9
10 * Loading data -----
11 Table    SAM(u,v)          social accounting matrix
12         BRD      MLK      CAP      LAB      HOH
13 BRD
14 MLK
15 CAP      5          20
16 LAB      10         15
17 HOH
18
19
20
21 * Loading the initial values -----
22 Parameter X0(i)            household consumption of the i-th good
23           F0(h,j)          the h-th factor input by the j-th firm
24           Z0(j)            output of the j-th good
25           FF(h)            factor endowment of the h-th factor
26
27
28 X0(i) =SAM(i,"HOH");
29 F0(h,j)=SAM(h,j);
30 Z0(j) =sum(h, F0(h,j));
31 FF(h) =SAM("HOH",h);
32 Display X0, F0, Z0, FF;
33 * Calibration -----
34
35 Parameters      alpha(i)      share parameter in utility function
36                 beta(h,j)     share parameter in production function
37                 b(j)          scale parameter in production function
38
39 alpha(i)=X0(i)/sum(j, X0(j));
40 beta(h,j)=F0(h,j)/sum(k, F0(k,j));
41 b(j) =Z0(j)/prod(h, F0(h,j)**beta(h,j));
42 Display alpha, beta, b;
43 * -----
44
45 * Defining model system -----
46 Variable        X(i)          household consumption of the i-th good
47                 F(h,j)        the h-th factor input by the j-th firm
48                 Z(j)          output of the j-th good
49                 px(i)         demand price of the i-th good
50                 pz(j)         supply price of the i-th good
51                 pf(h)         the h-th factor price
52
53                 UU            utility [fictitious]
54
55

```

List 5.1 (Continued)

```

56 Equation      eqX(i)      household demand function
57              eqpz(i)      production function
58              eqF(h,j)     factor demand function
59              eqpx(i)      good market clearing condition
60              eqpf(h)      factor market clearing condition
61              eqZ(i)       price equation
62
63              obj          utility function [fictitious]
64 ;
65
66 eqX(i) ..      X(i)      =e= alpha(i)*sum(h, pf(h)*FF(h))/px(i);
67 eqpz(j) ..      Z(j)      =e= b(j)*prod(h, F(h,j)**beta(h,j));
68 eqF(h,j) ..      F(h,j)   =e= beta(h,j)*pz(j)*Z(j)/pf(h);
69 eqpx(i) ..      X(i)      =e= Z(i);
70 eqpf(h) ..      sum(j, F(h,j)) =e= FF(h);
71 eqZ(i) ..      px(i)      =e= pz(i);
72
73 obj ..          UU        =e= prod(i, X(i)**alpha(i));
74 * -----
75
76 * Initializing variables -----
77 X.l(i) =X0(i);
78 F.l(h,j)=F0(h,j);
79 Z.l(j) =Z0(j);
80 px.l(i) =1;
81 pz.l(j) =1;
82 pf.l(h) =1;
83 * -----
84
85 * Setting lower bounds to avoid division by zero -----
86 X.lo(i) =0.001;
87 F.lo(h,j)=0.001;
88 Z.lo(j) =0.001;
89 px.lo(i)=0.001;
90 pz.lo(j)=0.001;
91 pf.lo(h)=0.001;
92 * -----
93 pf.fx("LAB")=1;
94
95 * Defining and solving the model -----
96 Model splcge /all/;
97 Solve splcge maximizing UU using nlp;
98 * -----
99 * end of model -----
100 * -----

```

5.4.2 Declaration and definition of sets

Like the program for the household utility maximization problem, the input file usually starts with a declaration and definition of sets. Lines 4–7 declare and define the sets used in the model. As the model is now going to encompass the SAM, we need a set which includes the row and column labels of the SAM as its elements. For such a set, we define a (universal) set U . The Set directive in line 4 can be interpreted as a

Table 5.2 Abbreviations used in the SAM and program

	Abbreviations
Bread	BRD
Milk	MLK
Capital	CAP
Labour	LAB
Household	HOH

definition of set U :

$$u \in U = \{BRD, MLK, CAP, LAB, HOH\}$$

Line 7 reads that `Alias (u, v)` declares a new set V , which is made by duplicating U . In the statement `Alias (u, v)`, the indices v and u can be used interchangeably. Incidentally, ‘alias’ means ‘otherwise called’. We discuss later why the `Alias` directive is needed in this program.

Returning to lines 5–7, these lines define the sets I for goods and H for factors as the subsets of U . To indicate that these subsets belong to the universal set U , we attach the index ‘(u)’ after ‘i’ or ‘h’; e.g., ‘i(u)’ or ‘h(u)’. As the statement ‘`Alias (u, v), (i, j), (h, k);`’ appears in line 7, the indices j and k are identical to the indices i and h respectively. Lines 5–7 can be interpreted as follows:

$$i, j \in I = \{BRD, MLK\} \subset U$$
$$h, k \in H = \{CAP, LAB\} \subset U$$

A universal set and its subsets might look complicated but can simplify the data-handling process such as installing the SAM (lines 11–18) and retrieving data from the SAM (lines 28–31).

5.4.3 Installation of the SAM

Lines 11–18 declare the matrix of constants in the SAM denoted by ‘`SAM(u, v)`’ and define its elements. The `Table` directive indicates that the data should appear in matrix form; in other words, they should appear exactly as they do in the SAM. It should be noted that the order of the indices in ‘`SAM(u, v)`’ has significance. The first index ‘ u ’ corresponds to the row labels of the SAM (which appear in the leftmost column in

the SAM), while the second index ‘v’ corresponds to the column labels (which appear in the top row). When we use the `Table` directive, as explained in detail in Subsection 3.3.2, we have to place values correctly below their column labels (not spread values below two or more column labels) to avoid confusion in the data input. The mathematical expression corresponding to lines 11–18 is:

$$\mathbf{SAM} = \{SAM_{u,v}\} = \begin{bmatrix} 0 & 0 & 0 & 0 & 15 \\ 0 & 0 & 0 & 0 & 35 \\ 5 & 20 & 0 & 0 & 0 \\ 10 & 15 & 0 & 0 & 0 \\ 0 & 0 & 25 & 25 & 0 \end{bmatrix}$$

(The row and column labels in the SAM are omitted above.)

The usefulness of the `Alias` directive is important to understand. The row labels and the column labels of $\{SAM_{u,v}\}$ are identical. Even so, the same index cannot be assigned to one matrix such as $SAM_{u,u}$. If we do so, the matrix is interpreted as a diagonal matrix with elements: $SAM_{BRD,BRD}$, $SAM_{MLK,MLK}$, ..., $SAM_{HOH,HOH}$. Therefore, different indices must be used there; the `Alias` directive works. Another useful property of the `Alias` directive is explained later.

5.4.4 Retrieval of data from the SAM

Lines 22–31 define the initial equilibrium values of the endogenous variables as well as those of the exogenous variables to be retrieved from the SAM. The objective of this retrieval is twofold. The first is to check whether the retrieved data can reproduce the initial equilibrium values that the SAM depicts. If they cannot, an error possibly occurred in the installation process (discussed in this subsection and Section 5.5). The second is to ascertain the initial equilibrium values of certain variables, which are utilized for calibration (discussed in Subsection 5.4.5) and for initialization of computation (discussed in Subsection 5.4.7).

The initial equilibrium and the base run equilibrium are worth describing here. The initial equilibrium is the one depicted by the SAM and derived directly from the SAM without passing through any solution process. In contrast, in our definition, the base run equilibrium is a solved equilibrium of the model, which is constructed on the basis of the constants and exogenous variables calibrated to the SAM data without any changes or shocks. Therefore, the base run equilibrium is numerically identical to the initial equilibrium indicated by the SAM data. It is important to confirm that the base run equilibrium solution is identical to the

figures in the SAM and the initial equilibrium values. This matter will be discussed in Section 5.5.

Returning to the input file, the `Parameter` directive in lines 22–26 declares and defines the constants indicating the initial equilibrium values of various variables.¹¹ Lines 28–31 specify the cells of the SAM from which the initial equilibrium values are retrieved.¹² The cells to be retrieved are illustrated in the bottom line of Table 5.1. The formulae for this retrieval process are:

$$\begin{aligned} p_i^{x0} X_i^0 &= SAM_{i,HOH} \quad \forall i \\ p_h^{f0} F_{h,j}^0 &= SAM_{h,j} \quad \forall h, j \\ p_j^{z0} Z_j^0 &= \sum_h p_h^{f0} F_{h,j}^0 \quad \forall j \\ p_h^{f0} FF_h &= SAM_{HOH,h} \quad \forall h \end{aligned}$$

Among these formulae, the third one for the values of $p_j^{z0} Z_j^0$ retrieves its data not directly from the SAM but from the firm's zero-profit condition (5.2), as discussed in Subsection 5.3.1, using $F_{h,j}^0$ derived in the second formula above. This simple CGE model does not include any indirect taxes; we can set all the prices in the initial equilibrium at unity as discussed in Section 5.2. Therefore, we do not have to declare explicitly the initial equilibrium prices p_i^{x0} , p_j^{z0} and p_h^{f0} in the program.

The `Display` directive in line 32 indicates that the values computed by these formulae are printed in the output file. We include this statement to verify the consistency between the retrieved values and those in the original SAM. Although this line is simply for such verification and is optional, it is useful when we debug any errors. As for the uses of the `Display` directive, we must not attach any indices to the symbols even if they have them. For a discussion of the detailed uses of the `Display` directive, refer to Subsection A.4.1 in Annex A.

5.4.5 Calibration

Lines 35–38 declare the coefficients α_i , $\beta_{h,j}$ and b_j to be calibrated. Lines 39–41, which define the calibration formulae for these coefficients as discussed in Section 5.3, correspond to:

$$\alpha_i = \frac{p_i^{x0} X_i^0}{\sum_j p_j^{x0} X_j^0} \quad \forall i \quad (5.1)$$

$$\beta_{h,j} = \frac{p_h^{f0} F_{h,j}^0}{\sum_k p_k^{f0} F_{k,j}^0} \quad \forall h, j \quad (5.3)$$

$$b_j = \frac{Z_j^0}{\prod_h F_{h,j}^{\beta_{h,j}}} \quad \forall j \quad (5.4)$$

As all the prices are set at unity in the initial equilibrium, the values of the prices do not explicitly appear in the input file.

It should be noted that the summation (sum) in line 39 occurs over j , not i . If we rewrite it as follows:

```
alpha(i) = x0(i)/sum(i, x0(i));
```

an error occurs.¹³ Because the meaning of i 's in sum is different from that of i 's in $\alpha(i)$ and $x0(i)$, we must use different symbols of i and j to distinguish them explicitly as follows:

```
alpha(i) = x0(i)/sum(j, x0(j));
```

For this purpose, we have defined j as an alias of i in line 7. For the same reason, we cannot use h but must use its alias k in sum in line 40. To use i and j (or h and k) interchangeably, we need the `Alias` directive.

Line 42 with the `Display` directive is included to print the calibrated values of these coefficients in the output file (List 5.2). It is strongly recommended that the validity of the calibrated coefficients be verified. Particularly, we should verify whether the sum of the expenditure share coefficients is equal to unity.

List 5.2 Calibrated coefficients in the output file

```

----      42 PARAMETER alpha  share parameter in utility function
BRD 0.300,      MLK 0.700

----      42 PARAMETER beta   share parameter in production function
              BRD              MLK
CAP          0.333            0.571
LAB          0.667            0.429

----      42 PARAMETER b     scale parameter in production function
BRD 1.890,      MLK 1.980

```

5.4.6 Specifying and solving the CGE model

After completing the calibration, we specify and solve the CGE model, which occurs from line 46 to the end of the input file (List 5.1). The solution strategy for this model is the same as the one we used for the household utility maximization model in Chapter 3. We also use the household utility function as a fictitious objective function. The components of the specification and solution part of the input file, mentioned as the fifth step in Subsection 5.4.1, are as follows:

- (5-1) Declaration and definition of endogenous variables (lines 46–54)
- (5-2) Declaration of equation names (lines 56–64)
- (5-3) Definition of equations (lines 66–73)
- (5-4) Setting initial values for endogenous variables to ease computation (lines 77–82)
- (5-5) Setting of lower bounds (lines 86–91)
- (5-6) Setting of a numeraire price (line 93)
- (5-7) Declaration of a model name and definition of the model (line 96)
- (5-8) Solve statement (line 97).

Comparing these components with those of the household utility maximization model, we find that steps (5-4) and (5-6) are new.

Regarding (5-4), as the model size increases, we often set initial points of the endogenous variables to start the computation (we call this process ‘initialization’) to make our numerical computation easier. Thanks to this technique, we can reduce computational loads, get a solution of the model more quickly and reduce the chance of encountering infeasibility.

As for (5-6), a numeraire is necessary because of the very nature of the general equilibrium theory as discussed in Section 2.5. We must choose a numeraire, with which the prices of all the other goods and factors are compared, and explicitly fix its price at unity. (The choice of unity is for consistency with the prices assumed in the initial equilibrium for calibration.) Incidentally, unlike the present model, all the prices are exogenously assumed in the household utility maximization model; thus, the numeraire is not needed. Some technicality regarding the initialization of variables, and the setting of lower bounds and a numeraire, are explained in detail below.

5.4.7 Initial values for numerical computation

Lines 77–82 set the initial values of the endogenous variables, which give starting points for the numerical computation. Usually, the initial

equilibrium values derived directly from the SAM are used to initialize the endogenous variables. For example, in line 77, 'X.l(i)' initializes (i.e., sets the computational starting point of) the LEVEL value of the endogenous variable X_i to 'X0(i)', which has been computed from the installed SAM in line 28. (It should be noted that the index following the dot in 'X.l(i)' is the letter 'l', which indicates the LEVEL value of the endogenous variable, not the number '1'. This is GAMS syntax.)

While we can set any initial values, we usually use the initial equilibrium values such as X_i^0 . This is because unless we assume a huge shock in coefficients and exogenous variables, we can expect that the new equilibrium will be located 'close' to the initial equilibrium. In our exercise, following this common practice, we use the values computed directly from the SAM, shown in lines 28–31, as the initial values. The initialization is not needed from an economics viewpoint; however, there are certain situations in which we cannot numerically solve the model without proper initialization even if the model and its calibration are perfectly correct.¹⁴

5.4.8 Uses of lower bounds

The uses of lower bounds of endogenous variables have been explained in connection with the household utility maximization model in Chapter 3. Lines 86–91 impose lower bounds on endogenous variables. For example, consider the demand price $px(i)$, which is in the denominator on the right-hand side of the household demand function in line 66. If the price becomes zero in the equilibrium or during the numerical computation process, a computational error may occur because of division by zero. To avoid such an error, we set artificial lower bounds on the endogenous variables at an arbitrary level for computational convenience.

This practice is appropriate as long as the solved value is higher than the artificially set lower bound. However, if this does not hold, the solution values may be incorrect. To examine this, we must check the VAR block carefully. Consider the example output file shown below, and we see that the solved value (LEVEL) of $pz(i)$ for BRD is equal to its LOWER value. In this case, even if the output file contains a message such as '**Optimal solution' in the SOLVE SUMMARY, the solution is incorrect unless the real solution is exactly equal to 0.001. (The input file shown in List 5.1 does not generate a problem in this regard.)

```
---- VAR pz supply price of the i-th good
      LOWER  LEVEL  UPPER  MARGINAL
BRD    0.001  0.001  +INF    EPS
MLK    0.001  1.100  +INF    EPS
```

When we face this problem, there are two possibilities: (1) the artificially set lower bound is too high, or (2) the model is erroneous. If the former is the case, we must lower the lower bound and solve the model again. If we cannot obtain any reasonable solution, we should then suspect the latter case and check the input file for any typos or reconsider the entire model.

5.4.9 Choice of numeraire

This model is homogeneous of degree zero in prices. Because of Walras's law, we cannot solve the model for all prices (in absolute terms) but only for relative prices (see Section 2.5 for a discussion of Walras's law.) We have to choose one good or factor as a numeraire and fix its price at some level, say unity. In this model, the price of labour or the wage rate is chosen as the numeraire price and is fixed at unity by using the suffix '*.fx*' in line 93.

Regarding the GAMS syntax, we should note that when we use the suffix '*.fx*' to fix endogenous variables, this statement must be placed after the statement regarding their lower or upper bounds. This is because the suffixes of '*.fx*', '*.lo*' and '*.up*' override others placed in the previous lines.¹⁵ That is, for example, if we include statements to set a lower bound (at, say, 0.001) and an upper bound (at, say, 0.5) of the wage rate *after* the statement to fix the wage rate, the latter statement is overridden by the former. In such a case, the intended numeraire wage rate of 1 is not attained.

5.5 Solution of the simple CGE model

Finally, we must finish preparation of the input file to compute the general equilibrium solution of the simple CGE model. As line 97 in List 5.1 has the *Solve* statement, we save this input file as '*sp1cge.gms*', click



the icon on the GAMS IDE and obtain the solution.¹⁶ The output file is automatically generated with the extension '*.lst*'. A part of this output file is shown in List 5.3. As demonstrated in the output file (List 3.2) of the household utility maximization model, the echo print of the input file appears headed with line numbers.¹⁷ If no syntax error in that part is identified by GAMS, the computational results for the endogenous variables as well as for the constants instructed by the *Display* directives are printed in the area following the echo print.

List 5.3 Output file of the simple CGE model (splcge.lst)

```

GAMS Rev 143 Intel/MS Windows 03/26/08 16:17:43 Page 1
A Simple CGE Model in Ch. 5 (SPLCGE,SEQ=275)

2
3 * Definition of sets for suffix -----
4 Set      u      SAM entry      /BRD, MLK, CAP, LAB, HOH/
5          i(u)    goods          /BRD, MLK/
6          h(u)    factor         /CAP, LAB/;
...
97 Solve splcge maximizing UU using nlp;
98 * -----
99 * end of model -----
100 * -----

...
----      32 PARAMETER X0  household consumption of the i-th good
BRD 15.000,      MLK 35.000

----      32 PARAMETER F0  the h-th factor input by the j-th firm
BRD      MLK
CAP      5.000      20.000
LAB      10.000     15.000

----      32 PARAMETER Z0  output of the j-th good
BRD 15.000,      MLK 35.000

----      32 PARAMETER FF  factor endowment of the h-th factor
CAP 25.000,      LAB 25.000

----      42 PARAMETER alpha  share parameter in utility function
BRD 0.300,      MLK 0.700

----      42 PARAMETER beta  share parameter in production function
BRD      MLK
CAP      0.333      0.571
LAB      0.667      0.429

----      42 PARAMETER b  scale parameter in production function
BRD 1.890,      MLK 1.980

...
          S O L V E      S U M M A R Y
MODEL    splcge          OBJECTIVE  UU
TYPE     NLP              DIRECTION  MAXIMIZE
SOLVER   CONOPT           FROM LINE  97

**** SOLVER STATUS      1 NORMAL COMPLETION
**** MODEL STATUS      2 LOCALLY OPTIMAL
**** OBJECTIVE VALUE          27.1441

RESOURCE USAGE, LIMIT      0.000      1000.000
ITERATION COUNT, LIMIT     4          10000
EVALUATION ERRORS          0          0

C O N O P T 3 Intel/MS Windows version 3.14M-016-055
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Bagsvaerdvej 246 A
DK-2880 Bagsvaerd, Denmark

```

List 5.3 (Continued)

Using default options.

```

** Optimal solution. There are no superbasic variables.
...
---- EQU eqX  household demand function
      LOWER    LEVEL    UPPER    MARGINAL
BRD      .      .      .      EPS
MLK      .      .      .      EPS

---- EQU eqpz  production function
      LOWER    LEVEL    UPPER    MARGINAL
BRD      .      .      .      0.543
MLK      .      .      .      0.543

---- EQU eqF  factor demand function
      LOWER    LEVEL    UPPER    MARGINAL
CAP.BRD  .      .      .      EPS
CAP.MLK  .      .      .      EPS
LAB.BRD  .      .      .      EPS
LAB.MLK  .      .      .      EPS

---- EQU eqpx  good market clearing condition
      LOWER    LEVEL    UPPER    MARGINAL
BRD      .      .      .      0.543
MLK      .      .      .      0.543

---- EQU eqpf  factor market clearing condition
      LOWER    LEVEL    UPPER    MARGINAL
CAP      25.000  25.000  25.000  0.543
LAB      25.000  25.000  25.000  0.543

---- EQU eqZ  PRICE equation
      LOWER    LEVEL    UPPER    MARGINAL
BRD      .      .      .      EPS
MLK      .      .      .      .

      LOWER    LEVEL    UPPER    MARGINAL
---- EQU obj      .      .      .      1.000
obj utility function [fictitious]

---- VAR X  household consumption of the i-th good
      LOWER    LEVEL    UPPER    MARGINAL
BRD      0.001  15.000  +INF      .
MLK      0.001  35.000  +INF      .

---- VAR F  the h-th factor input by the j-th firm
      LOWER    LEVEL    UPPER    MARGINAL
CAP.BRD  0.001  5.000  +INF      .
CAP.MLK  0.001  20.000  +INF      .
LAB.BRD  0.001  10.000  +INF      .
LAB.MLK  0.001  15.000  +INF      .

---- VAR Z  output of the j-th good
      LOWER    LEVEL    UPPER    MARGINAL
BRD      0.001  15.000  +INF      .
MLK      0.001  35.000  +INF      .

```

List 5.3 (Continued)

```

---- VAR px  demand PRICE of the i-th good
      LOWER    LEVEL    UPPER    MARGINAL
BRD      0.001      1.000    +INF      .
MLK      0.001      1.000    +INF      .

---- VAR pz  supply PRICE of the i-th good
      LOWER    LEVEL    UPPER    MARGINAL
BRD      0.001      1.000    +INF      .
MLK      0.001      1.000    +INF      .

---- VAR pf  the h-th factor PRICE
      LOWER    LEVEL    UPPER    MARGINAL
CAP      0.001      1.000    +INF      .
LAB      1.000      1.000      1.000    EPS

---- VAR UU                                LOWER    LEVEL    UPPER    MARGINAL
      -INF      27.144    +INF      .

UU  utility [fictitious]

```

Before proceeding to the interpretation of the solutions, we must look at the SOLVE SUMMARY block to check whether or not the message ‘**Optimal solution’ is printed. If this message is not printed there, it indicates that the model is not solved correctly and that the model has some errors other than syntax errors. To correct the errors, refer to Annex B.

As stated previously, it is important to confirm that the base run equilibrium solution is identical to the figures in the SAM and the initial equilibrium. The first step is to examine whether the SAM data have been retrieved correctly as the initial equilibrium. By the Display directive in line 32 of List 5.1, these figures are printed before the SOLVE SUMMARY header. For example, ‘X0’ appears in the output file (List 5.3) as follows:¹⁸

```

---- 32 PARAMETER X0 household consumption of the i-th good
BRD 15.000, MLK 35.000

```

This is followed by the figures for factor inputs ‘F0’ and output ‘Z0’ in the initial equilibrium as well as the factor endowments ‘FF’.

Now we move to the solution phase of the model and verify whether or not the solution is consistent with that shown by the SAM. As for the price variables, all the prices should be unity because there is no indirect tax and they are set at unity in the calibration process. The price solutions are in the VAR block, where the three variables px (demand price), pz (supply price) and pf (factor prices) are shown. Their LEVEL

columns show values of 1 for these price variables, indicating that all the price variables are correctly solved.

The quantity variables are X (consumption), F (factor input) and Z (production). The LEVEL column of the VAR X block shows that consumption is 15 units of bread and 35 units of milk. We can find the same numbers in the 'BRD'-'HOH' and the 'MLK'-'HOH' cells in the SAM in Table 5.1 (we can alternatively use the information about 'X0' printed in the output file by the Display directive to verify the solution.)

In the VAR F block in List 5.3, we find that the row labels are two-dimensional delimited with a period '.'; e.g., 'CAP.BRD' and 'CAP.MLK'. Because the endogenous variable F has two indices 'h' and 'j' as defined in line 47 of List 5.1, its values are printed with these two indices in the VAR F block. The labels 'CAP' and 'LAB' are placed first, followed by the labels 'BRD' and 'MLK'. This order of indices is consistent with that used in the variable declaration. The figures in the LEVEL column of the VAR F block show the same values shown in the cells of 'Factors'-'Activity' in Table 5.1.

Finally, let us check the production Z. The LEVEL column of the VAR Z block indicates that production is 15 units of bread and 35 units of milk. As discussed previously, bread and milk production are not explicitly shown in the SAM in Table 5.1 but can be computed as 15 and 35 units respectively with the zero-profit condition (5.2) shown in Subsection 5.3.1. The solved values are all consistent with those originally reported in the SAM.

As mentioned in Chapter 3, there are two more points to be made. First, the endogenous variable UU shown at the end of List 5.3 is a fictitious objective variable. It is introduced only for the purpose of solving the model originally formulated as a system of simultaneous equations by using a solver for nonlinear programming problems such as CONOPT. As in the household utility maximization problem discussed in Chapter 3, we set the household utility function as a fictitious objection function for the present model. Although the objective function is fictitious, the solution of UU indicates the utility level of the household in this economy, which can be used as a measurement of the economy's welfare. This problem is discussed in Chapter 8.

Second, as for the EQU block, the MARGINAL columns, in general, report solutions of the Lagrange multipliers for the nonlinear programming problems. However, as this model is formulated as a fictitious optimization problem as stated above, the values of the Lagrange multipliers have no meaning either. We can disregard the EQU block.¹⁹

CGE models rely greatly on coefficients and exogenous variables calibrated on the basis of the SAM. The validity of CGE models critically depends on whether the SAM is installed properly in the program and the calibration is conducted correctly. As discussed repeatedly in this section, it is necessary for us to verify the model by examining whether or not the SAM data are exactly produced as a base run equilibrium solution. If they are not, we have to correct both computational and economics errors in the models. Once we confirm the reproduction of the SAM data by calibration, we can begin the simulations, where counterfactual values are assumed for some of the constants in the model to examine the impact of exogenous or policy shocks on the economy. This is discussed in Chapter 8.

Notes

1. Despite such a difficulty, Jorgenson and Yun (1990) conducted a study where a CGE model was econometrically estimated.
2. Note that boldface type represents vectors/matrices.
3. For example, imagine a model that determines the value of consumption, production and their prices, given endowments of factors and coefficients for expenditure and input share.
4. While the calibration method illustrated here is to estimate a model in such a way that the initial equilibrium is exactly reproduced, the calibration method in general encompasses a broader scope of parameter estimation methods. A similar calibration technique is used for estimation of macroeconomic models known as real business cycle models or dynamic stochastic general equilibrium models (McCandless 2008). In these models, the calibrated model does not exactly reproduce a reference equilibrium path but a similar path. Dawkins et al. (2001) discuss details of this calibration method as well as the merits and demerits of calibration methods in general. In this book, however, 'calibration' refers to a model estimation method that exactly reproduces the initial equilibrium of the estimated model.
5. In this discussion, we choose one (yen) for the definition of a new physical unit. This cannot guarantee that this price realizes in equilibria other than the initial equilibrium. When we give a shock to the model, the resulting equilibrium prices may well deviate from unity.
6. Alternatively, we can set the tax-inclusive price at unity and compute the quantity. There is no significant difference.
7. As explained in Section 2.4, the index i indicating the good is interchangeable with j . In the derivation of (5.1), the index i in the left-hand side of (2.b) is replaced with j . This replacement is not necessarily required for this mathematical manipulation but is recommended to avoid possible errors in GAMS programming, which will be shown in Subsection 5.4.5. A similar replacement of interchangeable indices is made between h and k indicating the factors involved in the derivation of (5.3).

8. When we alternatively set the demand price of the i -th good p_i^{x0} at unity, the corresponding supply price p_i^{z0} becomes $1/(1 + \tau_i^z)$ rather than unity.
9. We discuss the elasticity parameters in Section 6.5.
10. The first line starting with “\$Title...” is included only for file management purposes in the GAMS Model Library and is thus meaningless. This line can be omitted.
11. The uses of the `Parameter` directive demonstrated here are different from that in the household utility maximization model in Chapter 3. While we placed the directive to declare and to define the constants in one statement in List 3.1, these two procedures are done separately in two statements in List 5.1. For example, the declaration of `X0(i)` is done in line 22; its values are given in line 28. Following this practice, `F0(h, j)` is declared in line 23 and its value is given in line 29. When we express constants in two separate statements, the `Parameter` directive, not the `Table` directive, must be used even if they have two (or more) indices like `F0(h, j)`. Details about the uses of the `Parameter` directive are provided in Section A.2 in Annex A.
12. We indicate the rows and columns of the cells in the SAM using indices such as i, j, h and k , or individual elements such as ‘HOH’. In the latter case, we enclose the element with double (or single) quotation marks.
13. This error message in the output file is ‘***** 125 Set is under control already’.
14. Given the model in List 5.1, we can solve the model either by initializing the variables or by setting lower bounds on the endogenous variables (discussed later) when we use the CONOPT solver. In contrast, when we use the MINOS solver, we cannot solve the model without variable initialization. When the solvers are updated, this result may differ.
15. The suffix ‘.fx’ sets both the lower and the upper bounds of the domain of an endogenous variable at the same value. Therefore, when the ‘.fx’ is followed by ‘.lo’ or ‘.up’ for the same endogenous variable, the latter statement takes effect and resets the lower or upper bound respectively.
16. For further information about the uses of GAMS IDE, see the online help or its printable file available at www.gams.com/dd/docs/tools/gamside.pdf.
17. There are some exceptions regarding the line numbers. Lines beginning with a ‘\$’ symbol (e.g., line 1 in List 5.1) are not printed in the output files. For a detailed discussion of their uses, see Appendix C in *GAMS – A User’s Guide*. If GAMS finds a syntax error, the line with an error is indicated by four asterisks ‘*****’ with an error code inserted in the echo print part. For a discussion about syntax errors, see Section B.1 in Annex B of this book.
18. When the `Display` directive is put after the `Solve` statement in the program, the printout by the `Display` directive appears after the `SOLVE SUMMARY` header, and vice versa.
19. See Appendix III for details about the Lagrange multipliers. See Appendix IV for details about the fictitious objective function for computation of a system of simultaneous equations.

6

The Standard CGE Model

The ‘simple CGE model’ presented in Chapter 2 is equipped with only the essential features of a very basic macroeconomic model, and thus it cannot be used for empirical analyses. Here, we extend the model by incorporating the following four features. First, we introduce intermediate inputs into the production process. Second, we introduce a government into the model, where its consumption, and direct and indirect taxes revenues including import tariffs, are considered. Third, we also introduce investment and savings. Finally, we extend the model to an open economy model, where international trade is considered. We call this model the ‘standard CGE model’, whose social accounting matrix (SAM) was previously shown in Table 4.2 in Chapter 4.¹

Section 6.1 provides an overview of the standard CGE model. Intermediate inputs are included in the model in Section 6.2, a government in Section 6.3, investment and savings in Section 6.4 and international trade in Section 6.5. Section 6.6 explains the market-clearing conditions. Section 6.7 provides the complete system of simultaneous equations for the standard CGE model. In Section 6.8, we apply the computer program, explained in Chapter 5, covering installation of the SAM through to calibration in order to solve the standard CGE model. By understanding the structure of the model and the computer program presented in this chapter, readers can develop and solve their own models for their research objectives.

6.1 Overview of the standard CGE model

Figure 6.1 provides an overview of the standard CGE model from the viewpoint of the flows of goods and factors in an economy.² As the standard CGE model is an extension of the ‘simple CGE model’ described in

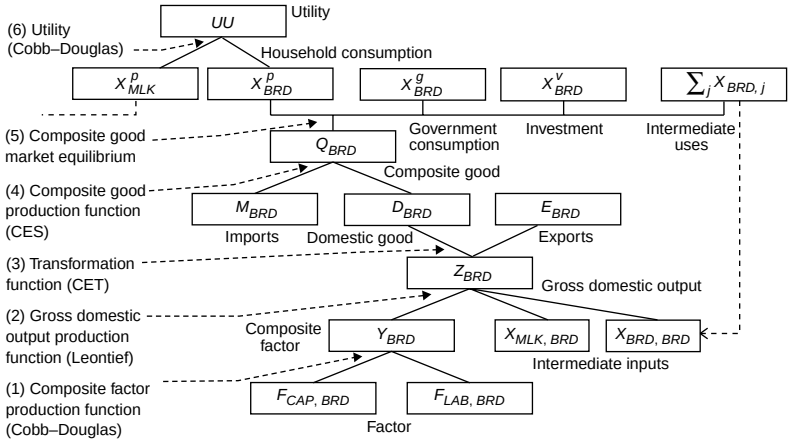


Figure 6.1 Overview of the standard CGE model

Note: Assumed functional forms are indicated in parentheses.

Figure 2.1, some variables reappear in Figure 6.1. These are UU , X_{BRD}^P and X_{MLK}^P of the utility function shown in the northwest corner (the latter two were denoted by X_{BRD} and X_{MLK} in Figure 2.1), and $F_{CAP, BRD}$, $F_{LAB, BRD}$ and Y_{BRD} (the last was Z_{BRD} in Figure 2.1) of the production function shown at the bottom.

Next, we explain the flows of goods and factors at each stage where they are combined for either production or consumption. The flows are explained from the bottom to the top in Figure 6.1, taking the bread sector as an example.

- (1) Capital $F_{CAP, BRD}$ and labour $F_{LAB, BRD}$ are aggregated into the composite factor Y_{BRD} using the composite factor production function.
- (2) This composite factor Y_{BRD} is combined with the intermediate inputs of bread $X_{BRD, BRD}$ and milk $X_{MLK, BRD}$ to produce the gross domestic output Z_{BRD} using the gross domestic output production function.
- (3) The gross domestic output Z_{BRD} is transformed into the exports E_{BRD} and the domestic good D_{BRD} using the gross domestic output transformation function.
- (4) The domestic good D_{BRD} is combined with the imports M_{BRD} to produce the composite good Q_{BRD} with the composite good production function.
- (5) The composite good Q_{BRD} is distributed among household consumption X_{BRD}^P , government consumption X_{BRD}^G , investment X_{BRD}^V and intermediate uses by the bread and milk sectors $\sum_j X_{BRD, j}$.³

- (6) Household utility UU is generated by consumption X_{BRD}^p and X_{MLK}^p as the utility function indicates.

Details of the composite factors, the composite goods and the functions newly mentioned here are explained below.

6.2 Intermediate inputs

In the simple CGE model, only capital and labour are assumed to be used for the production of goods. Here, in contrast, we make the model more realistic by assuming that firms use intermediate inputs in their production process. Following this extension, the behaviour of firms becomes more complicated; we divide the production process (or firms) into two stages.

In the first stage, capital and labour are used for the production of a composite factor (or value added). The production process of the composite factor can be regarded as the behaviour of a virtual factory, which maximizes its profit by choosing its output (composite factor) level and inputs (capital and labour) use, depending on their relative prices subject to its technology. In the second stage, the composite factor is combined with intermediates to produce the gross domestic output, as indicated by the gross domestic output production function.

As for the technology in this two-stage production process, we assume a Cobb–Douglas-type production function for the first stage and a Leontief-type production function for the second stage. They are both homogeneous of degree one and thus characterized as constant-returns-to-scale.⁴ The Cobb–Douglas-type production function allows us to describe substitution between inputs, while the Leontief-type production function does not. As empirical CGE models are developed on the basis of the input–output (IO) tables, distinguishing dozens of sectors/goods, the number of endogenous variables, particularly for intermediate inputs, increases in accordance with the square of the number of sectors/goods. In this regard, the Leontief-type production function significantly reduces the complexity of the model and thereby the computational load.

The profit-maximization problems for the j -th firm can be written as follows:⁵

– For the first stage:

$$\underset{Y_j, F_{h,j}}{\text{maximize}} \quad \pi_j^y = p_j^y Y_j - \sum_h p_h^f F_{h,j}$$

subject to

$$Y_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad (6.1)$$

– For the second stage:⁶

$$\text{maximize } \pi_j^z = p_j^z Z_j - \left(p_j^y Y_j + \sum_i p_i^q X_{i,j} \right)$$

subject to

$$Z_j = \min \left(\frac{X_{BRD,j}}{ax_{BRD,j}}, \frac{X_{MLK,j}}{ax_{MLK,j}}, \frac{Y_j}{ay_j} \right) \quad (6.5')$$

Notations are:

- π_j^y : profit of the j-th firm producing composite factor Y_j in the first stage,
- π_j^z : profit of the j-th firm producing gross domestic output Z_j in the second stage,
- Y_j : composite factor, produced in the first stage and used in the second stage by the j-th firm,
- $F_{h,j}$: the h-th factor used by the j-th firm in the first stage,
- Z_j : gross domestic output of the j-th firm,
- $X_{i,j}$: intermediate input of the i-th good used by the j-th firm,
- p_j^y : price of the j-th composite factor,
- p_h^f : price of the h-th factor,
- p_j^z : price of the j-th gross domestic output,
- p_i^q : price of the i-th composite good,
- $\beta_{h,j}$: share coefficient in the composite factor production function,
- b_j : scaling coefficient in the composite factor production function,
- $ax_{i,j}$: input requirement coefficient of the i-th intermediate input for a unit output of the j-th good,
- ay_j : input requirement coefficient of the j-th composite good for a unit output of the j-th good.

In each stage of production, the objective value is the profits of the firm. In the first-stage profit function, the first term on the right-hand side represents the sales of the composite factor; the second term represents the input costs of capital and labour used for its production. The constraint (6.1) represents the technology of the composite factor production described by a Cobb–Douglas-type production function.

In the second-stage profit function, the first term on the right-hand side is the sales of the gross domestic output, which consists of ordinary

goods such as bread and milk in this model; the second and third terms are the costs of the composite factor input and those of the intermediate inputs used in the second-stage production respectively. The constraint (6.5') is a Leontief-type production function for production of the gross domestic output with the composite factor and intermediate inputs.

By solving these two problems, we obtain:⁷

$$Y_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j \quad (6.1)$$

$$F_{h,j} = \frac{\beta_{h,j} p_j^y}{p_h^f} Y_j \quad \forall h, j \quad (6.2)$$

$$X_{i,j} = ax_{i,j} Z_j \quad \forall i, j \quad (6.3)$$

$$Y_j = ay_j Z_j \quad \forall j \quad (6.4)$$

$$Z_j = \min \left(\frac{X_{BRD,j}}{ax_{BRD,j}}, \frac{X_{MLK,j}}{ax_{MLK,j}}, \frac{Y_j}{ay_j} \right) \quad \forall j \quad (6.5')$$

The production function (6.5') generates rectangular isoquants (iso-output curves) as shown in Figure V.1 in Appendix V. The kinks in the isoquants often cause difficulty in numerical computations.⁸ To work around such a computational problem, we replace (6.5') with a zero-profit condition, which should always hold, as we explained with (2.7) in Section 2.5:⁹

$$\pi_j^z = p_j^z Z_j - \left(p_j^y Y_j + \sum_i p_i^q X_{i,j} \right) = 0 \quad \forall j$$

We can include this zero-profit condition in the model, but it is more convenient to transform it into a simpler expression of a unit cost function. Using (6.3) and (6.4), we can eliminate $X_{i,j}$ and Y_j to obtain:

$$p_j^z Z_j - \left(ay_j p_j^y Z_j + \sum_i ax_{i,j} p_i^q Z_j \right) = 0 \quad \forall j$$

and again by eliminating Z_j , we get the following unit cost function:

$$p_j^z = ay_j p_j^y + \sum_i ax_{i,j} p_i^q \quad \forall j \quad (6.5)$$

Replacing (6.5') with (6.5), we can describe the firms' behaviour with (6.1)–(6.5).

6.3 Government

CGE models are often used for policy analysis. The main concern is the consequences of changes in government policy devices – typically tax rates. Thus, any realistic CGE model must include a government. In this section, we discuss how to model government behaviour.

In our CGE model, the government is supposed to collect taxes and consume goods. It should be noted that there is no single perfect way of modelling these government activities from the viewpoint of micro-foundations, while we have modelled the behaviour of the household and the firms firmly based on their microfoundations. Therefore, we must develop our CGE models with a government, depending on the purpose of our analysis, the availability of data or sometimes the preference of the modeller. The model of the government presented here is one example among various possible specifications.

We assume that the government levies a direct tax on household income at the tax rate τ^d , an ad valorem production tax (an indirect tax) on gross domestic output at the tax rate τ_j^z and an ad valorem import tariff on imports at the rate τ_i^m . (Details about imports as well as exports are discussed later in Section 6.5.) At the same time, we assume that (1) the government spends all tax revenues on their consumption, and that (2) the government consumes each good (i.e., bread or milk) in fixed proportions in total government expenditure. For example, the government spends 40% of its total revenues on the purchase of bread and 60% on the purchase of milk.

The above assumptions can be written as follows:

$$T^d = \tau^d \sum_h p_h^f F F_h \quad (6.6)$$

$$T_j^z = \tau_j^z p_j^z Z_j \quad \forall j \quad (6.7)$$

$$T_i^m = \tau_i^m p_i^m M_i \quad \forall i \quad (6.8)$$

$$X_i^g = \frac{\mu_i}{p_i^g} \left(T^d + \sum_j T_j^z + \sum_j T_j^m \right) \quad \forall i \quad (6.9')$$

where:

- T^d : direct tax,
- T_j^z : production tax on the j -th good,
- T_i^m : import tariff on the i -th good,
- τ^d : direct tax rate,
- τ_j^z : production tax rate on the j -th good,
- τ_i^m : import tariff rate on the i -th good,
- FF_h : endowments of the h -th factor for the household,
- Z_j : gross domestic output of the j -th firm,
- M_i : imports of the i -th good,
- X_i^g : government consumption of the i -th good,
- p_j^z : price of the j -th gross domestic output,
- p_h^f : price of the h -th factor,
- p_i^m : price of the i -th imported good,
- p_i^q : price of the i -th composite good,
- μ_i : share of the i -th good in government expenditure ($0 \leq \mu_i \leq 1$, $\sum_i \mu_i = 1$).

(The composite good is explained in Section 6.5.)

Although we assume that government expenditure is allocated among goods for consumption proportionately as (6.9') indicates in this example, we can use other assumptions. For example, we can further simplify government behaviour by setting its consumption at the initial equilibrium level X_i^{g0} :¹⁰

$$X_i^g = X_i^{g0} \quad \forall i$$

When the government sells its assets, such sales appear as negative consumption in statistical databases such as IO tables. (A similar observation can also take place in the investment account, when a decrease in stocks occurs.) An application of the proportionate government expenditure behaviour suggested above might not be suitable for such a negative consumption case. We can alternatively develop a model that allows negative values for some government consumption. That is, we can set a negative value for the consumption of some goods and assume positive proportional expenditure for the other goods.

6.4 Investment and savings

6.4.1 Introduction of investment and savings

The CGE model that we are developing here is a static model. Thus, strictly speaking, introducing dynamic factors like investment and savings is inconsistent with its original setup as a static model.¹¹ However,

we cannot ignore investment because it has a significantly large share in final demand.¹² Although we cannot model investment in a manner that is perfectly consistent with economic theory, we have to incorporate it in some way.

Recall the discussion about the virtual investment agent in Subsection 4.1.2. The investment agent collects funds from the household, the government and the external sector and spends them on purchases of investment goods. Although the household and the government can make their own decisions about investment and savings, the present model assumes that a virtual agent absorbs all the savings of an economy and spends them on the purchase of goods proportionately with a constant share λ_i . We can describe its behaviour using the investment demand function (6.10). This is similar to the assumption about the government demand function for goods:

$$X_i^v = \frac{\lambda_i}{p_i^q} (S^p + S^g + \varepsilon S^f) \quad \forall i \quad (6.10)$$

Notations are:

S^p : household savings,

S^g : government savings,

S^f : current account deficits in foreign currency terms
(or equivalently foreign savings),

X_i^v : demand for the i -th investment good,

ε : foreign exchange rate (domestic currency/foreign currency),¹³

p_i^q : price of the i -th composite good,

λ_i : expenditure share of the i -th good in total investment
($0 \leq \lambda_i \leq 1$, $\sum_i \lambda_i = 1$).

The variables in parentheses on the right-hand side of (6.10) correspond to total savings consisting of savings by the household, the government and the external sector. It should be noted that, as the sum of the share parameter λ_i is equal to unity, (6.10) implies that the total savings in an economy are always equal to its total investment.

Then, let us assume that household and government savings are determined by constant average propensities for savings as follows:

$$S^p = ss^p \sum_h p_h^f F F_h \quad (6.11)$$

$$S^g = ss^g \left(T^d + \sum_j T_j^z + \sum_j T_j^m \right) \quad (6.12)$$

where:

ss^p : average propensity for savings by the household,

ss^g : average propensity for savings by the government.

In addition to S^p and S^g , the economy has other savings; namely, foreign savings S^f . In the present model, foreign savings S^f are assumed to be exogenous; however, we can alternatively assume any of these savings variables to be either endogenous or exogenous depending on our point of view of the real economy. This issue is closely related to the issue of the macro closure rules, which will be discussed in depth in the next chapter.

It should be noted that the investment determined by (6.10) implies abandoning goods, which contributes neither to household utility nor to firm production. In fact, the utility function is not dependent on the amount of investment X_i^y . Furthermore, the endowments of capital FF_{CAP} are predetermined in this economy and thus cannot be increased by the investment X_i^y in this static model.

6.4.2 Modification of household and government behaviour

The introduction of the government and investment and savings into the model requires us to modify the original model equations describing the behaviour of the household and the government. While we assume the same utility function, the household budget constraint has to be slightly modified. (Note that we have used X_i for household consumption in Chapter 2 but hereinafter use X_i^p for household consumption because other uses such as government consumption, investment uses and intermediate inputs are considered in the standard CGE model.) That is, the available funds for household consumption of goods are now reduced by the amount of household savings and direct tax payments; thus, the household problem is updated as follows:

$$\underset{X_i^p}{\text{maximize}} \quad UU = \prod_i X_i^{p^{\alpha_i}}$$

subject to

$$\sum_i p_i^q X_i^p = \sum_h p_h^f FF_h - S^p - T^d$$

where:

UU : utility,

X_i^p : household consumption of the i -th good,

FF_h : endowments of the h -th factor for the household,

S^p : household savings,

T^d : direct tax,

p_i^q : price of the i -th composite good,

p_h^f : price of the h -th factor,

α_i : share parameter in the utility function ($0 \leq \alpha_i \leq 1$, $\sum_i \alpha_i = 1$).

Solving this modified household problem in the same way as in Section 2.2, we obtain the household demand function for the i -th good:

$$X_i^p = \frac{\alpha_i}{p_i^q} \left(\sum_h p_h^f F F_h - S^p - T^d \right) \quad \forall i \quad (6.13)$$

The government demand function for the i -th good is modified analogously by incorporation of government savings in a similar manner:

$$X_i^g = \frac{\mu_i}{p_i^q} \left(T^d + \sum_j T_j^z + \sum_j T_j^m - S^g \right) \quad \forall i \quad (6.9)$$

6.5 International trade

6.5.1 Small-country assumption and balance of payments

The third major feature of the standard CGE model is the extension of the original closed economy model to an open economy model. For simplicity, we assume that this economy is so small that it does not have a significant impact on the rest of the world – even with an extreme activity such as export dumping.¹⁴ The essence of the small-country assumption is that the export and import prices quoted in foreign currency terms are exogenously given for this economy.

Regarding international trade, we must distinguish between two types of price variables. One is prices in terms of the domestic currency p_i^e and p_i^m ; the other is prices in terms of the foreign currency p_i^{We} and p_i^{Wm} . They are linked with each other as follows:

$$p_i^e = \varepsilon p_i^{We} \quad \forall i \quad (6.14)$$

$$p_i^m = \varepsilon p_i^{Wm} \quad \forall i \quad (6.15)$$

Furthermore, it is assumed that the economy faces balance of payments constraints, which can be described with export and import prices in foreign currency terms:

$$\sum_i p_i^{We} E_i + S^f = \sum_i p_i^{Wm} M_i \quad (6.16)$$

Notations are:

- p_i^{We} : export price in terms of foreign currency (exogenous),
- p_i^e : export price in terms of domestic currency,
- ε : foreign exchange rate (domestic currency/foreign currency),
- E_i : exports of the i -th good,
- p_i^{Wm} : import price in terms of foreign currency (exogenous),
- p_i^m : import price in terms of domestic currency,
- M_i : imports of the i -th good,
- S^f : current account deficit in terms of foreign currency
(or equivalently foreign savings; exogenous).

As mentioned in Subsection 6.4.1, the current account deficit in foreign currency terms S^f is an exogenous variable. While in the present model the balance of payments constraints are expressed in terms of foreign currency, that constraint can be alternatively expressed in terms of domestic currency by replacing p_i^{We} and p_i^{Wm} with p_i^e and p_i^m using (6.14) and (6.15).

6.5.2 Armington's assumption

When we extend the model to an open economy model, we have to consider differences (or similarities) between goods domestically produced/consumed and those imported/exported. In this section, we find that it is necessary to assume that they are *imperfectly* substitutable with each other. That is, domestic-made bread is supposed to be similar to but is slightly different from imported bread.

Suppose that all the exported goods are perfectly substitutable with the corresponding imported goods, such that there cannot be both exports and imports for the same goods simultaneously.¹⁵ It is nonsense to import 100 units of bread while exporting 20 units. Instead, we should import the net amount of 80 units. However, actual data often report both exports and imports for the same good. This is known as two-way trade or cross-hauling. To reconcile such a conflict between theory and practice, we distinguish an imported good from an exported one even when they are classified in the data as the same good. The degree of difference/similarity between them can be measured by a parameter such as the elasticity of substitution in constant elasticity of substitution (CES) functions. If they are significantly different from each other, the elasticity of substitution becomes small (i.e., inelastic) and vice versa.

In reality, it seems, however, that substitution (or competition) is more relevant between imports and domestic goods, and between exports and domestic goods, than between exports and imports. In CGE models, we

assume substitution between imports and domestic goods, and between exports and domestic goods in a pairwise manner.¹⁶ The assumption about imperfect substitution between imports and domestic goods is called Armington's (1969) assumption.

6.5.3 Substitution between imports and domestic goods

Armington's assumption implies that households and firms do not directly consume or use imported goods but instead a so-called 'Armington composite good', which comprises imports and the corresponding domestic goods. To describe the process of constructing Armington composite goods, we assume virtual firms that behave so as to maximize their profits by choosing a suitable combination of imported and domestic goods. The solution of their profit-maximization problem leads to their input demands for imports and that for domestic goods, and the output level by adjusting quantities of imported and domestic goods, depending upon all the prices involved.

For example, let us suppose that a firm mixes imported orange juice with domestic orange juice to produce orange juice bottled with its company's labels as we see at supermarkets. This production process can often be described by a CES function (6.17). This production function is an extension of the celebrated Cobb–Douglas and Leontief functions. The CES function is characterized by the parameter of elasticity of substitution, σ_i , which indicates the percentage changes in the input factor ratio caused by a 1% change in relative input prices. Graphically speaking, this parameter determines the curvature of isoquants. The larger the elasticity, the gentler the curvature of the isoquants or the more flexibly the input share is adjusted (see its isoquant in Figure 6.2; however, note that import tariff rates are omitted). The extreme case is the Leontief-type function, where $\sigma_i = 0$.

The optimization problem for the i -th Armington-composite-good-producing firm can be written as follows:

$$\underset{Q_i, M_i, D_i}{\text{maximize}} \quad \pi_i^q = p_i^q Q_i - \left[(1 + \tau_i^m) p_i^m M_i + p_i^d D_i \right]$$

subject to

$$Q_i = \gamma_i \left(\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i} \right)^{\frac{1}{\eta_i}} \quad (6.17)$$

Notations are:

π_i^q : profit of the firm producing the i -th Armington composite good,
 p_i^q : price of the i -th Armington composite good,

p_i^m : price of the i -th imported good in terms of domestic currency,
 p_i^d : price of the i -th domestic good,
 Q_i : the i -th Armington composite good,
 M_i : the i -th imported good,
 D_i : the i -th domestic good,
 τ_i^m : import tariff rate on the i -th good,
 γ_i : scaling coefficient in the Armington composite good production function,

$\delta m_i, \delta d_i$: input share coefficients in the Armington composite good production function ($0 \leq \delta m_i \leq 1$, $0 \leq \delta d_i \leq 1$, $\delta m_i + \delta d_i = 1$),
 η_i : parameter defined by the elasticity of substitution,
 $(\eta_i = (\sigma_i - 1)/\sigma_i, \eta_i \leq 1)$,
 σ_i : elasticity of substitution in the Armington composite good production function, $\left(\sigma_i = -\frac{d(M_i/D_i)}{M_i/D_i} \bigg/ \frac{d(p_i^m/p_i^d)}{p_i^m/p_i^d}\right)$.

The first-order conditions for the optimality of the above problem imply the following demand functions for imports and the domestic good:¹⁷

$$M_i = \left[\frac{\gamma_i^{\eta_i} \delta m_i p_i^q}{(1 + \tau_i^m) p_i^m} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (6.18)$$

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i p_i^q}{p_i^d} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (6.19)$$

It should be noted that the Armington-composite-good-producing firm faces tariff-inclusive import prices $(1 + \tau_i^m)p_i^m$ rather than the tariff-exclusive import prices p_i^m ; therefore, the tariff rate τ_i^m appears in the definition of its profit π_i^q . Consequently, $(1 + \tau_i^m)p_i^m$ is also in the derived import demand function (6.18).

6.5.4 Transformation between exports and domestic goods

Let us consider the supply side; that is, exports and the domestic goods. We assume that the firms transform the gross domestic output into goods sold in international markets and in domestic markets. In this transformation process, we also assume imperfect substitution (strictly speaking, it should be called imperfect transformation) between exports and the domestic good supply.¹⁸

For example, electronic appliances are commonly used all over the world but are often customized by country considering the preferences of

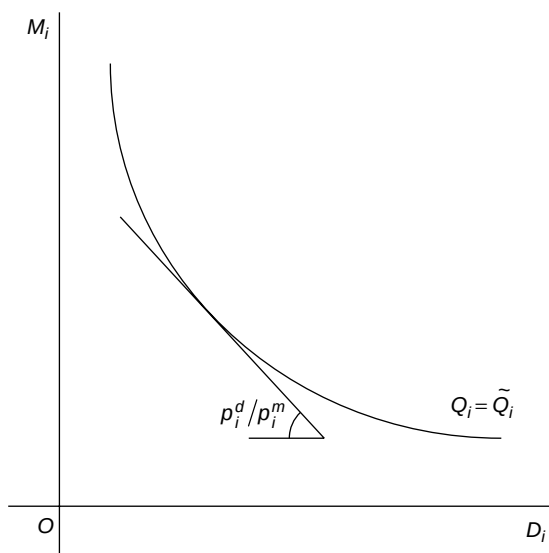


Figure 6.2 Isoquant of the CES function for the Armington composite good

targeted users. Those supplied to Japan are likely to have many functions in a small body, while those exported to the international markets are rather simple in function and of a larger size. For example, automobiles for domestic sales in Japan are often equipped with luxurious options, whereas those for exports have only essential ones.

Suppose that a firm (or the final stage of a production process in a firm), which is involved in shipping of the gross domestic output to international markets and to the domestic market, decides the supply ratio between these two markets and customizes its output to be suitable for these targeted markets. We express such a transformation process with a constant elasticity of transformation (CET) function. The isoquants (iso-input curves) for this transformation shown in Figure 6.3 are the mirror images of the isoquants (iso-output curves) of the CES function in Figure 6.2. Depending on the relative price between exports and domestic goods, the supply ratio changes. The larger the elasticity of transformation ψ_i , the gentler the curvature of the isoquants. That is, with a large elasticity of transformation, the export–domestic supply ratio tends to be more sensitive to a change in relative prices.

The profit-maximization problem for the i -th firm transforming the gross domestic output into exports and domestic goods can be expressed

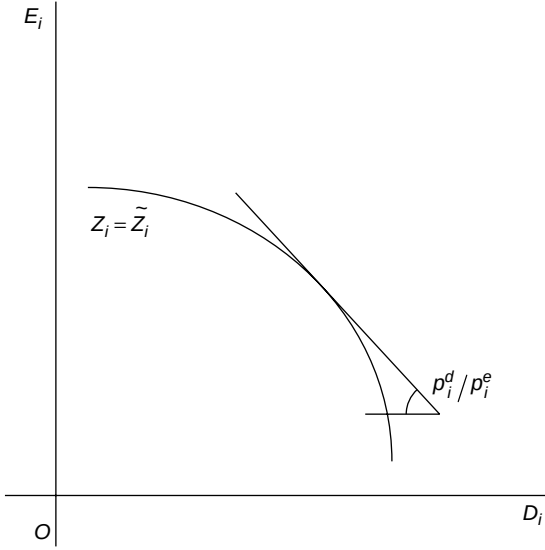


Figure 6.3 Isoquant of the CET function

as follows:

$$\underset{Z_i, E_i, D_i}{\text{maximize}} \pi_i = (p_i^e E_i + p_i^d D_i) - (1 + \tau_i^z) p_i^z Z_i$$

subject to

$$Z_i = \theta_i \left(\xi e_i E_i^{\phi_i} + \xi d_i D_i^{\phi_i} \right)^{\frac{1}{\phi_i}} \quad (6.20)$$

Notations are:

- π_i : profit of the firm engaged in the i-th transformation,
- p_i^e : price of the i-th export good in terms of domestic currency,
- p_i^d : price of the i-th domestic good,
- p_i^z : price of the i-th gross domestic output,
- E_i : exports of the i-th good,
- D_i : supply of the i-th domestic good,
- Z_i : gross domestic output of the i-th good,
- τ_i^z : production tax on the i-th gross domestic output,
- θ_i : scaling coefficient of the i-th transformation,¹⁹

$\xi e_i, \xi d_i$: share coefficients for the i-th good transformation,
 $(0 \leq \xi e_i \leq 1, 0 \leq \xi d_i \leq 1, \xi e_i + \xi d_i = 1),$

ϕ_i : parameter defined by the elasticity of transformation,

$$(\phi_i = (\psi_i + 1)/\psi_i, \psi_i \geq 1),$$

ψ_i : elasticity of transformation of the i -th good transformation,

$$\left(\psi_i = \frac{d(E_i/D_i)}{E_i/D_i} \bigg/ \frac{d(p_i^e/p_i^d)}{p_i^e/p_i^d} \right).$$

By solving this maximization problem, we get the following supply functions for exports and for domestic goods:

$$E_i = \left[\frac{\theta_i^{\phi_i} \xi e_i (1 + \tau_i^z) p_i^z}{p_i^e} \right]^{\frac{1}{1-\phi_i}} Z_i \quad (6.21)$$

$$D_i = \left[\frac{\theta_i^{\phi_i} \xi d_i (1 + \tau_i^z) p_i^z}{p_i^d} \right]^{\frac{1}{1-\phi_i}} Z_i \quad (6.22)$$

Because the production tax τ_i^z is imposed on the gross domestic output Z_i , which is used as the input in this transformation process, τ_i^z appears in the equation defining the profit π_i and consequently also in the numerators of the above two supply functions.

6.6 Market-clearing conditions

We have described the behaviour of economic agents, such as the household, the firms, the government, the investment agent and the external sector, with a set of equations. Our final step of this modelling process is imposing the market-clearing conditions so that demand meets supply in all markets as follows:

$$Q_i = X_i^p + X_i^g + X_i^v + \sum_j X_{i,j} \quad \forall i \quad (6.23)$$

$$\sum_j F_{h,j} = FF_h \quad \forall h \quad (6.24)$$

The market-clearing condition for the Armington composite goods is described by (6.23). As discussed in Subsection 6.5.3, the composite good Q_i is used by the household, the government and the investment agent as well as for intermediate input; we apply the same price p_i^q to all of them. Equation (6.24) is the factor market-clearing condition, which also appeared as (2.5) in the simple CGE model.

Recall that we have imposed the constraint (2.6) to equilibrate the supply price of each good with its demand price (p_i^z and p_i^x in Chapter 2), because we have assumed that the household and the firm must face the same price in each good market. In the present model, however, the price p_i^q , which the household faces, is not directly linked to the price p_i^z , which the firm faces. The CES and CET structures, which represent substitution between imports and domestic goods, and transformation between exports and domestic goods respectively, bring about equality between the demand and supply of goods by these agents but do not make a direct link between p_i^q and p_i^z . Therefore, we do not impose price equalization constraints between p_i^q and p_i^z such as (2.6) in this model.

6.7 Model system

As discussed above, we have developed a system of simultaneous equations for the standard CGE model consisting of (6.1)–(6.24).

– Domestic production:

$$Y_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j \quad (6.1)$$

$$F_{h,j} = \frac{\beta_{h,j} p_i^y}{p_h^f} Y_j \quad \forall h, j \quad (6.2)$$

$$X_{i,j} = a x_{i,j} Z_j \quad \forall i, j \quad (6.3)$$

$$Y_j = a y_j Z_j \quad \forall j \quad (6.4)$$

$$p_j^z = a y_j p_j^y + \sum_i a x_{i,j} p_i^q \quad \forall j \quad (6.5)$$

– Government:

$$T^d = \tau^d \sum_h p_h^f F F_h \quad (6.6)$$

$$T_j^z = \tau_j^z p_j^z Z_j \quad \forall j \quad (6.7)$$

$$T_i^m = \tau_i^m p_i^m M_i \quad \forall i \quad (6.8)$$

$$X_i^g = \frac{\mu_i}{p_i^q} \left(T^d + \sum_j T_j^z + \sum_j T_j^m - S^g \right) \quad \forall i \quad (6.9)$$

– Investment and savings:

$$X_i^v = \frac{\lambda_i}{p_i^q} \left(S^p + S^g + \varepsilon S^f \right) \quad \forall i \quad (6.10)$$

$$S^p = s s^p \sum_h p_h^f F F_h \quad (6.11)$$

$$S^g = s s^g \left(T^d + \sum_j T_j^z + \sum_j T_j^m \right) \quad (6.12)$$

– Household:

$$X_i^p = \frac{\alpha_i}{p_i^q} \left(\sum_h p_h^f F F_h - S^p - T^d \right) \quad \forall i \quad (6.13)$$

– Export and import prices and the balance of payments constraint:

$$p_i^e = \varepsilon p_i^{We} \quad \forall i \quad (6.14)$$

$$p_i^m = \varepsilon p_i^{Wm} \quad \forall i \quad (6.15)$$

$$\sum_i p_i^{We} E_i + S^f = \sum_i p_i^{Wm} M_i \quad (6.16)$$

– Substitution between imports and domestic goods (Armington composite):

$$Q_i = \gamma_i \left(\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i} \right)^{\frac{1}{\eta_i}} \quad \forall i \quad (6.17)$$

$$M_i = \left[\frac{\gamma_i^{\eta_i} \delta m_i p_i^q}{(1 + \tau_i^m) p_i^m} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (6.18)$$

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i p_i^q}{p_i^d} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (6.19)$$

– Transformation between exports and domestic goods:

$$Z_i = \theta_i \left(\xi e_i E_i^{\phi_i} + \xi d_i D_i^{\phi_i} \right)^{\frac{1}{\phi_i}} \quad \forall i \quad (6.20)$$

$$E_i = \left[\frac{\theta_i^{\phi_i} \xi e_i (1 + \tau_i^z) p_i^z}{p_i^e} \right]^{\frac{1}{1-\phi_i}} Z_i \quad \forall i \quad (6.21)$$

$$D_i = \left[\frac{\theta_i^{\phi_i} \xi d_i (1 + \tau_i^z) p_i^z}{p_i^d} \right]^{\frac{1}{1-\phi_i}} Z_i \quad \forall i \quad (6.22)$$

– Market-clearing conditions:

$$Q_i = X_i^p + X_i^g + X_i^v + \sum_j X_{i,j} \quad \forall i \quad (6.23)$$

$$\sum_j F_{h,j} = FF_h \quad \forall h \quad (6.24)$$

The above system of simultaneous equations consists of 24 sets of equations/48 single equations and the same number of endogenous variables. (More precisely, the number of single equations and endogenous variables is $(h + i + 17) \cdot j + h + 4$.) The endogenous variables in this model are:

$$Y_j, \quad F_{h,j}, \quad X_{i,j}, \quad Z_j, \quad X_i^p, \quad X_i^g, \quad X_i^v, \quad E_i, \quad M_i, \quad Q_i, \quad D_i, \quad p_i^f, \\ p_j^v, \quad p_j^z, \quad p_i^q, \quad p_i^e, \quad p_i^m, \quad p_i^d, \quad \varepsilon, \quad S^p, \quad S^g, \quad T^d, \quad T_j^z \text{ and } T_i^m.$$

The exogenous variables are:²⁰

$$FF_h, \quad S^f, \quad p_i^{We}, \quad p_i^{Wm}, \quad \tau^d, \quad \tau_j^z \text{ and } \tau_i^m.$$

Before concluding our modelling of the standard CGE model, we have to reconsider Walras's law. According to Walras's law, we do not have to impose the market-clearing conditions on all the n markets because the general equilibrium of this model economy holds with only the $n - 1$ market clearing conditions.²¹ One of the equations of the model is redundant; therefore, all the prices cannot be solved. We have to choose a numeraire and fix its price at a certain level, and express all the other

prices relative to the numeraire. As theory allows us to fix its price at an arbitrary level, we can fix it at unity for simplicity and for consistency with our calibration procedure, where all the (tax-exclusive) prices are set at unity. (We have not specified a numeraire in this mathematical model, but will choose a specific good or factor as the numeraire in the following GAMS programming model.)

6.8 **GAMS programming**

Here, we start development of a GAMS input file for the standard CGE model. As demonstrated above, we consider two goods (bread and milk) and two factors (capital and labour). We can express the model presented above in an input file as shown in List 6.1 (stdcge.gms).²²

Table 6.1 SAM for the standard CGE model

		Activity		Factor		Indirect Tax		Final Demand			External Total	
		BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	
Activity	BRD	21	8					20	19	16	8	92
	MLK	17	9					30	14	15	4	89
Factor	CAP	20	30									50
	LAB	15	25									40
Indirect Tax	IDT	5	4									9
	TRF	1	2									3
Final Demand	HOH			50	40							90
	GOV					9	3	23				35
External	INV							17	2			31
	EXT	13	11								12	24
Total		92	89	50	40	9	3	90	35	31	24	
		$p_i^{q0} X_{i,j}^{f0}$	$p_h^{f0} F_{h,j}^{m0}$	$p_j^{m0} M_j^{f0}$	$p_h^{f0} F_h^{z0}$	$\sum_j T_j^{z0}$	$\sum_j T_j^{m0}$	T^{d0}	S^{p0}	S^{g0}	$\epsilon^0 S^f$	
		T_j^{z0}	T_j^{m0}			$p_i^{q0} X_i^{p0}$	$p_i^{q0} X_i^{g0}$	$p_i^{q0} X_i^{v0}$	$p_i^{e0} E_i^0$			

Unit: yen
Note: The original SAM is shown in Table 4.2.

List 6.1 Input file for the standard CGE model (stdcge.gms)

```
1 $ Title A Standard CGE Model in Ch. 6 (STDCGE,SEQ=276)
2
3 * Definition of sets for suffix -----
4 Set      u          SAM entry      /BRD, MLK, CAP, LAB, IDT, TRF, HOH, GOV,
5                                     INV, EXT/
6         i(u)        goods          /BRD, MLK/
7         h(u)        factor         /CAP, LAB/;
8 Alias (u,v), (i,j), (h,k);
9 * -----
10
11 * Loading data -----
```

List 6.1 (Continued)

Table	SAM(u,v)		social accounting matrix		
	BRD	MLK	CAP	LAB	IDT
BRD	21	8			
MLK	17	9			
CAP	20	30			
LAB	15	25			
IDT	5	4			
TRF	1	2			
HOH			50	40	
GOV					9
INV					
EXT	13	11			
+	TRF	HOH	GOV	INV	EXT
BRD		20	19	16	8
MLK		30	14	15	4
CAP					
LAB					
IDT					
TRF					
HOH					
GOV	3	23			
INV		17	2		12
EXT					

```

36 ;
37 * Loading the initial values -----
38 Parameter      Y0(j)      composite factor
39                F0(h,j)    the h-th factor input by the j-th firm
40                X0(i,j)    intermediate input
41                Z0(j)      output of the j-th good
42                Xp0(i)     household consumption of the i-th good
43                Xg0(i)     government consumption
44                Xv0(i)     investment demand
45                E0(i)      exports
46                M0(i)      imports
47                Q0(i)      Armington's composite good
48                D0(i)      domestic good
49                Sp0        private saving
50                Sg0        government saving
51                Td0        direct tax
52                Tz0(j)     production tax
53                Tm0(j)     import tariff
54
55                FF(h)      factor endowment of the h-th factor
56                Sf         foreign saving in US dollars
57                pWe(i)     export price in US dollars
58                pWm(i)     import price in US dollars
59                tauz(i)     production tax rate
60                taum(i)     import tariff rate
61 ;
62 Td0 =SAM("GOV","HOH");
63 Tz0(j) =SAM("IDT",j);
64 Tm0(j) =SAM("TRF",j);
65
66 F0(h,j) =SAM(h,j);
67 Y0(j) =sum(h, F0(h,j));

```


List 6.1 (Continued)

```

68 X0(i,j) =SAM(i,j);
69 Z0(j)   =Y0(j) +sum(i, X0(i,j));
70 M0(i)   =SAM("EXT",i);
71
72 tauz(j) =Tz0(j)/Z0(j);
73 taum(j) =Tm0(j)/M0(j);
74
75 Xp0(i)  =SAM(i,"HOH");
76 FF(h)   =SAM("HOH",h);
77
78 Xg0(i)  =SAM(i,"GOV");
79 Xv0(i)  =SAM(i,"INV");
80 E0(i)   =SAM(i,"EXT");
81 Q0(i)   =Xp0(i)+Xg0(i)+Xv0(i)+sum(j, X0(i,j));
82 D0(i)   =(1+tauz(i))*Z0(i)-E0(i);
83 Sp0     =SAM("INV","HOH");
84 Sg0     =SAM("INV","GOV");
85 Sf      =SAM("INV","EXT");
86
87 pWe(i)  =1;
88 pWm(i)  =1;
89
90 Display Y0,F0,X0,Z0,Xp0,Xg0,Xv0,E0,M0,Q0,D0,Sp0,Sg0,Td0,Tz0,Tm0,
91         FF,Sf,tauz,taum;
92 * Calibration -----
93 Parameter      sigma(i)      elasticity of substitution
94                 psi(i)       elasticity of transformation
95                 eta(i)       substitution elasticity parameter
96                 phi(i)       transformation elasticity parameter
97 ;
98 sigma(i)=2;
99 psi(i)  =2;
100 eta(i)=(sigma(i)-1)/sigma(i);
101 phi(i)=(psi(i)+1)/psi(i);
102
103 Parameter      alpha(i)      share parameter in utility func.
104                 beta(h,j)    share parameter in production func.
105                 b(j)         scale parameter in production func.
106                 ax(i,j)      intermediate input requirement coeff.
107                 ay(j)        composite fact. input req. coeff.
108                 mu(i)        government consumption share
109                 lambda(i)     investment demand share
110                 deltam(i)     share par. in Armington func.
111                 deltad(i)     share par. in Armington func.
112                 gamma(i)     scale par. in Armington func.
113                 xld(i)        share par. in transformation func.
114                 xie(i)        share par. in transformation func.
115                 theta(i)      scale par. in transformation func.
116                 ssp           average propensity for private saving
117                 ssg           average propensity for gov. saving
118                 taud          direct tax rate
119 ;
120
121 alpha(i)=Xp0(i)/sum(j, Xp0(j));
122 beta(h,j)=F0(h,j)/sum(k, F0(k,j));
123 b(j)     =Y0(j)/prod(h, F0(h,j)**beta(h,j));

```

List 6.1 (Continued)

```

124
125 ax(i,j) =X0(i,j)/Z0(j);
126 ay(j)  =Y0(j)/Z0(j);
127 mu(i)  =Xg0(i)/sum(j, Xg0(j));
128 lambda(i)=Xv0(i)/(Sp0+Sg0+Sf);
129
130 deltam(i)=(1+taum(i))*M0(i)**(1-eta(i))
131           /((1+taum(i))*M0(i)**(1-eta(i)) +D0(i)**(1-eta(i)));
132 deltad(i)=D0(i)**(1-eta(i))
133           /((1+taum(i))*M0(i)**(1-eta(i)) +D0(i)**(1-eta(i)));
134 gamma(i)=Q0(i)/(deltam(i)*M0(i)**eta(i)+deltad(i)*D0(i)**eta(i))
135           ** (1/eta(i));
136
137 xie(i)=E0(i)**(1-phi(i))/(E0(i)**(1-phi(i))+D0(i)**(1-phi(i)));
138 xid(i)=D0(i)**(1-phi(i))/(E0(i)**(1-phi(i))+D0(i)**(1-phi(i)));
139 theta(i)=Z0(i)
140           /(xie(i)*E0(i)**phi(i)+xid(i)*D0(i)**phi(i))** (1/phi(i));
141
142 ssp      =Sp0/sum(h, FF(h));
143 ssg      =Sg0/(Td0+sum(j, Tz0(j))+sum(j, Tm0(j)));
144 taud     =Td0/sum(h, FF(h));
145
146 Display alpha,beta,b,ax,ay,mu,lambda,deltam,deltad,gamma,xie,
147         xid,theta,ssp,ssg,taud;
148 * -----
149
150 * Defining model system -----
151 Variable      Y(j)           composite factor
152               F(h,j)        the h-th factor input by the j-th firm
153               X(i,j)        intermediate input
154               Z(j)          output of the j-th good
155               Xp(i)         household consumption of the i-th good
156               Xg(i)         government consumption
157               Xv(i)         investment demand
158               E(i)          exports
159               M(i)          imports
160               Q(i)          Armington's composite good
161               D(i)          domestic good
162
163               pf(h)         the h-th factor price
164               py(j)         composite factor price
165               pz(j)         supply price of the i-th good
166               pq(i)         Armington's composite good price
167               pe(i)         export price in local currency
168               pm(i)         import price in local currency
169               pd(i)         the i-th domestic good price
170               epsilon       exchange rate
171
172               Sp            private saving
173               Sg            government saving
174               Td            direct tax
175               Tz(j)         production tax
176               Tm(i)         import tariff
177
178               UU            utility [fictitious]
179 ,

```

List 6.1 (Continued)

180	Equation	eqpy(j)	composite factor agg. func.
181		eqF(h,j)	factor demand function
182		eqX(i,j)	intermediate demand function
183		eqY(j)	composite factor demand function
184		eqpzs(j)	unit cost function
185			
186		eqTd	direct tax revenue function
187		eqTz(j)	production tax revenue function
188		eqTm(i)	import tariff revenue function
189		eqXg(i)	government demand function
190			
191		eqXv(i)	investment demand function
192		eqSp	private saving function
193		eqSg	government saving function
194			
195		eqXp(i)	household demand function
196			
197		eqpe(i)	world export price equation
198		eqpm(i)	world import price equation
199		eqepsilon	balance of payments
200			
201		eqpqs(i)	Armington function
202		eqM(i)	import demand function
203		eqD(i)	domestic good demand function
204			
205		eqpzd(i)	transformation function
206		eqDs(i)	domestic good supply function
207		eqE(i)	export supply function
208			
209		eqpqd(i)	market clearing cond. for comp. good
210		eqpf(h)	factor market clearing condition
211			
212		obj	utility function [fictitious]
213			
214	*[domestic production] ----		
215	eqpy(j) ..	Y(j)	=e= b(j)*prod(h, F(h,j)**beta(h,j));
216	eqF(h,j) ..	F(h,j)	=e= beta(h,j)*py(j)*Y(j)/pf(h);
217	eqX(i,j) ..	X(i,j)	=e= ax(i,j)*Z(j);
218	eqY(j) ..	Y(j)	=e= ay(j)*Z(j);
219	eqpzs(j) ..	pz(j)	=e= ay(j)*py(j) +sum(i, ax(i,j)*pq(i));
220			
221	*[government behavior] ----		
222	eqTd ..	Td	=e= taud*sum(h, pf(h)*FF(h));
223	eqTz(j) ..	Tz(j)	=e= tauz(j)*pz(j)*Z(j);
224	eqTm(i) ..	Tm(i)	=e= taum(i)*pm(i)*M(i);
225	eqXg(i) ..	Xg(i)	=e= mu(i)*(Td +sum(j, Tz(j)) +sum(j, Tm(j))
226			-Sg)/pq(i);
227	*[investment behavior] ----		
228	eqXv(i) ..	Xv(i)	=e= lambda(i)*(Sp +Sg +epsilon*Sf)/pq(i);
229			
230	*[savings] -----		
231	eqSp ..	Sp	=e= ssp*sum(h, pf(h)*FF(h));
232	eqSg ..	Sg	=e= ssg*(Td +sum(j, Tz(j)) +sum(j, Tm(j)));
233			
234	*[household consumption] --		
235	eqXp(i) ..	Xp(i)	=e= alpha(i)*(sum(h, pf(h)*FF(h)) -Sp -Td)

List 6.1 (Continued)

```

236 /pq(i);
237 *[international trade] ----
238 eqpe(i)..      pe(i)      =e= epsilon*pWe(i);
239 eqpm(i)..      pm(i)      =e= epsilon*pWm(i);
240 eqepsilon..    sum(i, pWe(i)*E(i)) +Sf
241              =e= sum(i, pWm(i)*M(i));
242
243 *[Armington function] -----
244 eqqqs(i)..      Q(i)      =e= gamma(i)*(deltam(i)*M(i)**eta(i)+deltad(i)
245              *D(i)**eta(i))**(1/eta(i));
246 eqM(i)..        M(i)      =e= (gamma(i)**eta(i)*deltam(i)*pq(i)
247              /( (1+taum(i))*pm(i) ) )**(1/(1-eta(i))) *Q(i);
248 eqD(i)..        D(i)      =e= (gamma(i)**eta(i)*deltad(i)*pq(i)/pd(i)
249              ** (1/(1-eta(i))) ) *Q(i);
250
251 *[transformation function] -----
252 eqqzd(i)..      Z(i)      =e= theta(i)*(xie(i)*E(i)**phi(i)+xid(i)
253              *D(i)**phi(i))**(1/phi(i));
254 eqE(i)..        E(i)      =e= (theta(i)**phi(i)*xie(i)*(1+tauz(i))*pz(i)
255              /pe(i))**(1/(1-phi(i))) *Z(i);
256 eqDs(i)..      D(i)      =e= (theta(i)**phi(i)*xid(i)*(1+tauz(i))*pz(i)
257              /pd(i))**(1/(1-phi(i))) *Z(i);
258
259 *[market clearing condition]
260 eqpqd(i)..      Q(i)      =e= Xp(i) +Xg(i) +Xv(i) +sum(j, X(i,j));
261 eqpf(h)..      sum(j, F(h,j)) =e= FF(h);
262
263 *[fictitious objective function]
264 obj..          UU          =e= prod(i, Xp(i)**alpha(i));
265 * -----
266
267 * Initializing variables -----
268 Y.l(j) =Y0(j);
269 F.l(h,j)=F0(h,j);
270 X.l(i,j)=X0(i,j);
271 Z.l(j) =Z0(j);
272 Xp.l(i) =Xp0(i);
273 Xg.l(i) =Xg0(i);
274 Xv.l(i) =Xv0(i);
275 E.l(i) =E0(i);
276 M.l(i) =M0(i);
277 Q.l(i) =Q0(i);
278 D.l(i) =D0(i);
279 pf.l(h) =1;
280 py.l(j) =1;
281 pz.l(j) =1;
282 pq.l(i) =1;
283 pe.l(i) =1;
284 pm.l(i) =1;
285 pd.l(i) =1;
286 epsilon.l=1;
287 Sp.l =Sp0;
288 Sg.l =Sg0;
289 Td.l =Td0;
290 Tz.l(j) =Tz0(j);
291 Tm.l(i) =Tm0(i);
292 * -----

```

List 6.1 (Continued)

```

293
294 * Setting lower bounds to avoid division by zero -----
295 Y.lo(j) =0.00001;
296 F.lo(h,j)=0.00001;
297 X.lo(i,j)=0.00001;
298 Z.lo(j) =0.00001;
299 Xp.lo(i)=0.00001;
300 Xg.lo(i)=0.00001;
301 Xv.lo(i)=0.00001;
302 E.lo(i) =0.00001;
303 M.lo(i) =0.00001;
304 Q.lo(i) =0.00001;
305 D.lo(i) =0.00001;
306 pf.lo(h)=0.00001;
307 py.lo(j)=0.00001;
308 pz.lo(j)=0.00001;
309 pq.lo(i)=0.00001;
310 pe.lo(i)=0.00001;
311 pm.lo(i)=0.00001;
312 pd.lo(i)=0.00001;
313 epsilon.lo=0.00001;
314 Sp.lo =0.00001;
315 Sg.lo =0.00001;
316 Td.lo =0.00001;
317 Tz.lo(j)=0.0000;
318 Tm.lo(i)=0.0000;
319 * -----
320 * numeraire ---
321 pf.fx("LAB")=1;
322 * -----
323 * Defining and solving the model -----
324 Model stdcge /all/;
325 Solve stdcge maximizing UU using nlp;

```

6.8.1 Declaration of sets, installation of the SAM and derivation of the initial equilibrium values

Lines 4–7 with the `Set` directive declare sets that include goods, agents and others as their elements, indicated by indices. The declaration of sets is followed by the `Alias` directive for duplicating these sets. New SAM entries such as the government, investment, the external sector, the production tax (an indirect tax) and import tariffs are abbreviated to `GOV`, `INV`, `EXT`, `IDT` and `TRF` respectively. The SAM shown in Table 6.1 appears in lines 12–36 with the `Table` directive. It should be noted that because the matrix is too wide to be placed in the input file, it is divided into two submatrices. A plus symbol ‘+’ is placed at the beginning of line 25 (i.e., the first row of the second submatrix) so that these two submatrices constitute one matrix of the SAM. When the matrix is much

larger and needs to be broken into three or more submatrices, we can use the '+' symbol in the same manner.

Lines 38–61 declare the names of constants, which consist of the initial equilibrium values of endogenous variables and some exogenously given variables. The initial equilibrium values of endogenous variables are indicated by their symbols followed by '0'.

The initial equilibrium values of the endogenous variables are split into two types. The first is those directly derived from the SAM. The second is those derived from the SAM through computation. The derivation of these values is shown in lines 62–85. The first type of derivation is straightforward. Consider, for example, line 68, which shows that the amount of intermediate inputs is retrieved directly from the SAM. Table 6.1 helps readers to understand from which cell of the SAM $X_{i,j}^0$ should be retrieved.

A little more explanation is needed about the second type: Z_j^0 , Q_i^0 and D_i^0 , which are not explicitly shown in the SAM. Line 69 shows that Z_j^0 is derived using the zero-profit condition discussed in Section 6.2, evaluated at the initial equilibrium:²³

$$p_j^{z0} Z_j^0 = p_j^{y0} Y_j^0 + \sum_i p_i^{q0} X_{i,j}^0 \quad \forall j$$

In line 81, Q_i^0 is derived using the market-clearing condition (6.23) in the initial equilibrium:²⁴

$$Q_i^0 = X_i^{p0} + X_i^{g0} + X_i^{v0} + \sum_j X_{i,j}^0 \quad \forall i$$

In line 82, D_i^0 is derived using the zero-profit condition of the gross domestic output transforming firm:

$$(1 + \tau_i^z) p_i^{z0} Z_i^0 = p_i^{e0} E_i^0 + p_i^{d0} D_i^0 \quad \forall i$$

We must pay attention to the order of these computational processes. For example, before computing D_i^0 in line 82, the production tax rate τ_j^z must have been computed in line 72 as follows:

$$\tau_j^z = \frac{T_j^{z0}}{p_j^{z0} Z_j^0} \quad \forall j$$

Similarly, the computation of τ_j^z in line 72 must be preceded by that of T_j^{z0} (line 63) and Z_j^0 (line 69), which appear on the right-hand side of the above formula.

Derivation of the initial equilibrium values of the endogenous variables is followed by that of the exogenously given variables. The international (tax-exclusive) prices of exports and imports, expressed in terms of US dollars, are set at unity (lines 87–88) to maintain consistency among all the prices in the initial equilibrium. In addition to these variables, the parameters of the CET and CES functions (lines 98–99) are exogenously given and are discussed later.

6.8.2 Calibration

Lines 93–101 derive the elasticity parameters η_i in the CES function and ϕ_i in the CET function. To derive η_i (line 100), the symbol for the elasticity of substitution σ_i – characterizing the curvature of the isoquant curves of the CES functions – is defined (line 93) and set equal to two (line 98). Similarly, to derive ϕ_i (line 101), the elasticity of transformation ψ_i – characterizing the curvature of the isoquant curves of the CET function – is defined (line 94) and set equal to two (line 99). (We will soon discuss the reason that we do not calibrate these elasticities σ_i and ψ_i but instead set their values exogenously.)

Lines 103–119 declare the symbols for the other coefficients and exogenous variables to be calibrated. In lines 121–144, we calibrate their values. While we employed only Cobb–Douglas-type functions in the simple CGE model in Chapter 2, we now have to calibrate coefficients in the Leontief and CES/CET functions as well as the direct tax rate and the savings rates. In the following, we need not explain the calibration processes for α_i , $\beta_{h,j}$ and b_j , because they were discussed in Subsection 5.4.5. Equally, the calibration processes for the expenditure share coefficients for government consumption μ_i and the investment uses λ_i are omitted, as they are identical to the calibration process for α_i as shown in lines 127–128.

Calibration of the Leontief-type function

Lines 125–126 calibrate the coefficients in the Leontief-type production function $ax_{i,j}$ and ay_j as follows:

$$ax_{i,j} = \frac{X_{i,j}^0}{Z_j^0} \quad \forall i, j$$

$$ay_j = \frac{Y_j^0}{Z_j^0} \quad \forall j$$

Recall that these formulae are derived from the demand functions for the intermediate inputs and the composite factor:

$$X_{i,j} = ax_{i,j}Z_j \quad \forall i, j \quad (6.3)$$

$$Y_j = ay_jZ_j \quad \forall j \quad (6.4)$$

As the initial equilibrium values $X_{i,j}^0$, Y_j^0 and Z_j^0 have been already determined, $ax_{i,j}$ and ay_j can be easily computed.

Calibration of the CES function

The Armington-composite-good-production function (6.17) contains four unknown coefficients δm_i , δd_i , γ_i and η_i , and these coefficients appear in three equations (6.17)–(6.19). The initial equilibrium values of the endogenous variables Q_i , M_i , D_i , p_i^q , p_i^m and p_i^d , which appear in those three equations, have been retrieved from the SAM or can be set at unity; in either case, they are all known. In sum, we have four unknown coefficients with three equations; thus, we cannot solve them using our calibration technique. Therefore, we have to assume a value for one of these four coefficients using data or facts outside the model and the SAM. We usually assume the value of the elasticity of substitution and the elasticity of transformation on the basis of survey work in previous studies or available databases.²⁵ Here, we assume that the elasticity of substitution σ_i is two for all the goods (line 98). Once the elasticity of substitution σ_i is decided, we can obtain the elasticity parameter η_i in the CES function (line 100) as $\eta_i = (\sigma_i - 1)/\sigma_i$.

After the elasticity of substitution σ_i is assumed to be two, the input share coefficients δm_i and δd_i are estimated in lines 130–133:²⁶

$$\delta m_i = \frac{(1 + \tau_i^m) p_i^{m0} M_i^{0(1-\eta_i)}}{(1 + \tau_i^m) p_i^{m0} M_i^{0(1-\eta_i)} + p_i^{d0} D_i^{0(1-\eta_i)}} \quad \forall i \quad (6.e)$$

$$\delta d_i = \frac{p_i^{d0} D_i^{0(1-\eta_i)}}{(1 + \tau_i^m) p_i^{m0} M_i^{0(1-\eta_i)} + p_i^{d0} D_i^{0(1-\eta_i)}} \quad \forall i \quad (6.f)$$

In calibrating δm_i and δd_i , it must be noted that the tariff-inclusive import price in the initial equilibrium $(1 + \tau_i^m) p_i^{m0}$ is *not* equal to unity. This tariff-inclusive import price, however, can be obtained because the initial equilibrium import price p_i^{m0} is set at unity and τ_i^m has been computed in line 73.

Having calibrated these share coefficients δd_i and δm_i , we can calibrate the scaling coefficient γ_i . Simply inverting the original Armington-composite-good-production function (6.17), we get:

$$\gamma_i = \frac{Q_i^0}{(\delta m_i M_i^{0\eta_i} + \delta d_i D_i^{0\eta_i})^{1/\eta_i}} \quad \forall i$$

Calibration of the CET function

The calibration procedures for the CET functions are almost the same as those for the CES functions discussed above. There are four unknown coefficients ξe_i , ξd_i , θ_i and ϕ_i but only three equations (6.20)–(6.22). Among these four unknowns, the elasticity of transformation ψ_i is exogenously set at two for all the goods (line 99); then, the elasticity parameter is computed as $\phi_i = (\psi_i + 1)/\psi_i$ (line 101). Once the elasticity parameter ϕ_i is determined, we can calibrate the unknown share coefficients ξe_i and ξd_i using the supply functions for exports (6.21) and the domestic good (6.22) in the following manner:²⁷

$$\xi e_i = \frac{p_i^{e0} E_i^{0(1-\phi_i)}}{p_i^{e0} E_i^{0(1-\phi_i)} + p_i^{d0} D_i^{0(1-\phi_i)}} \quad \forall i$$

$$\xi d_i = \frac{p_i^{d0} D_i^{0(1-\phi_i)}}{p_i^{e0} E_i^{0(1-\phi_i)} + p_i^{d0} D_i^{0(1-\phi_i)}} \quad \forall i$$

These equations are in lines 137–138. Finally, lines 139–140 use these coefficients to calibrate the scaling coefficient θ_i in the CET function (6.20) as follows:

$$\theta_i = \frac{Z_i^0}{(\xi e_i E_i^{0\phi_i} + \xi d_i D_i^{0\phi_i})^{1/\phi_i}} \quad \forall i$$

Estimation of savings rates and direct tax rate

Regarding savings rates and tax rates, there are three unknown coefficients remaining: the average savings propensity for the household ss^p and for the government ss^g , and the direct tax rate τ^d . These coefficients for the average savings propensity are derived from the savings functions

(6.11) and (6.12) in lines 142–143 as follows:

$$ss^p = \frac{S^{p0}}{\sum_h p_h^{f0} FF_h}$$

$$ss^g = \frac{S^{g0}}{T^{d0} + \sum_j T_j^{z0} + \sum_j T_j^{m0}}$$

The direct tax rate can be obtained by feeding the direct tax revenues and household income, which is supposed to be the tax base, derived from the initial equilibrium, into the following equation (line 144):

$$\tau^d = \frac{T^{d0}}{\sum_h p_h^{f0} FF_h}$$

Table 6.2 Equations in the standard CGE model in List 6.1

Line number in List 6.1	Equation number	Description of equations
215	(6.1)	Composite factor production function
216	(6.2)	Factor demand function
217	(6.3)	Intermediate input demand function
218	(6.4)	Composite factor demand function
219	(6.5)	Unit cost function for gross domestic output
222	(6.6)	Direct tax revenue function
223	(6.7)	Production tax revenue function
224	(6.8)	Import tariff revenue function
225–226	(6.9)	Government demand function
228	(6.10)	Investment demand function
231	(6.11)	Household savings function
232	(6.12)	Government savings function
235–236	(6.13)	Household demand function
238	(6.14)	Export price conversion equation
239	(6.15)	Import price conversion equation
240–241	(6.16)	Balance of payments constraint
244–245	(6.17)	Armington-composite-good-production function
246–247	(6.18)	Import demand function
248–249	(6.19)	Domestic good demand function
252–253	(6.20)	Gross-domestic-output-transformation function
254–255	(6.21)	Export supply function
256–257	(6.22)	Domestic good supply function
260	(6.23)	Market-clearing condition for the Armington composite good
261	(6.24)	Market-clearing condition for factor
264	–	Household utility function

6.8.3 Model solution

The procedures in and after line 151 are identical to those in List 5.1 for the simple CGE model and thus do not need to be repeated here. After

setting the `Solve` statement (line 325), we click the icon  on the GAMS IDE and obtain the solution; then, the output file is automatically generated with the extension 'lst'.²⁸ (The output file of the standard CGE model `stdcge.lst` is not shown in this book.)

As the model is expanded with many additional features, we are likely to encounter errors in our programming. Errors can be fixed by carefully examining the input file for missing parentheses and semicolons, typos of operators for multiplication and power, and so on (some typical errors are demonstrated in Annex B). As explained in Chapter 5, in the case that the errors are beyond syntactical ones, we should examine the model itself from the beginning. Table 6.2 provides the mapping between the model equations and their entries in the input file.

Notes

1. In this standard CGE model, we still assume only one representative household. A model with multiple and heterogeneous households is demonstrated in Section 10.1.
2. In this figure, while we describe the flow of goods, mainly bread, we can also do this for milk.
3. Note that we do not further decompose investment into private and public, or fixed capital formation and increases in stocks.
4. A homogeneous function of degree n $f(u, v)$ is defined as follows:

$$f(t \cdot u, t \cdot v) = t^n \cdot f(u, v)$$

This implies that when we increase the amount of all the inputs by a factor of t , output also increases by a factor of t . We have two examples of the homogeneity in our CGE model. One is that the production functions used in this book, except for those used in Section 10.6, are homogeneous of degree one. The other is that the CGE model system is homogeneous of degree zero in prices. Let the model system denote $\text{CGE}(\mathbf{p}, \mathbf{q}) = \mathbf{0}$, where $\mathbf{p}$ denotes a price vector and $\mathbf{q}$ a quantity vector in the system. This homogeneity implies that the following holds:

$$\text{CGE}(t\mathbf{p}, \mathbf{q}) = t^0 \text{CGE}(\mathbf{p}, \mathbf{q})$$

5. The function $\min(u, v)$ indicates the smallest value among its elements; for example, $\min(10, 20) = 10$.
6. We will discuss the composite good and its price in Section 6.5.

7. We can derive (6.1) and (6.2) in the same manner that we derived (2.2) and (2.3) by solving the profit-maximization problem in Chapter 2. However, we cannot solve the profit-maximization problem for the second stage by simply applying the Lagrange multiplier method because the Leontief-type production function (6.5') is not differentiable. To derive the intermediate input demand function (6.3) and the composite factor demand function (6.4), we have to use a constant elasticity of substitution (CES) function, which is a generalized functional form of the Leontief-type function, and derive these first-order conditions. When the elasticity of substitution in the CES function tends to zero, the derived first-order conditions approach the expressions of (6.3), (6.4) and (6.5'). To understand this graphically, see Appendix V.
8. In most numerical computations, we make use of differential coefficients of the concerned variables to search for optimal solutions. When we use functions with kinks such as the Leontief-type function, we cannot define the differential coefficients well and encounter computational difficulties.
9. The profit π_j^z is supposed to be zero in equilibrium because perfect competition does not allow any excess profits.
10. When we adopt this assumption, we also have to modify the government revenue part so that the government budget is balanced. For example, we can assume that the direct tax is a lump-sum tax on household income irrespective of its income and that the direct tax is flexibly adjusted so as to achieve a balanced government budget.
11. Investment and savings in a static model are, in a sense, 'alien bodies' (Wiltenbockel 1994: 60). This issue has something to do with the macro closure issue, which is discussed in Chapter 7.
12. In Japan, the share of gross fixed capital formation is about one-quarter of GDP.
13. This ε is different from that used as the Lagrange multiplier ε_h in Appendix II.
14. This situation is similar to our daily life where we do not have a significant impact on the Dow Jones or the Nikkei 225 average indices in stock markets.
15. This is caused partly because the classification is not detailed enough to classify similar but different goods in separate categories in trade statistics. For example, Japan exports small cars and imports large cars. The trade statistics, however, simply record that 'cars' are both exported and imported. Another reason involves seasonal variation in trade patterns. Harvesting seasons are opposite between the northern and the southern hemispheres. This explains why exports and imports of the same agricultural good occur in different seasons, such that both are recorded in the trade statistics for the same year.
16. We assume that the gross domestic output is transformed into two products shipped as exports and for domestic markets. The good shipped to domestic markets is called the 'domestic good' hereinafter. See Subsection 6.5.4 for details.
17. We can derive (6.18) and (6.19) in the following manner. Let us define a Lagrange function $L_i(\cdot)$ with the Lagrange multiplier θ_i for the above profit-maximization problem for the i -th firm as follows:

$$\begin{aligned}
 L_i(Q_i, M_i, D_i; \theta_i) = & p_i^q Q_i - [(1 + \tau_i^m) p_i^m M_i + p_i^d D_i] \\
 & + \theta_i [\gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{\frac{1}{\eta_i}} - Q_i]
 \end{aligned}$$

Its first-order conditions are:

$$\frac{\partial L_i}{\partial Q_i} = p_i^q - \theta_i = 0 \quad (6.a)$$

$$\frac{\partial L_i}{\partial M_i} = -(1 + \tau_i^m) p_i^m + \theta_i \frac{1}{\eta_i} \gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{\frac{1}{\eta_i}-1} \eta_i \delta m_i M_i^{\eta_i-1} = 0 \quad (6.b)$$

$$\frac{\partial L_i}{\partial D_i} = -p_i^d + \theta_i \frac{1}{\eta_i} \gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{\frac{1}{\eta_i}-1} \eta_i \delta d_i D_i^{\eta_i-1} = 0 \quad (6.c)$$

$$\frac{\partial L_i}{\partial \theta_i} = \gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{\frac{1}{\eta_i}} - Q_i = 0 \quad (6.d)$$

As the transformation of (6.d) implies $\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i} = (Q_i/\gamma_i)^{\eta_i}$, we can eliminate the terms $(\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})$ in (6.b) and (6.c) using this expression and θ_i using (6.a) to obtain:

$$-(1 + \tau_i^m) p_i^m + p_i^q \gamma_i \left(\frac{Q_i}{\gamma_i} \right)^{\eta_i \frac{1-\eta_i}{\eta_i}} \delta m_i M_i^{\eta_i-1} = 0 \quad (6.b')$$

$$-p_i^d + p_i^q \gamma_i \left(\frac{Q_i}{\gamma_i} \right)^{\eta_i \frac{1-\eta_i}{\eta_i}} \delta d_i D_i^{\eta_i-1} = 0 \quad (6.c')$$

Rearranging (6.b') and (6.c') for M_i and D_i , we get the expressions for the demand functions (6.18) and (6.19).

18. Alternatively, we sometimes assume that exports and the domestic good are *perfectly* substitutable (or, strictly speaking, transformable) while maintaining Armington's assumption for imports and the domestic good demand. In this case, we replace (6.20)–(6.22) in the original model with the following equations:

$$Z_i = E_i + D_i \quad \forall i \quad (6.20')$$

$$(1 + \tau_i^z) p_i^z = p_i^e \quad \forall i \quad (6.21')$$

$$(1 + \tau_i^z) p_i^z = p_i^d \quad \forall i \quad (6.22')$$

These three equations correspond to a special case of the original equations (6.20)–(6.22) where the elasticity of transformation ψ_i is set at infinity.

19. This θ_i has nothing to do with the Lagrange multiplier θ_i in Note 17 in Chapter 6 and Appendix II.
20. In this model, we assume that domestic savings S^p and S^s are determined by (6.11) and (6.12), and that foreign savings S^f is exogenous in foreign currency terms, while the amount of investment X_i^y is determined by (6.10) so as to equilibrate total savings with total investment. That is, our model uses a savings-driven closure where savings are determined first and investment is made within the predetermined budget of total savings. Instead, we can assume that investment is determined first, followed by the determination of

savings. Similar issues arise regarding the choice of the assumption whether the exchange rate ε and foreign savings S^f should be endogenous or exogenous in our model. This is an issue of macro closure rules, which are discussed in the next chapter in detail.

21. Regarding Walras's law, see Section 2.5 or a textbook of microeconomics; for example, Varian (2006).
22. The input file (`stdcge.gms`) available in the GAMS Model Library contains more code than shown in List 6.1. These additional codes are explained in Sections 8.1–8.3, and A.5 in Annex A.
23. As all the prices are set at unity in the initial equilibrium, no price explicitly appears in line 69.
24. Instead, after estimating D_i^0 , we can compute Q_i^0 using the zero-profit condition of the Armington-composite-good-producing firm:

$$p_i^{q0} Q_i^0 = p_i^{d0} D_i^0 + (1 + \tau_i^m) p_i^{m0} M_i^0$$

Whichever option we choose, the results must match each other. Any inconsistency in these estimates indicates errors in the data of the SAM, model equations or estimation processes of the initial equilibrium values.

25. For example, we can use the GTAP database and a survey study by Stern et al. (1976) as well as the estimation results of the price elasticity of import demand. Alternatively, we can conduct our own original estimation of the elasticity of substitution. Liu et al. (2004) examine the validity of these parameters suggested in the GTAP database from an econometrics viewpoint. We can use the price elasticity of import demand as the elasticity of substitution in the CES function because the price elasticity of the (import) demand function derived from the CES function is approximately equal to the elasticity of substitution. Details are discussed in Section 10.2.
26. To derive (6.e) and (6.f), we divide both sides of the import demand function (6.18) by the domestic good demand function (6.19) as follows:

$$\frac{M_i}{D_i} = \left[\frac{\delta m_i}{\delta d_i} \frac{p_i^d}{(1 + \tau_i^m) p_i^m} \right]^{\frac{1}{1-\eta_i}} \quad \forall i$$

As the sum of these input share coefficients is always equal to unity, we can eliminate δd_i from the above equation to get:

$$\delta m_i = \frac{(M_i/D_i)^{1-\eta_i} (1 + \tau_i^m) p_i^m / p_i^d}{1 + (M_i/D_i)^{1-\eta_i} (1 + \tau_i^m) p_i^m / p_i^d} \quad \forall i$$

Multiplying both the denominator and the numerator on the right-hand side of the above expression by $p_i^d D_i^{1-\eta_i}$, we get (6.e). We can also get (6.f) in a similar manner.

27. These coefficients can be derived in exactly the same way that δm_i and δd_i are derived.
28. For an explanation of the uses of GAMS IDE, see the online help or its printable file available at www.gams.com/dd/docs/tools/gamside.pdf.

7

Macro Closure

Any empirical model depicts only part of the real economy; thus, it has a borderline distinguishing between the ‘inside’ of the model, which is represented by endogenous variables, and the ‘outside’, which is assumed to be given. In the ‘simple CGE model’ developed in Chapter 2, the supply of primary factors is assumed to be exogenously constrained by their endowments FF_h . Instead, their supply can be endogenized with a factor supply function. For example, we can incorporate consumption of time as leisure in the household utility function to generate voluntary unemployment or an inflow of foreign labour. In principle, we can freely set a borderline delineating the endogenous part from the exogenous part of the model, considering the purpose of the analysis with insights into the actual economic phenomena.

In the case of CGE modelling, the freedom to choose the ‘inside’ and ‘outside’ of the model is constrained by the model size, which largely determines the computability and the handling complexity. As CGE models tend to be large scale, the model size is a serious matter to be considered. We often use static models, rather than dynamic ones, mainly because of consideration of the model size. In static models, we cannot embed any microfoundations, or the optimization behaviour of the agents related to investment and savings. As a practical solution, we assume that either of (key parameters characterizing) investment or savings (or both of them) is exogenously given and that the other, as an endogenous variable, adjusts itself to establish equality between investment and savings. The issue of macro closure is, in short, the choice of exogenous variables among all the variables in the model.¹

Section 7.1 is devoted to a discussion of macro closure regarding investment and savings in a closed economy model. Section 7.2 discusses the same issue in an open economy, where the balance of payments

constraint is an additional factor of the macro closure. This is because the current account deficit in the balance of payments equals foreign savings, which adds another dynamic element to the model.

7.1 Investment and savings – macro closure in a closed economy model

In this book, we have developed only static CGE models. These models assume that all economic activities are conducted and completed within one period – usually one year. It implies that the end of this period means ‘the end of the world’, and no activities occur in the ‘next’ period. In contrast, actual economic activities are directed, to some extent, to generate benefits in the next period, and these activities are recorded in statistical data (Figure 7.1). Activities for the future include investment in fixed capital formation and stocks as well as savings to finance it.

More detailed discussion is needed on this issue. As presented in Chapter 5, calibration occurs on the basis of data for one period, which is also the time horizon of our comparative static analysis; however, real economic activities are conducted in multiperiod time horizons. Even in one-period data, there is always ‘something related to future periods’ such as investment and savings (or borrowings if negative). Therefore, if we use ‘something related to future periods’ in a one-period model framework (or, more generally, a finite-period model framework), there should be some logic to justify it.

Let us consider a closed economy model, where an agent reserves part of its income as savings, and other agents use these savings for investment, for simplicity. To attain the general equilibrium of this model economy, the investment–savings balance must hold. If we develop a dynamic model, we can incorporate investment and savings functions with an endogenous variable – for example, an interest rate – into the

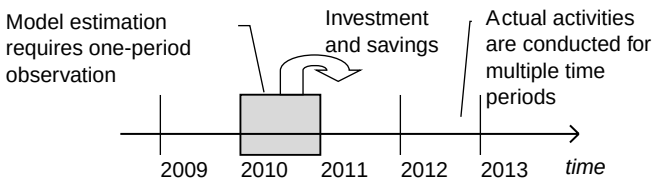


Figure 7.1 Differences in time horizon between actual economic activities and data used in model estimation

model, where the interest rate is assumed to adjust itself so as to establish the investment–savings balance.

In contrast, the static CGE model, which we are developing here, is not equipped with such investment and savings functions based on micro-foundations of intertemporal decision-making; therefore, we cannot simply rely on the equality of the number of equations and endogenous variables related to investment and savings to attain balance.

More specifically, although the model has only one equation representing the investment–savings balance, there are two endogenous variables – namely, investment and savings. We cannot obtain solutions for all the endogenous variables in the model. As a result, we have to make one of these two endogenous variables an exogenous variable or we have to add one more equation constraining these variables. Unless we increase the number of equations, there are obviously two ways to solve this problem. The first is to fix investment I as an exogenous variable and to equalize savings S endogenously to this investment. This type of closure is called investment-driven closure. The second is to fix savings S as an exogenous variable. This closure is called savings-driven closure.

To exemplify further the macro closure problem, let us consider our standard CGE model. It has three variables, S^p , S^s and I , and savings functions (6.11) and (6.12) determine S^p and S^s . (We do not consider foreign savings in this closed economy exercise.) Furthermore, because of the necessity of the investment–savings balance, investment I is endogenously solved to be equated to total savings consisting of S^p and S^s . Then, if a function to determine I is added, it will overdetermine the model. In such a case, we can equate the number of variables to the number of equations by adding, for example, a new endogenous variable such as transfers between the household and the government. With this specification, transfers between them work as an endogenous device to change the aggregate propensity for savings and to restore the investment–savings balance if the propensity is different between these two agents.

Now, let us compare savings-driven closure with investment-driven closure. In savings-driven closure, savings are first determined by income; then, investment is determined. Thus, an increase in income leads to an increase in both savings and investment. In the CGE models presented in this book, investment is simply a waste of resources; more investment implies lower welfare. The increase in welfare, generated by an increase in income, is partly cancelled out by an increase in investment. In contrast, in investment-driven closure, investment is

exogenous and independent of income changes. An increase in income, through an increase in consumption, leads to a full increase in welfare. Savings-driven closure would be suitable for quantifying the impact of certain policy changes or exogenous shocks, through changes in savings, exerted on other economic variables. Investment-driven closure would be useful for analysis that quantifies the impact of those shocks through changes in investment on welfare. We have to choose macro closure rules that reflect our research objectives and focus.

The above discussion suggests that there is no perfect choice of macro closure. This is because dynamic factors are incorporated awkwardly into our static CGE model simply for consistency with empirical data. To solve the contradiction of investment and savings in a static model fundamentally, we must rely on a dynamic model with a multiperiod time horizon.

7.2 Current account balance – macro closure in an open economy model

An open economy model has two additional variables: foreign savings S^f and the foreign exchange rate ε , and one additional equation; namely, the balance of payments constraint. Even after we solve the macro closure problem concerning the domestic investment–savings balance as discussed in Section 7.1, there is still one endogenous variable left undetermined. To resolve this problem, again we have to choose one of these two variables as an exogenous variable or we have to add a new equation to the model. Unless we increase the number of equations, we have to make S^f exogenous and ε endogenous, or we have to make ε exogenous and S^f endogenous. If the availability of foreign savings is actually limited as it is in most developing countries, S^f must be made exogenous. If the foreign exchange rate is pegged (to a price index associated with a numeraire) and foreign savings can be flexibly adjusted, it might be appropriate to make ε exogenous.

Several issues need to be considered before selecting the macro closure rules for an open economy model. First, the selected macro closure should be compatible with the objective of the analyses. It is self-evident that a model with a constant foreign savings S^f cannot be used for analyses of policies aimed at improvement of current account deficits. Equally, a model with a constant foreign exchange rate ε cannot be used to measure how a policy affects this foreign exchange rate and how changes in a certain economic variable affect the economy through changes in the

exchange rate. We have to choose macro closure rules suitable for the objective of analyses.

Second, we have to be aware that macro closure rules characterize the model structure and thus crucially affect simulation results and their implications. For example, suppose that we set foreign savings S^f as an endogenous variable and the foreign exchange rate as an exogenous variable in a static model. It must be noted, however, that foreign savings are not really savings or loans but merely a transfer from the external sector (i.e., a foreign country). Even if the domestic agent gets funds from the external sector as loans, they have no 'next period' to pay them back to the external sector. Welfare, measured by domestic household consumption, is dependent not only on its income from domestic sources (i.e., factor income plus some reimbursement of indirect tax revenues, if any) but also on such a de facto transfer from the external sector. If the contribution of the foreign transfer is considerable, this assumption about the model structure gives misleading results such as that the welfare of an economy would be improved by relying on foreign transfers that are free from repayment obligations.

Third, the influence of the selected numeraire on the simulation results must be taken into account. Fixing the exogenous nominal foreign exchange rate as a numeraire, accompanied by endogenous foreign savings, causes the loss of the homogeneity of degree zero in prices in the model. In contrast to the macro closure of the standard CGE model, if we choose the other numeraire and fix the nominal exchange rate, it means that we fix the real exchange rate – the fixed nominal exchange rate deflated by the numeraire price. In such a case, the homogeneity of degree zero in prices is not maintained. When there is no such homogeneity, we need to justify the choice of a numeraire in view of, say, the nature of the economy and/or to demonstrate to what extent the chosen numeraire affects the simulation results. The latter issue can be examined by conducting sensitivity analysis, which is discussed in detail in Section 8.4.

Finally, the relevance of the current account balances must be considered. In addition to a discussion about whether the current account deficits should be endogenous or exogenous, we have to decide whether the current account deficits should be balanced in the base run equilibrium and/or in the counterfactual equilibrium with policy changes or some other exogenous shocks. If we define an equilibrium in a sense that no current account deficits are allowed simply because this is an 'equilibrium' model, we face the following problem. As actual trade statistics data and input–output tables rarely report current account balances of

zero for any country, we are forced to adjust a social accounting matrix (SAM) to eliminate the deficits artificially to satisfy such a requirement. However, as suggested in Section 4.4, it should be noted that the matrix adjustments are costly and can cause more or less unknown bias to the original data.

It is an open question whether or not the current account balances should be zero, at least in the base run equilibrium. It is unlikely that we will reach a consensus about this question in the future. However, from the authors' point of view, it makes little sense to discuss the definition of equilibrium only in the context of current account deficits. If we eliminate the current account deficits in the SAM, we should also eliminate the domestic investment and savings, which are also recognized as a leakage of resources from (current-period) consumption to a useless purpose. The elimination of the current account deficits as well as domestic investment and savings would make the CGE models too unrealistic.

7.3 Other closure rules

We have discussed various macro closure rules concerning dynamic variables. In addition, there are more macro closure rules related to other variables, which determine the boundary of the inside and the outside of our CGE model. Consider the wage rate and the labour market. All the models demonstrated in this book are of the neoclassical class, where the wage rate is flexibly adjusted to achieve zero unemployment in the labour market. This model specification implies that the wage rate is endogenous while the unemployment rate is exogenous and fixed at zero. Instead, as Keynes pointed out, we can exchange the roles of these two variables to simulate an economy in depression.² The choice of the macro closure is again dependent on our views about the reality that we want to describe.

Notes

1. Macro closure is also discussed by Dewatripont and Michel (1987), Robinson (1989) and Willenbockel (1994).
2. However, again it should be noted that fixing the wage rate at a certain level means fixing the price of labour relative to the price of the numeraire that we choose. Depending on its choice, the simulation results will differ quantitatively and qualitatively, as Hosoe (2000) demonstrates using numerical examples.

8

Simulating General Equilibria

The purpose of developing CGE models is primarily for policy analysis and evaluation. We simulate various policy options to quantify their impact, and therefore we have to simulate various equilibria by solving the same model under several scenarios involving different assumptions regarding the coefficients and exogenous variables; then, the respective equilibria should be compared to evaluate the assumed policies. At the same time, the robustness of the simulation results must be examined to see whether the results are unreasonably affected by the calibrated parameters and those with the a priori determined values. This chapter consists of two parts. The first part presents an efficient way to conduct comparative statics for policy analyses, and the second part discusses sensitivity analysis in CGE modelling.

In the first part, we consider large-scale CGE models. It is difficult to compare the simulation results generated by different policy scenarios, because so many variables are involved. While the two-good and two-factor model presented in Chapter 6 includes 48 endogenous variables, a model of the same structure but with, say, five goods includes as many as 126 variables. Furthermore, to evaluate policies, we must analyse many indicators, which are computed from the solved values of the CGE model. Computational efficiency is extremely important. Section 8.1 demonstrates techniques for solving several models to simulate policy changes and exogenous shocks using different values for coefficients and exogenous variables in one program. Section 8.2 shows how to compute various indicators useful for our analysis from the solved values. Section 8.3 is devoted to the measurement of social welfare, which indicates the overall macroeconomic impact of the policies being analysed.

In the second part, we discuss sensitivity analysis in CGE modelling. As opposed to econometric models, in which the significance of estimated parameters can be tested statistically, CGE models cannot be subject to such statistical tests. The simulation results of CGE models, which are conducted on the basis of the calibration method, depend heavily on the calibrated coefficients and exogenous variables, including assumed parameter values such as the elasticities of substitution and transformation. Therefore, it is important to conduct sensitivity analysis on the robustness of the simulation results with respect to the key calibrated or assumed parameter values. Section 8.4 is devoted to a discussion about sensitivity analysis for this purpose.

8.1 Multiple runs in one input file

When we deal with a small-scale model like the ‘household utility maximization model’ in Chapter 3, we can easily examine the impact of a shock (such as a change in the price of a good) by comparing solutions for the two equilibria (with and without the price change) printed in separate output files. However, for large-scale models, it is too complicated to compare solved variables located in separate output files. We need to develop a computer program that can solve the different models consecutively and print the solutions in one output file.

Consider, for example, the ‘standard CGE model’ described by the input file shown in List 6.1. As stated in Subsection 5.4.4, because the model encompasses the coefficients and exogenous variables calibrated to a given social accounting matrix (SAM) and is solved without any shocks, the solution of this model is called the base run equilibrium, which can be considered a reference equilibrium for the comparative statics. In contrast to the base run equilibrium solution, the solutions of the same model with different shocks applied to certain coefficients or exogenous variables are called counterfactual equilibrium solutions. Here, we demonstrate how to compute the base run and various counterfactual equilibrium solutions using one input file.

Suppose that we analyse the effect of the import tariff rates τ_i^m being reduced to zero from the calibrated rates used in the base run. We can get a counterfactual equilibrium solution by adding the following two lines immediately after the last line of the original input file in List 6.1.¹

```
taum(i) = 0;
Solve stdcge maximizing UU using nlp;
```

The first line (re)sets the import tariff rates at zero for all goods:

$$\tau_i^m = 0 \quad \forall i$$

The second line solves the model again after resetting the import tariff rates τ_i^m as shown above. While we add these two lines to the original input file, the other lines remain unchanged.

The programming technique demonstrated above operates in the same way as the use of the `Parameter` directive (explained in Subsection 5.4.4; more details are provided in Section A.2 in Annex A), which assigns values to already declared constants (i.e., coefficients and exogenous variables). The GAMS syntax allows us to reset the values of constants as many times as needed. We should note that once a value is reset, the new value becomes effective in the subsequent lines of the input file. That is, the parameter values reset most recently before the `Solve` statement are used for computation of the model. As the new input file contains two `Solve` statements, there are two `SOLVE SUMMARY` blocks generated in its output file. These solutions are printed in order of these `Solve` statements.

We can freely make changes to coefficients and exogenous variables to simulate alternative scenarios; however, we are likely to encounter difficulty in computing counterfactual equilibria of a large-scale model with large shocks. This may occur even if there is no (syntax or logical) error in the model or in the simulation scenarios. To cope with such computational difficulty, it is recommended that the coefficients and exogenous variables be changed gradually from their base run values to the target values assumed in the counterfactual equilibrium. Suppose that the base run tariff rate is 10%. Instead of simulating the tariff removal immediately, we iteratively reduce the tariff rate by, say, 1 percentage point from 10% to 0% in the following manner:

```
taum(i) = 0.09;
Solve stdcge maximizing UU using nlp;
taum(i) = 0.08;
Solve stdcge maximizing UU using nlp;
...
taum(i) = 0.01;
Solve stdcge maximizing UU using nlp;
taum(i) = 0.00;
Solve stdcge maximizing UU using nlp;
```

Alternatively, to reduce the amount of code, we can use the `Loop` directive as follows:

```
Set t /1*10/;
Loop(t,
taum(i) = 0.10 - 0.10 * ord(t) / card(t);
Solve stdcge maximizing UU using nlp;
);
```

The first line declares a new set $T = \{1, 2, \dots, 10\}$ to define the number of iterations (or the grid size). The second and fifth lines are the beginning and the end of this iteration process using the `Loop` directive. The lines enclosed by these two lines are processed to reset the tariff rate repeatedly for ten times defined by the set ' t '. The third line updates the import tariff rate, through an automatic process that reduces the rate by 1 percentage point from 10% to 0%. The function ' $\text{ord}(t)$ ' indicates the order of the element t in all the elements in T (i.e., 1, 2, 3, ..., 9 or 10 in the above example); the function ' $\text{card}(t)$ ' indicates the number of elements in T (i.e., it is 10 in the above example). The code ' $\text{Loop}(t, \dots \text{ord}(t)/\text{card}(t) \dots)$ ' in the above example implies 1/10, 2/10, 3/10, ..., 9/10 or 10/10 over all iterations of t .

8.2 Computation of indicators from solved values

Apart from the process that prints various equilibrium solutions in one output file, we often need indicators computed from various runs; for example, change ratios of the values of variables and coefficients between the counterfactual run and its base run. To do this, we insert the following codes in List 8.1 into the original input file in List 6.1.²

List 8.1 Indicators for changes in endogenous variables

```
1 * List8.1: Display of changes
2 Parameter
3 dY(j), dF(h,j), dX(i,j), dZ(j), dXp(i), dXg(i), dXv(i),
4 dE(i), dM(i), dQ(i), dD(i), dpf(h), dpy(j), dpz(i), dpq(i),
5 dpe(i), dpm(i), dpd(i), depsilon, dTd, dTz(i), dTm(i), dSp, dSg;
6
7 dY(j)  =(Y.1(j)  /Y0(j)  -1)*100;
8 dF(h,j)=(F.1(h,j)/F0(h,j)-1)*100;
9 dX(i,j)=(X.1(i,j)/X0(i,j)-1)*100;
10 dZ(j)  =(Z.1(j)  /Z0(j)  -1)*100;
```


List 8.1 (Continued)

```

11 dXp(i) =(Xp.l(i) /Xp0(i) -1)*100;
12 dXg(i) =(Xg.l(i) /Xg0(i) -1)*100;
13 dXv(i) =(Xv.l(i) /Xv0(i) -1)*100;
14 dE(i) =(E.l(i) /E0(i) -1)*100;
15 dM(i) =(M.l(i) /M0(i) -1)*100;
16 dQ(i) =(Q.l(i) /Q0(i) -1)*100;
17 dD(i) =(D.l(i) /D0(i) -1)*100;
18 dpf(h) =(pf.l(h) /1 -1)*100;
19 dpy(j) =(py.l(j) /1 -1)*100;
20 dpz(j) =(pz.l(j) /1 -1)*100;
21 dpq(i) =(pq.l(i) /1 -1)*100;
22 dpe(i) =(pe.l(i) /1 -1)*100;
23 dpm(i) =(pm.l(i) /1 -1)*100;
24 dpd(i) =(pd.l(i) /1 -1)*100;
25 depsilon=(epsilon.l/1 -1)*100;
26 dTd =(Td.l /Td0 -1)*100;
27 dTz(j) =(Tz.l(j) /Tz0(j) -1)*100;
28 dTm(i) =(Tm.l(i) /Tm0(i) -1)*100;
29 dSp =(Sp.l /Sp0 -1)*100;
30 dSg =(Sg.l /Sg0 -1)*100;
31
32 Display
33 dY, dF, dX, dZ, dXp, dXg, dXv, dE, dM, dQ, dD, dpf, dpy, dpz,
34 dpq, dpe, dpm, dpd, depsilon, dTd, dTz, dTm, dSp, dSg;

```

The coding in List 8.1 is prepared to compute percentage changes of all the endogenous variables between the counterfactual run and the base run. With use of the `Parameter` directive, the symbols 'dY(j)', 'dF(h, j)', etc. indicate the percentage changes of 'Y(i)', 'F(h, j)', etc.³ While there are many lines included in List 8.1, their computation is simple. For example, dXp(i) measures the percentage change of X_i^p as follows:

$$dX_i^p = \left(\frac{X_i^p}{X_i^{p0}} - 1 \right) \cdot 100$$

This corresponds to line 11 in List 8.1:⁴

$$dXp(i) = (Xp.l(i) / Xp0(i) - 1) * 100;$$

where $Xp.l(i)$ is the solved `LEVEL` value of the counterfactual equilibrium solution, while $Xp0(i)$ is the initial equilibrium value shown in

the SAM and is supposed to indicate the base run equilibrium solution. It should be noted that $Xp.l(i)$, not $Xp(i).l$, is used in this computation. Remember that the output file reports four types of solutions (LOWER, LEVEL, UPPER and MARGINAL).⁵ Among these four, we use the LEVEL values for this computation, which are indicated by the suffix '.l'. (Be aware that this is the letter 'l'.)

While the solutions of endogenous variables are all automatically printed in the output file, indicators defined and computed in a custom-made manner by modellers are not. Therefore, we use the `Display` directive so that GAMS writes the values of these indicators in the output file. The last three lines in List 8.1 are included for this purpose.⁶ When we use the `Display` directive, we write '`Display dXp;`' without the suffix '(i)'.

8.3 Measurement of economic welfare

If the ultimate goal of economic policies is maximizing social welfare, then the impact of policies implied by the simulation results must be examined from this point of view. However, we have to determine the best way of measuring social welfare. Without such a measure, our discussion about the simulation results does not make any sense. One may consider utility as the welfare indicator and another may support GDP, or some may think that employment is important. Whether or not these indicators are reasonable and useful depends on the objective of the analysis, the issues that the economy faces, and the social priority set.

A direct measure of economic welfare is, almost by definition, utility. In fact, in our CGE model, the household makes its decisions to maximize its utility. Furthermore, utility measures the degree of 'happiness' of individuals and thus can be the first candidate for a measure of social welfare. However, utility has shortcomings as a welfare measure because of its ordinal nature.

First, using utility, we can compare the welfare associated with two equilibria to judge which is better; however, we cannot quantify the difference. In other words, utility is an ordinal measure of economic welfare, not an absolute measure. Second, when we have two or more households (with heterogeneous preferences) in an economy, we cannot, in general, judge the economy's welfare status using utility. If a policy results in higher utility to one household but lower utility to the other, we cannot immediately conclude that the policy is beneficial overall. Because utility is an ordinal measure, we cannot use utility to compare the degree of happiness of these two households or sum their utility to

assess total welfare. Only when no households are worse off and at least one is better off, we can conclude that social welfare is improved from the viewpoint of Pareto efficiency. Otherwise, no judgment about the welfare status of this economy is possible.

To overcome this shortcoming, an attempt has been made to convert utility to a (change in) utility level evaluated in monetary terms. With this conversion, we can quantitatively evaluate improvements in welfare and sum the gains and losses of individual households to measure the overall welfare impact for an economy.

Using the concept of the expenditure function (8.1) and using the standard CGE model in Chapter 6 as an example, we demonstrate how to convert a utility level into an expenditure level as follows:⁷

$$ep(\mathbf{p}^q, UU) = \min_{\mathbf{X}^p} \{ \mathbf{p}^q \cdot \mathbf{X}^p \mid UU(\mathbf{X}^p) = UU \} \quad (8.1)$$

Notations are:

$ep(\cdot)$: expenditure function,

$\mathbf{X}^p$: consumption vector (or $\mathbf{X}$ for the simple CGE model in Chapter 2),

$\mathbf{p}^q$: price vector (or $\mathbf{p}^x$ for the simple CGE model in Chapter 2),

UU : utility level (given),

$UU(\cdot)$: utility function.

The expenditure function $ep(\mathbf{p}^q, UU)$ indicates the minimum expenditure level $\mathbf{p}^q \cdot \mathbf{X}^p$ that satisfies the given utility level UU under the price vector $\mathbf{p}^q$. The above expression converts a utility level into the corresponding minimum expenditure level. By comparing these expenditure levels, we can make implications about welfare.

When we interpret the expenditure function (8.1), it should be noted that we cannot simply compare the expenditure levels between two equilibria because different equilibrium prices prevail. A fair comparison requires a constant price such as the Laspeyres price index to control for the effects of price changes. That is, we consistently use the base run price vector $\mathbf{p}^{q0}$ when comparing the utility levels of the counterfactual run UU^1 and the base run UU^0 . In this case, we can define a welfare change indicator, called Hicksian equivalent variation (EV), as follows:⁸

$$EV = ep(\mathbf{p}^{q0}, UU^1) - ep(\mathbf{p}^{q0}, UU^0) \quad (8.2)$$

This welfare change indicator quantifies the changes in the utility level in monetary terms. Furthermore, dividing EV by GDP, expressed as

$\sum_h p_h^f FF_h$ following the notations in this book, we can measure the welfare changes relative to GDP. The indicator EV/GDP is useful, particularly when comparing welfare impact between different countries.

A decomposition of the change in the utility level may help readers to grasp the essence of this welfare indicator. The change in the utility level can be decomposed into the income effect and the substitution effect. The EV captures the former effect. Figure 8.1 provides a graphical presentation for the two-good case. The base run equilibrium is indicated by E^0 , and the counterfactual one by E^1 . Given the relative price in the base run equilibrium $p^{q0}(=p_1^{q0}/p_2^{q0})$, the virtual equilibrium $\hat{E}^1$ achieves the same utility level as E^1 . The first term of (8.2) represents the expenditure level of $\hat{E}^1$. The expenditure level, divided by p_2^{q0} , corresponds to the intercept on the X_2^p axis of the price lines passing through $\hat{E}^1$. The second term represents the expenditure level of the base run equilibrium. Its expenditure level, divided by p_2^{q0} , corresponds to the intercept on the X_2^p axis of the price lines passing through E^0 . The EV is the gap between these intercepts.

We should be aware that the EV compares E^0 with $\hat{E}^1$, not with E^1 . Given the Laspeyres price index of $\mathbf{p}^{q0}$, UU^1 can be achieved by expenditure $\hat{E}^1$, not by E^1 . When we use E^1 for this comparison with the Laspeyres price index $\mathbf{p}^{q0}$, the estimate of the welfare impact is biased toward overestimation.⁹

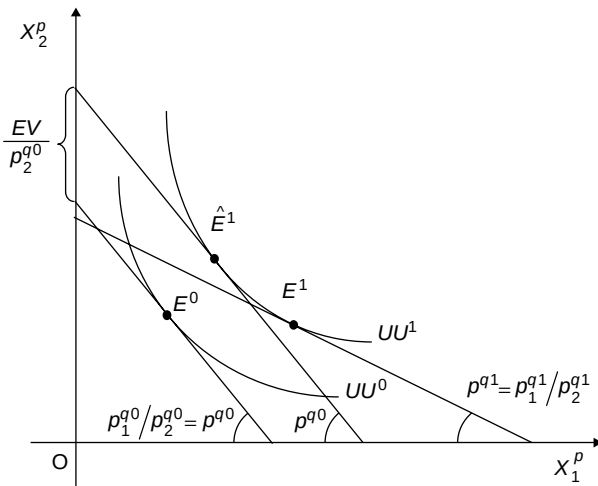


Figure 8.1 Equivalent variation

The welfare index defined above is incorporated in the input file, shown in List 8.2 and explained later. As we assume a Cobb–Douglas-type utility function, the expenditure function (8.1) can be obtained as a solution of the following expenditure-minimization problem for a given utility level UU :

$$\text{minimize}_{X_i^p} \quad ep = \sum_i p_i^q X_i^p$$

subject to

$$\prod_i X_i^{p\alpha_i} = UU$$

From the first-order conditions for this optimization problem we get:

$$ep = \frac{UU}{\prod_i (\alpha_i / p_i^q)^{\alpha_i}}$$

The series of codes shown in List 8.2 is to be attached to the end of the original input file to compute EV.¹⁰ List 8.2 declares symbols used for computation of EV with the `Parameter` directive, such as the base run utility level UU^0 , the base run value of the expenditure function ep^0 , its counterfactual run value ep^1 , and the welfare change indicator of EV. In contrast, the utility level in the counterfactual run (indicated by the indifference curve UU^1 in Figure 8.1), which appears in the computation of the minimum expenditure $ep(\cdot)$, is not declared in the `Parameter` directive. Instead, the solved `LEVEL` value of the utility '`UU.l`', included as a fictitious objective function in the input file, is used for the utility level in the counterfactual run.¹¹ As the base run equilibrium price p_i^{q0} is

List 8.2 Computation of equivalent variations

```
* Welfare measure: Hicksian equivalent variations -----
Parameter      UU0          utility level in the Base Run Eq.
                ep0          expenditure func. in the Base Run Eq.
                ep1          expenditure func. in the C-f Eq.
                EV           Hicksian equivalent variations
;
UU0      =prod(i, Xp0(i)**alpha(i));
ep0      =UU0 /prod(i, (alpha(i)/1)**alpha(i));
ep1      =UU.l/prod(i, (alpha(i)/1)**alpha(i));
EV       =ep1-ep0;

Display EV;
```

set at unity, '1' appears in the denominator. As we need to have only EV appear in the output file, we use the `Display` directive to print it in the last line.

8.4 Sensitivity analysis

In the development of CGE models, we mainly use the calibration method rather than econometric methods. While the calibration method allows us to estimate the model with only one period of data, it is often criticized because we cannot objectively test the robustness of the parameter estimates and thus the simulation results.

To address this critique of CGE analysis, we often conduct sensitivity analysis to check robustness. Sensitivity analysis is conducted for two purposes. One purpose is to test the robustness of our simulation results with respect to the assumed values for some key parameters. The other purpose is to provide a kind of 'confidence interval' for the simulation results.

8.4.1 The concept of sensitivity analysis

Sensitivity analysis is used to demonstrate the robustness of simulation results by varying parameters that may significantly affect the results. As the CGE models encompass structures describing various economic activities in an economy, many parameters (coefficients and exogenous variables) are included. It is unrealistic to examine the robustness of simulation results with respect to all the assumed parameter values in the model; instead, we have to focus on certain parameters relevant to the objective of our CGE analysis and the key model features that are expected to affect the results most significantly. For example, if we conduct a trade analysis, it is well known that the Armington constant elasticity of substitution (CES) and constant elasticity of transformation (CET) structures should be of primary importance in conducting sensitivity analysis. If a domestic tax reform is simulated, we should examine the part of the model that is closely related to the tax bases covered by the reform and/or fiscal schemes that redistribute the tax revenues.

Suppose we are simulating the abolition of import tariffs and analysing its impact, particularly on sectoral output. In the case of trade analysis, it is widely known that the elasticity of substitution/transformation in the CES/CET functions is one of the key parameters that can significantly affect the simulation results. By increasing or decreasing the value of this parameter, we can examine the range over which output changes.

After specifying which parameter is to be examined, we have to establish criteria, based on which we can judge the robustness of our simulation results. To evaluate the robustness of the simulation results, we set, for example, the following two criteria:

Criterion 1: whether the signs of the sectoral output changes are unchanged in all cases,

Criterion 2: whether the ordering of the output changes among sectors is maintained in all cases.

When the results of the sensitivity analysis satisfy both viewpoints, we may well conclude that simulation results are robust. If only one of the two is satisfied, robustness is less evident.

8.4.2 Example of sensitivity analysis with the standard CGE model

Next, we conduct an illustrative sensitivity analysis with the standard CGE model. The SAM in Table 4.2 indicates that the base run import tariff rates are 7.7% for bread and 18.2% for milk.¹² Consider a simulation that abolishes these import tariffs and quantifies the impact on the output of bread and milk. Hereafter, the names of the goods and sectors producing those goods are used synonymously. Thus a two-good model is interchangeably called a two-sector model – the bread sector and the milk sector. The same applies for a three-sector model with three kinds of goods.

The elasticities of substitution/transformation in the CES/CET functions accommodating international trade are σ_i embedded in η_i in (6.17) and ψ_i embedded in ϕ_i in (6.20). Their values are set at two in lines 98–99 in List 6.1. Assume these parameter values as the baseline case. Then, let us define a higher-elasticity case with 20% higher values for σ_i and ψ_i than those assumed in the baseline case. A lower-elasticity case is one with 20% lower values for these parameters. The simulation results indicate that if the assumed elasticity values are well within the confidence interval whose boundaries are set by the 20% higher and lower values, we can conclude that the import tariff abolition would lead to an increase in bread and milk output of about 2% and –1.5%, respectively (Table 8.1).¹³ Based on these figures, we can safely suggest that the bread sector would increase its output while the milk sector will decrease its output.

The results show that the abolition of import tariffs would lead to an increase of bread output and a decrease of milk output in all three

Table 8.1 Impact of import tariff abolition on sectoral output

Output of:	Elasticity of substitution/transformation		
	Baseline Case	Higher-elasticity Case	Lower-elasticity Case
Bread	2.2	2.8	1.6
Milk	-1.4	-1.8	-1.0

Unit: changes from the base run values in %

cases, which indicates that the simulation results are robust with respect to Criterion 1. As for robustness with respect to Criterion 2, we must consider the fact that there is a high correlation of satisfaction with Criterion 1 and with Criterion 2 in the case of a two-sector model because an expansion of output in one sector needs more factor inputs, which can only be achieved by mobilizing factors from the other sector. Therefore, the simulation results that are robust with respect to Criterion 1 are very likely to be robust with respect to Criterion 2. To elucidate this problem, we examine a three-sector model in Subsection 8.4.3.

8.4.3 Sensitivity analysis with a three-sector model

Let us consider two three-sector models. One model (named Model 1) encompasses three goods, Good A, Good B and Good C, which are included in the SAM in Table 8.2. The other model (named Model 2) encompasses Good D, Good E and Good F, which are included in the SAM in Table 8.3. These two SAMs share the same structure, but different values are assigned to some of their cells. We develop and use two different models by calibrating to these SAMs. The structure of these two models is the same as that of the standard CGE model except for the extension to encompass three sectors. Based on the SAMs given in Tables 8.2 and 8.3, we develop two CGE models calibrating the exogenous variables and coefficients in each model separately. Assuming that the elasticities of substitution/transformation in the CES/CET functions in these models all equal 2, we calibrate the import tariff rates to 7.7% for Good A, 18.2% for Good B and 25.0% for Good C in Model 1, and 7.7% for Good D, 18.2% for Good E and 41.7% for Good F in Model 2.

Table 8.2 SAM for Model 1

	Good A	Good B	Good C	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	Total
Good A	21	17	5					20	19	16	8	106
Good B	17	9	5					30	14	15	14	104
Good C	14	6	10					20	10	20	10	90
CAP	20	30	10									60
LAB	15	25	30									70
IDT	5	4	5									14
TRF	1	2	5									8
HOH				60	70							130
GOV						14	8	25				47
INV								35	4		12	51
EXT	13	11	20									44
Total	106	104	90	60	70	14	8	130	47	51	44	

Unit: yen

Table 8.3 SAM for Model 2

	Good D	Good E	Good F	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT	Total
Good D	21	8	5					20	19	16	8	97
Good E	17	9	5					30	14	15	5	95
Good F	5	6	10					20	10	20	11	82
CAP	20	30	10									60
LAB	15	25	30									70
IDT	5	4	5									14
TRF	1	2	5									8
HOH				60	70							130
GOV						14	8	25				47
INV								35	4		12	51
EXT	13	11	12									36
Total	97	95	82	60	70	14	8	130	47	51	36	

Unit: yen

Then, using the CGE models, Model 1 and Model 2, we quantify the impact of import tariff abolition on the sectoral output and conduct the same sensitivity analysis considering higher- and lower-elasticity cases with respect to the elasticity of substitution/transformation in the CES/CET functions. The results of the sensitivity analysis are summarized in Table 8.4 for Model 1 and in Table 8.5 for Model 2. In the following part, we first examine the robustness of the simulation results derived by Model 1 and then by Model 2.

The result of the sensitivity analysis with Model 1 shown in Table 8.4 indicates that the simulation results are robust with respect to Criterion 1

Table 8.4 Impact of import tariff abolition on sectoral output with Model 1

Output of:	Elasticity of substitution/transformation		
	Baseline Case	Higher-elasticity Case	Lower-elasticity Case
Good A	1.3	2.0	0.6
Good B	0.9	1.1	0.7
Good C	-2.4	-3.3	-1.5

Unit: changes from the base run values in %

Table 8.5 Impact of import tariff abolition on sectoral output with Model 2

Output of:	Elasticity of substitution/transformation		
	Baseline Case	Higher-elasticity Case	Lower-elasticity Case
Good D	2.3	3.4	1.2
Good E	-0.8	-0.7	-0.8
Good F	-1.0	-2.0	0.1

Unit: changes from the base run values in %

but not with respect to Criterion 2. That is, in all three cases with different assumed elasticity values, the output of Goods A and B would increase while that of Good C would decrease. However, as for the relative size of the change in production, the order would be Good A, Good B and Good C in the baseline and higher-elasticity cases, and Good B, Good A and Good C in the lower-elasticity case. These results mean that although the abolition of import tariffs results in an increase in the output of Goods A and B at the sacrifice of Good C, it is not clear which of Good A or Good B experiences the larger increase in output. If the relative benefits of Good A or Good B are a matter of concern, more reliable estimates of these elasticity values must be obtained from other sources.

The sensitivity analysis for Model 2, as shown in Table 8.5, indicates that the simulation results satisfy neither of the two criteria stated above; thus, robustness does not exist. More precisely, while the output of Goods D and E would always be affected in the same direction by the import tariff abolition, the output of Good F would decrease in the baseline and higher-elasticity cases but would increase in the lower-elasticity case. We cannot reach any conclusion about whether the import tariff abolition would bring about a preferable or adverse impact on the output

of Good F. If the sector producing Good F has little significance in the economy (for example, because its output share is so small), we can trust the simulation result as a whole for the reason that the results regarding the output of Goods D and E are robust. In contrast, if Good F plays a very important role in the economy, we cannot reach any conclusion about the effect of the import tariff abolition without reliable empirical support for the assumed elasticity parameters.

In the above discussions about sensitivity analysis, we used a hypothetical simulation analysis that quantifies the impact of import tariff abolition on domestic sectoral output. Given different research objectives, we will focus on a different economic phenomenon and need other criteria in our sensitivity analysis than those suggested here. For a policy simulation on the impact of eco-taxes in the context of the Kyoto protocol agreements, the criteria of our sensitivity analysis about the robustness of the simulation results should focus on energy sector output and whether the Kyoto protocol goals can be achieved using a particular set of eco-taxes under different parameter setups.

Further suggestions can be made regarding sensitivity analysis. While we have used three-point estimates of the critical parameters corresponding to the baseline, higher and lower cases, we can instead use their confidence intervals and distributions if this information is available. If we have information that statistically characterizes the critical parameter values, we do not have to use the a priori assumption of 20% higher and lower elasticity values in the sensitivity analysis. Using such statistical inference about the parameter values, we can obtain confidence intervals and distributions in the solutions of all the endogenous variables. When the number of critical parameters is large, we can use a technique called systematic sensitivity analysis to minimize the number of alternative simulations.¹⁴

Notes

1. These codes correspond to lines 329–330 in `stdcge.gms` included in the GAMS Model Library.
2. The codes in List 8.1 are included in `stdcge.gms` in the GAMS Model Library.
3. We attached no memos to the declared symbols for those indicators in the `Parameter` statement because their meanings are self-evident.
4. List 8.1 might be a bit carelessly prepared. Given the model in List 6.1, the codes in List 8.1 work perfectly. However, in general, the codes in List 8.1 can cause a division-by-zero error. We have to add restrictions to avoid division by zero for the indicators whose initial equilibrium values are zero, as we demonstrate in Section A.2 in Annex A.

5. These four represent the LOWER bound value, the LEVEL value, the UPPER bound value and the MARGINAL value of endogenous variables and equations. For the models demonstrated in this textbook, only the LEVEL values indicate the equilibrium solutions.
6. Instead of printing solutions in the output file, we can alternatively print them in a separate file for spreadsheet software such as Excel. Details are given in Section A.5 in Annex A.
7. The right-hand side of the expenditure function (8.1) is the solution of the objective value in the following expenditure minimization problem:

$$\underset{X^P}{\text{minimize}} \quad \mathbf{p}^q \cdot \mathbf{X}^P$$

subject to

$$UU(X^P) = UU$$

8. When we instead use the Paasche price index, the welfare measure is called compensating variation. There is no significant difference between these two welfare indicators; we only use EV in the following discussions.
9. This comparison always leads to overestimation. This is because the definition of the expenditure function $ep(\mathbf{p}^q, UU^1)$ implies the *minimum* expenditure that achieves UU^1 as long as the utility function implies that the indifference curves are convex to the origin.
10. The codes in List 8.2 have already been included in lines 368–379 as a part of the input file `stdcge.gms` in the GAMS Model Library.
11. As discussed in Section 3.1, the utility function is used as a fictitious objective function to formulate the model as a kind of optimization problem, purely for numerical computation purposes. Although any kind of objective function can be used as a fictitious objective function, we have to use the utility function if we plan to use the utility value of the solution as demonstrated in List 8.2.
12. In Table 4.2, the import tariff revenues for bread and milk are one and two yen respectively, while the values of imports are 13 yen for bread and 11 yen for milk.
13. The range of 20% above and below the central value is assumed in an ad hoc manner by the authors. Generally, the selected range depends on the modeller's confidence in the elasticity estimates used in the model.
14. For a discussion of systematic sensitivity analysis, see Harrison et al. (1993).

9

Interpretation of Simulation Results

We have explained how to develop a CGE model, prepare simulation scenarios and run a simulation to analyse the impact of policy or other exogenous shocks on an economy. The simulation results in the output files show changes of various economic variables, such as sectoral output, consumption, prices and welfare, induced by such shocks. However, a mere listing of these changes does not constitute an ‘analysis’. Interpretation of the simulation results, although usually the final step, is the most important task in CGE analysis. A discussion of the simulation results can only be made persuasive by explaining them using intuitive reasoning. To do so, we have to understand how the model structure combined with the simulation scenarios leads to the numerical solutions.

As the size of a model increases, it is likely to become increasingly complicated and to become a black box not only for readers but also for the modellers. To interpret the simulation results correctly, it is necessary to grasp the key structures of the CGE model, which may differ depending on the objective of the research. We must understand what kind of shocks (i.e., changes in coefficients or exogenous variables) assumed in simulation scenarios and what kind of outcomes (i.e., changes in endogenous variables) are most important, and how they are interconnected. The interpretation of simulation results of CGE models thus should focus on the key structure of the model.

In this chapter, we conduct a trade policy analysis and show how to interpret the simulation results of our open-economy CGE model, which is very similar to our ‘standard CGE model’ in Chapter 6. We try to interpret the simulation results as graphically and intuitively as possible. Our interpretation follows the main issues of typical trade policy analysis: whether or not trade liberalization and other trade-related reforms or shocks promote competition between domestic production and foreign

supply and improve the economies' welfare. For this purpose, we develop two types of prototype CGE models for trade analysis: a one-sector model in Section 9.1 and a two-sector model in Section 9.2.

This chapter largely relies on Devarajan et al. (1990) and de Melo and Tarr (1992), particularly their graphical presentations and analyses of trade-related issues with one-sector models. In Section 9.1, we follow their explanations with some adjustments to fit our models in the preceding chapters. The trade-related issues considered in Section 9.1 include abolition of import tariffs, transfers from abroad and terms-of-trade shocks. In Section 9.2, an analysis is undertaken with the two-sector model expounded by Hosoe (1999, Ch. 2), which is an extension of the one-sector model framework of Devarajan et al. (1990) and de Melo and Tarr (1992). We analyse a two-sector situation and how the impact of a shock in one sector is transmitted to the other sector.

9.1 One-sector model

9.1.1 A small open economy

We simplify the CGE model presented in the previous chapters to trace graphically how the model structure, subject to a given shock, determines the endogenous variables while maintaining the important features of open-economy CGE models. We assume that the economy is small and analyse the impact of shocks induced by import tariff abolition in Subsection 9.1.2, by transfers from abroad in Subsection 9.1.3, and by a change in the terms of trade in Subsection 9.1.4.

Suppose that there is one good (and thus one sector) and one household in this economy. For simplicity, we omit a government, investment and savings, and intermediate inputs in the production process but keep the Armington constant elasticity of substitution/transformation (CES/CET) structure as shown in Figure 9.1. The assumption of one good simplifies the model in the following manner:

- (1) No choice between goods is available for the household; thus, the CES-type production function (9.1), which shows that the Armington composite good Q is made from imports M and the domestic good D , can be regarded as the utility function for the household.^{1,2}
- (2) All the factors available are used by the one sector in the model; thus, we do not have to consider market-clearing conditions for the factors explicitly.
- (3) Total inputs (i.e., factors) and output are constant; thus, they can be regarded as exogenous variables.³

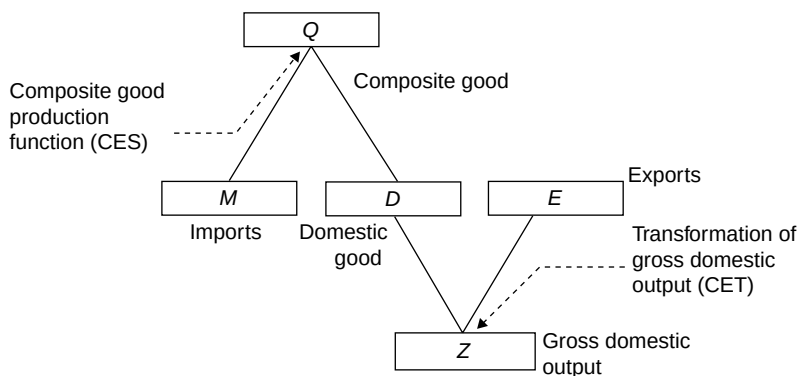


Figure 9.1 Structure of the one-sector open economy model

- (4) The gross domestic output Z is transformed into the domestic good D and exports E using the CET-type transformation function (9.2).
- (5) The household budget constraint overlaps the balance of payments constraint; thus, only the latter constraint (9.3) is imposed. In the above assumptions, the essence of the CES/CET structure explained in Subsections 6.5.3–6.5.4 is maintained.

To simplify further the model presentation, we assume the following:

- (1) The revenue of the import tariff, which is assumed to be an ad valorem tax, is reimbursed to the household in a lump-sum fashion.
- (2) The exchange rate is chosen as the numeraire price and fixed at unity; this assumption reduces the model size because we do not have to distinguish export/import prices in the domestic currency terms from those in the foreign currency terms. Consequently, the export price p^e and the import price p^m are exogenous for this small economy.

The whole model system is shown below:⁴

$$CES(D, M) = Q \quad (9.1)$$

$$CET(D, E) = Z \quad (9.2)$$

$$p^e E + S^f = p^m M \quad (9.3)$$

$$M = ces^m((1 + \tau^m)p^m, p^q, Q) \quad (9.4)$$

$$D = ces^d(p^d, p^q, Q) \quad (9.5)$$

$$E = cet^e(p^e, p^z, Z) \quad (9.6)$$

$$D = cet^d(p^d, p^z, Z) \quad (9.7)$$

Endogenous variables are:

D : domestic good,

E : exports,

M : imports,

Q : Armington composite good (or utility level),

p^d : price of domestic good,

p^z : price of gross domestic output,

p^q : price of Armington composite good.

Exogenous variables are:

Z : gross domestic output,

S^f : current account deficits (or transfers from abroad),

p^e : export price,

p^m : import price,

τ^m : import tariff rate.

The one-sector prototype model can be drawn in Figure 9.2. The interpretation of this graph is as follows:

- (1) The fourth quadrant shows the production possibility frontier derived from the CET-type transformation function (9.2), together with a line of the export price relative to the domestic price p^e/p^d .⁵ Each point on the frontier indicates an output bundle of exports E and domestic good D . The optimal production bundle, indicated by B , is determined by the relative price line tangent to the production possibility frontier.
- (2) The first quadrant shows a straight line representing the balance of payments constraint (9.3), whose slope is determined by the export price relative to the import price p^e/p^m . The graph is drawn to show the current account deficit, whose value denominated by the import price S^f/p^m , is indicated by the vertical-axis intercept of the balance of payments constraint. Exports, determined by the production possibility frontier and the relative price p^e/p^d in the fourth quadrant,

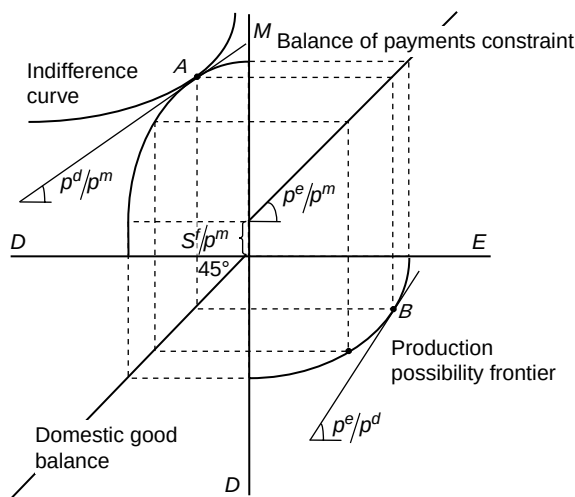


Figure 9.2 One-sector model

Source: Devarajan et al. (1990), modified by the authors.

plus the allowed current account deficits, are projected onto the vertical axis through the balance of payments constraint to determine attainable imports.

- (3) The third quadrant shows demand and supply of domestic goods. A 45-degree line represents the demand–supply balance in the domestic good market. The supply of the domestic good D in the production possibility frontier in the fourth quadrant is translated into its attainable demand D in the second quadrant, through the demand–supply balance line.
- (4) The second quadrant shows household preferences for the domestic good D and the imported good M , together with the consumption possibility frontier. Household preferences are expressed by the indifference curve, which is derived from the utility function (or the iso-output curve derived from the Armington composite good production function). The indifference curve is convex toward the origin as usual. The consumption possibility frontier is formed by attainable imports M translated from the first quadrant and attainable domestic good D translated from the fourth quadrant through the third quadrant.
- (5) In the second quadrant, the optimal consumption bundle is represented by A , on which the consumption possibility frontier is tangent

to the indifference curve. The slope of the line tangent to these curves corresponds to the relative price p^d/p^m .

It is widely recognized that the transformation between exports and the domestic good represented by the CET function and the substitution between imports and the domestic good represented by the CES function are the key structures of the trade analysis CGE models. The former structure is drawn by the production possibility frontier in the fourth quadrant (or its projection onto the second quadrant shown as the consumption possibility frontier); the latter is drawn by the indifference curves generated by the utility function (or the iso-output curves of the Armington composite good production function). Their curvatures are determined by the elasticities of transformation and substitution respectively. The larger the elasticity, the gentler their curvature and vice versa.⁶

9.1.2 Impact of import tariffs⁷

Simulating import tariff abolition (or reduction) is one of the most popular CGE analyses. In Figure 9.3 we describe two equilibria with and without import tariffs. The implications of the import tariff abolition can be derived by comparing these two equilibria.

For the simplicity of our analysis, we assume that the current account deficits are always equal to zero (i.e., $S^f = 0$) as shown in Figure 9.3. We introduce the following notations:

A^1, B^1 : consumption and production equilibrium with import tariffs,
 A^0, B^0 : consumption and production equilibrium without import tariffs,

p^{d1} : domestic good price in the equilibrium with import tariffs,

p^{d0} : domestic good price in the equilibrium without import tariffs.

The implications of import tariff abolition are explained in two steps. First, we describe the two equilibria with and without tariffs in Figure 9.3.

- (1) In the second quadrant, the import tariffs τ_i^m cause a gap between the demand price of imports relative to the domestic good price $p^{d1}/[(1 + \tau^m)p^m]$ and the corresponding supply price p^{d1}/p^m .
- (2) As the household faces the demand price $p^{d1}/[(1 + \tau^m)p^m]$, the consumption equilibrium is A^1 , where the budget line with the slope $p^{d1}/[(1 + \tau^m)p^m]$ is tangent to the indifference curve.

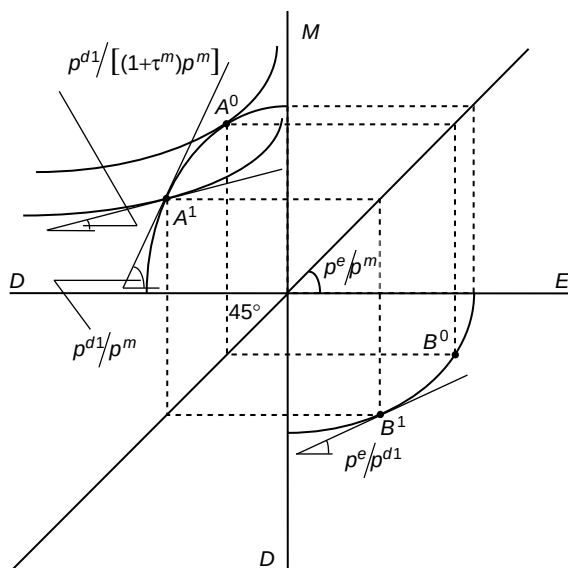


Figure 9.3 One-sector model – impact of import tariffs

Source: de Melo and Tarr (1992), modified by the authors.

- (3) A^1 , which indicates an equilibrium consumption bundle of the domestic good D and imports M , corresponds to B^1 for the production bundle in the fourth quadrant. A^1 and B^1 constitute the equilibrium with import tariffs.
- (4) At A^1 , the consumption possibility frontier is not tangent to the indifference curve; instead, the former intersects the latter. The point where the consumption possibility frontier is tangent to an indifference curve is A^0 , which corresponds to B^0 in the fourth quadrant. A^0 and B^0 constitute the equilibrium without import tariffs.

Second, we determine the impact of import tariff abolition by comparing the two equilibria stated above. It should be noted that the import tariff abolition is represented by a move from A^1 and B^1 to A^0 and B^0 in Figure 9.3.

- (1) In the second quadrant, the new price line p^{d0}/p^m is steeper than the original price line $p^{d1}/[(1+\tau^m)p^m]$. (Note that the price line p^{d0}/p^m is tangent to both the indifference curve and the consumption possibility frontier at A^0 but is not shown in Figure 9.3.) With a cheaper

line in the first quadrant upward; as a result, the balance of payments constraint (9.3) has a positive vertical-axis intercept.

The second difference is in the consumption possibility frontier in the second quadrant. The intercept of the frontier on the horizontal axis remains the same. This is because that intercept indicates the maximum domestic good supply, which is solely determined by the domestic production technology with the production possibility frontier and the resource endowments irrespective of the availability of transfers from abroad. In contrast, the intercept on the vertical axis shifts upward, because of an increase of the attainable imports induced by the transfers from abroad. It leads to an expansion of the consumption possibility set by an upward shift of the original frontier, as Figure 9.4 indicates.

In this setting, the household achieves a better consumption bundle A^2 with more of the domestic good and more imports compared with the original bundle A^0 . As A^2 is located on an indifference curve farther from the origin than A^0 , the household can enjoy a higher utility level thanks to the transfers from abroad.

Recall the import tariff abolition case in Figure 9.3. An increase in imports resulting from the tariff abolition requires an increase in exports to earn more foreign currency. This is done by sacrificing the domestic good supply, which is in sharp contrast to this transfers case.

In this transfers case, as the balance of payments constraint is relaxed by the transfers from abroad, imports can be increased, accompanied by a decrease in exports. Thus, the domestic good supply is increased. This is confirmed by the fact that the new production equilibrium B^2 is located to the southwest of the original equilibrium B^0 .

9.1.4 Impact of terms of trade shock

Finally, we consider an improvement in the terms of trade (TOT) p^e/p^m for this economy (Figure 9.5). For simplicity, we omit the import tariffs and assume that the current account deficit S^f is always equal to zero. We compare A^3 and B^3 of the post-TOT improvement case with A^0 and B^0 of the pre-TOT improvement case.

- (1) As the TOT is shown as the slope of the balance of payments line, an improvement in the TOT makes this line steeper in the first quadrant. This leads to a vertical enlargement of the consumption possibility frontier in the second quadrant.
- (2) Given this new consumption possibility frontier, the household updates its consumption bundle to A^3 . Assuming (a likely case)

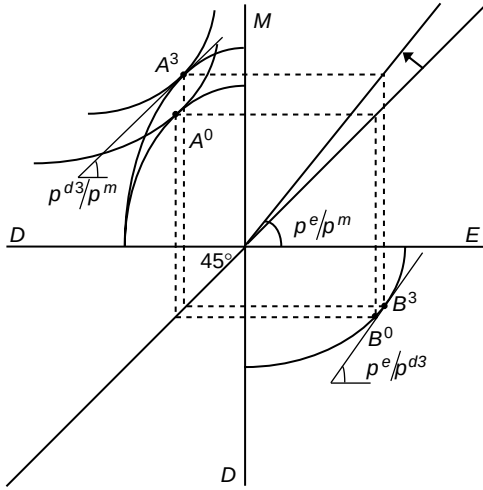


Figure 9.5 One-sector model – impact of terms of trade shock

Source: Devarajan et al. (1990), modified by the authors.

that the elasticity of substitution σ in the Armington-composite-good-production function is larger than unity, A^3 is located to the northeast of A^0 .⁹ In such a case, the household reduces domestic good consumption but enjoys more imports. Because of the TOT improvement, the price of the imported good relative to domestic good falls; thus, p^d/p^m rises to p^{d3}/p^m . This coincides with the fact that A^3 is located to the northeast of A^0 .

- (3) Because the consumption of the domestic good decreases and the export price relative to the domestic good price p^e/p^d rises to p^e/p^{d3} , the resource is mobilized to increase exports. This is indicated by the move from the original equilibrium B^0 to the new equilibrium B^3 in the fourth quadrant.
- (4) The move of the consumption point from A^0 to A^3 indicates that the household attains a higher utility level thanks to the improved TOT.

9.2 Two-sector model

We have discussed the structure of the prototype trade analysis CGE model with one sector and illustrated its behaviour. The CGE models usually encompass several sectors/goods; the behaviour of the CGE models under a multisectoral setup can be different not only quantitatively

but also qualitatively from the one under the one-sector setting. Next we build a two-sector model and illustrate its behaviour, focusing on interactions between these two sectors from the viewpoints of substitution and complementarity.

Suppose that there are two sectors, agriculture (AGR) and manufacturing (MAN), and that these two sectors use only capital and labour for their production (without any intermediate inputs). To deal with substitution and complementarity, we need to have some mechanism connecting the two sectors in the model. There are two connecting mechanisms assumed here. The first is substitution of goods in consumption as described in the 'simple CGE model' and the 'standard CGE model'. (Note that the Armington composite goods are consumed by the household in this two-sector model.) The second is that both sectors use capital and labour, whose endowments are fixed, and capital and labour are mobile between these two sectors.

The following explanation requires the use of two figures: one for AGR (Figure 9.6.A) and the other for MAN (Figure 9.6.B). These figures are similar to those used in previous sections; however, there are two major differences. First, because of factor mobility, the 'production possibility frontiers' in the fourth quadrants can be shifted inward and outward. Thus, we cannot call them 'frontiers', but we will call them 'iso-input curves'.

The second difference is the lines in the first quadrants. They no longer represent the balance of payments constraint, but instead their slopes indicate the terms of trade between exported and imported goods, and their vertical-axis intercepts indicate the net imports of each sector. For

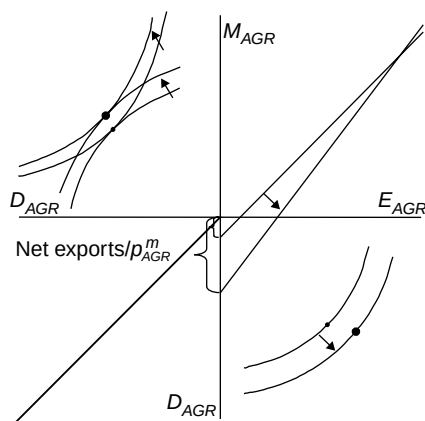


Figure 9.6.A Two-sector model – agricultural sector

simplicity of discussion, we assume that the current account balance in this whole economy is always equal to zero, even though there may be imbalances in each sector.

In the following exercise, we use a simple scenario assuming a price rise only for the agricultural export good, which makes the agricultural sector's terms of trade line steeper in the first quadrant of Figure 9.6.A. This leads not only to an increase in agricultural exports but also to an increase in net exports. An increase in domestic agricultural output also occurs, because factors move from the manufacturing sector to the agricultural sector. This is indicated by the southeast shift of the isoquant curve in the fourth quadrant. This result is in sharp contrast with that of the one-sector model where output stays constant and an increase in exports reduces the domestic good supply, as demonstrated in Subsection 9.1.4.

The shift of the terms of trade line of the agricultural sector implies a kind of income effect brought about by the abundance of foreign currency because of the agricultural export boom. The isoquant in the second quadrant of the agricultural sector shifts outward resulting in an increase in agricultural imports and domestic good consumption.

Now turn to the manufacturing sector (Figure 9.6.B). On the supply side, the manufacturing sector can use fewer factors because of the increased factor demand by the agricultural sector mentioned above; thus, the isoquant curve of the manufacturing sector shifts inward. However, it is not clear which of the exports or domestic good supply decreases more sharply; this is the compromising result of the income

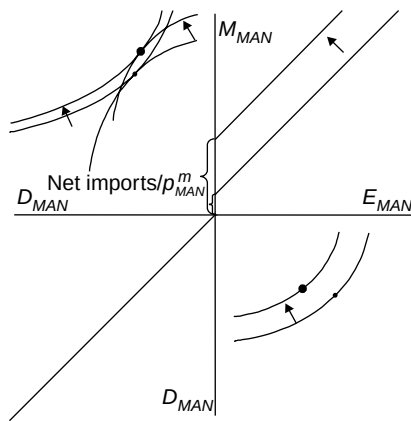


Figure 9.6.B Two-sector model – manufacturing sector

effect and substitution effect observed in the second quadrant. While the income effect increases demand of both the imports and the domestic good, the substitution effect increases demand of the imports and decreases demand of the domestic good. The situation depends on assumed values of the relevant parameters, which are calibrated to a social accounting matrix (SAM) in empirical models. The features in Figure 9.6.B represent just one case example.

On the demand side of the manufacturing sector, the foreign currency earned by the increased net agricultural exports also exerts an income effect on the manufacturing sector. The isoquant in its second quadrant shifts outward, making it possible to spend more on manufacturing imports. Fewer exports and more imports of the manufacturing good in such a case as Figure 9.6.8 lead to an increase in its net imports. This is shown as a parallel upward shift of the terms-of-trade line in the first quadrant, which generates a positive vertical-axis intercept.

In sum, we find:

- (1) The magnitudes of the shifts in net agricultural exports/imports and net manufacturing exports/imports must be equal but of opposite sign, because of the assumed zero current account deficits.¹⁰
- (2) The increases of both agricultural and manufacturing consumption immediately imply an improvement of the utility level.

Before concluding the analysis regarding the behaviour of the two-sector prototype model, we make two remarks. The first involves the number of sectors dealt with in a model, and the second relates to the model assumptions about market structure and scale economies. As for the first point, an increase in the number of sectors makes the interrelations of sectors more complicated. We have seen that the results in a two-sector model are different from those in a one-sector model. In a two-sector model, the sectors always compete with each other for factors, whose endowments are given. An expansion of one sector always leads to a contraction of the other. This is illustrated by the opposite shifts of isoquants in the fourth quadrants between Figures 9.6.A and 9.6.B. In a three-sector model, in addition to competition in the factor markets, we must also take account of the more complicated intersectoral linkages of sectoral output described by the input–output structure. As for the second point, our arguments so far are based on the assumptions of perfectly competitive markets and constant-returns-to-scale technology. The assumptions of imperfect competition and increasing-returns-to-scale can lead to different outcomes, with implications for policy.¹¹

Notes

1. For details of the domestic good, see Subsection 6.5.2.
2. Assume a utility function, which is dependent on consumption of the one good, as follows:

$$UU = UU(Q)$$

Because of monotonicity of utility functions, changes in Q have the same effect as changes in UU . Therefore, the iso-input curves of the Armington composite good production functions are regarded as the indifference curves implied by the utility function.

3. We can assume a production function with inputs equal to the entire factor endowments in this economy $Z(FF)$. As FF is exogenous, production Z becomes constant.
4. $CES(\cdot)$ in (9.1) and $CET(\cdot)$ in (9.2) are abbreviations of the functions corresponding to (6.17) and (6.20). $ces^m(\cdot)$, $ces^d(\cdot)$, $cet^e(\cdot)$ and $cet^d(\cdot)$ in (9.4)–(9.7) are abbreviations of the demand functions (6.18) and (6.19), and the supply functions (6.21) and (6.22).
5. Generally, it should be called an iso-input curve, but it can now be regarded as the production possibility frontier as discussed above.
6. See also Subsections 6.5.3 and 6.5.4 for details about the CES/CET functions, and Section 8.4 for a numerical example demonstrating how much the simulation results differ depending on these elasticities.
7. Although de Melo and Tarr (1992) analysed the imposition of tariffs, the essence is the same.
8. The main idea of this section is derived from the analysis on foreign capital inflows in Devarajan et al. (1990). As ‘Dutch disease’ was the main issue in their article, the changes in exchange rates were analysed in detail; however, the exchange rate problems are omitted in our analysis for the sake of simplicity.
9. In contrast, when the elasticity of substitution σ is less than unity, A^3 is located to the northwest of A^0 .
10. As the prices of imports are all unity, the balance of payments constraint equilibrating net agricultural exports with net manufacturing imports leads to the same magnitude of intercept shifts graphically. When the import prices diverge from unity, the magnitudes of these intercept shifts might not be equal to each other but still have opposite signs.
11. Takeda (2010) simulated trade liberalization using world trade CGE models with different assumptions regarding these features, and compared their welfare impacts.

10

Model Extension

Almost all the objectives of this book were accomplished by the end of Chapter 9. That is, we have developed a ‘standard CGE model’ and explained how to compute it with GAMS and interpret the model solutions. The model is equipped with minimum standard features for empirical analysis. By understanding the model structure and computer program, readers are expected to be able to develop and solve their own models. In this chapter, to model a more complicated economy we extend the standard CGE model by replacing some of its basic assumptions. This chapter is aimed mainly at advanced CGE modellers.

In Section 10.1, we present a multihousehold model, in contrast to the simple and the standard CGE models, which have only one household. In Section 10.2, we relax the small-country assumption and develop a large-country model to describe the international economy better. In Section 10.3, we extend the one-country model to a two-country model with some suggestions for multicountry models. In Section 10.4, the assumption of perfect competition is replaced by that of monopoly and oligopoly. In Section 10.5, rather than import tariffs, we introduce a quantitative restriction on imports into the standard CGE model. The chapter concludes with Section 10.6, where economies of scale are considered.

10.1 Multihousehold model

In the simple CGE model and the standard CGE model, we assumed that there is only one representative household. It should be noted that this assumption does not mean that the economy has a population of just one but that it has only one *type* of household. Although this assumption simplifies the model, we cannot ignore the reality that there are

several types of individuals with heterogeneous preferences, and differences in family structure, which lead to different household preferences. For example, when we analyse a regional disparity issue for China, we can hardly assume that households in coastal areas have the same preferences as those in inland areas. Similarly, when we develop a world economy model, we have to distinguish between households in Japan and those in the US and assume different utility functions for each household group.

In this section, we consider a model in which there are two households with different utility functions. This two-household model can be extended in two ways. First, having learned the two-household model, we can easily extend it to models with three or more households. Second, we can convert the two-household model into a two-country model, where we must naturally assume that households' preferences differ between countries. A two-country model is shown in Section 10.3.

By modifying the social accounting matrix (SAM) in Table 4.2, we provide a new SAM with two households, as shown in Table 10.1. The label 'HOH' in Table 4.2 is now subdivided into 'HOH1' and 'HOH2'. The rows and columns of 'HOH1' and 'HOH2' are shaded in Table 10.1. No changes are made in the other parts of the original SAM.

As we now have two households, the variables related to households, such as X_i^p , T^d , S^p and FF_h in Chapter 6, must be distinguished by household type; we introduce a new suffix, $l = 1, 2$, to indicate the household type. All the other variables, such as production, indirect taxes and international trade, stay the same. As a result, the following equations are introduced in the model in place of the original ones:

– Government:

$$T_l^d = \tau_l^d \sum_h p_h^f FF_{h,l} \quad \forall l \quad (10.1), \text{ corresponding to (6.6)}$$

$$X_i^g = \frac{\mu_i}{p_i^q} \left(\sum_l T_l^d + \sum_j T_j^z + \sum_j T_j^m - S^g \right) \quad \forall i \quad (10.2), \text{ corresponding to (6.9)}$$

– Investment and savings:

$$X_i^v = \frac{\lambda_i}{p_i^q} \left(\sum_l S_l^p + S^g + \varepsilon S^f \right) \quad \forall i \quad (10.3), \text{ corresponding to (6.10)}$$

$$S_l^p = s_l^p \sum_h p_h^f FF_{h,l} \quad \forall l \quad (10.4), \text{ corresponding to (6.11)}$$

Table 10.1 SAM with two households

Unit: yen															
	Activity				Factor			Indirect Tax			Final Demand			External	Total
	BRD	MLK	CAP		LAB	IDT	TRF	HOH1	HOH2	GOV	INV	EXT			
Activity	BRD	21	8					10	10	19	16	8			
	MLK	17	9					20	10	14	15	4			
Factor	CAP	20	30												
	LAB	15	25												
Indirect Tax	IDT	5	4												
	TRF	1	2												
Final Demand	HOH1	45	5					15	8	2					
	HOH2							10	7						
External	GOV							10	7						
	INV							8							
Total	EXT							35							
								55							

$$S^g = ss^g \left(\sum_l T_l^d + \sum_j T_j^z + \sum_j T_j^m \right) \quad (10.5), \text{ corresponding to (6.12)}$$

– Household:

$$X_{i,l}^p = \frac{\alpha_{i,l}}{p_i^q} \left(\sum_h p_h^f FF_{h,l} - S_l^p - T_l^d \right) \quad \forall i, l \quad (10.6), \text{ corresponding to (6.13)}$$

– Market-clearing condition:

$$Q_i = \sum_l X_{i,l}^p + X_i^g + X_i^v + \sum_j X_{i,j} \quad \forall i \quad (10.7), \text{ corresponding to (6.23)}$$

$$\sum_j F_{h,j} = \sum_l FF_{h,l} \quad \forall h \quad (10.8), \text{ corresponding to (6.24)}$$

Using (10.1)–(10.8) in place of their corresponding equations in the original model summarized in Section 6.7, we can develop the two-household CGE model.¹ Readers can write the computer program for this two-household model by making the adjustments stated above to the standard CGE model, whose input file is shown as List 6.1.

10.2 Large-country model

The standard CGE model in Chapter 6 is based on the small-country assumption that the export and import prices quoted in terms of the foreign currency are given exogenously. This assumption is invalid not only for developed countries with a large economy but also for small developing countries that have significantly large shares in some specific commodity markets. Think about oil-producing countries. Even if their economy is small, they can affect the international oil price by controlling their production. Aside from oil-producing countries as an extreme case, many developing countries are monoculture economies with some primary commodities and have significant shares in their international markets. Such staples create a large amount of national income and government revenues; they often attract our research attention. We need to develop a CGE model with a large country that has power to influence international markets.

In this section, we extend the small-country model to a large-country model. Although we mention two countries – the home country and the foreign country – the present model is still a one-country model intrinsically, because we include the foreign country only to the extent that its

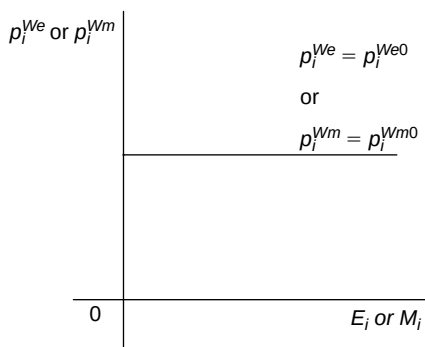


Figure 10.1 Export demand and import supply curves for a small country

imports/exports enter the model. In fact, the foreign country is nothing but the external sector in the one-country CGE model framework.

The difference between a large-country model and the standard model of the small-country assumption is that the former has smaller price-elasticity for imports/exports. In the original small-country model, the supply and demand curves of traded goods are assumed to be horizontal, as shown in Figure 10.1. In other words, regardless of the quantities of the home country's exports and imports, the international prices stay constant.

In contrast, in the large-country model, the curve of the international export price p_i^{We} with respect to E_i is downward sloping, as shown in Figure 10.2. This means that if the large home country increases its exports E_i , the international price decreases. This is consistent with the property of a standard demand function.

The curve of the international import price p_i^{Wm} with respect to M_i is upward-sloping. If the home country increases its imports M_i , the international price goes up. The curve of the international import price p_i^{Wm} with respect to M_i represents the foreign country's supply for goods imported by the home country. These downward- and upward-sloping curves indicate the home country's influential power in international markets.

To develop one of the simplest large-country models, we assume constant price elasticities to obtain a downward-sloping foreign country's demand for goods exported by the home country and an upward-sloping foreign country's supply for goods imported by the home country:

$$\frac{E_i}{E_i^0} = \left(\frac{p_i^{We}}{p_i^{We0}} \right)^{-\sigma_i} \quad \forall i \quad (10.9)$$

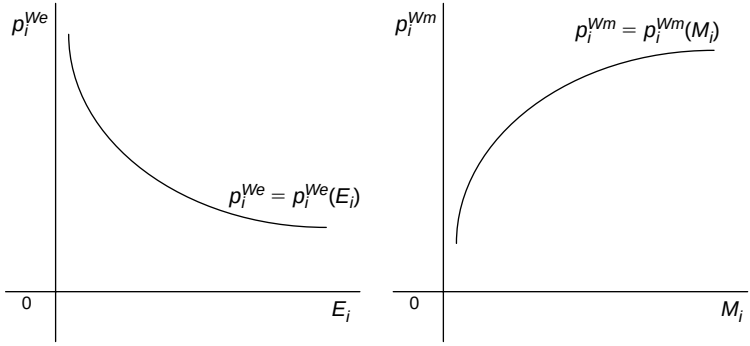


Figure 10.2 Export demand and import supply curves for a large country

$$\frac{M_i}{M_i^0} = \left(\frac{p_i^{Wm}}{p_i^{Wm0}} \right)^{\psi_i} \quad \forall i \quad (10.10)$$

where:

E_i : home country exports of the i -th goods to the foreign country,

M_i : home country imports of the i -th goods from the foreign country,

p_i^{We} : export price of the i -th goods in terms of the foreign currency,

p_i^{Wm} : import price of the i -th goods imported in terms of the foreign currency.

Now that the international prices p_i^{We} and p_i^{Wm} are endogenous variables, rather than exogenous variables, as they are affected by the behaviour of the home country (as is the case with our notation, symbols with a superscript '0' denote the initial equilibrium values of the corresponding endogenous variables). Note that the curves describing the functions of export and import prices converge to the horizontal ones as the price elasticities σ_i and ψ_i tend to infinity.

The price elasticities in (10.9) and (10.10), denoted by σ_i and ψ_i , indicate that they have a linkage with the elasticities of substitution/transformation in constant elasticity of substitution/transformation (CES/CET) functions, as discussed in Section 6.5. This reflects our assumptions that both the home and the foreign countries have the same CES/CET structure and the same elasticities of substitution/transformation (Figure 10.3). As the price elasticities of demand/supply functions derived from the corresponding CES/CET functions are approximately equal to the elasticities of substitution/transformation in CES/CET functions, the price elasticities of the foreign country's demand/supply for exports/imports by the home country are

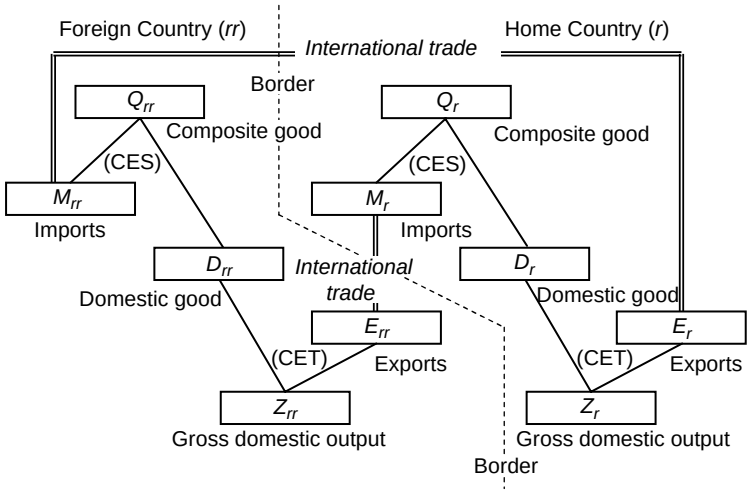


Figure 10.3 International trade between two countries and CES/CET structure

approximately equal to the elasticities of substitution/transformation in CES/CET functions of the home country.² The same is the case for the price elasticities of the home country's demand/supply for exports/imports by the foreign country.

Using the Armington structure in the foreign country (*rr*), we define and solve the profit-maximization problem with the composite good production function for the foreign country by extending the original setup for one country in Chapter 6 but using a common elasticity of substitution σ_i for simplicity and thus a common elasticity parameter η_i between these two countries:

$$Q_{i,rr} = \gamma_{i,rr} (\delta m_{i,rr} M_{i,rr}^{\eta_i} + \delta d_{i,rr} D_{i,rr}^{\eta_i})^{\frac{1}{\eta_i}} \quad \forall i, rr$$

The first-order conditions for the profit-maximization problem lead to the foreign country's demand function for imports from the home country:³

$$M_{i,rr} = \left(\frac{\gamma_{i,rr}^{\eta_i} \delta m_{i,rr} p_{i,rr}^q}{p_{i,rr}^m} \right)^{\frac{1}{1-\eta_i}} \quad Q_{i,rr} = \left(\frac{\gamma_{i,rr}^{\frac{\sigma_i-1}{\sigma_i}} \delta m_{i,rr} p_{i,rr}^q}{p_{i,rr}^m} \right)^{\sigma_i} Q_{i,rr} \quad \forall i, rr$$

Here, its price elasticity can be derived in the following two ways. First, if the import price $p_{i,rr}^m$ affects neither the price of composite good $p_{i,rr}^q$

nor its quantity $Q_{i,rr}$, the import price elasticity ρ_i is computed as follows:

$$\rho_i \equiv - \frac{\partial M_{i,rr} / M_{i,rr}}{\partial p_{i,rr}^m / p_{i,rr}^m} \bigg|_{p_{i,rr}^q, Q_{i,rr}: \text{constant}} = \sigma_i \quad \forall i$$

Second, we relax the assumption stated above and assume instead that the import price $p_{i,rr}^m$ affects the price of composite good $p_{i,rr}^q$ but not its quantity $Q_{i,rr}$. To consider this case, using the foreign country's composite good production function shown above, we derive a unit cost function as follows:

$$p_{i,rr}^q = \frac{1}{\gamma_{i,rr}} (\delta m_{i,rr}^{\sigma_i} p_{i,rr}^m (1-\sigma_i) + \delta d_{i,rr}^{\sigma_i} p_{i,rr}^d (1-\sigma_i))^{-\frac{1}{1-\sigma_i}} \quad \forall i$$

With this unit cost function, we can obtain an alternative price elasticity $\tilde{\rho}_i$ as follows:

$$\tilde{\rho}_i \equiv - \frac{\partial M_{i,rr} / M_{i,rr}}{\partial p_{i,rr}^m / p_{i,rr}^m} \bigg|_{Q_{i,rr}: \text{constant}} = \sigma_i \left(1 - \frac{\delta m_{i,rr}^{\sigma_i} p_{i,rr}^m (1-\sigma_i)}{\delta m_{i,rr}^{\sigma_i} p_{i,rr}^m (1-\sigma_i) + \delta d_{i,rr}^{\sigma_i} p_{i,rr}^d (1-\sigma_i)} \right) \quad \forall i$$

If the foreign country's import share parameter $\delta m_{i,rr}$ is sufficiently small, we can omit the second term in parentheses. In this case, the alternative elasticity $\tilde{\rho}_i$ converges to ρ_i .⁴ The same method applies in the derivation of the price elasticity ψ_i of supply functions from the CET functions.

In sum, the large-country CGE model encompasses the foreign country's less price-elastic supply functions of goods imported by the home country and the foreign country's less price-elastic demand functions of goods exported from the home country, implicitly assuming the same CES/CET structure in the foreign country. For the price elasticities of those supply and demand functions (10.9) and (10.10), we use the same symbols of σ_i and ψ_i as used in Chapter 6, because those price elasticities are approximately equal to the elasticities of substitution/transformation in CES/CET functions, assuming the same CES/CET structure and the same elasticity parameters between the home and the foreign countries. If, however, more detailed information about the elasticities by country is available, we can use different magnitudes for these elasticities by country.

Assuming the same elasticity parameters for the home and foreign countries for simplicity in this example, we can develop a large-country

model that consists of new equations for (10.9) and (1.10) in addition to (6.1)–(6.24). Now, in addition to the original endogenous variables, the new endogenous variables for the export price p_i^{We} and the import price p_i^{Wm} are included in the model; the endogenous variables are as follows:

$$Y_j, F_{h,j}, X_{i,j}, Z_j, X_i^p, X_i^s, X_i^v, E_i, M_i, Q_i, D_i, p_h^f, p_j^v, p_j^z, p_i^q, p_i^e, p_i^m, p_i^d, \varepsilon, S, S^s, T^d, T_j^z, T_i^m, p_i^{We} \text{ and } p_i^{Wm}.$$

Exogenous variables are:

$$FF_h, S^f, p_i^{We0}, p_i^{Wm0}, \tau^d, \tau_j^z \text{ and } \tau_i^m.$$

We prepare an input file for this model as shown in List 10.1. As this large-country model (lrgcge.gms) is developed on the basis of

List 10.1 Large-country model (lrgcge.gms)

```

38 Parameter      Y0(j)      composite factor
...
56              Sf          foreign saving in US dollars
57              pWe0(i)     export price in US dollars
58              pWm0(i)     import price in US dollars
...
85 Sf            =SAM("INV","EXT");
86
87 pWe0(i) =1;
88 pWm0(i) =1;
...
150 Variable      Y(j)          value added
...
169              epsilon       exchange rate
170              pWe(i)         world export price
171              pWm(i)         world import price
...
181 Equation       eqpy(j)       composite factor aggregation func.
...
200              eqepsilon     balance of payments
201              eqfe(i)        foreign export demand function
202              eqfm(i)        foreign import supply function
...
245 eqepsilon..    sum(i, pWe(i)*E(i)) +Sf =e= sum(i, pWm(i)*M(i));
246 eqfe(i)..      E(i)/Z0(i) =e= (pWe(i)/pWe0(i))*(-sigma(i));
247 eqfm(i)..      M(i)/M0(i) =e= (pWm(i)/pWm0(i))*psi(i);
...
292 epsilon.l=1;
293 pWe.l(i)=1;
294 pWm.l(i)=1;
...
321 epsilon.lo=0.00001;
322 pWe.lo(i)=0.00001;
323 pWm.lo(i)=0.00001;
...

```

the small-country model (`stdcge.gms`), some modifications are introduced in lines 57–58, 87–88, 170–171, 201–202, 246–247, 293–294 and 322–323 of List 10.1.

10.3 World trade model

The CGE models that have been presented so far are one-country models with and without extensions. We can use these one-country models for analysis focusing on economic activities and issues within a country but might not be able to derive meaningful implications, for example, in analysis regarding international trade issues. In this section, we extend the one-country model to a world trade model with two or more countries.

We have to take into account that the foreign country (or countries) affects the home country's imports, exports and current account deficits, through which the home country's whole economy is influenced. Developing a world trade model, we must describe the internal structure of the foreign country (or countries) in as much detail as for the home country. Although the model is often called a world trade model, it is not realistic for us to describe all the individual countries using the same level of detail. Instead, we often choose only several important countries (or aggregated regions) in the world and aggregate the other countries as a region representing the rest of the world (ROW).

In this section, we first present a two-country model. Then, we discuss a multicountry model, focusing on its differences from the two-country model.

10.3.1 Two-country model

Here, we develop a two-country model for, say, Japan and the US. Although we assume a symmetric model structure for both countries, the data in their SAMs differ; thus, calibration of constants based on the SAMs, such as the share coefficients in the utility and production functions, tax rates and factor endowments, can be done with the same calibration formulae but will generate different estimates of coefficients and exogenous variables. The elasticities of substitution/transformation for the CES/CET functions related to international trade between the two countries are assumed to be the same in the present model.

To develop a two-country model, we prepare two one-country models and join them using the variables related to international trade, as shown in Figure 10.3. Note that the home country's exports/imports

are the foreign country's imports/exports. Similarly, the home country's current account surplus/deficit is the foreign country's current account deficit/surplus. The present two-country model is also a static model; the current account deficit is still, in fact, international transfers.

There is a conspicuous difference between the one-country model in Chapter 6 and the present model. In the one-country model based on the small-country assumption, the international prices of exports and imports in terms of foreign currency are exogenous variables. In contrast, in this two-country model, these international prices are endogenous variables, which are adjusted to clear the international markets.

The model equations are listed below:

– Domestic production:

$$Y_{j,r} = b_{j,r} \prod_h F_{h,j,r}^{\beta_{h,j,r}} \quad \forall j, r \quad (10.11)$$

$$F_{h,j,r} = \frac{\beta_{h,j,r} p_{j,r}^y}{p_{h,r}^f} Y_{j,r} \quad \forall h, j, r \quad (10.12)$$

$$X_{i,j,r} = a x_{i,j,r} Z_{j,r} \quad \forall i, j, r \quad (10.13)$$

$$Y_{j,r} = a y_{j,r} Z_{j,r} \quad \forall j, r \quad (10.14)$$

$$p_{j,r}^z = a y_{j,r} p_{j,r}^y + \sum_i a x_{i,j,r} p_{i,r}^q \quad \forall j, r \quad (10.15)$$

– Government:

$$T_r^d = \tau_r^d \sum_h p_{h,r}^f F F_{h,r} \quad \forall r \quad (10.16)$$

$$T_{j,r}^z = \tau_{j,r}^z p_{j,r}^z Z_{j,r} \quad \forall j, r \quad (10.17)$$

$$T_{i,r}^m = \tau_{i,r}^m p_{i,r}^m M_{i,r} \quad \forall i, r \quad (10.18)$$

$$X_{i,r}^g = \frac{\mu_{i,r}}{p_{i,r}^q} \left(T_r^d + \sum_j T_{j,r}^z + \sum_j T_{j,r}^m - S_r^g \right) \quad \forall i, r \quad (10.19)$$

– Investment and savings:

$$X_{i,r}^v = \frac{\lambda_{i,r}}{p_{i,r}^q} (S_r^p + S_r^g + \varepsilon_r S_r^f) \quad \forall i, r \quad (10.20)$$

$$S_r^p = s s_r^p \sum_h p_{h,r}^f FF_{h,r} \quad \forall r \quad (10.21)$$

$$S_r^g = s s_r^g (T_r^d + \sum_j T_{j,r}^z + \sum_j T_{j,r}^m) \quad \forall r \quad (10.22)$$

– Household:

$$X_{i,r}^p = \frac{\alpha_{i,r}}{p_{i,r}^q} \left(\sum_h p_{h,r}^f FF_{h,r} - S_r^p - T_r^d \right) \quad \forall i, r \quad (10.23)$$

– Export and import prices, and balance of payments constraint:

$$p_{i,r}^e = \varepsilon_r p_{i,r}^{We} \quad \forall i, r \quad (10.24)$$

$$p_{i,r}^m = \varepsilon_r p_{i,r}^{Wm} \quad \forall i, r \quad (10.25)$$

$$\sum_i p_{i,r}^{We} E_{i,r} + S_r^f = \sum_i p_{i,r}^{Wm} M_{i,r} \quad \forall r \quad (10.26)$$

– Substitution between imports and domestic goods:

$$Q_{i,r} = \gamma_{i,r} (\delta m_{i,r} M_{i,r}^{\eta_i} + \delta d_{i,r} D_{i,r}^{\eta_i})^{\frac{1}{\eta_i}} \quad \forall i, r \quad (10.27)$$

$$M_{i,r} = \left[\frac{\gamma_{i,r}^{\eta_i} \delta m_{i,r} p_{i,r}^q}{(1 + \tau_{i,r}^m) p_{i,r}^m} \right]^{\frac{1}{1-\eta_i}} Q_{i,r} \quad \forall i, r \quad (10.28)$$

$$D_{i,r} = \left[\frac{\gamma_{i,r}^{\eta_i} \delta d_{i,r} p_{i,r}^q}{p_{i,r}^d} \right]^{\frac{1}{1-\eta_i}} Q_{i,r} \quad \forall i, r \quad (10.29)$$

– Transformation between exports and domestic goods:

$$Z_{i,r} = \theta_{i,r} (\xi e_{i,r} E_{i,r}^{\phi_i} + \xi d_{i,r} D_{i,r}^{\phi_i})^{\frac{1}{\phi_i}} \quad \forall i, r \quad (10.30)$$

$$E_{i,r} = \left[\frac{\theta_{i,r}^{\phi_i} \xi e_{i,r} (1 + \tau_{i,r}^z) p_{i,r}^z}{p_{i,r}^e} \right]^{\frac{1}{1-\phi_i}} Z_{i,r} \quad \forall i, r \quad (10.31)$$

$$D_{i,r} = \left[\frac{\theta_{i,r}^{\phi_i} \xi d_{i,r} (1 + \tau_{i,r}^z) p_{i,r}^z}{p_{i,r}^d} \right]^{\frac{1}{1-\phi_i}} Z_{i,r} \quad \forall i, r \quad (10.32)$$

– Market-clearing conditions:

$$Q_{i,r} = X_{i,r}^p + X_{i,r}^g + X_{i,r}^v + \sum_j X_{i,j,r} \quad \forall i, r \quad (10.33)$$

$$\sum_j F_{h,j,r} = FF_{h,r} \quad \forall h, r \quad (10.34)$$

$$p_{i,r}^{We} = p_{i,rr}^{Wm} \quad \forall i, r \neq rr \quad (10.35)$$

$$E_{i,r} = M_{i,rr} \quad \forall i, r \neq rr \quad (10.36)$$

The new symbols used in this two-country model are as follows:

r, rr : country (JPN = Japan, USA = the US),

ε_r : the foreign exchange rate of the r -th country (domestic currency per unit of foreign currency).⁵

Like the standard CGE model, this system of simultaneous equations is solved as an optimization problem in GAMS. While a utility function was used as the fictitious objective function in the one-country models, the fictitious objective function in this two-country model is SW , which is the sum of the utility levels in the two countries:⁶

$$SW = \sum_r UU_r = \sum_r \prod_i X_{i,r}^{p_{i,r}^{\alpha_i}} \quad (10.37)$$

where:

SW : social welfare (fictitious objective function).

In comparison with the standard CGE model, there are two new equations (10.35) and (10.36). The others are the same as those in the standard

CGE model. These new equations are international market-clearing conditions being imposed in a skew-symmetric manner. Equation (10.35) ensures that the export price in one country's currency terms of the r -th country is equal to the corresponding import price in the same country's currency terms of the rr -th country. Equation (10.36) imposes a similar skew-symmetry condition on quantities. (Recall the structure of the two-country model as shown in Figure 10.3.)

Attention must be paid to Walras's law and numerares. As in the standard CGE model, Walras's law, which holds for each country, leads to indeterminacy of the price level of these two countries. We have to choose a numeraire for each country.⁷ This two-country model has another redundant equation: one of the two balance of payments constraints (10.26). This is because when one of these two holds, the other must automatically hold. Redundancy regarding the balance of payments constraints leads to the indeterminacy of one of the foreign exchange rates ε_r . Therefore, we have to set a value for one of the exchange rate variables, which is arbitrarily chosen, as an exogenous variable.⁸

Using the SAM in Table 10.2, we present the input file of the two-country CGE model as List 10.2 (*twocge.gms*).⁹ No changes are made in lines 4–10; however, changes are made to lines 8 and 10, which declare and define a new set of ' r ' and ' rr ' to indicate countries. Lines 14–62 install the SAM (Table 10.2). As the model is extended to a two-country model with an additional index of r , the SAM has three dimensions (u, v, r). Each element of the three dimensions is expressed by one *row-name*, followed by the pair of *column-name* and *country-name*, while a period '.' is used to delimit these two. To incorporate a large SAM, we divide it into four submatrices. The '+' symbol is attached in the northwest corner of the subsequent submatrices to indicate a continuing table. The process of calibration and the presentation of equations and variables appear in the same manner as for the standard CGE model.

Before calibrating the model, we have to examine whether the SAM has been properly installed, particularly paying attention to whether the skew-symmetry holds with respect to trade values and current account deficits as the model expects. If the initial equilibrium indicated in the SAM does not satisfy the above international market-clearing conditions for goods and funds, the model will be calibrated on the basis of an erroneous SAM and thus cannot reproduce the initial equilibrium.

Table 10.2 SAM for the two-country model

Japan												
Activity			Factor			Indirect Tax			Final Demand			External
BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT			Total
BRD	21	8				20	19	16			8	92
MLK	17	9				30	14	15			4	89
CAP	20	30										50
LAB	15	25										40
IDT	5	4										9
TRF	1	2										3
HOH			50									90
GOV				9	3	23	2				12	35
INV						17						31
EXT												24
Total	92	89	50	40	3	90	35	31				24
US												
Activity			Factor			Indirect Tax			Final Demand			External
BRD	MLK	CAP	LAB	IDT	TRF	HOH	GOV	INV	EXT			Total
BRD	40	1				30	20	20			13	124
MLK	17	29				30	14	15			11	116
CAP	33	30										63
LAB	15	31										46
IDT	10	20										30
TRF	1	1										2
HOH			63									109
GOV				30	2	29	27				-12	61
INV						20						35
EXT												12
Total	124	116	63	46	2	109	61	35				12

List 10.2 Two-country model (twocge.gms)

```

...
4 Set      u          SAM entry      /BRD, MLK, CAP, LAB, IDT, TRF, HOH, GOV,
5          i(u)       goods          /BRD, MLK/
6          h(u)       factor         /CAP, LAB/
7          r          country        /JPN, USA/;
8
9
10 Alias (u,v), (i,j), (h,k), (r,rr);
11 * -----
12
13 * loading data -----
14 Table    SAM(u,v,r)                social accounting matrix
15          BRD.JPN MLK.JPN CAP.JPN LAB.JPN IDT.JPN
16 BRD      21      8
17 MLK      17      9
18 CAP      20      30
19 LAB      15      25
20 IDT      5       4
21 TRF      1       2
22 HOH              50      40
23 GOV              9
24 INV
25 EXT      13      11
26
27 +        TRF.JPN HOH.JPN GOV.JPN INV.JPN EXT.JPN
28 BRD      20      19      16      8
29
30 ...
174 Variable Y(j,r)                  composite factor
31
32 ...
193          epsilon(r)              exchange rate
194          pWe(i,r)                 export price in US dollars
195          pWm(i,r)                 import price in US dollars
33
34 ...
203          UU(r)                   utility
204          SW                       Social Welfare [fictitious obj. func.]
35
36 ...
207 Equation eqpy(j,r)               composite factor aggregation func.
37
38 ...
236          eqpw(i,r,rr)             international price equilibrium
237          eqw(i,r,rr)              international quantity equilibrium
39
40 ...
242          eqUU(r)                  utility function
243          obj                       social welfare function [fictitious]
41
42 ...
300 *[international market clearing condition]
301 eqpw(i,r,rr).. (pWe(i,r) -pWm(i,rr))$(ord(r) ne ord(rr)) =e= 0;
302 eqw(i,r,rr).. (E(i,r) -M(i,rr))$(ord(r) ne ord(rr)) =e= 0;
303
304 *[fictitious objective function]
305 eqUU(r)..      UU(r)      =e= prod(i, Xp(i,r)**alpha(i,r));
306 obj..          SW        =e= sum(r, UU(r));
43
44 ...

```

List 10.2 (Continued)

```

328 epsilon.l(r)=1;
329 pWe.l(i,r)=1;
330 pWm.l(i,r)=1;
...
356 epsilon.lo(r)=0.00001;
357 pWe.lo(i,r)=0.00001;
358 pWm.lo(i,r)=0.00001;
...
366 * numeraire ---
367 pf.fx("LAB",r)=1;
368 * fixing the redundant variable
369 epsilon.fx("USA")=1;
...

```

As the new international market-clearing conditions (10.35) and (10.36) are introduced to ensure skew-symmetry between two individual countries, they have to be imposed for transactions only between different countries. Therefore, these equations are applied only in the case that $r \neq rr$. This selective application of the equations can be done with conditional expressions of GAMS as '\$(ord(r) ne ord(rr))\$' in lines 301–302.¹⁰ Line 306 shows the social welfare function, which is the sum of the utility levels of households in Japan and the US and is used as the fictitious objective function. Line 367 pins down the labour price at unity in each country to indicate the numeraire. Line 369 fixes the foreign exchange rate for the US ε_{USA} at unity because of its redundancy.

10.3.2 Multicountry model

When we develop a model with three or more countries, we can use the same structure as that of the two-country model, but we have to modify the model regarding exports and imports. All exports are categorized as one group, creating a contrast to the domestic goods with respect to the elasticity of substitution, as stated below. Imports are treated in the same manner. Because more than two countries are involved, let r denote the home country, and let r' and r'' denote its counterpart countries. While in the one- and two-country models, domestic goods were combined with imports to form composite goods (Figure 6.1), in a multicountry model, we set two nested stages to produce the composite goods (Figure 10.4). In the first stage, we assume that the r -th (or the home) country's composite imports QM_r are made up of imports from all the counterpart countries, i.e., the r' -th country's $M_{r',r}$ and the r'' -th country's $M_{r'',r}$, using a CES production function. In the second stage, the composite imports QM_r

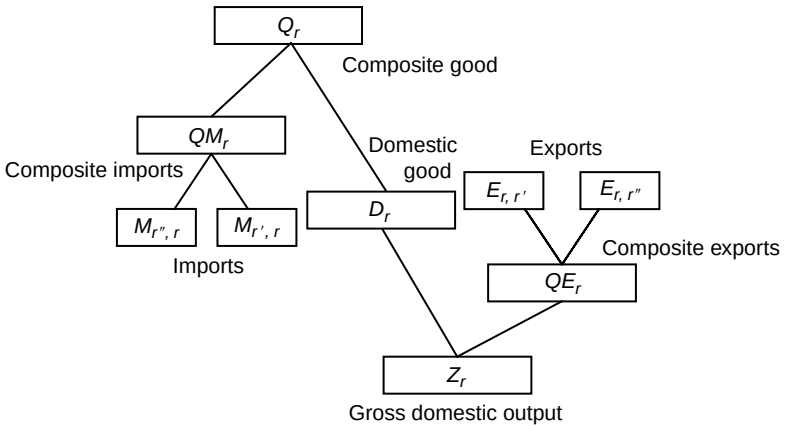


Figure 10.4 Nested CES/CET structure in the r -th country

are combined with the domestic good D_r to produce the composite good Q_r , as in the standard CGE model.

In this process, we often assume that the elasticity of substitution among imports from the counterpart countries is larger than that between composite imports and domestic goods. This indicates an assumption that imports from various origins are highly substitutable with each other, compared with the choice between the domestic good and imported ones. For example, substitutability between US and Australian beef is probably higher than that between imported and Japanese beef for Japanese consumers. Using this nested CES structure, we can assume a different magnitude for the elasticity of substitution between imports from different origins and for the elasticity of substitution between composite imports and domestic goods.

We also assume a similar nested CET structure, where gross domestic output Z_r is transformed into domestic goods D_r and composite exports QE_r using the CET function. Then, the composite exports QE_r are further broken into exports to various destinations, i.e., r' -th country $E_{r,r'}$ and r'' -th country $E_{r,r''}$, using the CET function.

The extended model with the nested CES/CET structure can be calibrated in the same manner as we demonstrated in Subsection 6.8.2. However, there is a problem regarding the elasticity of substitution among imports from individual country origins. While the elasticity of substitution between composite imports and domestic goods is generally available, the former elasticity is rarely available. Regarding this point,

we often follow GTAP practice. Although the GTAP database provides various elasticity parameters, which are originally reported in the meta-analysis about the elasticities in the SALTER project, the GTAP group does not actually undertake any (econometric) estimation of the elasticities of substitution among different countries. Instead, they assume its values to be twice as large as those of the elasticity of substitution between domestic goods and composite imports.^{11,12}

10.4 Imperfect competition model

10.4.1 Monopoly model

All the models developed so far are perfect competition models with constant-returns-to-scale technology. When many agents exist in the market, sellers offering a slightly higher/lower price than the one offered by others can attract no/all buyers in the market. As a result, market competition leads to an equilibrium price equal to marginal cost, which is the lowest price that sellers can offer. In reality, however, we usually face a situation in which there is only one seller (monopoly) or only a few sellers (oligopoly) in the market. We have to modify the standard CGE model demonstrated in Chapter 6 to reflect such a situation.

First, consider the case of a monopoly. When the i -th sector has only one monopoly supplier, the gap between marginal costs MC_i and supply price p_i is dependent on the price elasticity of demand Γ_i (> 1):

$$\left(1 - \frac{1}{\Gamma_i}\right)p_i = MC_i$$

Here, we incorporate the monopoly structure in the domestic good D_i market in the standard CGE model. Let us equate the supply price of the i -th domestic good p_i^d to the marginal cost. Recall that the price elasticity of the demand function (Γ_i in the above equation) for domestic goods is approximately equal to the elasticity of substitution σ_i of the Armington CES function, explained in Section 10.2, and that as $\eta_i = (\sigma_i - 1)/\sigma_i$, the markup rate is $(1 - \eta_i)/\eta_i$. As the demand price diverges from this supply price by the markup rate $(1 - \eta_i)/\eta_i$, the demand price becomes $(1/\eta_i)p_i^d$.

Next, by replacing the denominator of the original demand function for domestic goods (6.19) with the above marked-up demand price, we get the demand function under monopoly:

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i p_i^q}{(1/\eta_i)p_i^d} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (10.38)$$

When we introduce the monopoly structure in the model, we have to specify which agent finally earns the monopoly rents RT_i . Without an equation specifying who captures the monopoly rents, the model becomes ‘not closed’. We assume that the monopoly rent is first captured by the firms, then transferred to the holder of the factors; i.e., the household.

As the monopoly rent constitutes a certain portion of the values of the domestic goods specified by the markup rate $(1 - \eta_i)/\eta_i$, like the production tax specified by the tax rate, we can compute the monopoly rent in the same manner as the production tax:

$$RT_i = \frac{1 - \eta_i}{\eta_i} p_i^d D_i \quad \forall i \quad (10.39)$$

As all rents are transferred to the household, the rent computed above must be added to household income, which constitutes part of the household budget constraint; then, the household demand function, comparable to (6.13) in the original model, becomes:

$$X_i^p = \frac{\alpha_i}{p_i^q} \left(\sum_h p_h^f FF_h + \sum_j RT_j - S^p - T^d \right) \quad \forall i \quad (10.40)$$

(If we alternatively assume that the government [or a state-owned enterprise] earns the monopoly rent, we include the monopoly rent as part of government revenue.)

Furthermore, because the direct tax and household savings are tied to household income, the following modifications of the direct tax revenue function (6.6) and the household savings function (6.11) are also needed:

$$T^d = \tau^d \left(\sum_h p_h^f FF_h + \sum_j RT_j \right) \quad (10.41)$$

$$S^p = ss^p \left(\sum_h p_h^f FF_h + \sum_j RT_j \right) \quad (10.42)$$

Using (10.38) and (10.40)–(10.42) in place of (6.19), (6.13), (6.6) and (6.11), and adding (10.39) with a new endogenous variable RT_i to the original model system in Chapter 6, we complete the monopoly model.

Because of the above modifications in the model, we have to modify the calibration processes. Assume that we use the SAM in Table 4.2. In this SAM, the figures corresponding to the factor payment cells of $SAM_{h,j}$ are presumed to contain the monopoly rent. Therefore, to derive

the net initial equilibrium value of factor input $F_{h,j}^0$, we must subtract from the values $SAM_{h,j}$ in the 'Factor'-'Activity' cells the amount of the monopoly rent supposed to be included in these cells (Figure 10.5). In the calculation of the monopoly rent contained in each of these cells, we assume that it is in proportion to the amount of factor income. Then we can compute the amount of monopoly rent in each factor income, by distributing the total monopoly rents with the weight $SAM_{h,j}/\sum_k SAM_{k,j}$. Here, $SAM_{h,j}$ stands for the value of the cell of the 'Factor' rows indexed by h and the 'Activity' columns indexed by j , while $\sum_k SAM_{k,j}$ is the total income derived from all the factors in the j -th good production. The net factor income (after the monopoly rent is subtracted) is computed as follows:

$$p_h^{f0} F_{h,j}^0 = SAM_{h,j} - \frac{SAM_{h,j}}{\sum_k SAM_{k,j}} RT_j^0 \quad \forall h, j \quad (10.43)$$

The computational process of the standard CGE model presented in Section 6.8 is illustrated in the left panel of Figure 10.6 and appears in the input file of List 6.1. The computation of $F_{h,j}^0$ in the standard CGE model was straightforward because it did not include monopoly rent, and the factor price in the initial equilibrium is set at unity. It could be directly derived from the corresponding cell of the SAM (in line 66 of List 6.1).

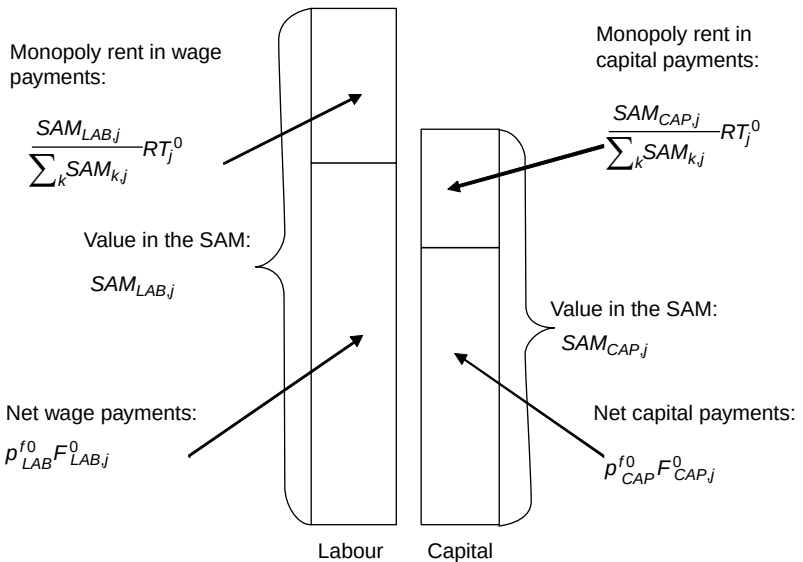


Figure 10.5 Monopoly rents mixed in factor payments

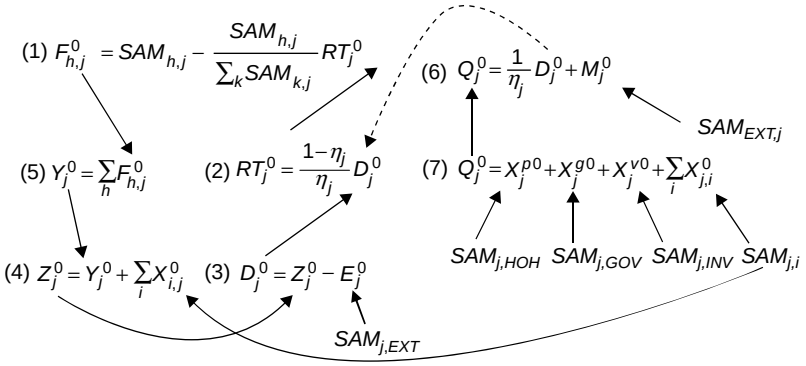
Data Loading Process in Chapter 6New Data Loading Process

Figure 10.6 Data loading processes for variables

Note: For simplicity of the figure, indirect taxes and prices in the initial equilibrium are omitted. To maintain the consistency of suffixes, i and j are sometimes replaced with each other.

However, in the monopoly case, if we follow the process expressed in List 6.1, we face a loop problem. The reasons for the loop and a solution are explained below:

- (1) Assuming that RT_j^0 is known, $F_{h,j}^0$ is obtained in (10.43); however, the derivation of RT_j^0 must go through a long string of computations shown below.
- (2) To obtain RT_j^0 in (10.39), we must know the initial equilibrium amount of domestic good input D_j^0 .
- (3) To obtain D_j^0 , we must know Z_j^0 and E_j^0 (line 82 in List 6.1).
- (4) In the above computation of D_j^0 , while E_j^0 is directly retrieved from the SAM (line 80 of List 6.1), Z_j^0 is computed by summing $X_{i,j}^0$ and Y_j^0 (line 69 of List 6.1).
- (5) While Y_j^0 is computed by summing $F_{h,j}^0$ (in line 67), $F_{h,j}^0$ cannot be derived without knowing RT_j^0 ; the computation process is looped.

To work around this loop, we change the calibration process, following the right panel of Figure 10.6. The main idea is to switch from the zero-profit condition of the firms engaged in transformation of the gross domestic output Z_j into exports E_j and domestic goods D_j as used in (3):

$$p_j^{z0} Z_j^0 = p_j^{d0} D_j^0 + p_j^{e0} E_j^0 \quad \forall j$$

to the zero-profit condition of the firms engaged in production of the Armington composite good Q_j :¹³

$$p_j^{q0} Q_j^0 = \frac{1}{\eta_j} p_j^{d0} D_j^0 + (1 + \tau_j^m) p_j^{m0} M_j^0 \quad \forall j$$

The new calibration process appears in the input file shown in List 10.3:

- (1) M_j^0 is derived from the SAM figure (line 77).
- (2) Q_j^0 is derived from the SAM figures of X_j^{p0} , X_j^{g0} , etc. (line 83).

List 10.3 Monopoly model (moncge.gms)

```

...
37 * Loading the initial values -----
38 Parameter      sigma(i)      elasticity of substitution
39                psi(i)        elasticity of transformation
40                eta(i)         substitution elasticity parameter
41                phi(i)         transformation elasticity parameter
42 ;
43 sigma(i)=3;
44 psi(i)  -2;
...
48 Parameter      Y0(j)         composite factor
...
63                Tm0(j)        import tariff
64                RT0(j)         monopoly rent
...
77 M0(i)  =SAM("EXT",i);
...
83 Q0(i)  =Xp0(i)+Xg0(i)+Xv0(i)+sum(j, X0(i,j));
84 D0(i)  =(Q0(i)-(1+taum(i))*M0(i))/(1/eta(i));
85 RT0(j) =(1-eta(j))/eta(j)*D0(j);
86 F0(h,j) =SAM(h,j)-SAM(h,j)/sum(k, SAM(k,j))*RT0(j);
87 FF(h)  =sum(j, F0(h,j));
...
129 deltam(i)=(1+taum(i))*M0(i)**(1-eta(i))
130           /( (1+taum(i))*M0(i)**(1-eta(i))
131             + (1/eta(i))*D0(i)**(1-eta(i)) );
132 deltad(i)=(1/eta(i))*D0(i)**(1-eta(i))
133           /( (1+taum(i))*M0(i)**(1-eta(i))
134             + (1/eta(i))*D0(i)**(1-eta(i)) );
...
142 ssp     =Sp0/(sum(h, FF(h))+sum(j, RT0(j)));
143 ssg     =Sg0/(Td0+sum(j, Tz0(j))+sum(j, Tm0(j)));
144 taud    =Td0/(sum(h, FF(h))+sum(j, RT0(j)));
...

```

List 10.3 (Continued)

```

151 Variable      Y(j)          composite factor
...
176             Tm(i)          import tariff
177             RT(j)          monopoly rent
...
181 Equation      eqpy(j)       composite factor aggregation func.
...
196             eqXp(i)        household demand function
197             eqRT(j)        monopoly rent function
...
223 *[government behavior] ----
224 eqTd..         Td           =e= taud*(sum(h, pf(h)*FF(h))+sum(j, RT(j)));
...
233 *[savings] -----
234 eqSp..         Sp           =e= ssp*(sum(h, pf(h)*FF(h))+sum(j, RT(j)));
235 eqSg..         Sg           =e= ssg*(Td +sum(j, Tz(j))+sum(j, Tm(j)));
236
237 *[household consumption] --
238 eqXp(i)..      Xp(i)        =e= alpha(i)*(sum(h, pf(h)*FF(h)) -Sp -Td
239                               +sum(j, RT(j)))/pq(i);
240 eqRT(j)..      RT(j)        =e= (1-eta(j))/eta(j)*pd(j)*D(j);
...
248 *[Armington function] ----
...
253 eqD(i)..       D(i)         =e= (gamma(i)**eta(i)*deltad(i)*pq(i)
254                               /((1/eta(i))*pd(i))**(1/(1-eta(i)))*Q(i);
...
296 Tm.l(i) =Tm0(i);
297 RT.l(j) =RT0(j);
...

```

- (3) D_j^0 is derived from the above zero-profit condition (line 84). (Note that while the marginal cost of producing D_j is still p_j^d in this monopoly model, the demand price becomes $(1/\eta_j)p_j^d$ because of the markups.)
- (4) RT_j^0 is computed using (10.39) (line 85).
- (5) $F_{h,j}^0$ is computed using (10.43) (line 86).

The monopoly model also requires a modification in estimation of the factor endowments FF_h . We cannot simply use the values of $SAM_{HOH,h}$ in the 'HOH'-'Factor' block for the values of FF_h because the monopoly rents are (assumed to be) mixed into these cells. Instead, we sum $F_{h,j}^0$, which is derived using (10.43), for all j to compute FF_h as shown in line 87.

The subsequent processes in List 10.3, which is the input file of this monopoly model, are almost the same as List 6.1.¹⁴ As the model is

slightly modified by the markup, the calibration processes for δm_i , δd_i , ss^p and τ^d in lines 129–131, 132–134, 142 and 144 are different from those used in the model in Chapter 6.¹⁵

10.4.2 Oligopoly model

Once we understand the monopoly CGE model, we can easily develop an oligopoly model applying a standard technique suggested by many microeconomics textbooks. For simplicity, we assume Cournot competition with n_i symmetric oligopoly firms in the i -th sector (note that n_i stands for the number of such firms and is exogenous). Given the number of firms, we can compute the demand price with markup as follows:

$$\left(1 - \frac{1}{n_i \Gamma_i}\right) p_i = MC_i \quad \forall i$$

Assuming that the market for D_i is the oligopoly market, the price elasticity of demand Γ_i can be replaced with the elasticity of substitution in the Armington composite good production function σ_i . The markup rate $(1 - \eta_i)/(-1 + n_i + \eta_i)$ implies the demand price $[n_i/(-1 + n_i + \eta_i)]p_i^d$.

The demand function of the oligopoly market, which corresponds to (6.19) or (10.38), becomes:

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i p_i^d}{[n_i/(-1 + n_i + \eta_i)]p_i^d} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i$$

Based on the monopoly rent function (10.39), we can obtain the oligopoly rents as follows:

$$RT_i = \frac{1 - \eta_i}{-1 + n_i + \eta_i} p_i^d D_i \quad \forall i$$

Replacing (10.38) and (10.39) of the monopoly model by these two equations, we can develop the oligopoly CGE model. Incidentally, note that the number of oligopoly firms n_i has to be declared as an exogenous variable in the oligopoly model.

10.5 Quantitative restrictions

To protect domestic markets from imports, the government can manipulate import tariff rates to limit imports to any level. In reality, however,

a more direct intervention device, i.e., a quantitative restriction, has been adopted in many cases. Abolition of quotas under the Multi-Fibre Arrangements (MFA), applied to exports of textile and apparel products mainly from developing countries to developed countries, was one of the major issues in the Uruguay Round trade negotiations. Many countries have even resorted to import quotas/bans (or sometimes voluntary export restraints set by their trade partners) to protect their domestic industries, such as metal products, supercomputers, automobiles and agricultural products. Food safety regulations also provide grounds to ban imports from specific countries. In addition, safeguard measures available under the WTO agreement can be used legitimately to protect domestic industries from an influx of imports, albeit temporarily.

In general, when the quota is set so as to yield marginal rent as high as the tariff rate, these quantitative restrictions can bring about exactly the same effect on volumes and prices as the tariffs do. While we discuss how to model the imposition of quotas and tariffs on imports in this section, a similar modelling approach is applicable to other quantitative restrictions and indirect taxes on other aspects of economic activities, such as exports, domestic production and so on. It should also be noted that we conduct below an analysis on import quotas and import tariffs in a general equilibrium framework; these are often also analysed in a partial equilibrium framework. The important point of the general equilibrium analysis is that we need to specify the agent (or agents) that capture the quota rents in order to close the model and to measure their income effects.

Figure 10.7 (although a familiar figure in partial equilibrium analysis) illustrates the cases with and without an import quota. When there is no quota imposed on imports, the equilibrium price and quantity are p^{m*} and M^* respectively. When we impose a ceiling on imports M^{quota} smaller than M^* , the quantity M is forced to equal M^{quota} ; therefore, the supply and demand prices become p^m and $(1 + \chi)p^m$, and quota rent $\chi p^m M^{quota}$ is generated. Note that M^{quota} is an exogenous variable, while quota rent rate χ is endogenous, which is flexibly adjusted to achieve market equilibrium.

In contrast, if the quota ceiling is set at a high level, such as $\tilde{M}$ in Figure 10.7, the equilibrium M^* is realized. The quota constraint turns out to be nonbinding, and quota rents are zero. This is equivalent to the case of a model without any quantitative restrictions.

Returning to the case of a quota M^{quota} , we can use the complementarity conditions to model both binding and nonbinding import quotas as

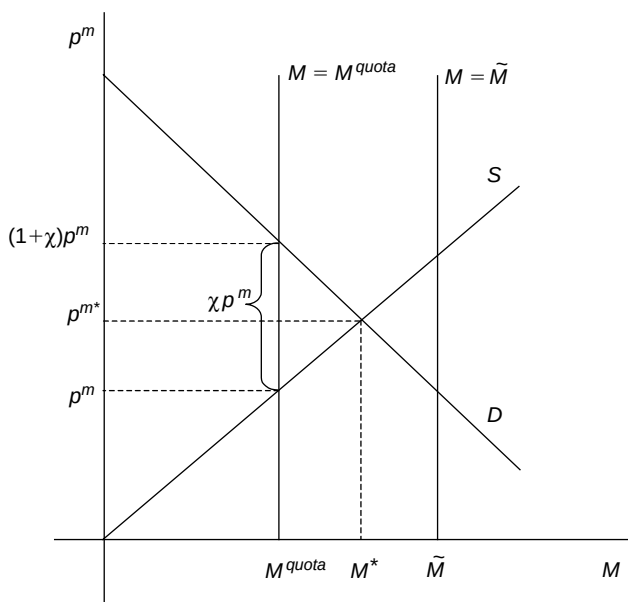


Figure 10.7 Effects of import quota

follows:

$$\chi \cdot (M^{quota} - M) = 0, \quad M^{quota} - M \geq 0, \quad \chi \geq 0$$

The implication of this complementarity condition is as follows. On the one hand, if the second constraint holds with strict equality (i.e., the amount of imports without any intervention hits its ceiling to make the quota constraint binding: $M^{quota} - M = 0$), quota rent rate χ , which is the shadow price of this quota constraint, can be zero or strictly positive to satisfy the first and the third constraints. On the other hand, if the second constraint holds with strict inequality (i.e., the amount of imports without any intervention is less than the ceiling; thus, the quota constraint is not binding: $M^{quota} - M > 0$), the third constraint has to hold with strict equality (i.e., $\chi = 0$) to satisfy the first constraint.

Here, the standard CGE model developed in Chapter 6 can be extended to include this import quota module. Suppose that the quota ceiling is set on imports of the i -th good so that it yields a quota rent rate χ_i and that the quota rent RT_i is eventually captured by the household, through

transfers from the sellers or the buyers. We introduce the quota rent rate χ_i in the import demand function (6.18) so that it operates in the same manner as the import tariff rate τ_i^m :

$$M_i = \left[\frac{\gamma_i^{\eta_i} \delta m_i p_i^q}{(1 + \chi_i + \tau_i^m) p_i^m} \right]^{\frac{1}{1-\eta_i}} Q_i \quad \forall i \quad (10.44)$$

The rents from the import quotas amount to:

$$RT_i = \chi_i p_i^m M_i \quad \forall i \quad (10.45)$$

As the rents are assumed to accrue to household income, we modify the equations where the household budget appears in the same manner as when we introduced the monopoly rents in household income in Subsection 10.4.1. We replace the household demand function (6.13) with (10.40), the direct tax revenue function (6.6) with (10.41) and the household savings function (6.11) with (10.42).

The complementarity condition for the quota on imports of the i -th import good is:

$$\chi_i \cdot (M_i^{quota} - M_i) = 0 \quad (10.46)$$

$$M_i^{quota} - M_i \geq 0 \quad (10.47)$$

$$\chi_i \geq 0$$

Among these three equation/inequalities, we do not have to include the last condition in the input file explicitly as long as the **Positive Variable** directive is used to declare the quota rent rate χ_i in the GAMS program. This import quota model is equipped with new endogenous variables χ_i for the quota rent rate and RT_i for the quota rents with new equations (10.45)–(10.47). Finally, we have the model system consisting of (6.1)–(6.5), (6.7)–(6.10), (6.12), (6.14)–(6.24), (10.40)–(10.42) and (10.44)–(10.47).

Now consider the input file of the import quota model (`quocge.gms`) presented in List 10.4. The ceiling on imports M_i^{quota} is declared in line 61. As we assume that the import quota is not imposed (or not binding) in the initial equilibrium, we set its value large enough – 100 times as large as the initial equilibrium imports M_i^0 in line 75. We declare the

List 10.4 Import quota model (quocge.gms)

```

38 Parameter      Y0(j)          composite factor
...
60              taum(i)          import tariff rate
61              Mquota(i)        import quotas
...
74 taum(j) =Tm0(j)/M0(j);
75 Mquota(i)=M0(i)*100;
...
153 Variable      Y(j)          value added
...
178              Tm(i)           import tariff
179              RT(i)           rent accruing from import quotas
180              UU              utility [fictitious]
181;
182 Positive Variable
183              chi(i)          marginal quasi-rent
184;
185 Equation       eqpy(j)        composite factor aggregation func.
...
200              eqXp(i)         household demand function
201              eqRT(i)         quasi-rent function
...
208              eqD(i)          domestic good demand function
209              eqchil(i)        import quota complementarity condition
210              eqchi2(i)        import quota complementarity condition
...
228 *[government behavior] ----
229 eqTd..         Td            =e= taud*(sum(h, pf(h)*FF(h))+sum(j, RT(j)));
...
238 *[savings] -----
239 eqSp..         Sp            =e= ssp*(sum(h, pf(h)*FF(h))+sum(j, RT(j)));
240 eqSg..         Sg            =e= ssg*(Td +sum(j, Tz(j))+sum(j, Tm(j)));
241
242 *[household consumption] --
243 eqXp(i)..      Xp(i)         =e= alpha(i)*(sum(h, pf(h)*FF(h)) +sum(j, RT(j))
244                      -Sp -Td)/pq(i);
245 eqRT(i)..      RT(i)         =e= chi(i)*pm(i)*M(i);
...
253 *[Armington function] ----
...
256 eqM(i)..      M(i)          =e= (gamma(i)**eta(i)*deltam(i)*pq(i)
257                      /((1+chi(i)+taum(i))*pm(i)))
258                      *(1/(1-eta(i)))*Q(i);
...
261 eqchil(i)..    chi(i)*Mquota(i) -M(i) =e= 0;
262 eqchi2(i)..    Mquota(i) -M(i) =g= 0;
...
304 Tm.1(i) =Tm0(i);
305 RT.1(i) =0;
306 chi.1(i)=0;
...
338 * Defining and solving the model -----
339 Model quocge /all/;
340 Solve quocge maximizing UU using nlp;
341
342 * -----
343 * Simulation Runs: Imposition of Quotas on Bread Imports
344 Mquota("BRD")=M0("BRD")*0.9;
345 Solve quocge maximizing UU using nlp;
...

```

new endogenous variables of the quota rent RT_i in line 179 and the quota rent rate χ_i in line 183. (Note that the Greek letter χ is expressed using its English name 'chi' in the input file.) The equation names for the complementarity conditions (10.46) and (10.47) are declared in lines 209–210; they are specified in lines 261–262. The rent RT_i appears as part of household income in lines 243–244, and its total amount is computed in line 245. The rent RT_i also affects household savings and direct tax revenues, as shown in lines 239 and 229, respectively. The quota rent rate χ_i increases the demand price of imports as shown in lines 256–258. Lines 305–306 set initial values for the rents and the quota rent rate. Both are zero in the base run equilibrium because we assume that the quota constraint is not binding in the initial equilibrium. Line 340 is included to compute the base run equilibrium.

Moving to the counterfactual run, we set the import quota of bread at 90% of the initial equilibrium imports of bread M_{BRD}^0 in line 344. This ceiling is lower than the initial equilibrium imports, and thus it is expected to make the import quota binding. The second `Solve` statement is included to compute the counterfactual equilibrium with import quotas.

When we solve this model, we find that the import quota imposition causes welfare gain in this country. This seems counterintuitive from the viewpoint of standard trade theory. To interpret this result properly, we have to take note of the assumption made in Section 6.4. While the import quotas would cause a distortion in resource allocation and tend to reduce all of X_i^p , X_i^g and X_i^v in the economy, the model assumes that the quota rent is captured only by the household to increase X_i^p . This means the rent works as transfers to the household from other agents. If we modify the model so that government and investment uses X_i^g and X_i^v are set to be constant; thus, possible transfers from X_i^g and X_i^v to the household are shielded, the simulation result becomes consistent with our intuition. This has something to do with the macro closure rules discussed in Chapter 7.

10.6 Increasing-returns-to-scale model

We now develop a CGE model with increasing-returns-to-scale (IRTS) in production technology, while we have so far assumed only constant-returns-to-scale technology. Among a variety of specifications to incorporate the scale economies in production, we choose fixed costs in this

example. We assume that the firms make payments of FC_j for use of some resource (such as land) for their operation irrespective of their production, in addition to the usual input costs of intermediate input $X_{i,j}$ and composite factor Y_j to produce gross domestic output Z_j . The first costs are the fixed costs for the firms, while the other costs are variable costs. The resource for which the firms pay a fixed cost is assumed to be provided by the household. It is further assumed that each sector has only one firm. To incorporate this new feature, the original unit cost function (6.5) is modified – assuming average cost pricing – as follows:

$$p_j^z = a_{vj}p_j^y + \sum_i a_{xi,j}p_i^q + \frac{FC_j}{Z_j} \quad \forall j \quad (10.48)$$

Following the assumption that FC_j is paid by firms to the provider of the above-mentioned resource, i.e., the household, we need to modify the direct tax revenue function (6.6), the household savings function (6.11) and the household demand function for goods (6.13) in the following manner:

$$T^d = \tau^d \left(\sum_h p_h^f FF_h + \sum_j FC_j \right) \quad (10.49)$$

$$S^p = ss^p \left(\sum_h p_h^f FF_h + \sum_j FC_j \right) \quad (10.50)$$

$$X_i^p = \frac{\alpha_i}{p_i^q} \left(\sum_h p_h^f FF_h + \sum_j FC_j - S^p - T^d \right) \quad \forall i \quad (10.51)$$

We again use the SAM in Table 4.2 as the database to which the IRTS CGE model is calibrated. In building an empirical model, we have to assume in which cells of the SAM the payments of funds corresponding to FC_j are included and their values. Here, by relying on prior information about firms' fixed costs from previous studies, we assume that a fixed share, say, v_j (note that this is the Greek letter *nu*) of the total capital service payment $SAM_{CAP,j}$ is allotted to the firms' fixed cost payment FC_j . Thus, FC_j is computed as follows:

$$FC_j = v_j SAM_{CAP,j} \quad \forall j$$

The firms pay FC_j through the factor management agent to the household. Consequently, the net use of the capital service $F_{CAP,j}^0$ is:

$$F_{CAP,j}^0 = (1 - v_j)SAM_{CAP,j} \quad \forall j$$

The modified zero-profit condition indicates the gross domestic output in the base run equilibrium as follows:

$$p_j^{z0} Z_j^0 = p_j^{y0} Y_j^0 + \sum_i p_i^{q0} X_{i,j}^0 + FC_j \quad \forall j$$

In calibrating τ^d and ss^p , we make use of (10.49) and (10.50), and we use a similar calibration process to those used in the monopoly model to estimate the factor endowments FF_h . The input file for the IRTS CGE model is shown in List 10.5 (irscge.gms).

List 10.5 Increasing-returns-to-scale model (irscge.gms)

```

...
37 * Loading the initial values -----
38 Parameter      nu(i)          share of fixed costs in capital costs;
39 nu(i)=0.1;
40
41 Parameter      Y0(j)          composite factor
42                F0(h,j)        the h-th factor input by the j-th firm
43                FC(j)          the fixed costs in the j-th firm
44
...
70 F0(h,j) =(1-nu(j))*SAM(h,j)*(ord(h) eq 1)
71          +SAM(h,j)*(ord(h) ne 1);
72 FC(j)    =nu(j)*SAM("CAP",j);
73 Y0(j)    =sum(h, F0(h,j));
74
...
81 Xp0(i)   =SAM(i,"HOH");
82 FF(h)    =sum(j, F0(h,j));
83
...
109 Parameter    alpha(i)        share parameter in utility func.
...
147 ssp         =Sp0/(sum(h, FF(h))+sum(j, FC(j)));
148 ssg         =Sg0/(Td0+sum(j, Tz0(j))+sum(j, Tm0(j)));
149 taud        =Td0/(sum(h, FF(h))+sum(j, FC(j)));
...
220 *[domestic production] ----
...
225 eqpzs(j)..   pz(j)           =e= ay(j)*py(j) +sum(i, ax(i,j)*pq(i))
226               +FC(j)/Z(j);
227 *[government behavior] ----
228 eqTd..        Td             =e= taud*(sum(h, pf(h)*FF(h)) +sum(j, FC(j)));
...
235 *[savings] -----
236 eqSp..        Sp            =e= ssp*(sum(h, pf(h)*FF(h)) +sum(j, FC(j)));
237 eqSg..        Sg            =e= ssg*(Td +sum(j, Tz(j))+sum(j, Tm(j)));
238 *[household consumption] --
239 eqXp(i)..     Xp(i)         =e= alpha(i)*(sum(h, pf(h)*FF(h))
240               +sum(j, FC(j)) -Sp -Td)/pq(i);

```

Notes

1. We also have to use a fictitious objective function to solve this model with GAMS.
2. This argument is made on the basis of Shoven and Whalley (1992, Ch. 5).
3. In the following discussion, we omit import tariffs for simplicity.
4. The assumption of a small share parameter δm_i implies that the impact of p_i^m on p_i^q is so small that we can omit it. We have already made this assumption in the derivation of the former elasticity ρ_i . Therefore, these two elasticities ρ_i and $\tilde{\rho}_i$ naturally converge to each other.
5. Note that only ε_{JPN} is an endogenous variable; ε_{USA} is exogenous because, by definition, $\varepsilon_{JPN} \cdot \varepsilon_{USA} = 1$ always holds.
6. Again, note that we can use any kind of objective functions to solve our CGE models. We call *SW* social welfare here simply because it is a commonly used name. Alternatively, we could call it whatever we like.
7. Some readers might feel uneasy with two numeraire in one model. Alternatively, we can fix a (nominal) foreign exchange rate (or more generally three among $p_{LAB,JPN}^f$, $p_{LAB,USA}^f$, ε_{JPN} and ε_{USA}) to make both countries use the same currency with only one numeraire, then we obtain the same equilibrium solution as long as the current account is always balanced. However, when the current account is not balanced, the choice of the numeraire is not neutral. This is because a change in foreign exchange rates causes a change in the current account deficits evaluated in terms of the domestic currency.
8. The foreign exchange rate is adjusted so that the balance of payments constraint is satisfied. That is, (the inverse of) the former is the shadow price of the latter. This can be proved by formulating the CGE model as a nonlinear complementarity programming problem, as Hashimoto (1998) demonstrates.
9. The input file of `twocge.gms`, which is available in the GAMS Model Library, contains additional code to simulate import tariff abolition and to compute the resultant changes from the base run solution.
10. See Section 8.1 for information about the `ord` function. See Section A.2 in Annex A for an explanation of conditional expressions.
11. A global trade CGE model was developed in the *SALER* project conducted by the Industry Commission, which is part of the Australian government. Details of assumptions about elasticity parameters in the GTAP database are provided by Huff et al. (1997).
12. This practice is called the 'rule of two'. Its validity is examined by Liu et al. (2004). They find that in general, the rule cannot be statistically rejected.
13. We can derive the zero-profit condition by setting the profit π_i^q at zero in Subsection 6.5.3 with a slight modification regarding the markups because of the monopolistic supply of D_i .
14. In this monopoly model, however, we assume a larger value (i.e., three) for the elasticity of substitution σ_i in line 43 of List 10.3 than its original value (i.e., two) in the standard CGE model. This is because the original elasticity value leads to estimates of RT_i^0 in (10.39) that are too large and, thus, we end up with a negative or zero value of $p_h^{f0} F_{h,j}^0$ in (10.43), which indicates an odd base run equilibrium.
15. The input file `moncge.gms`, which is available in the GAMS Model Library, contains additional code to simulate import tariff abolition and to compute changes from the base run solution in and after line 334.

11

Concluding Remarks

Many innovations by pioneering CGE modellers and economists have made CGE models one of the indispensable empirical tools for macro-economic analysis. CGE models have been developed to analyse a wide range of policy issues. In addition to the minimal data requirements for model parameter estimation in view of the model size, the significant flexibility in incorporating various features useful for empirical analysis facilitates the widespread use of CGE models. In response to the increasing need to deal with impending policy issues, CGE models have been extended from basic versions to those better suited to modelling the complexities of a modern economy. Such extensions have been made in two directions. One is within the CGE model framework, and the other is outside of it.

In this concluding chapter, we first discuss these two types of extensions of CGE models. We then state our philosophy on CGE modelling that underlies this book, particularly regarding the most frequent critique of CGE models – being a ‘black box’.

11.1 Extensions inside the CGE models

Some extensions within the CGE model framework have been demonstrated in this book. We can easily include special features often considered in the fields of applied economics, such as international trade, distortional policies like taxes and quotas, imperfect competition and scale economies. These features are included as modules within the CGE models.

As extensions applied to trade analysis in CGE models, we can mention two notable examples, both of which encompass increasing-returns-to-scale (IRTS) technology. First, Dixit and Stiglitz (1977) developed an

important theoretical foundation for the subsequent explorations about the 'new trade theory' or 'new economic geography' by introducing IRTS with respect to the number of input varieties. These models are suited to describing the impact of economic integration, which provides more choice for countries with lower trade barriers.

The second innovative study by Melitz (2003) opened a new frontier for deeper consideration and wider application of heterogeneity among firms. This may create a breakthrough in conventional CGE analysis, which mostly assumes a fiction of homogeneous or symmetric firms; that is, all the firms in each sector have the same production technology. Introduction of this heterogeneity into CGE models may provide more accurate estimates of the impact of economic integration even in the sectors that experienced little international trade before the integration. This phenomenon has been widely observed in the North American Free Trade Agreement integration but the conventional CGE studies poorly predicted it in CGE studies, as Kehoe (2005) pointed out. We believe that readers who have read this book can incorporate these new features into the CGE models. These and further extensions are made possible by exploiting the flexibility of CGE models.

11.2 Extensions outside the CGE models

While CGE models can incorporate various features internally, they can also be linked to other models or modules – outside the CGE model. We provide three examples of such extensions, while, needless to say, many more can be developed by creative modellers in the future.

First, one of the recent concerns in developing countries is poverty alleviation along the line of the millennium development goals (MDGs) promoted by international organizations under the umbrella of the United Nations. To assess policies and programmes for the MDGs, macroeconomic frameworks that link CGE models to poverty-focused microsimulation models have been developed. While CGE models, which can model the entire economy, are powerful for macro- or semi-macroeconomic analysis, microsimulation models provide information on the behaviour of dozens of different household types using data from living standards surveys. The CGE models linked to the microsimulation models, therefore, can describe household-level activities and transactions in greater detail than CGE models alone. Although these two models are not perfectly consistent with each other in their underlying assumptions and thus their results, they do overcome the individual weaknesses of each model.

Second, as often mentioned in many studies on the global warming problem, we need to measure the consequences of economic activities on environmental outcomes, such as carbon dioxide emission levels, and the rises of average temperatures and sea levels. The technical side of environmental problems can first be captured outside of the CGE models. Consider two cases of CGE models linked to outside environmental models. If the environmental externalities are expected to cause little feedback to economic activities, we can solve the CGE models simply by ignoring the impact of the externalities. Then, the solved levels of economic activity are fed into the environmental models to compute environmental indicators such as carbon dioxide emissions. Alternatively, if such environmental outcomes are expected to affect economic activities significantly, we have to integrate the environmental modules into the CGE model and derive the impact of externalities on, say, sectoral productivity, cultivatable acreage and so on.

Third, CGE models are used for the analysis of water resources, which are attracting growing attention under the impact of exploding population and increasing need for food in developing countries. Using the CGE model framework, combined with water balance models, we can quantify the volume of water resources indirectly consumed through trade of goods. Such modelling helps us to visualize how distant the places the water resources supporting our daily life are located.

In sum, we can easily incorporate new features into the CGE framework or externally attach these features to conventional CGE models in order to analyse emerging economic problems. Future developments in CGE modelling will no doubt expand the range of issues that can be analysed using such extensions.

11.3 Concluding remarks for better CGE modelling

To diffuse the uses of CGE models, many organizations and researchers have invented and disclosed their modelling devices. In particular, the GTAP team and the International Food Policy Research Institute (IFPRI) provide sophisticated ready-to-use CGE models with accompanying social accounting matrices. Those new to CGE modelling can immediately become users of CGE models with these ready-made models to simulate policies and exogenous shocks such as tax rates, sectoral productivity and so on.

The ready-to-use modelling approach may seem to be a very good starting point, particularly for beginners. Beginners can immediately run simulations with CGE models. Indeed, detailed documents and manuals

on the structure and operation of such CGE models are provided by, for example, Hertel (1997) and Löfgren et al. (2002), and they help users greatly. But they have a potential drawback. Users of these ready-made CGE models tend to run and solve the models without fully understanding the structure and modelling process of the CGE models. Thus, they may feel it difficult to change the models – equations, parameters, and so on – for their own purposes, if good guidance is unavailable. Furthermore, the readiness of those CGE models may retard users' learning of CGE modelling. To them, the CGE models might remain a black box for a long time. They might have little confidence in simulation results generated by such a black box and might feel uncomfortable in convincing other people of the results.

In contrast to the ready-to-use modelling approach, the aim of this book is to enable readers to develop and interpret their own CGE models. For this purpose, the book covers the setup of the economy, collection and compilation of data, estimation of model parameters, interpretations of simulation results and, above all, computer programming techniques. The book may be categorized as a 'how to'-type textbook, where the CGE models have been presented in a step-by-step manner. In particular, the book has focused on creating computer code and files from scratch and reading output written using the GAMS software. Exploiting its user-friendliness, we have only used GAMS as software for numerical computation throughout this book. Because of such a commitment to a single software package, some readers who prefer other software may feel at a disadvantage. Experience with GAMS, however, would help such readers to build CGE models using different software.

There have been a number of articles and books on frontier CGE analyses published. We believe that our book will function as a bridge that closes the gap between the entry point to CGE modelling and its frontiers. This is because we have placed emphasis on the very basics or fundamentals of CGE modelling, by explaining the nature of these models, which people often view as a black box. Readers of the book are expected to understand the basic structure and process of CGE modelling and beyond. We believe that this book will increase the number of CGE modellers. We sincerely hope that they will create further innovations in empirical CGE modelling, and ultimately contribute to better policy discussions based on CGE models.

Appendix I: Derivation of Household Demand Functions

The first-order condition (2.c) derived from the ‘household utility maximization model’ in Section 2.2 can be rewritten as follows:

$$\frac{\partial L}{\partial X_i} = \alpha_i \frac{UU}{X_i} - \varphi p_i^x = 0 \quad \forall i$$

Let us consider a two-good case. Eliminating the Lagrange multiplier φ , we can merge the above equations for the two goods *BRD* and *MLK* to obtain:

$$\frac{\alpha_{BRD}}{p_{BRD}^x} \frac{UU}{X_{BRD}} = \frac{\alpha_{MLK}}{p_{MLK}^x} \frac{UU}{X_{MLK}}$$

Eliminating UU and rearranging the above equation, we get:

$$p_{BRD}^x X_{BRD} = \frac{\alpha_{BRD}}{\alpha_{MLK}} p_{MLK}^x X_{MLK}$$

Recall the first-order condition (2.d):

$$\frac{\partial L}{\partial \varphi} = \sum_h p_h^f FF_h - \sum_i p_i^x X_i = 0 \quad (2.d)$$

Replacing $p_{BRD}^x X_{BRD}$ with $(\alpha_{BRD}/\alpha_{MLK})p_{MLK}^x X_{MLK}$, we can transform (2.d) for $i = MLK$ as follows:

$$\sum_h p_h^f FF_h = \left(1 + \frac{\alpha_{BRD}}{\alpha_{MLK}}\right) p_{MLK}^x X_{MLK}$$

Because $\sum_i \alpha_i = 1$, we get (2.1) for *MLK*. Similarly, we get (2.1) for *BRD*.

The above manipulation can be applied to the cases for $h = CAP, LAB$ to derive the factor demand functions (2.3).

Appendix II: Competitive Equilibrium vs Social Optimum

We could get the general equilibrium solution of an economy (i.e., the amounts of consumption and production, and their prices) by solving a system of simultaneous equations consisting of (2.1)–(2.6). Instead, we can get the same equilibrium by solving the following social welfare maximization problem:¹

$$\underset{X_i, Z_i, F_{h,j}}{\text{maximize}} \quad UU = \prod_i X_i^{\alpha_i}$$

subject to

$$Z_j = b_j \prod_h F_{h,j}^{\beta_{h,j}} \quad \forall j$$

$$X_i = Z_i \quad \forall i$$

$$\sum_j F_{h,j} = FF_h \quad \forall h$$

In the above maximization problem, we use the same notations as those in the system of the simultaneous equations. In this maximization problem, we regard the utility function as the social welfare function because only one household is assumed. The constraints of this maximization problem consist of the production function (2.2) and the market-clearing conditions of goods (2.4) and factors (2.5).

By defining a Lagrangian and deriving the first-order conditions for its optimality, we can prove the equivalence between the solution of the original simultaneous equations system and that of this social welfare maximization problem. We define the Lagrangian in the following manner:

$$\begin{aligned} L(X_i, Z_i, F_{h,j}; \delta_j, \theta_i, \varepsilon_h) = & \prod_i X_i^{\alpha_i} + \sum_j \delta_j \left(b_j \prod_h F_{h,j}^{\beta_{h,j}} - Z_j \right) \\ & + \sum_i \theta_i (Z_i - X_i) + \sum_h \varepsilon_h \left(FF_h - \sum_j F_{h,j} \right) \end{aligned}$$

Its first-order conditions, presuming interior solutions, are:

$$\frac{\partial L}{\partial X_i} = \alpha_i \frac{\prod_j X_j^{\alpha_j}}{X_i} - \theta_i = 0 \quad \forall i \quad (\text{II.1})$$

$$\frac{\partial L}{\partial Z_i} = -\delta_i + \theta_i = 0 \quad \forall i \quad (\text{II.2})$$

$$\frac{\partial L}{\partial F_{h,j}} = \delta_j \beta_{h,j} \frac{b_j \prod_k F_{k,j}^{\beta_{k,j}}}{F_{h,j}} - \varepsilon_h = 0 \quad \forall h, j \quad (\text{II.3})$$

$$\frac{\partial L}{\partial \delta_j} = b_j \prod_h F_{h,j}^{\beta_{h,j}} - Z_j = 0 \quad \forall j \quad (\text{II.4})$$

$$\frac{\partial L}{\partial \theta_i} = Z_i - X_i = 0 \quad \forall i \quad (\text{II.5})$$

$$\frac{\partial L}{\partial \varepsilon_h} = FF_h - \sum_j F_{h,j} = 0 \quad \forall h \quad (\text{II.6})$$

We interpret the Lagrange multipliers appearing above as follows:

$$\theta_i = p_i^x \quad \forall i$$

$$\delta_j = p_j^z \quad \forall j$$

$$\varepsilon_h = p_h^f \quad \forall h$$

In the following, we can show that the system of simultaneous equations with (II.1)–(II.6) above is consistent with the system consisting of (2.1)–(2.6), presented as the ‘simple CGE model’ in Chapter 2. First, transforming (II.1), we get:

$$X_i = \frac{\alpha_i}{p_i^x} \prod_j X_j^{\alpha_j} \quad \forall i$$

Summing up both sides of this equation over all i ’s, and considering $\sum_i \alpha_i = 1$, we obtain:

$$\sum_i p_i^x X_i = \prod_j X_j^{\alpha_j}$$

Combining the above two equations, we get:

$$X_i = \frac{\alpha_i}{p_i^x} \sum_j p_j^x X_j \quad \forall i$$

Replacing $\sum_j p_j^x X_j$ with $\sum_h p_h^f FF_h$ in the budget constraint (2.b), we get (2.1).

Finally, transformation of (II.3) leads to:

$$F_{h,j} = \frac{\beta_{h,j}}{p_h^f} p_j^z b_j \prod_k F_{k,j}^{\beta_{k,j}} \quad \forall h, j$$

Using (II.4), we can replace $b_i \prod_k F_{k,j}^{\beta_{k,j}}$ with Z_j in the above expression; then, we get (2.3). Equations (II.4), (II.5), (II.6) and (II.2) are exactly same as (2.2), (2.4), (2.5) and (2.6) respectively.

Note

1. However, note that when there are two or more heterogeneous households in the economy, we cannot simply apply this technique but instead have to introduce a social welfare function à la Negishi (1960).

Appendix III: Utility Maximization and Lagrange Multipliers

In Section 2.2, we presented the ‘household utility maximization model’ as a maximization problem; then, the original maximization model was reformulated into a system of simultaneous equations consisting of (2.1) in Chapter 3. This reformulation was made to solve the household utility maximization model as a system of simultaneous equations with GAMS for demonstration purposes.

Instead of relying on such an indirect method, we can directly solve the original household utility maximization model with GAMS as a (genuine) maximization problem, as we did in Subsection 3.1.1:

$$\underset{X_i}{\text{maximize}} \quad UU = \prod_i X_i^{\alpha_i} \quad (2.a)$$

subject to

$$\sum_i p_i^x X_i = \sum_h p_h^f FF_h \quad (2.b)$$

To solve the above maximization problem with GAMS, we have to modify lines 31 and 36 in the original input file in List 3.1, respectively, as follows:

```
Equation      phi marginal utility of income
...
phi..  sum(i, px(i)*X(i)) =e= sum(h, pf(h)*FF(h));
```

The SOLVE SUMMARY in the output file reports the same solutions in the VAR blocks as those of the simultaneous equations model but shows different solutions in the EQU blocks (cf. List 3.2):

	LOWER	LEVEL	UPPER	MARGINAL
---- EQU phi	40.000	40.000	40.000	0.348
---- EQU obj	.	.	.	1.000

As we have not mentioned anything about the information given in the EQU blocks in the main text, we discuss it here. In the EQU block, the MARGINAL column indicates the solution of the Lagrange multipliers. In this model, the MARGINAL value of the EQU phi is the solution of the Lagrange multiplier associated with the budget constraint (2.b), which indicates the marginal utility of income. That is, an additional one yen in household income increases its utility by 0.348 units. As for the MARGINAL value of EQU obj, which is the Lagrange multiplier of the objective function, it is always unity and cannot be interpreted.

When we formulate the model as an optimization problem, the output file reports meaningful solutions of the Lagrange multipliers. Therefore, it is convenient to declare the names for the Lagrange multipliers in an economically meaningful manner. In fact, phi is named the ‘marginal utility of income’ in line 31.

Appendix IV: Reformulation of a System of Simultaneous Equations into an Optimization Problem

In the main text, we reformulated the original model formulated as a system of simultaneous equations into an optimization problem with a fictitious objective function and all the equations of the original model system as its constraints. In this appendix, an attempt is made to justify this transformation, using a simple system of simultaneous equations. Suppose that this system consists of two linear equations:

$$\sum_i a_{ij}X_j = b_j \quad i, j = 1, 2$$

The values satisfying each of these equations can be expressed in one line, making two lines in total, shown in Figure IV.1. The solution of this system is located at the intersection of these two lines (X_1^*, X_2^*).

Instead of solving this system in the manner stated above, we alternatively set up an optimization problem as follows:

$$\underset{X_1, X_2}{\text{maximize}} \quad Y(X_1, X_2)$$

subject to

$$\sum_i a_{ij}X_j = b_j \quad \forall j$$

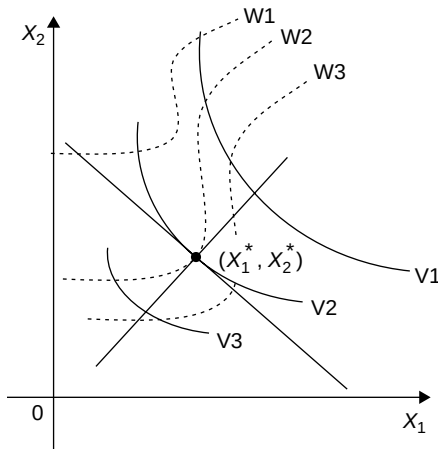


Figure IV.1 System of simultaneous equations and objective function

Here, we assume that the original simultaneous equations system has a unique solution. Because of this assumption, the feasibility set of this optimization problem is given at the unique point (X_1^*, X_2^*) , which is determined solely by the given constraints.¹ As this feasible point is unique, whatever objective function $Y(\cdot)$ (indicated by, for example, v_1, v_2 and v_3 ; or w_1, w_2 and w_3) we may use, or whether we maximize or minimize it, the solution cannot be other than the feasible point (X_1^*, X_2^*) . This justifies our reformulation of the CGE model system.

Note

1. When the feasibility set is not a unique point, the model can have multiple equilibria. See Mercenier (1995) for a discussion of multiple equilibria in CGE models.

Appendix V: Leontief-type Function and Optimization

In the main text, we have demonstrated an intuitive derivation of the demand functions as a solution of the optimization behaviour of firms with a Leontief-type production function. Here, we present their derivation in a graphical manner. In this derivation, we use a cost-minimization problem instead of a profit-maximization problem. For mathematical convenience, we assume that the firms use only two kinds of intermediate input $i, j = BRD, MLK$. The j -th firm cost minimization problem is:¹

$$\underset{X_{i,j}}{\text{minimize}} \quad C_j = \sum_i p_i^q X_{i,j} \tag{V.1}$$

subject to

$$\tilde{Z}_j = \min \left(\frac{X_{BRD,j}}{aX_{BRD,j}}, \frac{X_{MLK,j}}{aX_{MLK,j}} \right) \tag{V.2}$$

In this cost-minimization problem, the target amount of production $\tilde{Z}_j$ is exogenous. The solution of this problem can be graphically shown in Figure V.1. First, the isocost lines corresponding to the objective function (V.1) appear as downward-sloping lines in Figure V.1. The isocost lines located in the northeast represent higher costs than those located in the southwest, given the relative price of the goods. Second, the isoquant curve corresponding to the output level $\tilde{Z}_j$ under the Leontief-type production technology (V.2) is a right-angled curve.

The firm is supposed to choose an input bundle so as to minimize the costs; thus, the corner of the isoquant curve that touches the isocost line representing

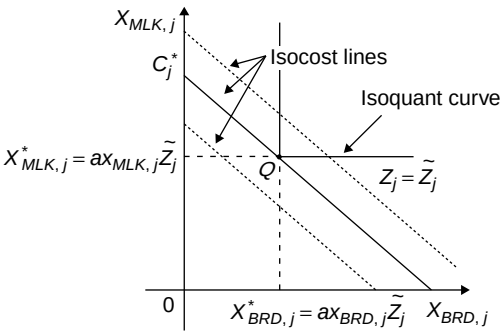


Figure V.1 Isoquants of Leontief-type production function and cost function

the lowest cost will be chosen. In Figure V.1, the cost-minimizing input bundle to attain the target output level $\tilde{Z}_j$ is represented by Q. Then:

$$X_{BRD,j}^* = ax_{BRD,j} \tilde{Z}_j$$

$$X_{MLK,j}^* = ax_{MLK,j} \tilde{Z}_j$$

As the optimum input bundle Q is located at the corner of the isoquant curve, the firm's optimal choice is always Q irrespective of the relative input prices.

Note

1. See Section 6.2 for an explanation of the $\min(\cdot)$ function in the constraint.

Annex A: Advanced Uses of GAMS

We have provided explanations of the uses of GAMS for CGE modelling, related to model building and solution interpretation, particularly in Chapter 3. Readers are expected to be familiar with the basic uses of GAMS. This annex presents advanced uses of GAMS for writing programs more efficiently and to avoid errors.

A.1 Set

A.1.1 Sequence in a set

To define a set $i = \{1, 2, 3, \dots, 10\}$, we usually write the following code:

```
Set    i      man      /1,2,3,4,5,6,7,8,9,10/;
```

However, for programming convenience, we can write this as follows:

```
Set    i      man      /1*10/;
```

A sequence can be attached also to a string such as $j = \{\text{Firm1}, \text{Firm2}, \dots, \text{Firm10}\}$:

```
Set    j      firm      /Firm1*Firm10/;
```

A.1.2 Alias of a set

Suppose the set $i = \{1, 2, 3, \dots, 10\}$ is already defined. We can define a set with the same elements $j = \{1, 2, 3, \dots, 10\}$ as follows:

```
Set    j      man      /1,2,3,4,5,6,7,8,9,10/;
```

However, to simplify the input file, we can write this as follows:

```
Alias(i, j);
```

In the latter case, we can use 'i' and 'j' interchangeably. Therefore, when we put the code in some subsequent part:

```
Parameter x(i);
```

GAMS recognizes that $x(i)$ and $x(j)$ are the same.

Although Alias is optional in general, there is a situation that requires it. Alias is used to avoid problems because of the overlapped use of the same index inside and outside of `sum(...)` or `prod(...)` in one equation. As explained in Subsection 5.4.5, the following is not allowed:

```
alpha(i) = x0(i) / sum(i, x0(i));
```

This must be replaced by the following:

```
alpha(i) = x0(i) / sum(j, x0(j));
```

A.1.3 Subset

We can define a subset $h = \{1, 2, 3\}$ of the set $i = \{1, 2, 3, \dots, 10\}$ as follows:

```
Set h(i) person /1,2,3/;
```

As the above statement shows, the suffix (i) attached to h indicates that the subset h belongs to the set i. We can also define a subset of the subset.

A.2 Setting values with formula

First, to set values for constants, we demonstrated how to declare their symbols with the `Scalar` or `Parameter` directive and to assign their values in a single statement in Subsection 3.3.2. Alternatively, we can achieve this using two separate statements as follows:

```
Parameter      x0(i);
x0(i)=2;
```

The `Scalar` directive can be used in the same manner. The `Table` directive, however, cannot be used in this manner, such as:

```
Table      z0(i,j);
z0(i,j)=3;
```

because the `Table` directive expects data input only in a matrix format.

Second, when we assign a solved value, e.g., the `LEVEL` solution of an endogenous variable 'y(i)' to a constant 'x0(i)', we write the following:

```
x0(i)= y.l(i);
```

To indicate the `LEVEL` value, we attach the '.l' suffix (letter 'l', not the numeral '1') to the symbol of the endogenous variable. Note the order of '.l' and '(i)'; '(i)' must be placed after '.l', although the endogenous variable is 'y(i)'. As for the other use of '.l' setting a certain value of an endogenous variable to initialize computation, see Subsection 5.4.7.

Third, when we use values of constants expressed in formulae, we should be careful not to conduct division by zero. Suppose, for example, to define 'dXp(i)' to compute a percentage change of an endogenous variable 'Xp(i)' from its base value 'Xp0(i)', we write the following code as suggested in Section 8.2:

```
dXp(i)=(Xp.l(i)/Xp0(i) - 1)*100;
```

However, this can cause an error in computation (not in syntax) because the constant vector 'Xp0(i)' sometimes has a zero element.

To avoid this, we should exclude cases where elements of 'Xp0(i)' equal zero using a conditional expression such as '\$(Xp0(i) ne 0)' (Table A.1). Further

Table A.1 GAMS syntax for conditional expressions

Mathematical Expression	GAMS Syntax
If $x_i^0 \neq 0$	$\$(x0(i) \text{ ne } 0)$, (or just: $\$x0(i)$)
If $x_i^0 = 0$	$\$(x0(i) \text{ eq } 0)$
If $x_i^0 \geq 0$, if $x_i^0 \leq 0$	$\$(x0(i) \text{ ge } 0)$, $\$(x0(i) \text{ le } 0)$
If $x_i^0 > 0$, if $x_i^0 < 0$	$\$(x0(i) \text{ gt } 0)$, $\$(x0(i) \text{ lt } 0)$

care must be taken in using the conditional expression. The following cannot resolve the problem of division by zero:

```
dXp(i)=(Xp.l(i)/Xp0(i) -1)*100$(Xp0(i) ne 0);
```

because, in the above statement, the conditional expression only reaches the term ‘*100’, which is located just before it. To avoid division by zero, we have to enclose, in parentheses, all the terms to be excluded as follows:

```
dXp(i)=((Xp.l(i)/Xp0(i) -1)*100)$(Xp0(i) ne 0);
```

We can also use the conditional expression in the equations; but no endogenous variables can appear in the conditions.

A.3 Large Table data input

When we put a large Table in an input file, we sometimes have to break it into several submatrices as follows:

Table	SAM(u,v)		Social Accounting Matrix		
	BRD	MLK	CAP	LAB	IDT
BRD	21	8			
MLK	17	9			
CAP	20	30			
LAB	15	25			
IDT	5	4			
TRF	1	2			
HOH			50	40	
GOV					9
INV					
EXT	13	11			
+	TRF	HOH	GOV	INV	EXT
BRD		20	19	16	8
MLK		30	14	15	4
CAP					
LAB					
IDT					

```

TRF
HOH
GOV      3      23
INV      17      2      12
EXT
;

```

It is important to put a '+' symbol at the northwest corner of each subsequent submatrix to indicate that it is the continuing submatrix, and to type the column and the row labels again. (The column labels naturally differ from those of the preceding submatrices.) Furthermore, as in this example, some rows (e.g., the 'CAP' row in the second submatrix) contain no data. We can omit such rows (or columns).

A.4 Output file

A.4.1 Printing variables and constants

While GAMS output files carry many kinds of information about a model and its solutions, they may not be sufficient for us. The GAMS reports the solutions (i.e., solved values of endogenous variables) of a model only to three decimal places as in List 3.2. Values of constants (i.e., exogenous variables and coefficients) are not printed automatically. If we need their print out to verify whether or not the calibration is conducted accurately, we use the `Display` and `Option` directives to print them out.

To print the coefficient, for example, `alpha(i)` in an output file, we write the following in an input file:

```
Display alpha;
```

While the `LEVEL` value of the endogenous variable of, for example, consumption `x(i)` is automatically printed in the `SOLVE SUMMARY` to three decimal places, we may want it printed with greater accuracy. To set the number of decimal places, we place the statement '`Option decimals=n;`' before the `Display` statement. (Note that in this statement, *n* is an integer between 0 and 8 while its default value is 3.) For example, we write the following:

```
Option decimals=8;
Display x.l;
```

after the `Solve` statement. Similarly, to print the marginal utility of income `phi`, i.e., the `MARGINAL` value of the Lagrange multiplier, we use '`.m`' in place of '`.l`':

```
Display phi.m;
```

In all cases of the `Display` directive, we must not attach any set suffix to the symbols even if they have one. That is, the following code causes a syntax error:

```
Display x.l(i);
```

A.4.2 Suppressing output

Although output files contain varied information related to the model and its solutions, many of them are not very important (as long as the model is working properly). By suppressing these currently useless parts of output files, we can avoid printing unnecessary pages and save disk space. Typically, we cut most of the information appearing after the echo print of the original input file and before the SOLVE SUMMARY in the output file, by stating the following:

```
$offsymxref offsymlist
Option limcol=0, limrow=0;
```

The details of SOLVE SUMMARY can be turned on (or off) by the following:

```
Option solprint=on (or off);
```

A.5 Communicating with spreadsheet software

In dealing with a large-scale model, we face difficulty in comprehending the large number of solved or calibrated values of variables, constants and other information in an output file. Similarly, it is cumbersome to transfer data in other files into an input file manually. To mitigate these problems, GAMS has been equipped with the GAMS Data Exchange (GDX) facilities and utilities, which help the GAMS system to communicate with spreadsheet software such as Excel and other database systems. By using the GDX facilities and utilities, we can transfer data in an input file or computed by the GAMS system into a separate file in the GDX format, and then we can unload these data from the GDX file into a spreadsheet file. We can also read data from a spreadsheet file through a GDX file and transfer these data from the GDX file for GAMS computation (Figure A.1). Here we demonstrate these processes with a few simple examples; for more elaborate uses of the GDX facilities, refer to the GDX manual by GAMS Development Corporation (2009).

In the following, by referring to the model (stdcge.gms in List 6.1) in Chapter 6, first we show how to transfer the values of variables and other information stored in an input file, or those generated by GAMS computation, into a spreadsheet. Second, we explain how to transfer data in a spreadsheet into an input file.

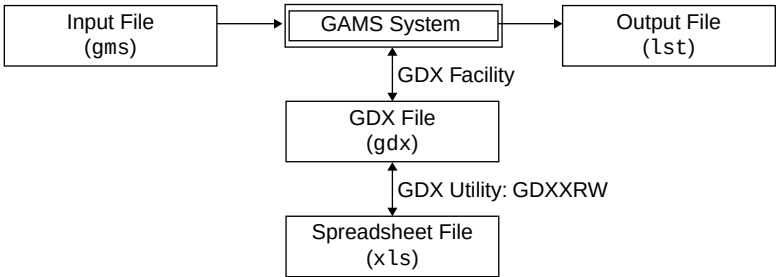


Figure A.1 Data exchange between files

A.5.1 Transfer of data in an input file into a spreadsheet

- (1) To store *all* the symbols and values used in an input file in a GDX file named, say, `output.gdx`, attach the following code to the end of `stdcge.gms`:

```
execute_unload "output.gdx";
```

In the above code, `output.gdx` indicates the name of the target GDX file, where data are stored. Instead of `output.gdx`, we can name the GDX file whatever we like. By omitting symbol names (i.e., endogenous variable and constant names, etc.), all symbols will be written to the GDX file. We can freely browse the data in this GDX file using GAMS IDE (Figure A.2). Using standard Windows operations, we can also copy the data from the GAMS IDE window and paste them into any other files such as spreadsheets.

While the code above inserts all the symbols and their values in an input file into the GDX file, the following code inserts only the values of `SAM(u, v)` (Figure A.3):

```
execute_unload "output.gdx", SAM;
```

The last part of the above statement (i.e., `SAM`) indicates the names of sets, constants, variables, etc. to be inserted into the GDX file.

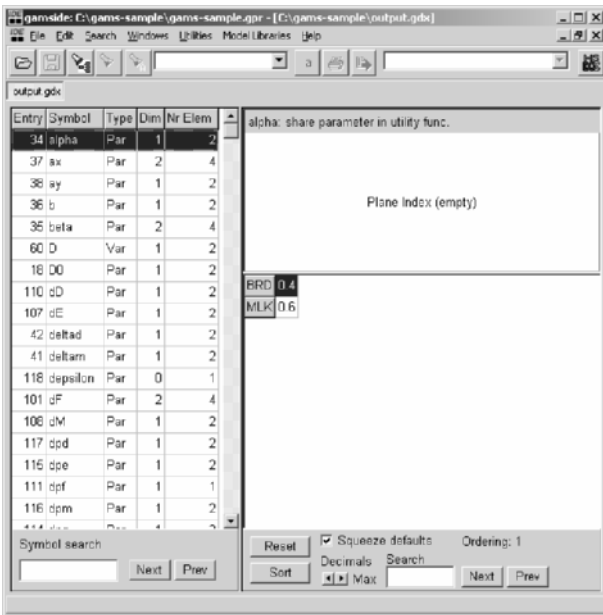


Figure A.2 A GDX file on the GAMS IDE



Figure A.3 Social Accounting Matrix data in a GDX file

- (2) The social accounting matrix (SAM) in Figure A.3, generated by the above statement, is not easy to read; we can rearrange the placement of the columns and rows of the SAM with our usual mouse operation by dragging the column and row labels (Figure A.4). The GAMS IDE documentation provides more details for changing the layout in the GDX viewer.
- (3) Finally, using the GDXXRW utility, we can create a spreadsheet file that stores data kept in the GDX file. Suppose we want to put the values of $SAM(u, v)$ kept in the GDX file into a spreadsheet file, say, `sam.xls`. After the code above, namely:

```
execute_unload "output.gdx", SAM;  
  
insert the following line:  
  
execute "gdxxrw output.gdx output=sam.xls par=sam";
```

Then, a spreadsheet file containing the SAM, `sam.xls`, is generated (Figure A.5). In the above code, ‘execute’ is the GAMS command to run the subsequent program enclosed by the double quotation marks. In other words, the GDXXRW utility is launched to open `output.gdx`, read the parameter values of $SAM(u, v)$ and generate a spreadsheet file `sam.xls`.

The values of variables and others generated by GAMS computation can be transferred into a spreadsheet in a similar manner to that stated above.

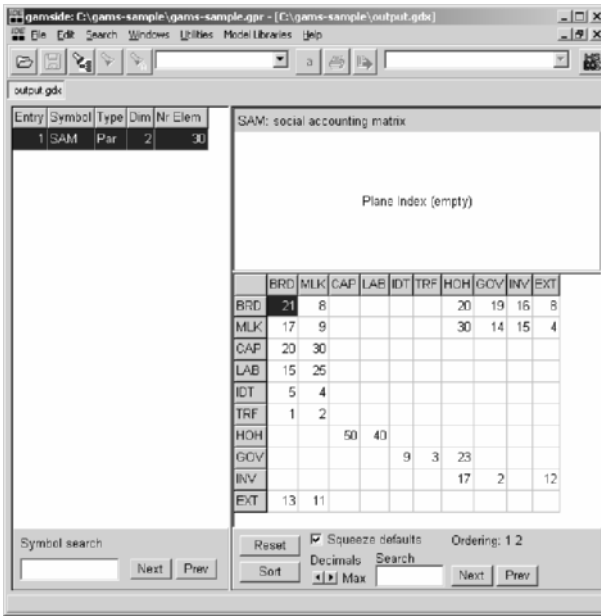


Figure A.4 A rearranged SAM in a GDX file

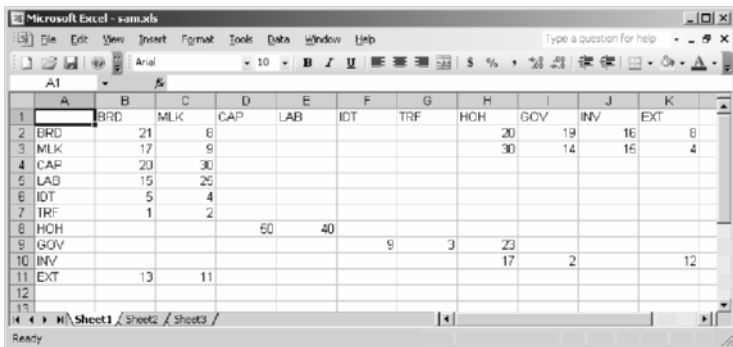


Figure A.5 A SAM in Excel generated by the GDXRW utility from a GDX file (sam.xls)¹

A.5.2 Transfer of data from a spreadsheet into a GDX file

Now we explain how to transfer data from a spreadsheet file into a GDX file and then read this file for GAMS computation. The process explained below is convenient, particularly when the SAM is large and/or when a dataset is shared by several modellers and users:

- (1) Given the SAM in `sam.xls`, we convert it into a GDX file, say, `sam.gdx`, using the following code in a *new* input file. Note that we should name the new file other than `stgcge.gms`, so as not to destroy the original input file.

```
execute "gdxrw sam.xls output=sam.gdx par=sam
rng=a1:k11 cdim=1 rdim=1";
```

(Note that the above code should be put in one line, even though it is shown as two lines here, due to page width limitations.)

In the code above, 'output', 'par', 'rng', 'cdim' and 'rdim' indicate the output filename, the name of parameters to be transferred into the GDX file, the range in the worksheet, and column and row dimensions of the data labels respectively.

- (2) Now that the SAM data are stored in a GDX file (`sam.gdx`), we can replace lines 12–36 in List 6.1 with the following:

```
Parameter SAM(u,v) social accounting matrix;

$gdxin sam.gdx
$loaddc sam
$gdxin
```

In the code above, '\$gdxin sam.gdx' initiates the retrieval of the SAM data from the GDX file `sam.gdx`. This is followed by '\$loaddc', which specifies the name of the parameters to be loaded from the GDX file; in this case, it is 'sam'. Finally, '\$gdxin' closes the file opened by the above code '\$gdxin sam.gdx'. Note that '\$gdxin' appearing in the last line is *not* followed by any filename.

In this manner, the spreadsheet data are transferred into a GDX file. Alternatively, we can generate a comma-separated values (CSV) file to show output in Excel directly through the put writing facility of GAMS. An example is shown in and after line 383 of `stdcge.gms`. Please refer to Chapter 15 of *GAMS – A User's Guide* for details of its syntax.

Note

1. This and earlier Microsoft product screen shots are reprinted with permission from Microsoft Corporation.

Annex B: How to Cope with Errors and Infeasibilities

In Chapter 3, we demonstrated how to use GAMS to solve CGE models and how to interpret their output files, presuming that the input files are correctly prepared and that the models are solved without any errors. However, in actual GAMS programming, we often encounter errors and/or fail to get reasonable results. Efficient programming requires a good understanding of errors and other problems printed in output files. Based on our long experience of working with CGE models using GAMS, here we provide tips for efficient GAMS programming and debugging.

Errors occur for several reasons. First, we have to identify the causes of errors; then, eliminate the cause of the errors in the input files. Figure B.1 helps us to identify the reasons for errors. There are six cases. Except for Case 6, where we successfully obtain a solution, all the cases indicate something erroneous in the input files. In the following section, we suggest possible causes of the errors and measures to fix them for Cases 1–5. Note that errors do not necessarily have only one cause as suggested in Table B.1.

B.1 Case 1: Compilation error

Compilation errors are caused by bad syntax in input files. We modify the input file in List 5.1 to create errors intentionally and to demonstrate how to fix these errors. List B.1 is part of the output file generated by GAMS with the erroneous input file. In the output file of List B.1, the following applies:

- (1) The symbol '****' appears just after the error line in the echo print of the original input file.
- (2) The line with the symbol '****' is followed by an error code such as '\$number', which indicates the type of syntax error. The position of the error code also indicates that the part of the statement just above it has something to do with the cause of the error.
- (3) After the echo print of the original input file, there are lines explaining the meaning of the error code cited above and providing hints for fixing them.

More detailed explanation is given using List B.1, in which two types of errors are shown. As for the first type, the error in line 67 is caused by the syntax error not in this line but by the missing semicolon ';' at the end of the preceding line 66. Without the semicolon ';', GAMS misinterprets line 67 beginning with 'eqpz(j)' as a continuing part of the statement in line 66. Thus, the error message after the echo print says that the 'eqpz(j)' is a '409 Unrecognizable item'. As for the second type, the error in line 73 is caused by the symbol 'ob'. While the symbol 'obj' is declared in line 63 as an equation name, the symbol 'ob' is erroneously typed in line 73. As there is no declaration of the symbol 'ob' in the input file, the error message says, '140 Unknown symbol'.

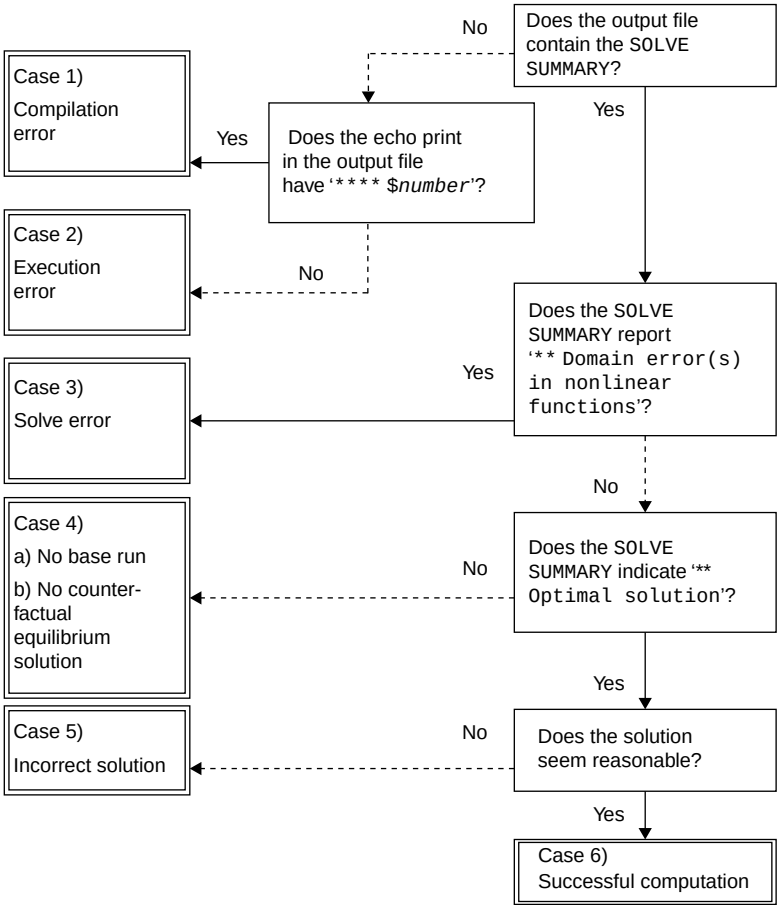


Figure B.1 Error and solution chart

Note: The messages suggested above may differ when using solvers other than CONOPT.

Table B.2 provides a summary of typical errors and their possible causes, immediately after the echo print in the output file. The numbers in Table B.2 indicate the error codes used in GAMS.

We often encounter many syntax and careless errors, particularly the first time the program is run. Most of these are caused by errors in an earlier part of the input file. For example, when we forget to declare the symbol of a coefficient, all subsequent statements that have something to do with that coefficient will produce error messages. Once the symbol is declared, the errors in the succeeding part disappear automatically. A large number of error messages with '****' in the output file should not be a cause for concern. Fix the errors one by one with patience, starting from the first part of the input file.

Reason		Case			
		Syntax error	Undefined mathematical operation	Inconsistency between calibration and model	Erroneous Misspecification of equations
Compilation error	Case 1	✓			
Execution error	Case 2	✓			
Solve error	Case 3		✓		
No base run equilibrium solution	Case 4a)		✓	✓	
No counterfactual equilibrium solution	Case 4b)			✓	
Incorrect solution	Case 5)			✓	✓

Note: Case 6 in Figure B.1 is a successful case that does not suffer from errors or any other computational problems. This case is not shown in this table.

List B.1 Example of errors indicated in an output file

```

56 Equation          eqX(i)          household demand function
57                  eqpz(i)          production function
58                  eqF(h,j)         factor demand function
59                  eqpx(i)          good market clearing condition
60                  eqpf(h)          factor market clearing condition
61                  eqZ(i)
62
63                  obj              utility function [fictitious]
64
65
66 eqX(i) ..          X(i)          =e= alpha(i)*sum(h, pf(h)*FF(h))/px(i)
67 eqpz(j) ..         Z(j)          =e= b(j)*prod(h, F(h,j)**beta(h,j));
**** $409
68 eqF(h,j) ..        F(h,j)        =e= beta(h,
69 eqpx(i) ..          X(i)          =e= Z(i);
70 eqpf(h) ..          sum(i, F(h,j)) =e= FF(h);
71 eqZ(i) ..           pz(i)         =e= pz(i);
72
73 ob ..              UU            =e= prod(i, X(i)**alpha(i));
**** $140
...
93 pf.fx("LAB")=1;
94
95 * Defining and solving the model -----
96 Model sploge /all/;
97 Solve sploge maximizing UU using nlp;
**** $257
98 * -----
99 * end of model -----
100 * -----
GAMS Rev 136 MS Windows
A Simple CGE Model in Ch. 5 (Solve Summary) 7:32:56 Page 2
Error Messages

140 Unknown symbol
257 Solve statement not checked because of previous errors
409 Unrecognizable item - skip to find a new statement
    looking for a ';' or a key word to get started again

**** 3 ERROR(S)    0 WARNING(S)

```

(1) Indicating lines with errors

(2) Indicating the place of erroneous parts with error codes

(3) Error codes and error message

B.2 Case 2: Execution error and Case 3: Solve error

The 'execution errors' and 'solve errors' are caused by illegal operations (e.g., division by zero) before and during processing the Solve statement of the model, respectively. The former case is indicated by an error message such as:

```
**** Exec Error at line 71: division by zero (0)
```

which indicates an illegal operation because of division by zero in line 71. The latter case is indicated by an error message in the SOLVE SUMMARY block as shown in List B.2.

The bottom part of List B.2 indicates illegal operations in the equations eqps(BRD) and obj and their possible causes. In this case, equations eqps

Table B.2 Typical syntax errors

7	'(' expected A left parenthesis is missing.
8	)' expected A right parenthesis is missing.
10	',' expected A comma “,” is missing.
37	'=l=' or '=e=' or '=g=' operator expected In the specification of equations, an operator either '=l=', '=e=' or '=g=' is missing. Sometimes, it indicates a typo with '=' instead of '=e='.
66	The symbol shown has not been defined or assigned A wild shot: You may have spurious commas in the explanatory text of a declaration. Check symbol reference list. While we declared a symbol (typically, indices and constants), we might have forgotten to assign a value to the symbol used in equations.
120	Unknown identifier entered as set The index used in a symbol is not defined.
125	Set is under control already The index used in the sum and prod functions overlaps the index outside these functions.
140	Unknown symbol The symbol is not defined.
141	Symbol neither initialized nor assigned A wild shot: You may have spurious commas in the explanatory text of a declaration. Check symbol reference list. While having been declared, a symbol (typically, indices and constants) is not provided with any values.
148	Dimension different – The symbol is referenced with more/less Indices as declared The number of indices used with a symbol is different from that in the declaration. For example, while declaring 'Parameter ax(i,j);', we use the constant 'ax(i)'.
149	Uncontrolled set entered as constant (1) The index used in the equation is not consistent with that of the equation name. For example, eqpx(h).. X(j) =e= Z(j); (2) The use of the index is inconsistent; the index in the equation name does not match that in the equation. For example, FF(h)=Z0(j);
171	Domain violation for set The index used in a symbol is different from that in declaration. For example, while declaring 'Variable Z(j);', it appears as 'eqpx(j).. X(j) =e= Z(h);'.

Table B.2 (Continued)

184	Domain list redefined
	Symbols are defined again with a different index domain that is already defined.
246	Objective variable is not a free variable
	The symbol of the objective variable must be declared with the Variable directive, not with the Positive Variable or Negative Variable directive.
257	Solve statement not checked because of previous errors
	Because of an error identified in a previous line, the model is not solved. Note that once all the bugs above this line are fixed, this error message disappears automatically.
408	Too many),} or]
	There are too many right parentheses or brackets.
409	Unrecognizable item – skip to find a new statement
	Looking for a ';' or a key word to get started again
	The statement cannot be recognized as a valid statement. This is typically caused by a missing ';' in the previous statement.

and obj suffer from a negative power value. There are several causes of illegal operations as follows:

- (a) division by zero
- (b) negative base in a logarithm
- (c) illegal operation in a power (like the square root of a negative number)
- (d) a number that is too large

Both constants and endogenous variables can cause these illegal operations during computation. In the iteration process to solve the model, the values of endogenous variables oscillate and may produce illegal operations. We have to cope with such illegal operations not only at the solutions but also during the computation process. We suggest the following four ways to avoid such illegal operations.

Setting upper and lower bounds on endogenous variables

Using the ' . up' and ' . lo' suffixes, we can set upper and lower bounds on endogenous variables to limit their domain and to avoid illegal operations. For example, to avoid division by zero during computation, we should set a lower bound at some slightly positive level for certain endogenous variables to exclude zero from their domain.

Initializing endogenous variables

We can set appropriate starting points of computation for endogenous variables. In CGE modelling, there is at least one trivial equilibrium – the initial equilibrium,

List B.2 Solve error

S O L V E		S U M M A R Y	
MODEL	Chapter02	OBJECTIVE	UU
TYPE	NLP	DIRECTION	MAXIMIZE
SOLVER	CONOPT	FROM LINE	102
**** SOLVER STATUS	5 EVALUATION ERROR LIMIT		
**** MODEL STATUS	6 INTERMEDIATE INFEASIBLE		
**** OBJECTIVE VALUE	0.0000		
RESOURCE USAGE, LIMIT	0.016	1000.000	
ITERATION COUNT, LIMIT	2	10000	
EVALUATION ERRORS	2	0	
C O N O P T 3	x86/MS Windows version 3.14S-017-061		
Copyright (C)	ARKI Consulting and Development A/S		
	Bagsvaerdvej 246 A		
	DK-2880 Bagsvaerd, Denmark		
Using default options.			
** Domain error(s) in nonlinear functions.			
Check bounds on variables.			
CONOPT time Total	0.000 seconds		
of which: Function evaluations	0.000 = 0.0%		
1st Derivative evaluations	0.000 = 0.0%		
Workspace	=	0.06 Mbytes	
Estimate	=	0.06 Mbytes	
Max used	=	0.02 Mbytes	
**** ERRORS/WARNINGS IN EQUATION eqps(BRD)			
1 error(s): vcPower: FUNC DOMAIN: x**c, x < 0			
**** ERRORS/WARNINGS IN EQUATION obj			
1 error(s): vcPower: FUNC DOMAIN: x**c, x < 0			

indicated in the social accounting matrix (SAM). Even the counterfactual equilibrium, which is created by applying shocks such as policy changes to constants in the model, is expected to be located somewhere 'close' to the initial equilibrium. Thus, we often use the initial equilibrium values to initialize the model with the '. l' suffix. (Note that this suffix is the letter 'l', not the numeral '1'.)

Reformulating equations

Recall the household demand function appearing in line 36 of List 3.1, where an endogenous variable $px(i)$ appears in the denominator of the right-hand side. If division by zero is caused by this term, we can reformulate the equation to avoid division by this endogenous variable. Therefore, instead of:

```
eqX(i).. X(i) =e= alpha(i)*sum(h, pf(h)*FF(h))/px(i);
```

use the following:

```
eqX(i).. px(i)*X(i) =e= alpha(i)*sum(h, pf(h)*FF(h));
```

Rescaling values

Because of the upper limit on the number of digits GAMS can handle, we may encounter errors associated with large values. If data in the SAM (defined as $SAM(u, v)$ in the input file) seem too large to compute, after installing the SAM we can rescale its values (for example, to one-thousandth of the original values) as follows:

```
SAM(u, v) = SAM(u, v)/1000;
```

B.3 Case 4a: No base run equilibrium solution

Our typical process to solve a CGE model is as follows. First, we obtain the initial equilibrium by retrieving a given SAM. Then, we build and solve a CGE model with coefficients and exogenous variables, calibrated on the basis of the SAM, and obtain the base run equilibrium solution. Therefore, in theory, the base run equilibrium solution must match the initial equilibrium. However, sometimes the former does not match the latter or cannot be obtained at all. Possible reasons are mistakes in the calibration process or in the specification of equations. Another possible reason is the model size exceeding the GAMS capacity of solvability. While the latter problem is discussed in Section B.6, this section discusses the former problem.

If the base run equilibrium solution that matches the initial equilibrium is not obtained, we have to reexamine the calibration process and, if no mistake is found there, the specification of the model equations. A way to identify those mistakes is to solve the model fixing all the endogenous variables at the initial equilibrium. More precisely, we solve the model shown in, say, List 5.1, pinning down all the endogenous variables (except for the objective variable) at the initial equilibrium using the *‘.fx’* suffix.¹

```
X.fx(i)      =X0(i);
F.fx(h, j)   =F0(h, j);
Z.fx(j)      =Z0(j);
px.fx(i)     =1;
pz.fx(i)     =1;
pf.fx(h)     =1;
```

If some equations contain something erroneous, they cannot satisfy the equality constraints of the model equations. In GAMS output files, they are marked with 'INFES' in the output file:

```
---- EQU eqX household demand function
```

	LOWER	LEVEL	UPPER	MARGINAL	
BRD	.	-0.600	.	EPS	INFES
MLK	.	-1.400	.	EPS	INFES

Then, it is necessary to remove bugs in, or related to, these equations so that we can obtain, at least, the base run equilibrium solution; but we may encounter the problem discussed in the next section.

B.4 Case 4b: No counterfactual equilibrium solution

When we succeed in computing a base run equilibrium that reproduces the initial equilibrium but fails to compute a counterfactual one, there are two main possible reasons. One is a mistake in the specification of equations that cannot be identified by the method mentioned in the previous section; the other is a mistake in the definition of the domains of the endogenous variables.

As for the mistakes in specification, they often occur because of typos that are associated with price variables. Consider the equation in lines 235–236 of List 6.1:

```
eqXp(i).. Xp(i) =e= alpha(i)*(sum(h, pf(h)*FF(h))
                    -Sp -Td)/pq(i);
```

Even if this is typed mistakenly as follows:

```
eqXp(i).. Xp(i) =e= alpha(i)*(sum(h, pf(h)*FF(h))
                    -Sp -Td)*pq(i);
```

no problem occurs in computation of the base run equilibrium, because all the solved prices, including $pq(i)$, are unity. However, in computation of a counterfactual equilibrium, the prices may well diverge from unity.² In fact, because of this typo, the property of the demand function seems very odd from an economics viewpoint – a price rise would lead to an increase in demand for this good. This upward-sloping demand function can prevent the price mechanism from having its excess demand converging toward zero.

Another example demonstrates the significance of typos using the same equation:

```
eqXp(i).. Xp(i) =e= alpha(i)*(sum(h, pf(h)*FF(h))
                    -Sp -Td)/pq(i);
```

Suppose that this equation is mistakenly typed as follows:

$$\text{eqXp}(i) \dots \text{Xp}(i) = \alpha(i) * (\text{sum}(h, \text{pf}(h) * \text{FF}(h)) - \text{Sp} - \text{Td}) / \text{py}(i);$$

In the base run equilibrium, both endogenous variables ‘ $\text{pq}(i)$ ’ and ‘ $\text{py}(i)$ ’ are unity; therefore, we can get the base run equilibrium even with this erroneous equation. However, we often encounter infeasibility in computation of a counterfactual equilibrium with this mistaken equation.

The second reason why we cannot obtain a counterfactual equilibrium solution is a mistake in the definition of the domains of the endogenous variables. When we set lower and upper bounds of endogenous variables using ‘. lo’, ‘. up’ or ‘. fx’, we must be careful not to exclude the expected solutions because of the lower and upper bounds. If we compute an equilibrium by setting such bounds and compute another equilibrium without changing the bounds, the solutions of some endogenous variables may be excluded from their domain in the latter computation; this leads to failure in computation or erroneous solution. For example, when we set a zero lower bound on an endogenous tax revenue variable and simulate a subsidy (i.e., a negative tax) through this tax variable, it is self-evident that we will encounter computational difficulties.

B.5 Case 5: Incorrect solution

When we find (seemingly) incorrect solutions even though the message ‘** Optimal Solution’ appears in the SOLVE SUMMARY, we should review the model for typos in equations and parameter settings and mistakes in the domain settings as discussed above. Then, if we cannot find anything erroneous there, all we have to do is to re-examine the model structure thoroughly from both economics and mathematics viewpoints. For example, if we erroneously assume values for the elasticity of substitution for the Armington’s constant elasticity of substitution (CES) functions that are too large or small, the solution may seem intuitively unacceptable while no error is reported in the GAMS output.

B.6 Capacity limitation of GAMS

If we still face computational difficulties after attempting to minimize the numerical computation load and to remedy erroneous or inconsistent economic and mathematical features of the model, we have to conclude that the model is too large or complicated to solve with GAMS. In this case, the output file will look like those of Cases 3–5. There is no almighty numerical computation software; we cannot completely resolve this problem, but you may try the following five techniques to work around the difficulty before considering numerical software other than GAMS.

Repeating the Solve statement

Simply repeat the same Solve statement several times. Even if GAMS aborts its computation in its first trial, GAMS tries to solve the model using the values of

endogenous variables found at the end of the previous round as their initial values in the next round. These values may provide better initial values for computation.

Reducing the assumed shock

If computation of a counterfactual equilibrium is found successful only with a small shock but not with a large one, we should solve it repeatedly by gradually increasing the magnitude of the shocks. An example of coding is provided in Section 8.1.

Using an option file

We can tune the behaviour of solvers using an option file (e.g., to increase the upper limit of iterations). Details are provided in Subsection B.3.31 in Appendix B of *GAMS – A User's Guide* and also in sections describing the options in the relevant solver manual.

Changing solvers

By using different solvers, we may be able to overcome computational difficulty. While we use CONOPT for our default computation setting in this book, we can alternatively use MINOS and other solvers for nonlinear programming problems. To change the solver used in our computation, reset the default solvers in GAMS IDE or insert the following statement before the `Solve` statement in the input file:³

```
Option nlp=solver-name;
```

Reducing the model size

If none of the above-mentioned techniques works, we have to reduce the model size. Typically, we can reduce the numbers of goods and agents in the model or, in the case of a dynamic analysis – which has not been dealt with in this book – the time horizon should be shortened.

Notes

1. However, note that this method is powerful but not perfect. There could be some other errors that are very hard to identify by this method. Remedies for these cases are discussed in Section B.4.
2. If a shock given to a constant is so small compared with the tolerance in the numerical computation, the model may yield a solution – though an erroneous one.
3. We can (re)set the default solvers in GAMS IDE using: File→Options→Solvers.

Annex C: Web Resources

(1) GAMS Development Corporation: www.gams.com

Provides a trial version of the GAMS system, capable of solving small models, while the full version of GAMS is commercially distributed software. The latest GAMS system can be downloaded from the following address:

www.gams.com/download

with installation notes (for PC) at the following address:

www.gams.com/dd/docs/gams/win-install.pdf

See the latest edition of *GAMS – A User's Guide* with the solver manuals for further details about GAMS, only a part of which has been explained in this book. The mailing lists for GAMS also help users who wish to exchange ideas about its highly technical uses and to be informed about the latest GAMS updates.

Our sample models developed in this book can be found at the GAMS Model Library:

www.gams.com/modlibs

Focusing on CGE modelling, the following web sites may be also informative.

(2) The Center for Global Trade Analysis, the Department of Agricultural Economics, Purdue University: www.gtap.org

Provides detailed information, technical notes, working papers, conference proceedings and other materials on the Global Trade Analysis Project (GTAP) Database and CGE models. It is a comprehensive global trade, production and consumption database. The current version 7 database covers 113 regions and 57 sectors. While its fully fledged database has to be purchased, its web site distributes several older-version datasets for public use.

(3) The Centre of Policy Studies, Monash University: www.monash.edu.au/policy

Provides substantial information on Australian CGE models developed by the Impact Project, as well as the GEMPACK software, which can also be used to run CGE models included in the GTAP database.

(4) The International Food Policy Research Institute (IFPRI): www.ifpri.org

Provides a CGE model for GAMS. SAMs for some developing countries are available upon request via the following address:

www.ifpri.org/datasets

The documentation for the IFPRI's CGE model by Löfgren et al. (2002) is also available in a PDF file at the following address:

www.ifpri.org/publication/standard-computable-general-equilibrium-cge-model-gams-0

(5) Professor Thomas F. Rutherford's web site: www.mpsge.org

Provides many useful sample models and utilities as well as notes for GAMS uses and CGE modelling. The MPSGE, which is also used for CGE modelling and works as a subsystem of GAMS, is explained along with sample models.

(6) The Statistics Bureau, the Ministry of Internal Affairs and Communications, the Government of Japan: www.stat.go.jp

Distributes Japan's statistical data, including input-output (IO) tables, and provides their information on their web site. The IO tables are available at the following address:

www.stat.go.jp/english/data/io

(7) The Economic and Social Research Institute, Cabinet Office, the Government of Japan: www.esri.go.jp

Another major source of economic data for Japan, particularly about its national account tables. They are available at the following address:

www.esri.cao.go.jp/en/sna

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